IGCSE 0607 Practice Questions

Total Questions: 663 — **Total Marks:** 3980 Marks

Table of Contents & Quick Navigation

- **E1.1** Types of number
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*
- **E1.2** Sets
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(2 Qs)*
- **E1.3** Powers and roots
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*
- **E1.4** Fractions, decimals and percentages
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*
- **E1.5** Ordering
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*
- **E1.6** The four operations
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*
- **E1.7** Indices I
  → **Paper 2 (Non-Calculator)** *(3 Qs)*
- **E1.8** Standard form
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*
- **E1.9** Estimation
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*
- **E1.10** Ratio and proportion
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*
- **E1.11** Rates
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*
- **E1.12** Percentages
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*
- **E1.13** Using a calculator
  → **Paper 4 (Calculator)** *(6 Qs)*
- **E1.14** Time
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*
- **E1.15** Money
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*
- **E1.16** Exponential growth and decay
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E1.17** Surds
  → **Paper 2 (Non-Calculator)** *(27 Qs)*

- **E2.1** Introduction to algebra
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E2.2** Algebraic manipulation
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E2.3** Algebraic fractions
  → **Paper 2 (Non-Calculator)** *(5 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E2.4** Indices II
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(1 Q)*

- **E2.5** Equations
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E2.6** Inequalities
  → **Paper 2 (Non-Calculator)** *(16 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

- **E2.7** Sequences
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E2.8** Proportion
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E3.1** Graphs of functions
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(15 Qs)*

- **E3.2** Sketching graphs on a calculator
  → **Paper 4 (Calculator)** *(6 Qs)*

- **E3.3** Functions
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E3.4** Finding a quadratic function using given information
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

- **E3.5** Asymptotes
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E3.6** Transforming graphs of functions
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E3.7** The logarithmic function
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(18 Qs)*

- **E4.1** Coordinates
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(1 Q)*

- **E4.2** Gradient of linear graphs
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E4.3** Length and midpoint
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E4.4** Equations of linear graphs
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E4.5** Parallel lines
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E4.6** Perpendicular lines
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E5.1** Geometrical terms
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E5.3** Similarity
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(15 Qs)*

- **E5.4** Symmetry
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E5.5** Angles
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E5.6** Circle theorems I
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

- **E5.7** Circle theorems II
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E6.1** Units of measure
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E6.2** Area and perimeter
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E6.3** Circles, arcs and sectors
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E6.4** Surface area and volume
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E6.5** Compound shapes and parts of shapes
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(16 Qs)*

- **E7.1** Pythagoras' theorem
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E7.2** Right-angled triangles
  → **Paper 4 (Calculator)** *(6 Qs)*

- **E7.3** Exact trigonometric values
  → **Paper 2 (Non-Calculator)** *(6 Qs)*

- **E7.4** Trigonometric functions
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E7.5** Non-right-angled triangles
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(26 Qs)*

- **E7.6** Pythagoras' theorem and trigonometry in 3D
  → **Paper 4 (Calculator)** *(3 Qs)*

- **E8.1** Transformations
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E8.2** Vectors in two dimensions
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

- **E8.3** Magnitude of a vector
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E9.1** Introduction to probability
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

- **E9.2** Relative and expected frequencies
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E9.3** Probability of combined events
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

- **E10.1** Classifying statistical data
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E10.2** Interpreting statistical data
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E10.3** Discrete and continuous data
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

- **E10.4** Averages and measures of spread
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E10.5** Averages on a calculator
  → **Paper 4 (Calculator)** *(6 Qs)*

- **E10.6** Statistical charts and diagrams
  → **Paper 2 (Non-Calculator)** *(2 Qs)* · **Paper 4 (Calculator)** *(4 Qs)*

- **E10.7** Scatter diagrams
  → **Paper 4 (Calculator)** *(6 Qs)*

- **E10.8** Cumulative frequency diagrams
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(5 Qs)*

1 **(a)** Write 84 as a product of its prime factors.

..... **[2]**

(b) Find the lowest common multiple (LCM) of 84 and 126.

..... **[2]**

2 Here is a list of numbers.

2 9 12 23 36 45 64 81

(a) Write down a number from the list which is:

(i) a multiple of 12,

..... **[1]**

(ii) a cube number,

..... **[1]**

(iii) both a prime number and a factor of 46.

..... **[1]**

(b) Find the lowest common multiple (LCM) of 36 and 64.

..... **[2]**

3 **(a)** Let x be a rational number and y be an irrational number.
State whether $x + y$ is rational or irrational.

..... **[1]**

(b) Give an example of an irrational number y to show that y^2 can be a rational number.

[1]

(c) The nth term of a sequence is given by the expression

$$n^2 + n + 41$$

(i) Work out the 3rd term of this sequence, and state whether it is a prime number.

..... **[2]**

(ii) Explain why the 41st term of this sequence is not a prime number.

..... **[2]**

4 Let $m = p^3q$ and $n = pq^3$, where p and q are distinct prime numbers.

(a) Express, in terms of p and q:

(i) the highest common factor (HCF) of m and n,

..... **[1]**

(ii) the lowest common multiple (LCM) of m and n.

..... **[1]**

(b) The HCF of m and n is 15.
It is given that $m > n$.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(c) Show algebraically that the product $m \times n$ is a square number.

[2]

(d) Using your values of p and q, find the smallest positive integer k such that $m \times k$ is a cube number.

$k =$ ..... **[2]**

5 Three numbers A, B and C are given as products of their prime factors.

$$A = 2^3 \times 3^2 \times 5 \qquad B = 2^2 \times 3^x \times 5^2 \qquad C = 2^y \times 3 \times 7$$

The highest common factor (HCF) of A and B is 180.
The lowest common multiple (LCM) of B and C is 12600.

Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

6 Number A and Number B are given as products of their prime factors.

$A = 2^2 \times 3^3 \times 5$

$B = 2^x \times 3 \times 7$

(a) The highest common factor (HCF) of A and B is 6.
Find the value of x.

$x =$ ..... **[2]**

(b) Number $C = 2^2 \times 3^y \times 5^2$.
The lowest common multiple (LCM) of A and C is 8100.
Find the value of y.

$y =$ ..... **[3]**

(c) Here is a list of numbers.

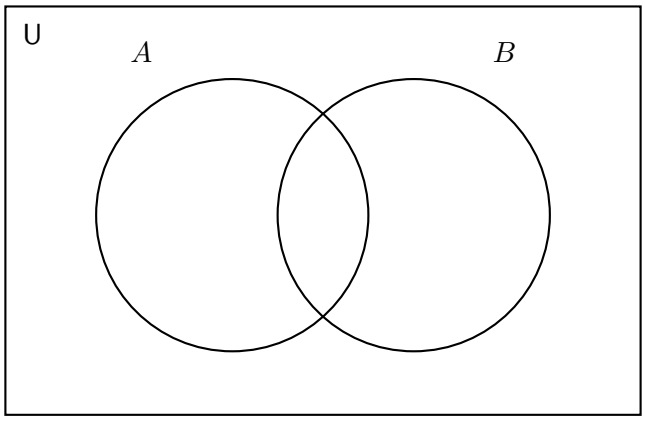
$$\sqrt{11} \qquad -3.5 \qquad 4\pi \qquad \frac{7}{9} \qquad \sqrt{144} \qquad 0.\dot{4}$$

Write down all the irrational numbers from this list.

..... **[2]**

7 Let U be the set of integers from 1 to 10 inclusive.
Let A be the set of prime numbers.
Let B be the set of odd numbers.

(a) Complete the Venn diagram to show the elements of these sets.



[2]

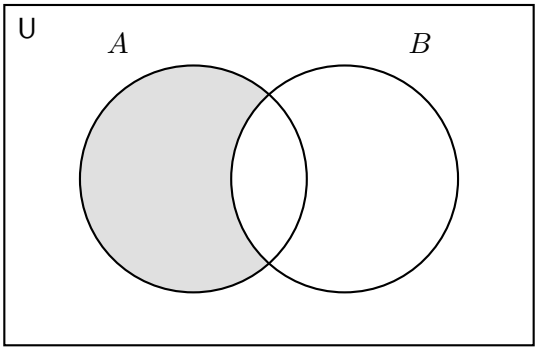
(b) List the elements of $A \cap B'$.

..... **[1]**

(c) Find $n(A \cup B)$.

..... **[1]**

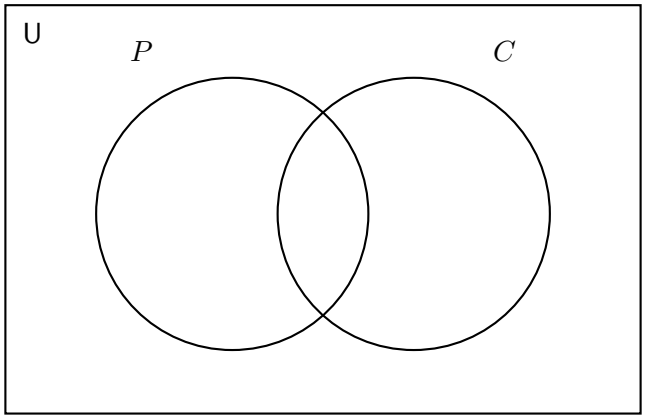
(d) Write down the set notation for the shaded region in the Venn diagram above.



..... **[1]**

8 There are 45 students in a class.
Let U be the set of students in this class.
Let P be the set of students who study Physics.
Let C be the set of students who study Chemistry.
$n(P) = 28$
$n(C) = 22$
$n(P \cap C) = x$
$n((P \cup C)') = 4$

(a) Complete the Venn diagram in terms of x.



[2]

(b) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[2]**

(c) Find $n(P \cap C')$.

..... **[1]**

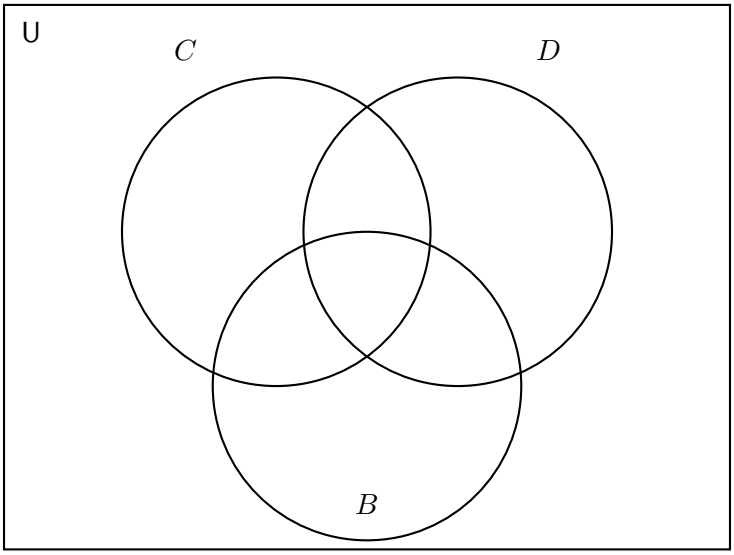
9 In a survey of 100 people, information was collected about the pets they own.

Let U be the set of people surveyed.
Let C be the set of people who own a cat.
Let D be the set of people who own a dog.
Let B be the set of people who own a bird.

- 8 people own all three pets.
- 12 people own a cat and a dog but not a bird.
- 7 people own a cat and a bird but not a dog.
- 10 people own a dog and a bird but not a cat.
- 3 people do not own any of these pets.

The number of people who own only a cat is x.
The number of people who own only a bird is also x.
The number of people who own only a dog is $2x$.

(a) Complete the Venn diagram to show this information.



[3]

(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find $n((C \cup D) \cap B')$.

..... **[2]**

10 **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

(b) Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

11 Solve the following equations.

(a)

$$\sqrt[3]{3x+4} = 4$$

$x =$ ..... **[3]**

(b)

$$x^2 = 4^3 + 6^2$$

$x =$ ..... , or $x =$ ..... **[2]**

(c)

$$\sqrt{x} + 2\sqrt{x} = 21$$

$x =$ ..... **[2]**

12 **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

(c) Find the largest integer n such that $n^4 < 2000$.

$n =$ ..... **[2]**

13 **(a)** Write $0.\dot{7}$ as a fraction.

..... **[1]**

(b) Write $0.\dot{3}\dot{6}$ as a fraction in its simplest form.

..... **[2]**

14 The length of a rectangle is increased by $33.\dot{3}\%$.
The width of the rectangle is decreased by $x\%$.
The overall area of the rectangle increases by $16.\dot{6}\%$.
Calculate the value of x.

$x =$ ..... **[5]**

15 In a school, $\frac{3}{8}$ of the students are in the Junior section, 45% are in the Middle section, and the rest are in the Senior section.
There are 140 students in the Senior section.

(a) Work out the total number of students in the school.

..... **[3]**

(b) The number of boys in the Junior section is 150% of the number of girls in the Junior section.
Work out the number of girls in the Junior section.

..... **[3]**

16 Write the following in order of size, starting with the smallest.

$$\frac{5}{8} \qquad 0.62 \qquad 61.5\% \qquad \frac{3}{5}$$

..... **[2]**

17 **(a)** Given that $0 < x < 1$, write the following expressions in order of size, starting with the smallest.

$$x \qquad x^2 \qquad \frac{1}{x} \qquad \sqrt{x}$$

..... **[2]**

(b) Given that $x < -1$, write the following expressions in order of size, starting with the largest.

$$x \qquad x^3 \qquad x^{-2} \qquad \frac{1}{x}$$

..... **[2]**

18 Write the following masses in order of size, starting with the largest.

5.2×10^{-4} kg 0.04 g 4.5×10^{3} mg 6×10^{-7} tonnes

..... **[3]**

19 **(a)** Work out the value of

$$15 - 3 \times (4 - (-2))$$

..... **[2]**

(b) Work out the value of

$$2\frac{1}{4} \div \frac{3}{8} - 1\frac{1}{2}$$

Give your answer in its simplest form.

..... **[3]**

20 Solve the equation, giving your answer as a fraction in its simplest form.

$$\frac{x+\frac{1}{3}}{2-0.4} = -1\frac{1}{4}$$

$x =$ ..... **[5]**

21 A water tank has a capacity of 85.5 litres.
It is initially completely empty.
A hose pumps water into the tank at a rate of $\frac{3}{4}$ of a litre per minute.
At the same time, a leak in the tank loses water at a rate of 0.45 litres per minute.

Work out how long it takes to completely fill the tank.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[4]**

22 **(a)** Work out the value of

$125^{\frac{1}{3}}$

..... **[1]**

(b) Work out the value of

6^{-2}

..... **[1]**

(c) Simplify.

$t^3 \times t^5$

..... **[1]**

23 **(a)** Find the exact value of

$16^{-\frac{3}{4}}$

..... **[2]**

(b) Simplify fully.

$(5x^3y^{-2})^2$

..... **[2]**

(c) Solve the equation.

$2^{3x-1} \times 4^x = 32$

$x =$ ..... **[3]**

24 **(a)** Simplify fully.

$$\left(\frac{16x^8}{y^4}\right)^{-\frac{1}{2}}$$

..... **[3]**

(b) Solve the equation.

$$4^x - 2^{2x+1} + 8 = 0$$

$x =$ ..... **[4]**

25 **(a)** Write 450000 in standard form.

..... **[1]**

(b) Write 3.08×10^{-4} as an ordinary number.

..... **[1]**

(c) Work out.

$(2 \times 10^3) \times (4 \times 10^5)(0)$

Give your answer in standard form.

..... **[1]**

26 Consider the following numerical values.

(a) Convert the following numbers as instructed.

(i) Write 0.00034 in standard form.

..... **[1]**

(ii) Write 5.2×10^4 as an ordinary number.

..... **[1]**

(b) Work out, giving your answer in standard form:

$(2 \times 10^3) \times (4 \times 10^5)$

..... **[2]**

27 The mass of the Earth is 5.97×10^{24} kg.
The mass of the Moon is 7.35×10^{22} kg.

(a) Calculate the difference between the mass of the Earth and the mass of the Moon.
Give your answer in standard form.

..... kg **[2]**

(b) The mass of the Sun is 1.99×10^{30} kg.
Find the value of k when

mass of the Sun = k ×mass of the Earth
Give your answer to 3 significant figures.

$k =$ ..... **[3]**

28 The mass of the Sun is 1.99×10^{30} kg.
The mass of Jupiter is 1.90×10^{27} kg.

(a) Calculate how many times greater the mass of the Sun is than the mass of Jupiter.
Give your answer in standard form correct to 3 significant figures.

..... **[3]**

(b) A comet has a mass of 4.5×10^{13} kg.
Calculate the mass of the comet in grams.
Give your answer in standard form.

..... g **[2]**

29 **(a)** Calculate, giving your answer in standard form.

$3.4 \times 10^7 + 2.1 \times 10^6$

..... **[2]**

(b) A computer can do a calculation in 4×10^{-9} seconds.
Work out how many of these calculations the computer can do in 5 minutes.
Give your answer in standard form.

..... **[3]**

30 A large cylindrical water tank has a base radius of 4.5×10^2 cm and a height of 1.2×10^3 cm.

(a) Calculate the volume of the water tank in cm^3.
Give your answer in standard form correct to 4 significant figures.

..... cm^3 **[3]**

(b) The empty tank is being filled with water at a constant rate of 1.5×10^{-4} m^3 per second.
Note that $1\ m^3 = 1 \times 10^6\ cm^3$.

Calculate the time taken to fill the tank completely.
Give your answer in hours, written in standard form correct to 3 significant figures.

..... hours **[4]**

31 Consider the number 47.5381.

(a) Write 47.5381 correct to:

(i) 2 decimal places.

..... **[1]**

(ii) 2 significant figures.

..... **[1]**

(iii) the nearest integer.

..... **[1]**

32 **(a)** Write 0.04758 correct to 2 significant figures.

..... **[1]**

(b) Write 384.6 correct to the nearest 10.

..... **[1]**

(c) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.7 \times 4.12}{19.6 - 2.87}$$

..... **[2]**

33 By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.4 \times 3.09}{0.187}$$

..... **[3]**

34 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.1^3}{0.192 \times 48.7}$$

You must show all your working.

..... **[3]**

(b) Given that $y = \dfrac{ax}{b-c}$, find an estimate for the value of y when
$a = 3.92$, $x = 0.0514$, $b = 1.08$ and $c = 0.812$.
Write all numbers correct to 1 significant figure in your calculation.

$y =$ ..... **[3]**

35 A rectangular field has length 125 m and width 74 m.

(a) Calculate the area of the field.
Give your answer correct to 2 significant figures.

..... m^2 **[2]**

(b) A different field is a square of side x metres.
The area of this square field is equal to the area of the rectangular field.
Calculate the value of x.
Give your answer correct to 1 decimal place.

$x =$ ..... **[3]**

36 A solid metal cylinder has a base radius of 4.15 cm and a height of 19.8 cm. The metal has a density of 8.13 grams per cubic centimetre.

(a) By writing each number correct to 1 significant figure, estimate the mass of the cylinder in grams.
Use $\pi \approx 3$ for your estimation.
Show your working clearly.

..... g **[3]**

(b) Calculate the actual mass of the cylinder.
Give your answer correct to 3 significant figures.

..... g **[3]**

37 Alex, Ben and Charlie share \$840 in the ratio 2 : 3 : 5.

(a) Calculate the amount of money Ben receives.

\$ ..... **[2]**

(b) Express Charlie's share as a fraction of the total amount, giving your answer in its simplest form.

..... **[2]**

38 **(a)** Write the ratio 45 minutes to 2 hours in its simplest form.

..... **[2]**

(b) A piece of wood of length 3.6 m is cut into three pieces in the ratio 2 : 3 : 7.
Calculate the length, in cm, of the longest piece.

..... cm **[3]**

39 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two towns on the map is 8.4 cm. Calculate the actual distance between the two towns in kilometres.

..... km **[3]**

(b) A forest has an actual area of 35 square kilometres.
Calculate the area of the forest on the map in square centimetres.

..... cm^2 **[3]**

40 A map is drawn to a scale of $1 : 250\,000$.

(a) The distance between two cities on the map is 16.8 cm.
Calculate the actual distance between the cities, in km.

..... km **[2]**

(b) A lake has an actual area of 56 km^2.
Calculate the area of the lake on the map, in cm^2.

..... cm^2 **[2]**

(c) On a different map of the same region, the area of the same lake is 14 cm^2.
Find the scale of this new map in the form $1 : n$.

$1 :$ ..... **[3]**

41 In a factory, the number of male workers to female workers is in the ratio 5 : 3.
The ratio of male workers who work full-time to male workers who work part-time is 3 : 2.
The ratio of female workers who work full-time to female workers who work part-time is 4 : 1.

(a) Find the ratio of the total number of full-time workers to the total number of part-time workers in the factory.
Give your answer in its simplest form.

..... **[4]**

(b) There are 42 more full-time male workers than part-time female workers.
Calculate the total number of workers in the factory.

..... **[3]**

42 Three partners, Alex, Ben and Chris, share a total profit of $\$P$. The ratio of Alex's share to Ben's share is 3 : 4. The ratio of Ben's share to Chris's share is 5 : 6.

(a) Find the ratio of Alex's share to Ben's share to Chris's share in its simplest form.

..... **[2]**

(b) Chris receives \$16 200 more than Alex. Calculate the value of P.

\$ ..... **[3]**

(c) Alex invests his share of the profit in a bank account that pays compound interest at a rate of 4.2% per year.
Calculate the value of Alex's investment at the end of 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

43 A train departs at 08 : 15 and arrives at its destination at 09 : 45.
The train travels a distance of 120 km.
Calculate the average speed of the train.

..... km/h **[2]**

44 A car travels 120 km in 1 hour and 30 minutes.

Calculate the average speed of the car in km/h.

..... km/h **[3]**

45 Water flows from a pipe into a rectangular tank at a rate of 45 litres per minute.

(a) Calculate the volume of water, in cubic centimetres (cm^3), that flows into the tank in 1 second.

..... cm^3 **[2]**

(b) The tank has a base measuring 1.5 m by 80 cm. Calculate the rate at which the water level is rising in the tank, giving your answer in cm/s.

..... cm/s **[3]**

46 A car uses fuel at an average rate of 6.4 litres per 100 km.

(a) Calculate the amount of fuel used for a journey of 350 km.

..... litres **[2]**

(b) The cost of fuel is 1.35 euros per litre.
The exchange rate is \$1 = 0.85 euros. Calculate the cost of the fuel for this journey in dollars (\$).
Give your answer correct to the nearest cent.

\$ ..... **[3]**

47 A cyclist travels from Town A to Town B, a distance of 48 km.
The cyclist travels the first 30 km at an average speed of x km/h. They travel the remaining distance at an average speed of $(x - 4)$ km/h. The total time taken for the journey is 2 hours and 30 minutes.

(a) Show that $5x^2 - 116x + 240 = 0$.

[4]

(b) Solve the equation $5x^2 - 116x + 240 = 0$ to find the possible values of x.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

48 A man runs 15 km at an average speed of x km/h.
He then walks 12 km at an average speed of $(x - 4)$ km/h.
The total time taken for the journey is 3.5 hours.

(a) Show that x satisfies the equation

$$7x^2 - 82x + 120 = 0$$

[3]

(b) **(i)** Solve the equation

$$7x^2 - 82x + 120 = 0$$

$x =$ ..... , or $x =$ ..... **[3]**

(ii) Find the time taken for the man to run the 15 km.

..... hours **[2]**

49 **(a)** Write 24 as a percentage of 60.

..... % **[2]**

(b) A coat has a normal price of \$80. In a sale, the price is reduced by 15%.
Work out the sale price of the coat.

\$ ..... **[2]**

50 A shop sells a jacket for \$120. During a sale, the shop offers a 15% discount on the price of the jacket.

(a) Calculate the sale price of the jacket.

\$ ..... **[2]**

(b) The original price of \$120 gives the shop a profit of 20% on the cost price.
Calculate the cost price of the jacket.

\$ ..... **[3]**

51 **(a)** In 2021, the population of a town was 45 000.
In 2022, the population increased by 4.2%.
Calculate the population in 2022.

..... **[2]**

(b) In a sale, the price of a television is reduced by 24%.
The sale price is $342.
Calculate the original price of the television.

$ ..... **[3]**

52 Amira invests \$4500 in a savings account that pays compound interest at a rate of 2.4% per year.

(a) Calculate the total amount in Amira's account at the end of 5 years.

\$ ..... **[3]**

(b) Ben invests \$4500 in a different savings account that pays simple interest at a rate of r% per year. At the end of 5 years, Ben has the same total amount in his account as Amira has in hers.

Calculate the value of r.

$r =$ ..... **[3]**

53 The population of a city increases by $x\%$ each year.
In 2015, the population was 1 250 000.
In 2018, the population was 1 367 631.

(a) Show that the value of x is approximately 3.

[3]

(b) Assume the population continues to increase by exactly 3% each year.
Calculate the year in which the population will first exceed double its 2015 value.

..... **[4]**

54 **(a)** Ali invests \$5000 at a rate of 2.5% per year compound interest. Calculate the value of his investment at the end of 3 years.
Give your answer correct to the nearest cent.

\$ ..... **[2]**

(b) Beatrice invests a sum of money at a rate of r% per year compound interest.
After 5 years, the value of her investment has increased by 30%.
Calculate the value of r.

$r =$ ..... **[3]**

(c) The value of a car depreciates by 12% each year.
After 3 years, the value of the car is \$17 036.80.
Calculate the original value of the car.

\$ ..... **[3]**

55 Work out the value of

$$\frac{\sqrt[3]{25.6 - 3.2^2}}{1.8 \times 0.45}$$

(a) Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

56 Consider the following calculation.

(a) Work out the value of

$$\frac{\sqrt{4.1^2 + 5.2}}{2.3^3 - 1.4}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Give your answer to part (a) correct to 3 significant figures.

..... **[1]**

57 A student uses a calculator to solve three different problems.

(a) The answer to a problem about money is displayed on the calculator as 8.4.
Write this amount appropriately in dollars and cents.

..... **[1]**

(b) The answer to a problem about time is displayed on the calculator as 5.15 hours.
Write this time appropriately in hours and minutes.

hours ..... , minutes ..... **[1]**

(c) A car travels 280 km at an average speed of 85 km/h. Calculate the time taken for this journey.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[3]**

58 A plumber charges a fixed fee of \$45 plus an hourly rate of \$32.50.

(a) The plumber works on a job for 3 hours and 40 minutes.
Calculate the total cost of this job.
Give your answer to the nearest cent.

\$ ..... **[3]**

(b) On another job, the total cost charged by the plumber is \$230.25.
Calculate the time the plumber worked, giving your answer in hours and minutes.

hours, minutes **[3]**

59 A water tank is in the shape of a sphere with radius 1.4 m.
The tank is completely full of water.

(a) Work out the volume of water in the tank.
Write down all the figures on your calculator display.

..... m^3 **[2]**

(b) Water leaks from the tank at a constant rate of 0.035 cubic metres per minute.
Calculate the time taken for the tank to become completely empty.
Do not round any intermediate values during your working.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[4]**

60 The volume, V, of a spherical cap is given by the formula

$$V = \frac{1}{3}\pi h^2(3R - h)$$

where R is the radius of the sphere and h is the height of the cap.

(a) Calculate V when $R = 8.45$ cm and $h = 3.2$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

(b) Rearrange the formula to make R the subject.

$R =$ ..... **[3]**

(c) A different spherical cap has volume $V = 500$ cm^3 and height $h = 5.6$ cm. Calculate the radius, R, of this sphere.
Give your answer correct to 3 significant figures.

$R =$ ..... cm **[2]**

61 A train timetable shows that a train leaves Paris at 09 55 and arrives in Lyon at 12 10.

(a) Work out the time taken for this journey in hours and minutes.

hours ..... , minutes ..... **[1]**

(b) Another train leaves Paris at 16 45.
This journey takes 2 hours and 35 minutes.
Work out the time this train arrives in Lyon.

..... **[2]**

62 A flight departs from London to Dubai at 21 : 45 local time.

The flight time is 6 hours and 50 minutes.
The local time in Dubai is 4 hours ahead of the local time in London.
Work out the local time in Dubai when the flight arrives.

..... **[3]**

63 A flight departs from Dubai at 14 30 local time and arrives in Sydney.
The flight time is 14 hours and 15 minutes.
The local time in Sydney is 7 hours ahead of the local time in Dubai.
Work out the local time in Sydney when the flight arrives.

..... **[3]**

64 A train travels from Station A to Station B. The total distance is 420 km.
The train departs Station A at 08 : 15 and arrives at Station B at 11 : 03.

(a) Calculate the time taken for the journey, in hours and minutes.

hours, minutes **[2]**

(b) Calculate the average speed of the train in km/h.

..... km/h **[3]**

65 A ship travels a distance of 920 km at an average speed of 24 km/h.

(a) Work out the time taken for the journey.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[3]**

(b) The ship left the port on Wednesday at 17 50.
Determine the day and the time it arrives at its destination.

Day ..... , Time ..... **[4]**

66 Consider the following applications of time, speed, and dates.

(a) A satellite completes one orbit of the Earth in exactly 92 minutes.
Calculate the number of full orbits it completes in exactly one non-leap year (365 days).

..... orbits **[4]**

(b) A plane flies from City P to City Q. The distance is 11 440 km and the average speed is 880 km/h.

(i) Calculate the flight time.

..... hours **[1]**

(ii) City P is in the time zone UTC−4.
City Q is in the time zone UTC+9.
The plane departs from City P on Monday at 22 : 45 local time.
Find the local day and time of arrival in City Q.

Day = ... , Time = ... **[3]**

67 John has \$400. He buys a new camera that costs 255 Euros (). The exchange rate is \$1 = 0.85.
Work out how much money John has left in dollars.

\$ ..... **[3]**

68 The exchange rate between pounds (£) and dollars ($) is £1 = $1.50.
Anna changes £300 into dollars.
She spends $210 on a tablet.
She changes the remaining dollars back into pounds at the same exchange rate.

Work out how many pounds she receives.

£ ..... **[3]**

69 A car dealer buys a car in Japan for 1 250 000 JPY.
The exchange rate is £1 = 145 JPY.

(a) Calculate the cost of the car in pounds (£).

£ ..... **[2]**

(b) The dealer pays shipping costs of £600.
He then sells the car in the UK for £12 500.

Calculate his percentage profit on the total cost of buying and shipping the car.

..... % **[3]**

70 Amira travels from the UK to Japan.
She changes 850 into Japanese Yen (JPY).
The bank charges a 2% fee on the amount exchanged.
The exchange rate is 1 = 180 JPY.

(a) Calculate the amount of JPY she receives.

..... JPY **[3]**

(b) While in Japan, she spends 115000 JPY.
She changes her remaining JPY back to pounds ().
The new exchange rate is 1 = 184 JPY.
The bank charges a fixed fee of 500 JPY for this transaction.
Calculate the amount of pounds she receives.

..... **[4]**

71 A trader buys 1000 items for y each.
He pays a shipping fee of 250.
He sells all 1000 items in the USA for \$4.20 each.

The exchange rate is 1 = \$1.20.
The trader makes a profit of 40% on his total cost in Euros.

Calculate the value of y.

$y =$ ..... **[5]**

72 In Year 1, the exchange rate between dollars ($) and euros (EUR) was $\$1 = x$ EUR.
In Year 2, the exchange rate changed to $\$1 = (x - 0.05)$ EUR.

In Year 1, a laptop cost 400 EUR. In Year 2, the same model laptop cost 450 EUR.

The cost of the laptop in dollars increased by $100 from Year 1 to Year 2.

(a) Write down an expression in terms of x for the cost of the laptop in dollars in Year 1.

$\$$ ..... **[1]**

(b) Form an equation in x and show that it simplifies to $20x^2 - 11x - 4 = 0$.

[4]

(c) Solve the equation to find the value of x.

$x =$ ..... **[3]**

73 A population of bacteria increases exponentially at a constant rate of $p\%$ per hour.
At 09 : 00, the population is 8 000.
At 12 : 00 on the same day, the population is 27 000.

(a) **(a)** Work out the value of p.

$p =$ ..... **[3]**

(b) Work out the population of the bacteria at 14 : 00 on the same day.

..... **[2]**

74 A rare stamp is bought for \$450. Its value increases exponentially at a rate of 6.5% per year. Calculate the value of the stamp after 12 years.

\$ ..... **[3]**

75 A machine is bought for \$32 000. It depreciates exponentially at a rate of 12% each year.

Calculate the value of the machine after 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

76 The population of a town was 85 000 on 1 January 2010.
The population increases exponentially at a constant rate of $r\%$ per year.
On 1 January 2022, the population was 112 000.

(a) **(a)** Calculate the value of r, correct to 3 significant figures.

$r =$ ..... **[3]**

(b) Assuming the rate of increase remains the same, calculate the expected population on 1 January 2030.
Give your answer correct to the nearest thousand.

..... **[2]**

77 Consider the depreciation of two different vehicles over time.

(a) A car is purchased for \$28 500. Its value depreciates exponentially at a rate of 12% per year. Calculate the value of the car after 5 years, giving your answer correct to the nearest dollar.

\$ ..... **[3]**

(b) A van, initially purchased for \$35 000, depreciates exponentially to a value of \$15 200 after 4 years.
Calculate the annual percentage rate of depreciation for this van.

..... % **[4]**

78 The population of a species of insect increases exponentially.
Initially, there are 2500 insects.
After 6 weeks, there are 9100 insects.

(a) Calculate the percentage increase in the population per week.

..... % **[4]**

(b) Calculate the insect population after 10 weeks, giving your answer correct to the nearest 100.

..... **[2]**

(c) Find the number of complete weeks it takes for the initial population to increase to over 50 000.

..... weeks **[3]**

79 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

80 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

81 Simplify fully.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

82 Simplify

$$\sqrt{75} - \sqrt{27} + \sqrt{12}$$

..... **[2]**

83 Simplify.

$\sqrt{75} + \sqrt{48} - \sqrt{27}$

..... **[2]**

84 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Simplify.

$\sqrt{20} + \sqrt{80}$

..... **[2]**

85 Simplify the following expression, leaving your answer in the form $k\sqrt{3}$.

$\sqrt{48} - \sqrt{12}$

..... **[3]**

86 **(a)** **(i)** Simplify.

$$\sqrt{45} + \sqrt{20}$$

..... **[2]**

(ii) Rationalise the denominator.

$$\frac{18}{\sqrt{3}}$$

..... **[2]**

87 **(a)** Simplify.

$\sqrt{45}$

..... **[1]**

(b) Work out the value of the following expression, giving your answer in the form $k\sqrt{2}$.

$\sqrt{72} + \sqrt{50} - \sqrt{18}$

..... **[2]**

(c) Rationalise the denominator.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

88 **(a)** Expand and simplify.

$(4 - \sqrt{3})(2 + \sqrt{3})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{21}{\sqrt{7}}$$

..... **[2]**

89 **(a)** Expand and simplify

$$(3 - \sqrt{5})^2$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

90 **(a)** Expand and simplify.

$(3 - 2\sqrt{5})(4 + \sqrt{5})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{14}{\sqrt{7}}$$

..... **[2]**

91 **(a)** **(a)** Expand and simplify

$$(3 - 2\sqrt{5})(4 + \sqrt{5})$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{15}{\sqrt{3}}$$

..... **[2]**

92 **(a)** Expand and simplify.

$$(2\sqrt{5}-3)(2\sqrt{5}+3)$$

..... **[2]**

(b) Rationalise the denominator.

$$\frac{14}{3-\sqrt{2}}$$

..... **[3]**

93 **(a)** Expand and simplify.

$$(3\sqrt{2}-1)(2\sqrt{2}+5)$$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{26}{4-\sqrt{3}}$$

..... **[3]**

94 **(a)** Rationalise the denominator and simplify fully.

$$\frac{15}{\sqrt{5}}$$

..... **[2]**

(b) Expand and simplify.

$$(2+\sqrt{3})(5-\sqrt{3})$$

..... **[3]**

95 **(a)** Expand and simplify.

$$(5 - 2\sqrt{7})(3 + \sqrt{7})$$

..... **[2]**

(b) Rationalise the denominator and simplify fully.

$$\frac{22}{5 - \sqrt{3}}$$

..... **[3]**

96 **(a)** Expand and simplify.

$(4 - 3\sqrt{2})(5 + 2\sqrt{2})$

..... **[2]**

(b) Rationalise the denominator and simplify.

$$\frac{11}{4 - \sqrt{5}}$$

..... **[3]**

(c) Solve the equation.

$x\sqrt{27} - 8 = x\sqrt{3} + 10$

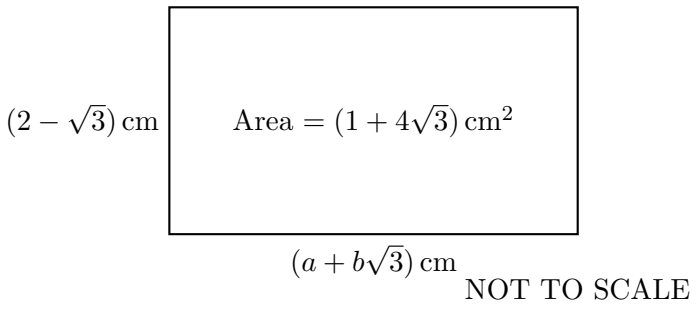
Give your answer in the form $k\sqrt{3}$ where k is an integer.

$x =$ ..... **[3]**

97 A rectangle has an area of $5 + \sqrt{7}$ cm^2 and a width of $3 - \sqrt{7}$ cm.
Calculate the length of the rectangle.
Give your answer in the form $a + b\sqrt{7}$ where a and b are integers.

..... *cm* **[4]**

98 The diagram shows a rectangle with width $(2 - \sqrt{3})$ cm and length $(a + b\sqrt{3})$ cm, where a and b are integers.
The area of the rectangle is $(1 + 4\sqrt{3})\,\text{cm}^2$.



NOT TO SCALE

Work out the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[5]**

99 **(a)** Rationalise the denominator and simplify.

$$\frac{5+\sqrt{3}}{2-\sqrt{3}}$$

..... **[3]**

(b) Solve the equation.
Give your answer in the form $a+b\sqrt{2}$, where a and b are integers.

$x(3-\sqrt{2})=14$

$x =$ ..... **[3]**

100 **(a)** Express $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ in the form $a+b\sqrt{3}$, where a and b are integers.

..... **[3]**

(b) A rectangle has an area of $12+4\sqrt{5}$ square centimetres and a width of $3+\sqrt{5}$ centimetres. Calculate the exact length of the rectangle, giving your answer in its simplest form.

Length = ... cm **[4]**

101 **(a)** Rationalise the denominator and simplify.

$$\frac{22}{5-\sqrt{3}}$$

..... **[3]**

(b) Solve the equation.

$x\sqrt{2} = x + \sqrt{8}$ Give your answer in the form $a + b\sqrt{2}$, where a and b are integers.

$x =$ ..... **[4]**

102 **(a)** A rectangle has an area of $(5 + 2\sqrt{3})$ cm^2 and a length of $(2 + \sqrt{3})$ cm.

Work out the width of the rectangle in the form $a + b\sqrt{3}$, where a and b are integers.

..... cm **[3]**

(b) Solve the equation.

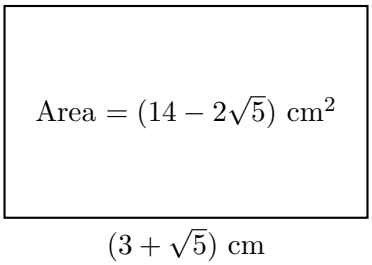
$$x\sqrt{5} - 2 = 2x + \sqrt{20}$$

Give your answer in the form $c + d\sqrt{5}$, where c and d are integers.

Show all your working.

$x =$ ..... **[4]**

103 A rectangle has length $(3 + \sqrt{5})$ cm and area $(14 - 2\sqrt{5})$ cm^2.



(a) Calculate the width of the rectangle.
Give your answer in the form $a + b\sqrt{5}$, where a and b are integers.

..... cm **[4]**

(b) The length of the diagonal of the rectangle is d cm.
Show that $d^2 = 308 - 124\sqrt{5}$.

[3]

104 A right-angled triangle has a base of length $(3+\sqrt{2})$ cm and an area of $(5+2\sqrt{2})$ cm^2.

(a) Show that the height of the triangle is $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ cm.

[2]

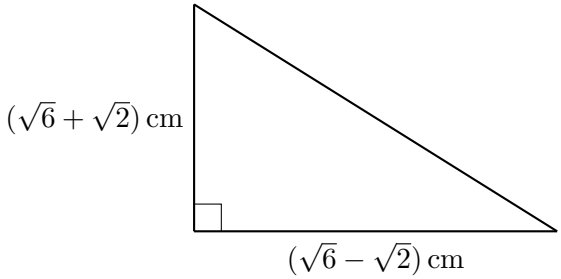
(b) Rationalise the denominator of $\dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ to express the height in the form $a+b\sqrt{2}$, where a and b are integers.

..... cm **[4]**

(c) Given instead that another expression for a length is $\dfrac{1}{\sqrt{7}-2}$, write this length in the form $\dfrac{\sqrt{7}+c}{d}$ where c and d are integers.

..... **[2]**

105 The diagram shows a right-angled triangle.



(a) Calculate the area of the triangle.

..... cm^2 **[3]**

(b) Show that the length of the hypotenuse is 4 cm.

[4]

(c) Another right-angled triangle is mathematically similar to this triangle.
The area of the similar triangle is 18 cm^2.
Calculate the length of its hypotenuse.

..... cm **[3]**

106 Find the value of $3a - 2b$ when $a = 4$ and $b = -5$.

..... **[2]**

107 Given the formula $v = u + at$.

Find the value of v when $u = 5$, $a = -2$, and $t = 3$.

$v =$ ..... **[3]**

108 **(a)** Work out the value of $5x - 2y$ when $x = 3$ and $y = -4$.

..... **[2]**

(b) Evaluate

$$\frac{a^2 - b^3}{2c}$$

when $a = -4$, $b = 2$ and $c = 3$.

..... **[2]**

109 **(a)** **(a)** The cost of hiring a plumber is calculated using the formula

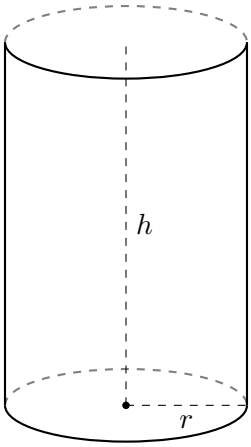
$C = 40 + 25h$ where C is the cost in dollars and h is the number of hours worked.
Work out the cost of hiring a plumber for 3.5 hours.

$C = \$ \ldots\ldots\ldots\ldots\ldots$ **[2]**

(b) Given that $v = u + at$, find the value of v when $u = -12$, $a = 9.8$ and $t = 3$.

$v = \ldots\ldots\ldots\ldots\ldots$ **[3]**

110 A cylindrical container of radius r and height h is open at the top.
The total exterior surface area of the container, A, consists of the curved surface area and the circular base.



(a) Show that the total surface area A can be written as

$$A = \pi r(2h + r)$$

[2]

(b) **(i)** Calculate the exact surface area A when $r = 4$ and $h = 6$.
Give your answer in terms of π.

$A =$ ..... **[2]**

(ii) The volume of the cylinder is V.
The formula for the volume is $V = \pi r^2 h$.
Express h in terms of V, π and r, and substitute this into the expression for A to show that

$$A = \frac{2V}{r} + \pi r^2$$

[3]

111 Consider the expression $P = \dfrac{2x^2 - 3y}{x - y}$.

(a) Calculate the value of P when $x = -2$ and $y = 4$.

$P =$ ..... **[3]**

(b) Rearrange the formula to make y the subject.

$y =$ ..... **[4]**

112 The cost, C dollars, of hiring a plumber for h hours is given by the formula:

$C = 45 + 35h$

(a) Calculate the cost of hiring the plumber for 3.5 hours.

$C =$ ..... **[2]**

(b) The plumber charges a total of \$325 for a job.
Calculate the number of hours the plumber worked.

$h =$ ..... hours **[3]**

113 The volume, V, of a spherical cap of height h and base radius a is given by the formula

$$V = \frac{1}{6}\pi h(3a^2 + h^2)$$

(a) Calculate the volume of a spherical cap with height $h = 4.2$ cm and base radius $a = 6.5$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

(b) Rearrange the formula to make a^2 the subject.

$a^2 =$ ..... **[3]**

(c) Find the base radius, a, of a spherical cap with volume $V = 150$ cm^3 and height $h = 3$ cm. Give your answer correct to 3 significant figures.

$a =$ ..... cm **[1]**

114 A formula used in physics is

$$s = ut + \frac{1}{2}at^2$$

(a) **(a)** Calculate the value of s when $u = 5.2$, $t = 4.5$ and $a = -9.8$.

$s =$ ..... **[2]**

(b) Calculate the positive value of t when $s = 40$, $u = 15$ and $a = -9.8$.

$t =$ ..... **[3]**

115 **(a)** Expand and simplify.

$3(2x - 5) - 2(x + 4)$

..... **[2]**

(b) Factorise completely.

$8x^2y - 12xy^2$

..... **[2]**

116 **(a)** Expand and simplify

$$3(2x - 5) - 4(x - 2)$$

..... **[2]**

(b) Factorise fully

$$12x^2y - 18xy^2$$

..... **[2]**

117 **(a)** Expand and simplify.

$3(2x - 5) - 4(x - 3)$

..... **[2]**

(b) Factorise fully.

$14x^2y + 21xy^2$

..... **[2]**

118 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$ac - 3bc + 2ad - 6bd$

..... **[2]**

(c) Factorise.

$16x^2 - 25y^2$

..... **[1]**

119 **(a)** Factorise.

$x^2 - 8x + 15$

..... **[2]**

(b) Factorise.

$2x^2 - 5x - 12$

..... **[2]**

(c) Factorise.

$ax - 3a + 2bx - 6b$

..... **[2]**

120 **(a)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

(b) Factorise completely.

$3x^3 + 10x^2 - 8x$

..... **[3]**

121 **(a)** Factorise

$$6ax - 2ay - 3bx + by$$

..... **[2]**

(b) Factorise completely.

(i)

$$50x^2 - 8y^2$$

..... **[3]**

(ii)

$$2x^3 + 5x^2 - 12x$$

..... **[3]**

122 **(a)** Expand and simplify

$$(x-3)(2x+1)(x+4)$$

..... **[3]**

(b) Factorise

$$2x^2 - 7x - 15$$

..... **[2]**

123 **(a)** Expand and simplify.

$(x-2)(x+3)(2x-1)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$2x^3 - 18x$

..... **[3]**

124 Write as a single fraction in its simplest form.

$$\frac{5}{x+2} + \frac{2}{3x-1}$$

..... **[3]**

125 Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{3}$$

..... **[3]**

126 **(a)** Simplify fully.

$$\frac{3x}{4} + \frac{2x}{5}$$

..... **[2]**

(b) Simplify fully.

$$\frac{5a^2b}{12} \div \frac{10ab^2}{9}$$

..... **[2]**

127 Simplify fully.

$$\frac{3x^2 - 12}{x^2 + 3x - 10}$$

..... **[4]**

128 **(a)** Simplify fully.

$$\frac{2x^2 - 5x - 3}{4x^2 - 9}$$

..... **[3]**

(b) Write as a single fraction in its simplest form.

$$\frac{3}{x-2} + \frac{4}{x+1} - \frac{2x}{x^2 - x - 2}$$

..... **[4]**

129 Simplify fully.

$$\frac{2x^2 - 9x + 9}{2x^2 - 18}$$

..... **[4]**

130 **(a)** Write as a single fraction in its simplest form.

$$\frac{x+3}{2x-1} - \frac{x-2}{x+4}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2-16}{x^2+4x}$$

..... **[2]**

131 **(a)** Write as a single fraction in its simplest form.

$$\frac{4}{x-3} - \frac{3}{2x-1}$$

..... **[3]**

(b) Simplify fully.

$$\frac{x^2-25}{2x^2-9x-5} \div \frac{x+5}{4x+2}$$

..... **[4]**

132 **(a)** Write as a single fraction in its simplest form.

$$\frac{3}{x-2} - \frac{x+5}{x^2-4}$$

..... **[3]**

(b) Simplify fully.

$$\frac{4x^2-25}{2x^2-5x} \times \frac{6x^3}{4x^2+10x}$$

..... **[4]**

133 **(a)** Simplify.

$$4x^2 \times 3x^{-5}$$

..... **[1]**

(b) Simplify.

$$\left(\frac{8x^9}{27}\right)^{\frac{1}{3}}$$

..... **[2]**

134 **(a)** Simplify.

$$w^4 \times w^7$$

..... **[1]**

(b) Simplify.

$$(3x^3)^2$$

..... **[2]**

(c) Simplify.

$$\frac{15y^8}{5y^{-2}}$$

..... **[2]**

135 **(a)** Simplify.

$p^4 \times p^7$

..... **[1]**

(b) Simplify.

$(3q^3)^2$

..... **[2]**

(c) Work out the value of $27^{\frac{2}{3}}$.

..... **[2]**

136 **(a)** Work out the exact value of

$$32^{-\frac{2}{5}}$$

..... **[2]**

(b) Solve the equation.

$$5^{x+2} = 125^{x}$$

$x =$ ..... **[3]**

137 **(a)** Simplify fully.

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

(b) Solve the equation.

$$3^{2x+1} = 27$$

$x =$ ..... **[2]**

(c) Simplify fully.

$$\left(\frac{8a^6}{b^3}\right)^{\frac{1}{3}}$$

..... **[2]**

138 **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

(b) Solve the equation.

$$2^{x+3} = 8^x$$

$x =$ ..... **[3]**

139 **(a)** Solve the equation.

$$16^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

(b) Simplify fully.

$$\left(\frac{81x^{12}}{16y^8}\right)^{-\frac{3}{4}}$$

..... **[3]**

140 **(a)** Simplify fully.

$$\frac{4x^{-\frac{1}{2}} \times 3x^2}{6x^{\frac{1}{3}}}$$

..... **[3]**

(b) Solve the equation.

$$9^{2x-1} \times 3^{x+1} = 27^x$$

$x =$ ..... **[4]**

141 **(a)** Simplify fully.

$$\frac{12x^{\frac{3}{2}}y^{-2}}{4x^{-\frac{1}{2}}y^{4}}$$

..... **[3]**

(b) Solve the equation.

$$4^{2x-1} \times 8^{x+1} = 32^{x}$$

$x =$ ..... **[4]**

142 Solve the equation.

$$\frac{3x-1}{4} = \frac{x+5}{3}$$

$x =$ ..... **[3]**

143 Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 22$

$x =$ ..... , $y =$ ..... **[3]**

144 **(a)** Solve the equation.

$$4(x - 3) = 2x + 10$$

$x =$ ..... **[2]**

(b) Rearrange the formula to make x the subject.

$$y = \frac{2x - 1}{5}$$

$x =$ ..... **[2]**

145 **(a)** Solve the equation.

$3(2x - 1) = 4x + 11$

$x =$ ..... **[2]**

(b) Solve the simultaneous equations.

$3x + 2y = 16$

$x - y = 2$

$x =$ ..... , $y =$ ..... **[3]**

146 **(a)** Solve

$$5x - 3 = 2(x + 6)$$

$x =$ ..... **[2]**

(b) Solve

$$\frac{x+2}{3} + \frac{2x-1}{4} = 3$$

$x =$ ..... **[3]**

147 **(a)** Solve the equation.

$$3(2x - 5) = 4x + 7$$

$x =$ ..... **[3]**

(b) Rearrange the formula to make x the subject.

$$y = \frac{2x + 3}{5}$$

$x =$ ..... **[2]**

148 **(a)** Solve the equation.

$$4(2x - 1) = 3x + 11$$

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$$y = \frac{3x - 5}{2}$$

$x =$ ..... **[3]**

149 Consider the following algebraic tasks.

(a) Solve the equation.

$$5(x-2) = 3x + 14$$

$x =$ ..... **[3]**

(b) Rearrange the formula to make b the subject.

$$A = \frac{h(a+b)}{2}$$

$b =$ ..... **[3]**

150 **(a)** Solve.

$4(x - 3) = 2x + 7$

$x =$ ..... **[3]**

(b) Solve the simultaneous equations.
You must show all your working.

$5x - 2y = 16$

$3x + 4y = 7$

$x =$ ..... , $y =$ ..... **[3]**

151 Rearrange the formula to make x the subject.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

152 **(a)** Solve the simultaneous equations.
You must show all your working.

$3x + 2y = 12$

$5x - 3y = 1$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation by factorisation.

$2x^2 - 7x - 15 = 0$

$x =$ ..... , or $x =$ ..... **[3]**

153 **(a)** Solve the simultaneous equations.
You must show all your working.

$3x - 2y = 19$

$4x + y = 18$

$x =$ ..... , $y =$ ..... **[3]**

(b) Rearrange the formula to make p the subject.

$$T = \frac{3p + 4}{p - 2}$$

$p =$ ..... **[4]**

154 **(a)** Solve the simultaneous equations.
You must show all your working.

$2x - y = 9$ $3x + 2y = 10$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation.

$$\frac{2x}{3} - \frac{x-4}{4} = \frac{5}{2}$$

$x =$ ..... **[4]**

155 Solve the following equations.

(a) Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 9$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation by factorisation.

$3x^2 - 10x - 8 = 0$

$x =$ ..... or $x =$ **[4]**

156 Solve the equation.

$$\frac{2}{x-1} + \frac{1}{x-2} = 1$$

Give your answers in the form $a \pm \sqrt{b}$, where a and b are integers.
You must show all your working.

$x =$ ..... **[6]**

157 **(a)** Rearrange the formula to make x the subject.

$$p = \frac{3x - 1}{x + 4}$$

$x =$ ..... **[4]**

(b) Solve the equation by factorising.

$$2x^2 - 5x - 12 = 0$$

$x =$ ..... , $x =$ ..... **[3]**

158 **(a)** A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm.
The perimeter of the rectangle is 42 cm.

Work out the value of x.

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$$p = \sqrt{\frac{x-3}{x}}$$

$x =$ ..... **[4]**

159 **(a)** Make x the subject of the formula.

$$y = \frac{2x + 5}{x - 3}$$

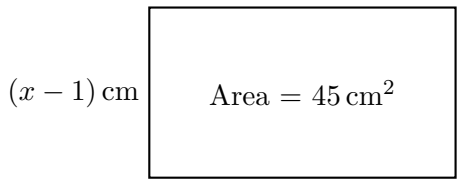
$x =$ ..... **[3]**

(b) Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{x + 1}{3} - \frac{2}{x - 2} = 1$$

$x =$ ..... , $x =$ ..... **[6]**

160 The diagram shows a rectangle with length $(2x + 3)$ cm and width $(x - 1)$ cm.
The area of the rectangle is $45\,\text{cm}^2$.



NOT TO SCALE

(a) Show that $2x^2 + x - 48 = 0$.

[2]

(b) Solve the equation $2x^2 + x - 48 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

(c) Make x the subject of the formula.

$$y = \frac{ax + 5}{x - 3}$$

$x =$ ..... **[4]**

161 Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{3}{x-2} + \frac{4}{2x+1} = 2$$

$x =$ ..... , $x =$ ..... **[6]**

162 Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{5}{x-2} - \frac{3}{2x+1} = 2$$

$x =$ ..... , $x =$ ..... **[7]**

163 **(a)** Show that $\dfrac{3}{x-1} - \dfrac{2}{x+3} = 1$ simplifies to $x^2 + x - 14 = 0$.

[4]

(b) Solve the equation $x^2 + x - 14 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

164 **(a)** A train travels 120 km at an average speed of x km/h.
It then travels a further 150 km at an average speed of $(x + 10)$ km/h.
The total time for the journey is 4 hours.
Write down an equation in x and show that it simplifies to $2x^2 - 115x - 600 = 0$.

[4]

(b) Solve the equation $2x^2 - 115x - 600 = 0$ to find the value of x.
Give your answer correct to 1 decimal place.

$x =$ ..... **[2]**

(c) Use your graphic display calculator to solve the equation.

$$2^x - 3 = \frac{5}{x}$$

Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[2]**

165 A bus travels 100 km at an average speed of x km/h.
A car travels 100 km at an average speed of $(x + 15)$ km/h.
The car takes 30 minutes less than the bus.

(a) Write down an equation in x and show that it simplifies to $x^2 + 15x - 3000 = 0$.

[3]

(b) Solve the equation $x^2 + 15x - 3000 = 0$.
You must show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

(c) Work out the time taken by the car to travel the 100 km.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

166 **(a)** Make x the subject of the formula.

$$y = \frac{3x - 2}{x + 4}$$

$x =$ ..... **[4]**

(b) Solve the equation.
Show all your working and give your answers correct to 2 decimal places.

$$\frac{5}{x - 2} - \frac{3}{2x + 1} = 2$$

$x =$ ..... , or $x =$ ..... **[6]**

167 Consider the following algebraic equations and formulas.

(a) Rearrange the formula to make x the subject.

$$y = \frac{3x - 1}{x + 2}$$

$x =$ ..... **[4]**

(b) Solve the equation.
Give your answers correct to 2 decimal places.
Show all your working.

$$\frac{2}{x - 1} + \frac{3}{x + 2} = 1$$

$x =$ ..... or $x =$ **[6]**

168 A train travels 150 km at an average speed of x km/h.

(a) Write down an expression, in terms of x, for the time taken in hours.

..... hours **[1]**

(b) A bus travels the same distance of 150 km at an average speed of $(x - 15)$ km/h.
The bus takes 30 minutes longer than the train.
Show that x satisfies the equation $x^2 - 15x - 4500 = 0$.

[4]

(c) Solve the equation $x^2 - 15x - 4500 = 0$ to find the average speed of the train.

..... km/h **[3]**

(d) Rearrange the formula to make q the subject.

$$P = \frac{2q + 3}{q - 1}$$

$q =$ ..... **[4]**

169 Solve the inequality.

$4(3x - 1) - 5x \geq 2(x + 8)$

..... **[3]**

170 **(a)** Solve the inequality

$$3(x-4) < 5x - 2$$

..... **[2]**

(b) Write down the smallest integer value of x that satisfies this inequality.

..... **[1]**

171 **(a)** Solve the inequality.

$$-5 < 3x + 1 \leq 13$$

..... **[2]**

(b) Write down all the integer values of x that satisfy this inequality.

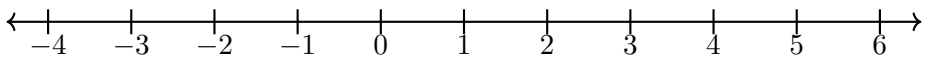
..... **[2]**

172 **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

(b) Show your solution to part (a) on the number line.



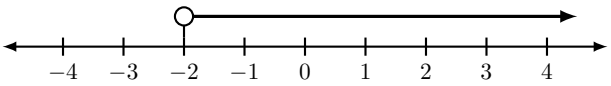
[2]

173 **(a)** Solve the inequality.

$4x - 3 < 2x + 9$

..... **[2]**

(b) Write down the inequality shown on the number line.



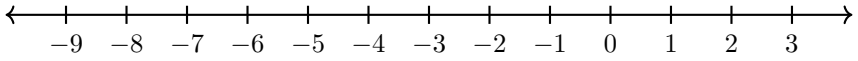
..... **[2]**

174 **(a)** Solve the inequality.

$$3(x-2) \leq 5x+8$$

..... **[3]**

(b) Represent the solution to part (a) on the number line.



[2]

175 **(a)** Solve the inequality.

$3x - 2 > 5x + 7$

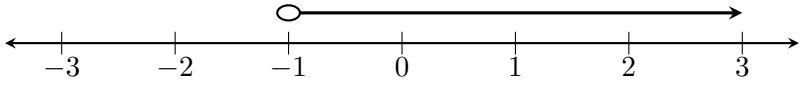
..... **[2]**

(b) Write down all the integer values of n that satisfy the inequality.

$-5 < 2n - 1 \leq 3$

..... **[3]**

(c) An inequality is shown on the number line above.
Write down the inequality.



..... **[1]**

176 **(a)** Solve the inequality.

$$4x - 3 \leq 7x + 12$$

..... **[2]**

(b) Solve the inequality.

$$-8 < 2x - 1 \leq 7$$

..... **[2]**

(c) Write down the largest integer and the smallest integer that satisfy the inequality in part (b).

Largest = ... , Smallest = ... **[2]**

177 **(a)** Solve the inequality.

$2 - 3x < 11$

..... **[2]**

(b) Solve the inequality.

$4(2x - 1) \leq 5x + 17$

..... **[3]**

(c) n is an integer such that $2 - 3n < 11$ and $4(2n - 1) \leq 5n + 17$.
Write down all the possible values of n.

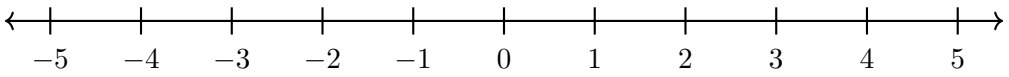
..... **[2]**

178 **(a)** Solve the inequality.

$3x - 5 \leq 7x + 3$

..... **[2]**

(b) On the number line, show the solution to the inequality in part (a).



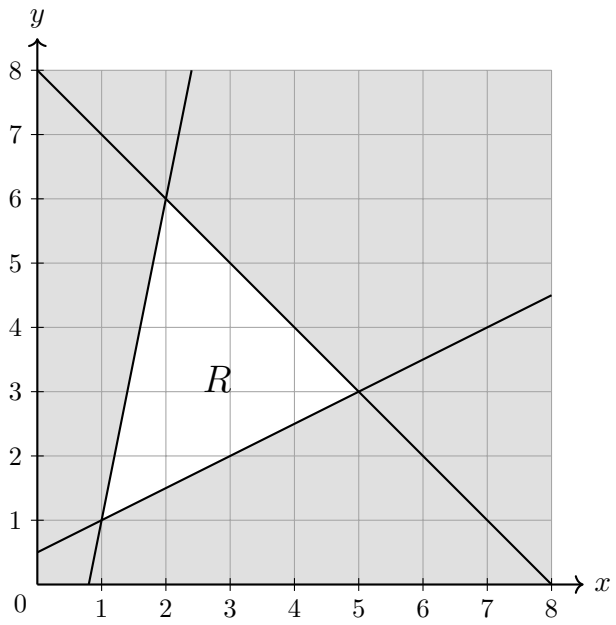
[2]

(c) List all the integers that satisfy the inequality.

$-2 < 3x - 5 \leq 4$

..... **[3]**

179 The diagram shows a region R, which is unshaded.
The unwanted regions are shaded.



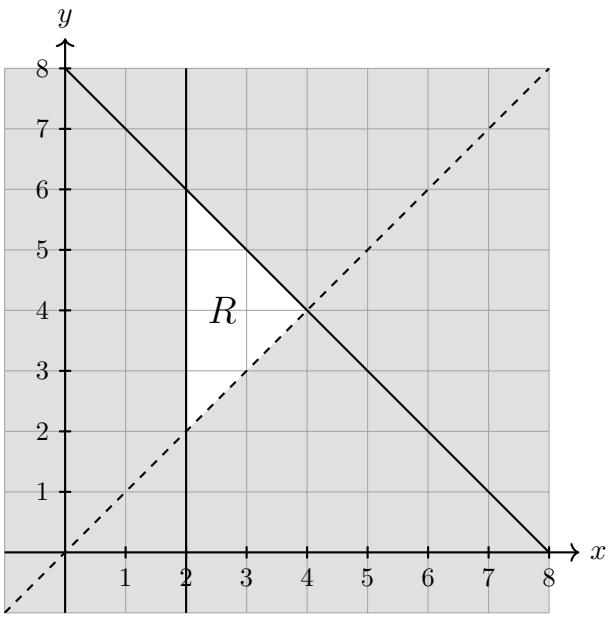
Write down the three inequalities that define region R.

.....

.....

..... **[3]**

180 The diagram shows a region R defined by three inequalities.
The unwanted regions are shaded.

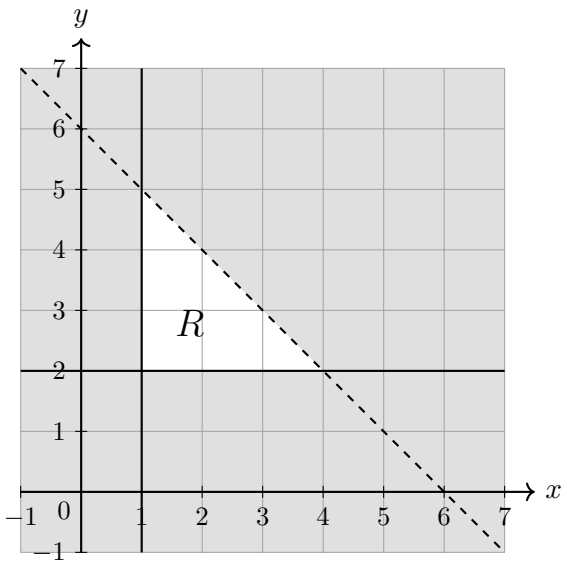


Write down the three inequalities that define the unshaded region R.

.....

..... **[3]**

181 The unshaded region R is defined by three inequalities.
The boundaries of the region are shown on the coordinate grid.



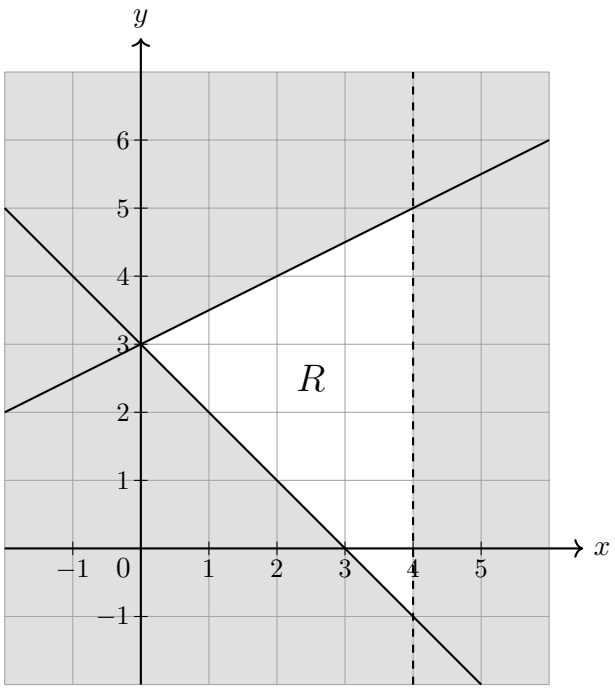
Write down the three inequalities that define the unshaded region R.

.....

.....

..... **[3]**

182 The unshaded region R is defined by three inequalities.



(a) Write down these three inequalities.

.....

.....

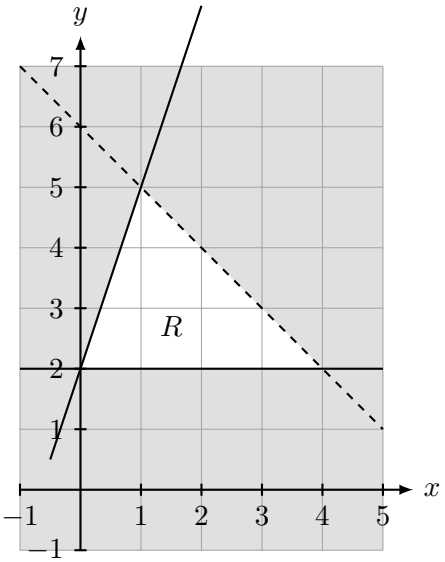
..... **[3]**

(b) The point (a, b), where a and b are integers, lies in the region R.
It is given that $a = 3$.

Find all possible values of b.

..... **[3]**

183 A region R is defined by three inequalities.



The region R, unshaded, is defined by three inequalities.
Write down these three inequalities.

.....

.....

..... **[6]**

184 **(a)** Solve the inequality.

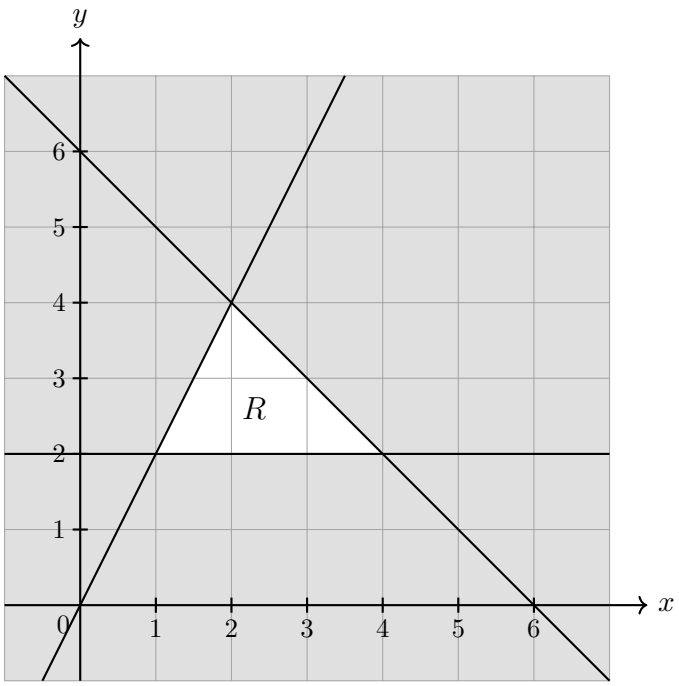
$$\frac{3x-2}{5} - \frac{2x-5}{3} \leq 2$$

..... **[4]**

(b) A region R is defined by the three inequalities $y > x$, $y < 4$, and $x + y > 2$.
Write down all the points in region R that have integer coordinates.

..... **[3]**

185 The diagram shows a region R unshaded.



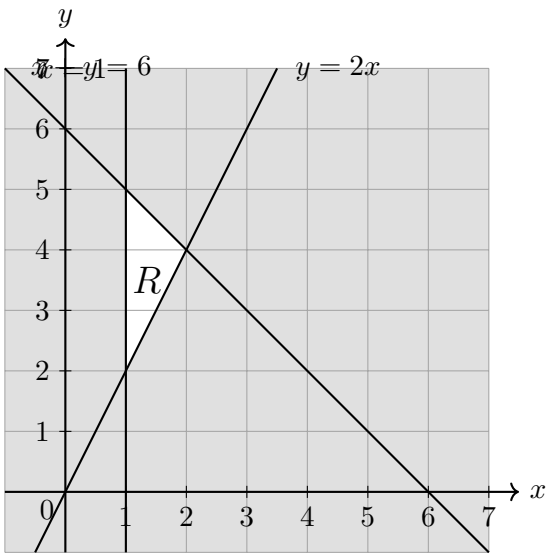
Write down the three inequalities that define the region R.

.....

.....

..... **[3]**

186 The diagram shows three straight lines: $x = 1$, $y = 2x$, and $x + y = 6$.
The unshaded region R is bounded by these lines.



Write down the three inequalities that define the unshaded region R.

.....

.....

..... **[3]**

187 **(a)** Solve the double inequality.

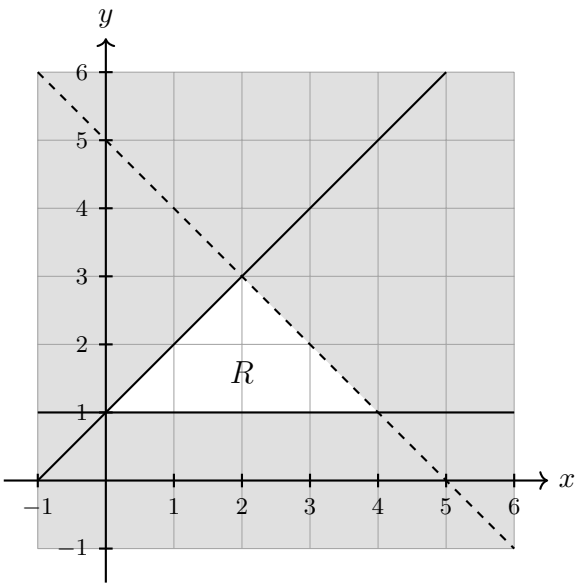
$$-7 \leq 3x - 1 < 11$$

..... **[3]**

(b) Write down all the integer values of x that satisfy the inequality in part (a).

..... **[2]**

188 The diagram shows a region R which is unshaded.
The unwanted regions are shaded.



(a) Write down the three inequalities that define region R.

.....

.....

..... **[4]**

(b) Find the point (x, y), with integer coordinates, inside region R which maximizes the value of $2y - x$.

((..... , ...)) **[2]**

189 **(a)** Solve the inequality.

$$-5 < 2x + 1 \leq 11$$

..... **[3]**

(b) Write down all the integers that satisfy the inequality.

$$-2.5 \leq y < 1$$

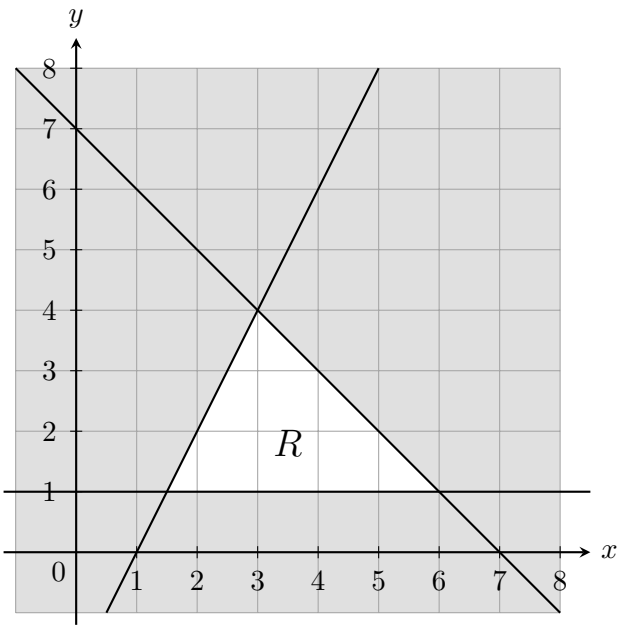
..... **[2]**

(c) Use your graphic display calculator to solve the inequality.

$$2x^2 - 7x + 3 > 0$$

..... **[2]**

190 The diagram shows a grid with three straight lines.



(a) The unshaded region R is defined by three inequalities.
The boundaries are included in the region R.
Write down the three inequalities.

.....

.....

..... **[5]**

(b) A point (x, y) lies in the region R.
Both x and y are integers.
Find the maximum value of $2x + y$.

..... **[2]**

191 A factory produces two types of bicycles: Mountain bikes and City bikes.
In a week, the factory produces x Mountain bikes and y City bikes.

- The factory produces at least 15 Mountain bikes.
- The factory can produce at most 40 bicycles in a week.
- The number of City bikes produced is at least half the number of Mountain bikes produced.

(a) Write down three inequalities, in terms of x and y, to show this information.

.....

.....

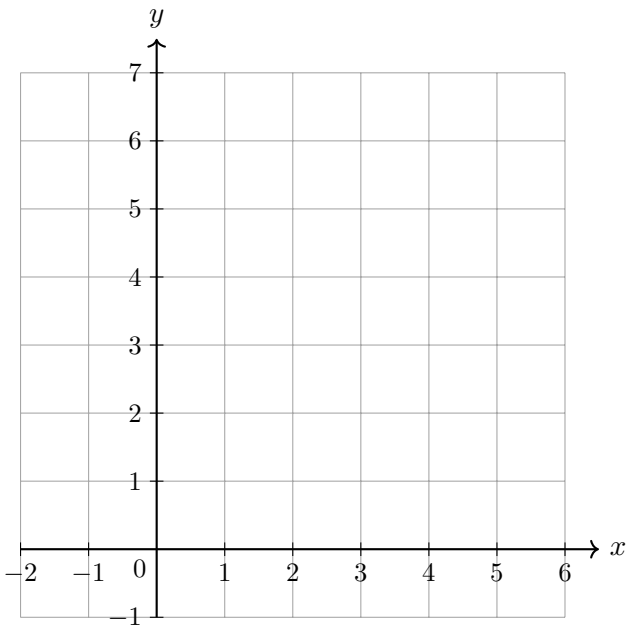
..... **[3]**

(b) The profit on each Mountain bike is \$400 and the profit on each City bike is \$150.
Work out the maximum possible profit the factory can make in a week.

\$ ..... **[3]**

192 **(a)** On the grid, draw the lines $y = 2$, $y = x + 2$, and $2x + y = 8$.
Shade the unwanted regions to show the region R defined by the inequalities:

- $y \geq 2$
- $y \leq x + 2$
- $2x + y \leq 8$

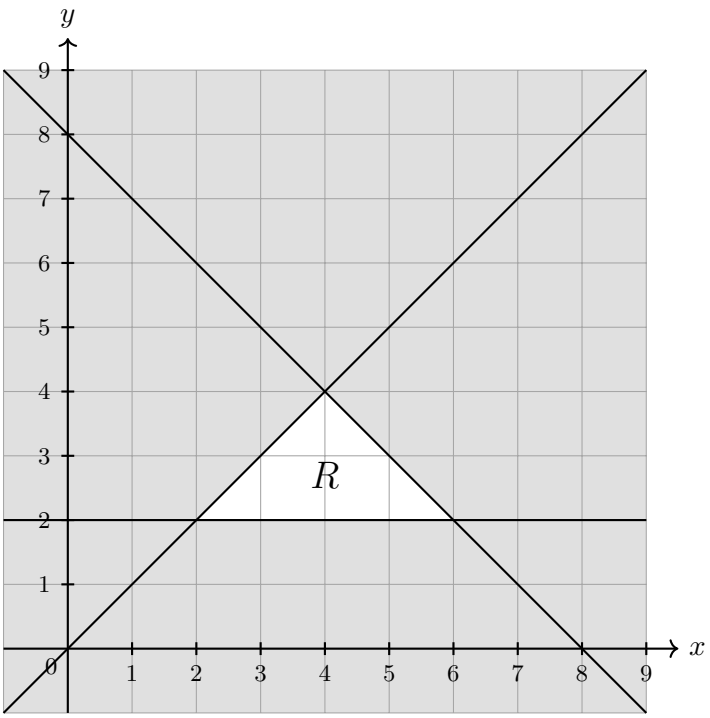


[4]

(b) Find the largest value of $3x + 2y$ for points (x, y) in the region R where both x and y are integers.

..... **[2]**

193 The diagram shows three straight lines on a coordinate grid.



(a) The region R is unshaded.
Write down the three inequalities that define the region R.

.....

.....

..... **[3]**

(b) Find the maximum value of $2x + 3y$ given that x and y are integers that satisfy these three inequalities.

..... **[3]**

(c) A point is chosen at random from the points with integer coordinates in region R.
Work out the probability that $x = y$.

..... **[2]**

194 **(a)** Solve the inequality.

$$-5 < 2x - 1 \leq 7$$

..... **[2]**

(b) Find the smallest integer x that satisfies the inequality.

$$\frac{2x-5}{3} - \frac{x-3}{4} > 1$$

$x =$ ..... **[3]**

(c) A taxi company, City Cabs, charges a fixed fare of \$3.20 plus \$1.50 per kilometre.
Another company, Metro Taxis, charges a fixed fare of \$5.00 plus \$1.10 per kilometre.
Construct and solve an inequality to find the minimum whole number of kilometres a passenger must travel for City Cabs to be more expensive than Metro Taxis.

..... kilometres **[3]**

195 Tickets for a community show cost \$5 for children and \$10 for adults.
The maximum number of tickets that can be sold is 500.
The organizers must sell at least 150 adult tickets.
The total amount of money raised from ticket sales must be at least \$3000.

Let x be the number of children's tickets sold and y be the number of adult tickets sold.

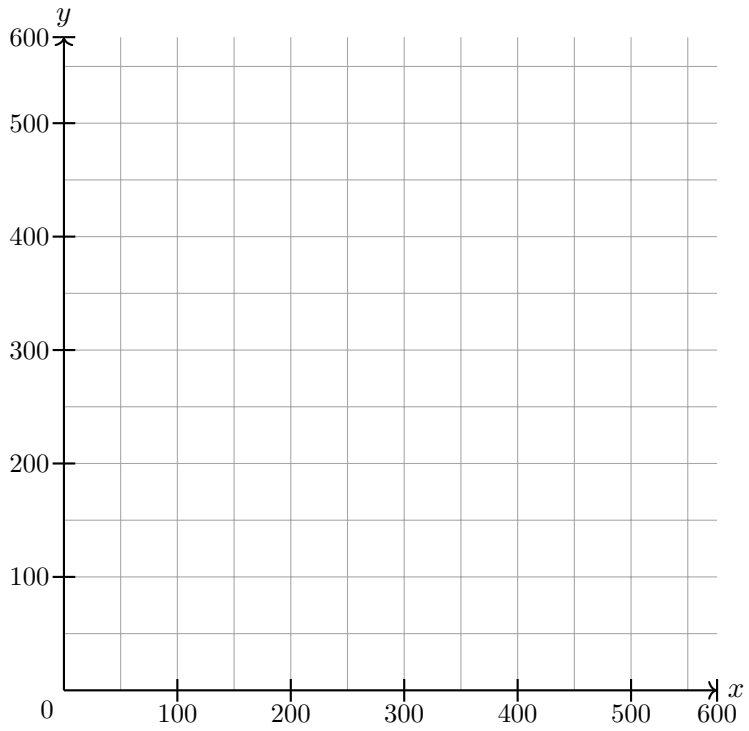
(a) Write down three inequalities in x and y to show this information.

.....

.....

..... **[3]**

(b) On the grid, draw three lines to show these inequalities.
Shade the unwanted regions to leave the region representing the possible number of tickets sold.



[4]

(c) The organizers want to maximize the number of children's tickets sold.
Find this maximum number of children's tickets and the corresponding number of adult tickets.

Children's tickets = ... , Adult tickets = ... **[2]**

196 A carpenter makes x tables and y chairs each week.
The total number of tables and chairs made each week must not exceed 60.
The number of tables made must be at least 10.
The number of chairs made must be at least half the number of tables made.

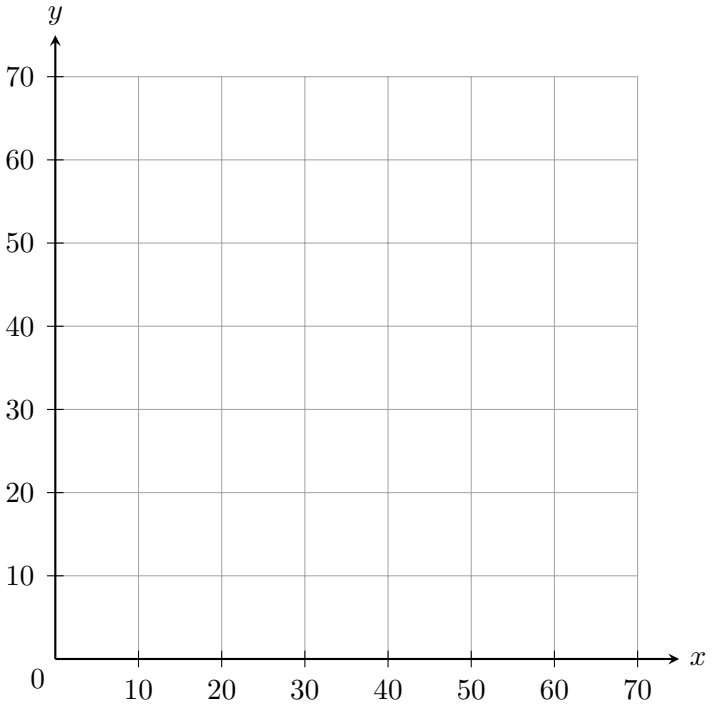
(a) Write down three inequalities, in terms of x and y, to show this information.

.....

.....

..... **[3]**

(b) On the grid, draw three straight lines and shade the unwanted regions to show the region which satisfies these inequalities.
Label the region R.



[4]

(c) The profit on each table is \$40 and the profit on each chair is \$25.
Work out the maximum possible weekly profit.

\$ ..... **[2]**

197 **(a)** Find the exact range of values of x for which

$$\frac{2x-3}{4} < \frac{x+5}{3} \leq \frac{3x-1}{2}$$

..... **[5]**

(b) A mobile phone company offers two plans.
Plan A costs \$12 per month plus \$0.07 per minute of calls.
Plan B costs \$25 per month for unlimited calls.

Let x be the number of minutes of calls in a month.

(i) Write down an inequality in terms of x to show the condition when Plan B is strictly cheaper than Plan A.

..... **[1]**

(ii) Solve your inequality to find the minimum number of whole minutes of calls required for Plan B to be cheaper than Plan A.

..... minutes **[3]**

198 **(a)** A factory makes x Mountain bicycles and y Road bicycles each day.

- The factory makes at most 60 bicycles in total each day.
- It makes at least 10 Road bicycles.
- It makes at least 15 Mountain bicycles.
- The number of Road bicycles is at most twice the number of Mountain bicycles.

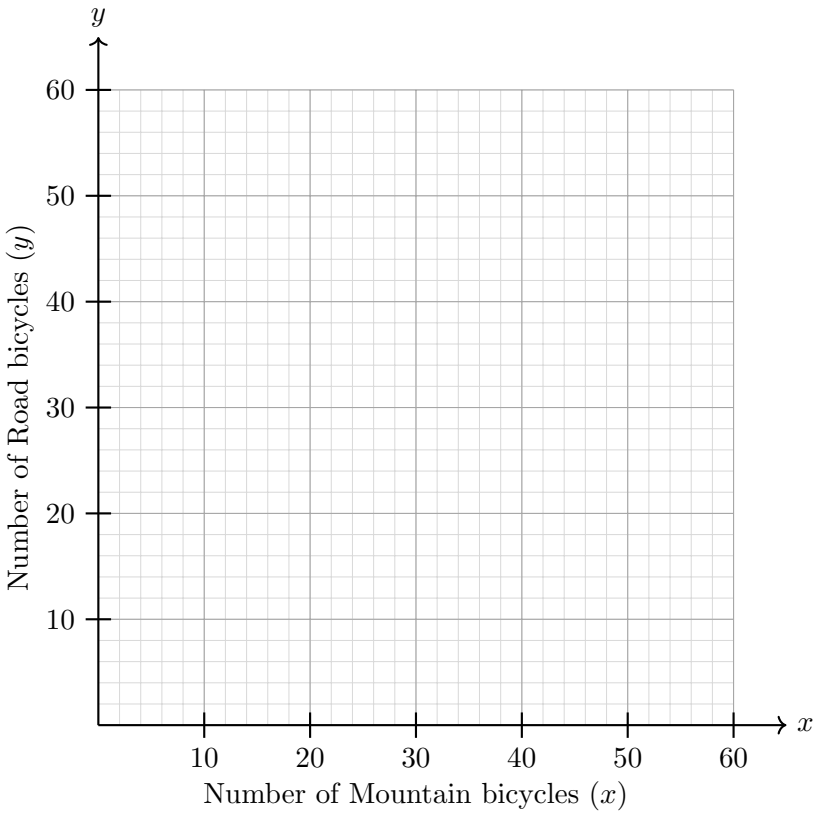
Write down four inequalities, in terms of x and y, to show this information.

.....

.....

..... **[4]**

(b) On the grid, draw four lines to show these inequalities.
By shading the unwanted regions, indicate the region R which satisfies all four inequalities.



[5]

(c) The profit on each Mountain bicycle is \$30 and the profit on each Road bicycle is \$50.
Find the maximum possible profit the factory can make in one day, and the number of each type of bicycle it must make to achieve this profit.

Maximum profit = \$ .....

Mountain bicycles = ...

Road bicycles = ... **[3]**

199 Here are the first four terms of a sequence.

5 12 19 26

(a) Write down the next term in the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

200 Here are the first four terms of a sequence.

5 11 17 23

(a) Find the next term in this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

201 **(a)** Sequence A is a linear sequence.

$$5 \quad 9 \quad 13 \quad 17 \quad \ldots$$

(i) Find the next two terms of sequence A.

..... **[1]**

(ii) Find the nth term of sequence A.

..... **[2]**

(b) Sequence B is a geometric sequence.

$$2 \quad 6 \quad 18 \quad 54 \quad \ldots$$

Find the nth term of sequence B.

..... **[2]**

202 The first five terms of a sequence, T_n, are given below.

$$4 \qquad 14 \qquad 34 \qquad 68 \qquad 120$$

(a) By finding the first, second and third differences, show that the nth term of the sequence is a cubic expression in n.

[2]

(b) Find an expression for T_n, the nth term of the sequence.

$T_n =$ ..... **[5]**

203 **(a)** The nth term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ... , 2nd term = ... , 3rd term = ...
[2]

(b) The first four terms of another sequence are

2 9 20 35

Find an expression for the nth term of this sequence.

..... **[3]**

204 The first four terms of a sequence are

$$-1 \qquad 2 \qquad 9 \qquad 20$$

(a) Find the nth term of this sequence.

..... **[3]**

(b) The nth term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

205 The first five terms of a sequence are

2 9 20 35 54 ...

(a) Find the next term.

..... **[1]**

(b) Find the nth term of the sequence.

..... **[3]**

(c) The kth term of the sequence is 464.
Find the value of k.

$k =$ ..... **[3]**

206 Consider the following three sequences, P, Q and R.
Sequence P:

$$2 \quad 9 \quad 28 \quad 65 \quad 126 \quad \ldots$$

Sequence Q:

$$3 \quad 8 \quad 15 \quad 24 \quad 35 \quad \ldots$$

Sequence R:

$$\frac{2}{3} \quad \frac{9}{8} \quad \frac{28}{15} \quad \frac{65}{24} \quad \frac{126}{35} \quad \ldots$$

(a) Find the nth term of sequence P.

..... **[2]**

(b) Find the nth term of sequence Q.

..... **[2]**

(c) Write down the nth term of sequence R.

..... **[1]**

(d) Find the 12th term of sequence R, giving your answer as a fraction in its simplest form.

..... **[2]**

207 **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the nth term of this sequence.

..... **[4]**

(b) An exponential sequence has nth term $P_n = 5 \times 3^{n-1}$.
The kth term of this sequence is 3645.
Work out the value of k.

$k =$ ..... **[2]**

(c) The nth term of sequence T is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that T_n approaches as n becomes very large.

..... **[1]**

208 y is directly proportional to the square of x.
When $x = 3$, $y = 36$.

(a) Find an equation connecting y and x.

$y =$ ..... **[2]**

(b) Find the value of y when $x = 5$.

$y =$ ..... **[1]**

209 y is directly proportional to x.
When $x = 5$, $y = 15$.

(a) Find an equation connecting y and x.

..... **[2]**

(b) Work out the value of y when $x = 12$.

$y =$ ..... **[2]**

210 y is directly proportional to $(x+2)^3$.
When $x = 1$, $y = 135$.

(a) Find an equation for y in terms of x.

$y =$ ..... **[2]**

(b) Find the value of y when $x = 0$.

$y =$ ..... **[1]**

(c) Find the value of x when $y = 320$.

$x =$ ..... **[2]**

211 P is directly proportional to the square root of Q.
Q is inversely proportional to R^3.
When $Q = 9$, $P = 12$.
When $R = 2$, $Q = 2$.

(a) Find an equation for P in terms of R.

$P =$ ..... **[5]**

(b) Calculate the value of P when $R = 4$.

$P =$ ..... **[2]**

212 y is inversely proportional to the square of x.
When $x = 4$, $y = 3$.

(a) Find an equation connecting y and x.

..... **[3]**

(b) Work out the possible values of x when $y = 0.75$.

$x =$ ..... **[3]**

213 M is inversely proportional to the square root of N.
When $N = 16$, $M = 15$.

(a) Find an equation connecting M and N.

$M =$ ..... **[2]**

(b) Find the value of M when $N = 100$.

$M =$ ..... **[2]**

(c) Find the value of N when $M = 1.2$.

$N =$ ..... **[3]**

214 P is inversely proportional to the square root of Q.
When $Q = 36$, $P = 4$.

(a) Find an equation for P in terms of Q.

$P =$ ..... **[2]**

(b) Calculate the value of P when $Q = 144$.

$P =$ ..... **[1]**

(c) Calculate the value of Q when $P = 1.5$.

$Q =$ ..... **[2]**

(d) The value of Q is increased by 300%.
Calculate the percentage decrease in the value of P.

..... % **[3]**

215 **(a)** Some corresponding values of x and y are given below.

$$x = 2,\ y = 75 \qquad x = 4,\ y = 18.75 \qquad x = 10,\ y = 3$$

Identify which of the following models best fits the data:

$$y \propto x \qquad y \propto x^2 \qquad y \propto \frac{1}{x} \qquad y \propto \frac{1}{x^2}$$

Hence, find the equation for y in terms of x.

$y =$ ..... **[3]**

(b) P is directly proportional to the cube of Q.
When Q is increased by 50%, work out the percentage increase in P.

..... % **[3]**

216 **(a)** The table shows some values of x and y.

x	2	4	5	8
y	32	8	5.12	2

(i) Identify the variation model and write down the equation for y in terms of x.

$y =$ ..... **[3]**

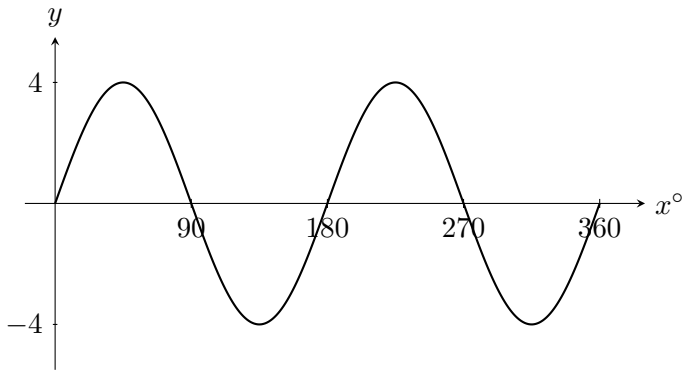
(ii) Calculate the value of y when $x = 10$.

$y =$ ..... **[1]**

(b) A is directly proportional to the cube of B.
When B is increased by $x\%$, the value of A increases by 72.8%.
Find the value of x.

$x =$ ..... **[3]**

217 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



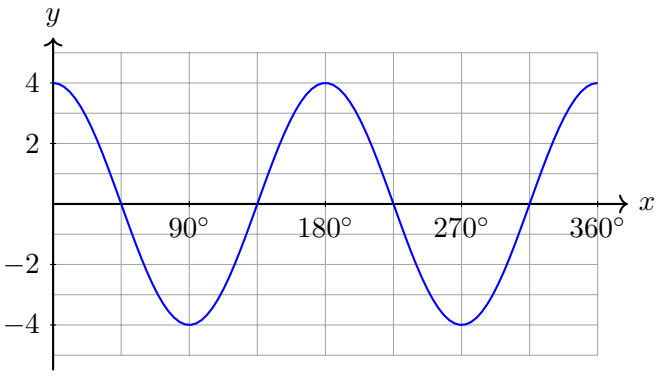
(a) Find the value of a.

$a =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[1]**

218 The diagram shows a sketch of the graph of $y = a\cos(bx)$ for $0° \leq x \leq 360°$.



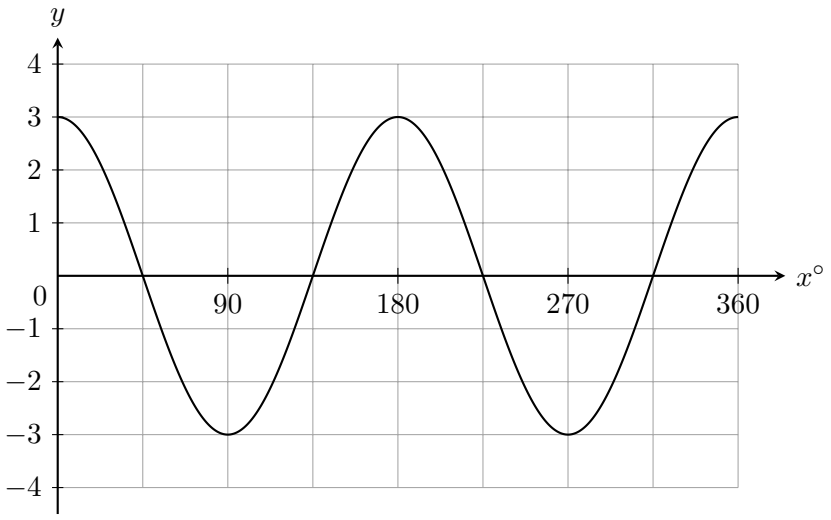
(a) Write down the value of a.

$a =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[2]**

219 The diagram shows the graph of the trigonometric function $f(x) = a\cos(bx)$ for $0° \leq x \leq 360°$.



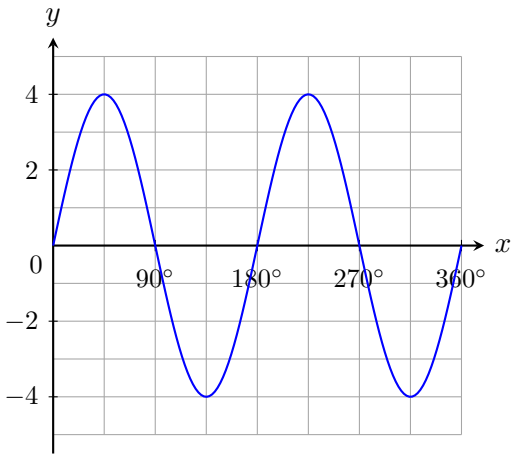
(a) Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the coordinates of the first minimum point where $x > 0$.

(..... , ...) **[1]**

220 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



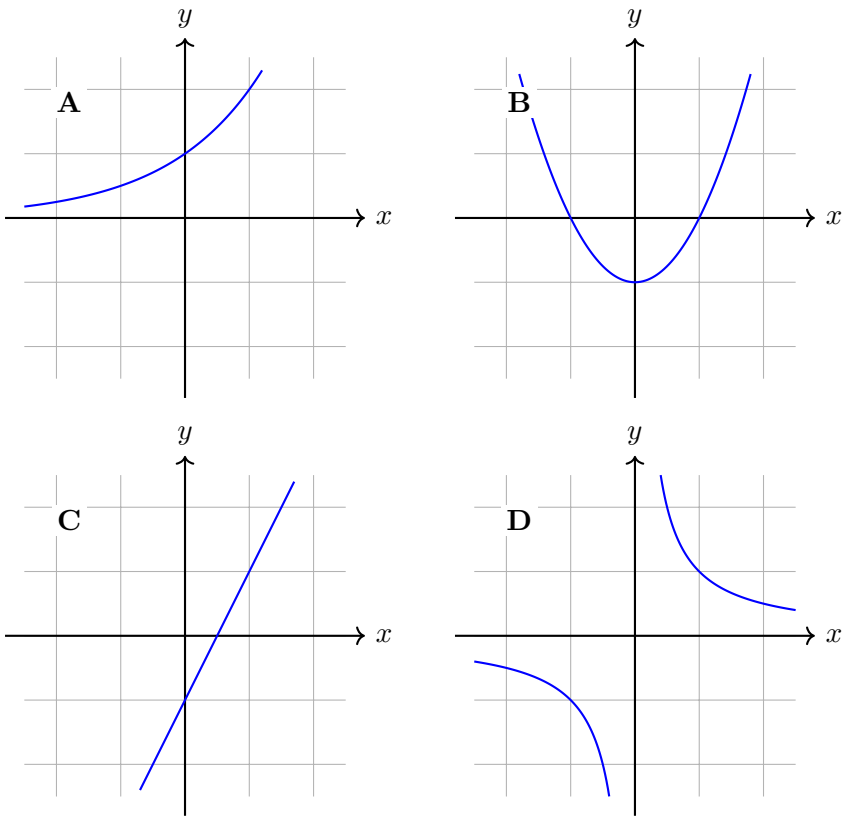
(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the period of the function.

..... ° **[1]**

221 Four graphs are shown below, labelled A, B, C, and D.



(a) Write down the letter of the graph that represents each function.

(i)

$$y = x^2 - 1$$

..... **[1]**

(ii)

$$y = 2^x$$

..... **[1]**

(iii)

$$y = \frac{1}{x}$$

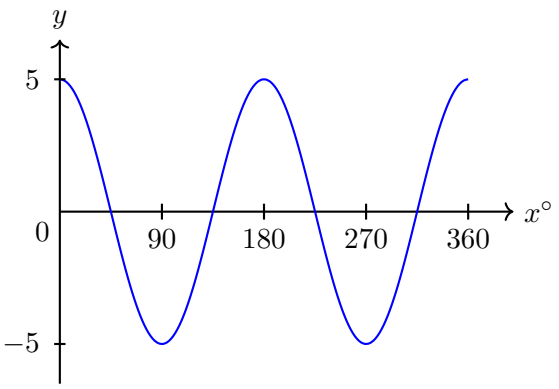
..... **[1]**

(iv)

$$y = 2x - 1$$

..... **[1]**

222 The diagram shows the graph of $y = a\cos(bx)$ for $0° \leq x \leq 360°$.



(a) Write down the amplitude of the graph.

..... **[1]**

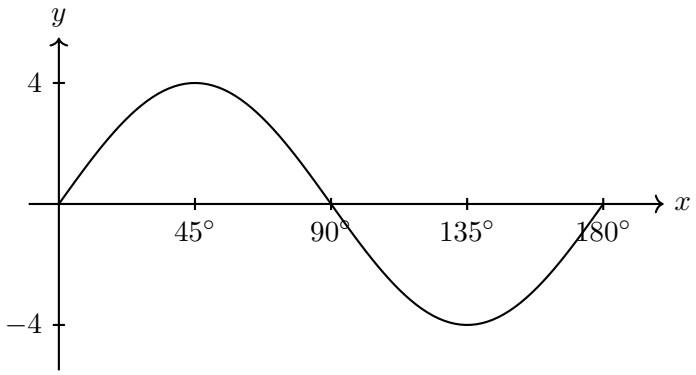
(b) Write down the period of the graph.

..... ° **[1]**

(c) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

223 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 180°$.



(a) State the amplitude.

..... **[1]**

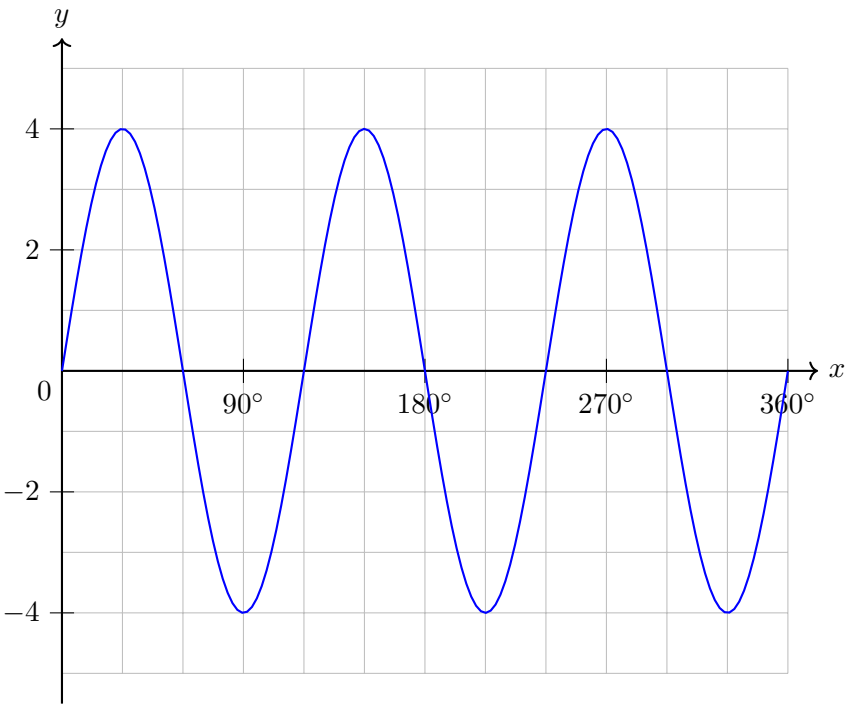
(b) State the period of the function.

..... ° **[1]**

(c) Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

224 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.
The values of a and b are positive integers.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

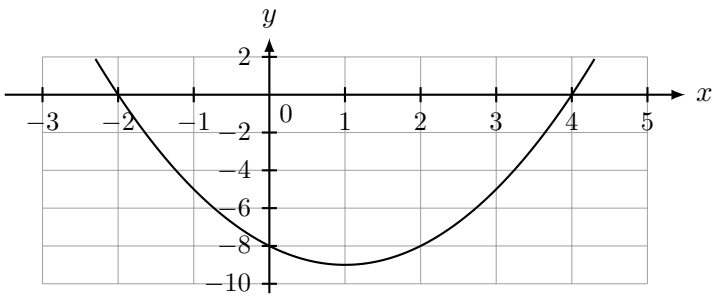
(b) Write down the period of the function.

..... ° **[1]**

(c) State the number of solutions to the equation $a\sin(bx) = 2$ in the interval $0° \leq x \leq 360°$.

..... **[1]**

225 The diagram shows the graph of the function $y = x^2 + bx + c$.



(a) Write down the value of c.

$c =$ ..... **[1]**

(b) Find the value of b.

$b =$ ..... **[2]**

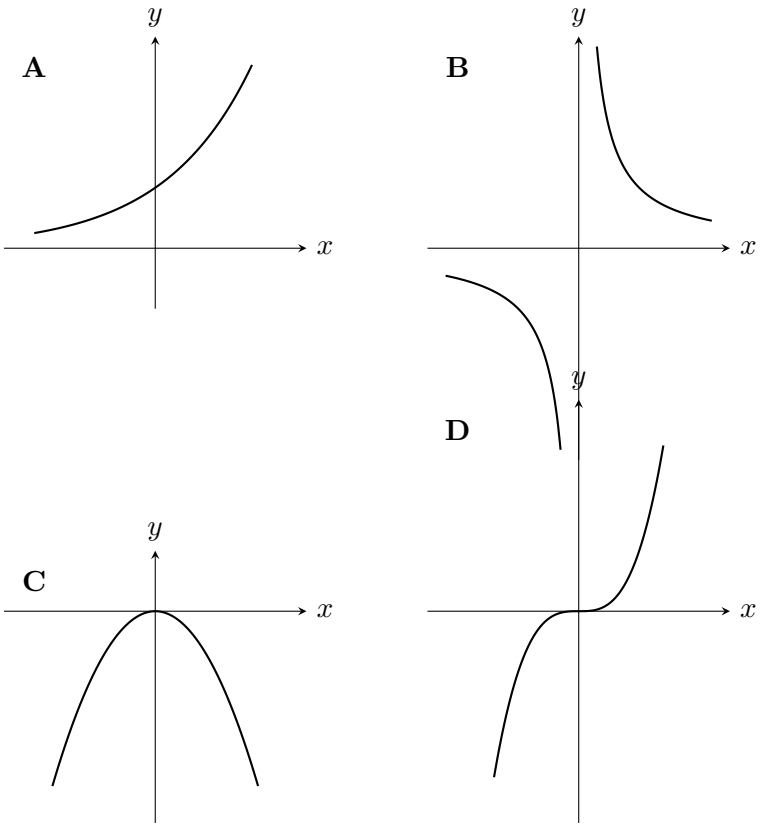
(c) Write down the coordinates of the minimum point of the graph.

(..... , ...) **[2]**

226 Six equations are shown below.

$$y = 2^x \qquad y = x^2 \qquad y = -x^2 \qquad y = x^3 \qquad y = \frac{1}{x} \qquad y = \cos x$$

The diagrams show four of these graphs, labelled A, B, C and D.

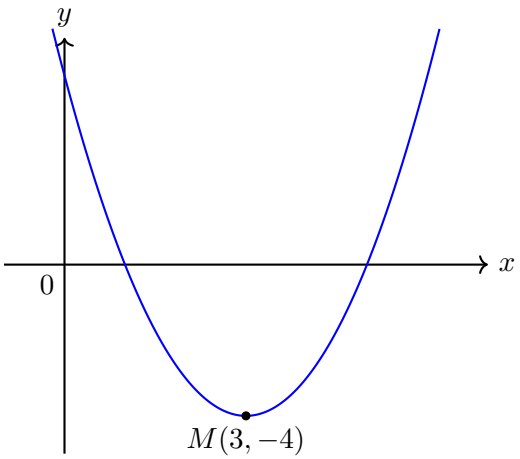


Write down the correct equation for each graph.

$A =$ ..... , $B =$ ..... , $C =$ ..... , $D =$ .....

[4]

227 The diagram shows a sketch of the graph of $y = x^2 + bx + c$.
The minimum point on the graph is $M(3, -4)$.



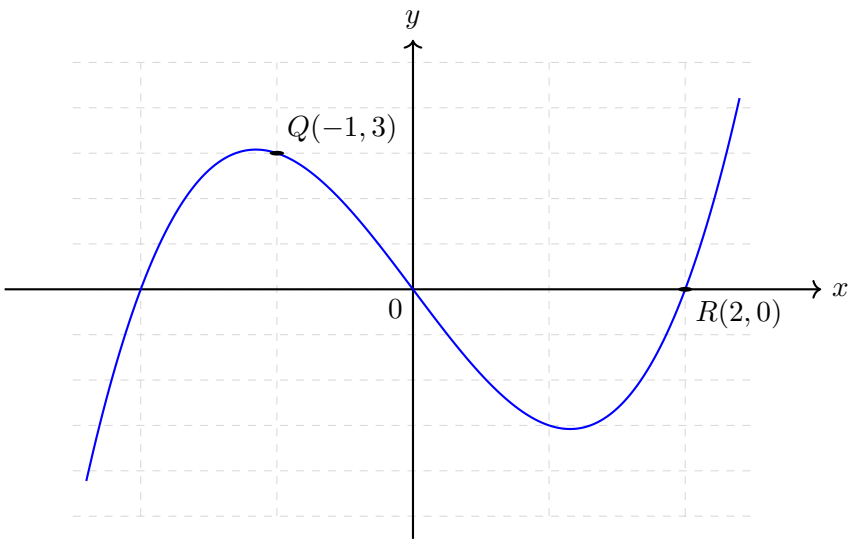
(a) Work out the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

(b) Find the coordinates of the points where the graph crosses the x-axis.

..... **[2]**

228 The diagram shows a sketch of the curve with equation $y = x^3 + ax^2 + bx$.
The curve passes through the points $Q(-1, 3)$ and $R(2, 0)$.



(a) Use the coordinates of Q and R to form two simultaneous equations in terms of a and b.

.....

..... **[2]**

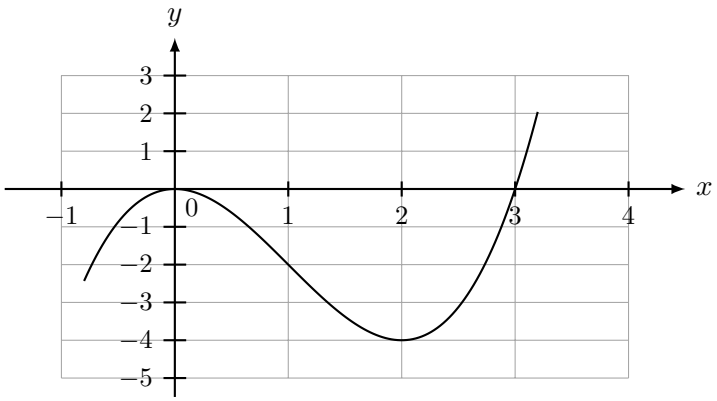
(b) Solve your equations to find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(c) Find the coordinates of the other two points where the curve crosses the x-axis.

..... **[3]**

229 The diagram shows the graph of $y = f(x)$ where $f(x) = ax^3 + bx^2$.
The graph passes through the origin $(0, 0)$, the point $(3, 0)$ and has a local minimum point at $(2, -4)$.



(a) Show that $3a + b = 0$.

[1]

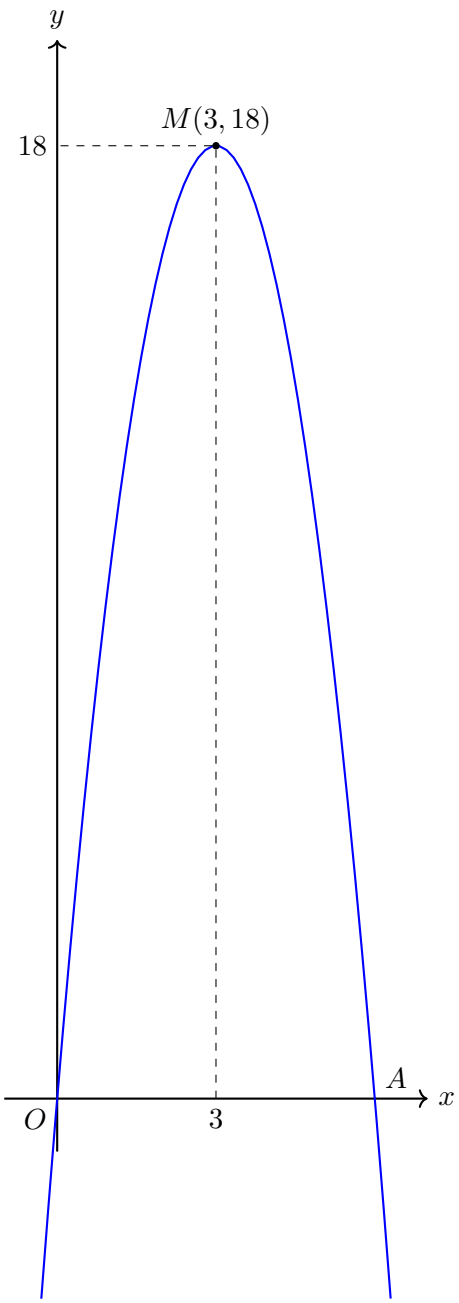
(b) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(c) Write down the range of values of k for which the equation $f(x) = k$ has exactly one solution.

..... **[2]**

230 The diagram shows a sketch of the graph of $y = f(x)$, where $f(x) = ax^2 + bx + c$.
The curve passes through the origin O and intersects the x-axis at the point A.
The point $M(3, 18)$ is the maximum point of the curve.



(a) Write down the value of c.

$c =$ ..... **[1]**

(b) Write down the coordinates of A.

(..... , ...) **[1]**

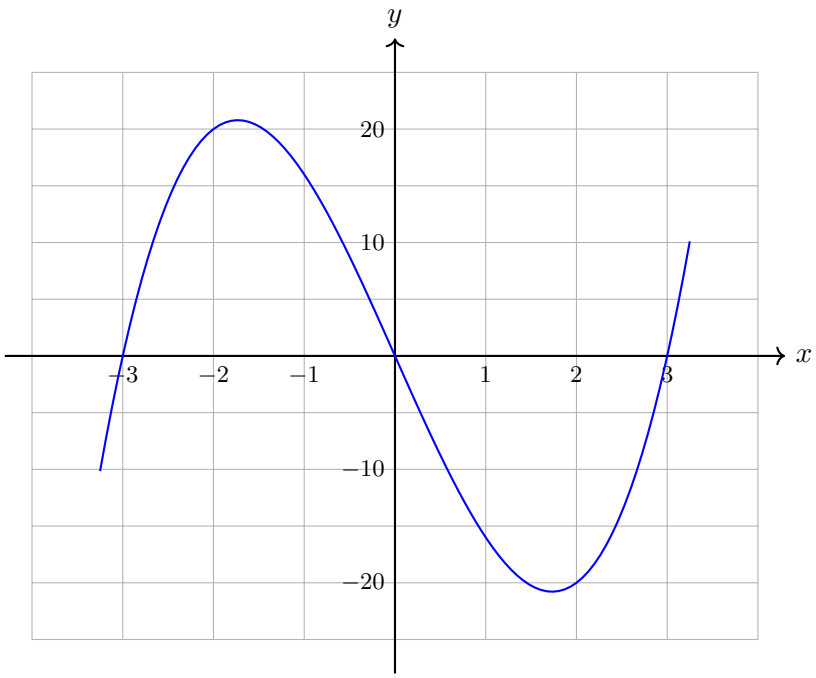
(c) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(d) Find the range of values of k for which the equation $f(x) = k$ has two distinct positive real roots.

..... **[2]**

231 The diagram shows the graph of the cubic function $y = ax^3 + bx^2 + cx + d$.



(a) Write down the value of d.

$d =$ ..... **[1]**

(b) The graph intersects the x-axis at $x = -3$ and $x = 3$.
Show that $b = 0$ and $9a + c = 0$.

[3]

(c) The graph passes through the point $(1, -16)$.
Find the value of a and the value of c.

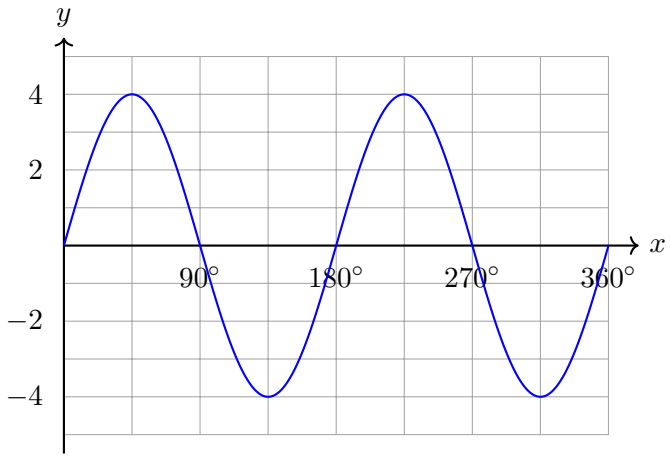
$a =$ ..... , $c =$ ..... **[3]**

(d) A different function $g(x) = px^2 + qx + r$ has a minimum point at $(0, -5)$ and passes through the point $(2, 7)$.

Find the values of p, q and r.

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

232 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



(a) Write down the value of a.

$a =$ ..... **[1]**

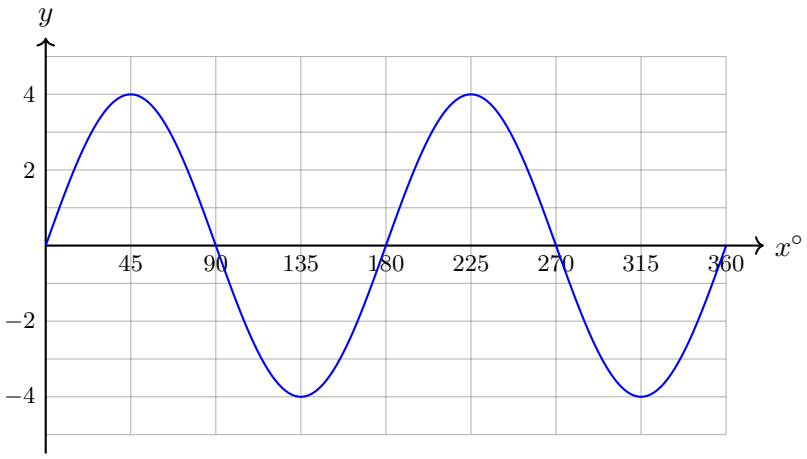
(b) Find the value of b.

$b =$ ..... **[1]**

(c) Calculate the value of y when $x = 30°$.
Give your answer correct to 3 significant figures.

$y =$ ..... **[2]**

233 The diagram shows the graph of $y = a\sin(bx)$ for $0° \leq x \leq 360°$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

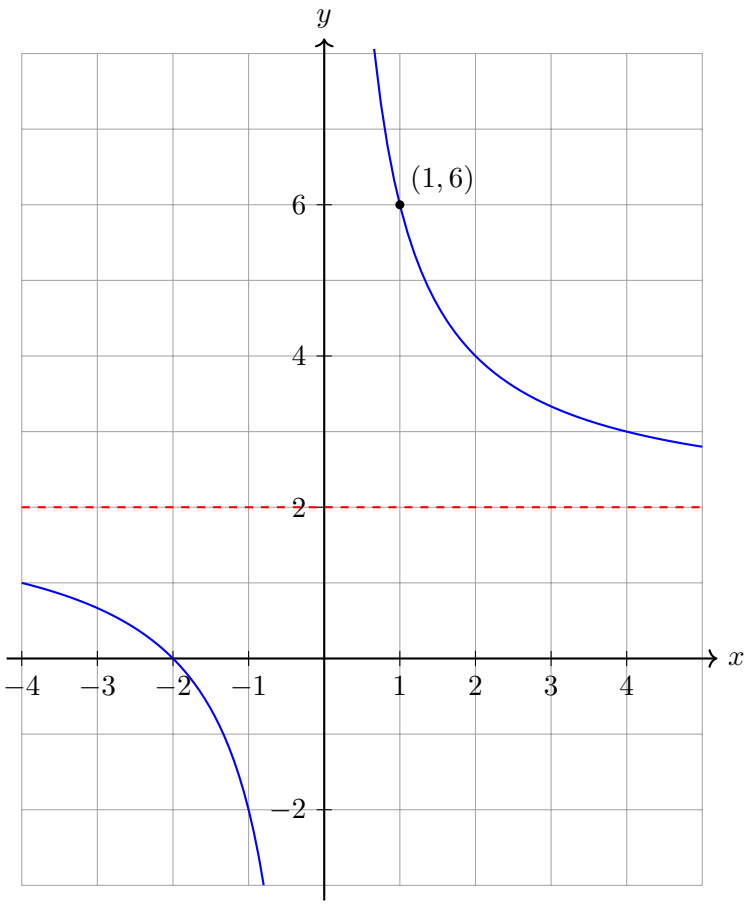
(b) The graph of $y = p^x$ passes through the point $(3, 125)$.
Find the value of p.

$p =$ ..... **[1]**

(c) The graph of $y = \dfrac{k}{x}$ passes through the point $(0.5, 8)$.
Find the value of k.

$k =$ ..... **[1]**

234 The diagram shows the graph of $y = \frac{a}{x} + b$.
The equation of the horizontal asymptote is $y = 2$.
The graph passes through the point $(1, 6)$.



(a) Write down the value of b.

$b =$ ..... **[1]**

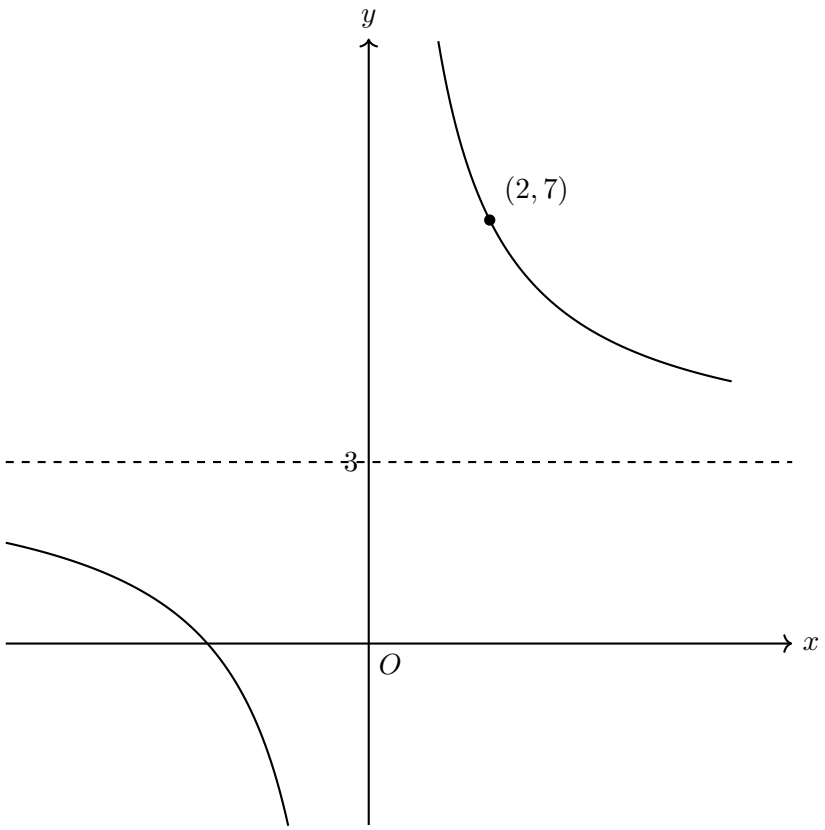
(b) Find the value of a.

$a =$ ..... **[2]**

(c) The graph of $y = k^x$, where $k > 0$, also passes through the point $(2, 4)$, which lies on the curve $y = \frac{a}{x} + b$.
Find the value of k.

$k =$ ..... **[2]**

235 The diagram shows a sketch of the graph of $f(x) = \frac{a}{x} + b$, where $xe0$.
The dashed line is a horizontal asymptote.
The graph passes through the point $(2, 7)$.



(a) Write down the value of b.

$b =$ ..... **[1]**

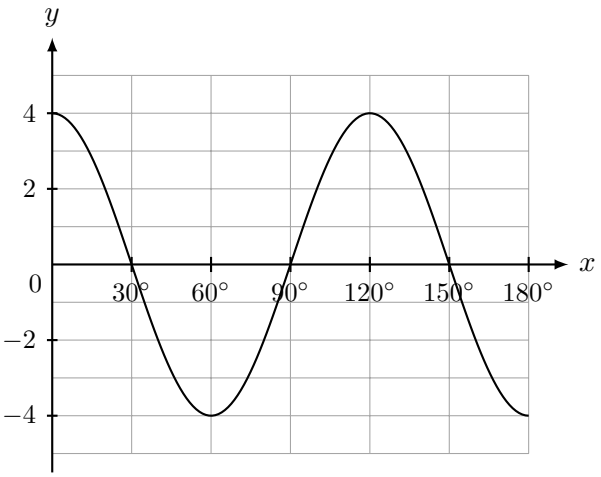
(b) Find the value of a.

$a =$ ..... **[2]**

(c) Find the coordinates of the point where the graph crosses the x-axis.

(..... , ...) **[2]**

236 The diagram shows a sketch of the graph of $f(x) = a\cos(bx)$ for $0° \leq x \leq 180°$.



(a) Write down the value of a.

$a =$ ..... **[1]**

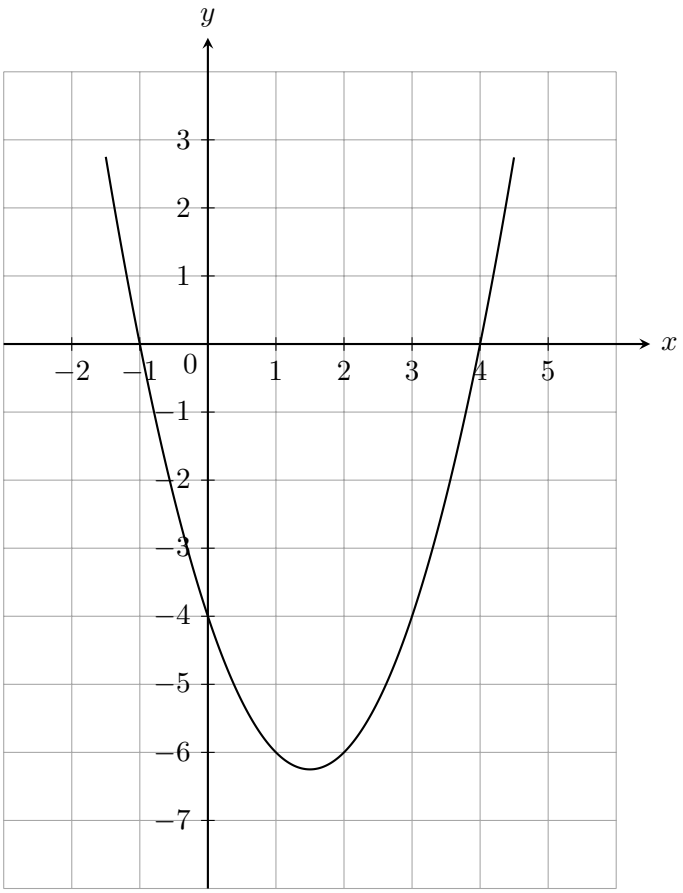
(b) Find the value of b.

$b =$ ..... **[2]**

(c) Determine the number of solutions to the equation $a\cos(bx) = 1.5$ in the interval $0° \leq x \leq 360°$.

..... **[2]**

237 The diagram shows the graph of a quadratic function $f(x) = ax^2 + bx + c$.
The graph crosses the y-axis at $(0, -4)$ and the x-axis at $(-1, 0)$ and $(4, 0)$.



(a) Write down the value of c.

$c =$ ..... **[1]**

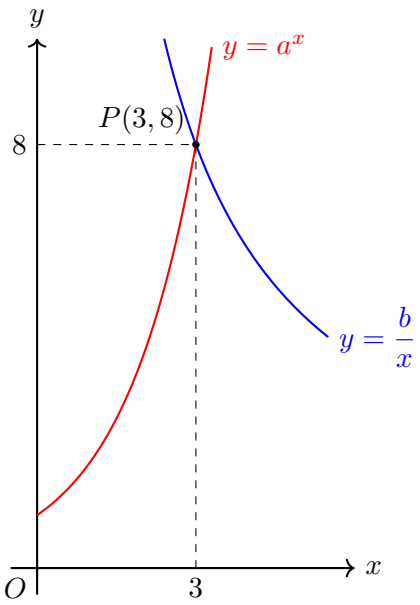
(b) Determine the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(c) Calculate the exact coordinates of the minimum point of the graph.

(..... , ...) **[2]**

238 The diagram shows a sketch of the graphs of $y = a^x$ and $y = \dfrac{b}{x}$ for $x > 0$.
The graphs intersect at the point $P(3, 8)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

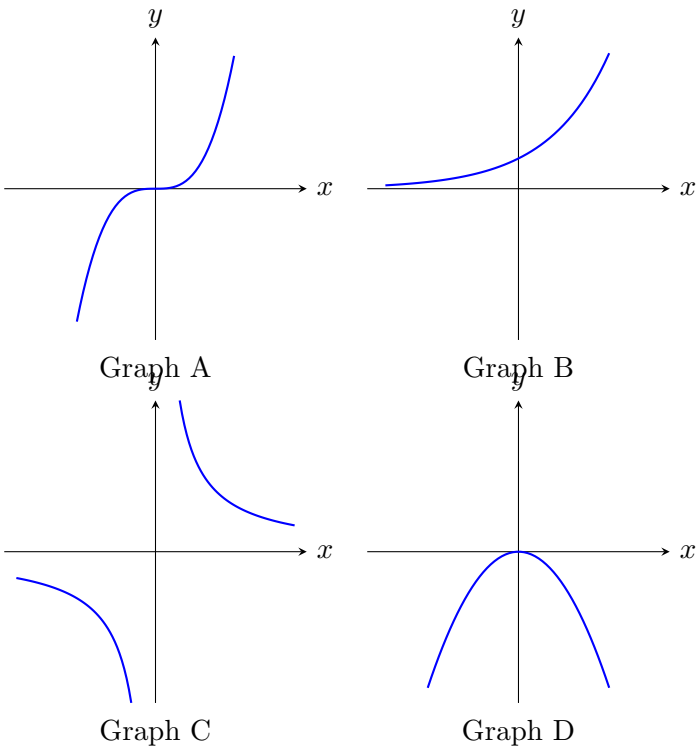
(b) The graph of $y = a^x$ intersects the y-axis at the point Q.
Write down the coordinates of Q.

(..... , ...) **[1]**

(c) Calculate the gradient of the straight line joining Q and P.

Gradient = ... **[2]**

239 Four graphs A, B, C and D are shown below.



(a) Write down the letter of the graph which could represent the function

(i) $f(x) = \dfrac{a}{x}$ where $a > 0$

..... **[1]**

(ii) $f(x) = a^x$ where $a > 1$

..... **[1]**

(iii) $f(x) = -ax^2$ where $a > 0$

..... **[1]**

(b) The equation of Graph B is $y = p^x$.
The graph passes through the point $(3, 125)$.
Find the value of p.

$p =$ ..... **[2]**

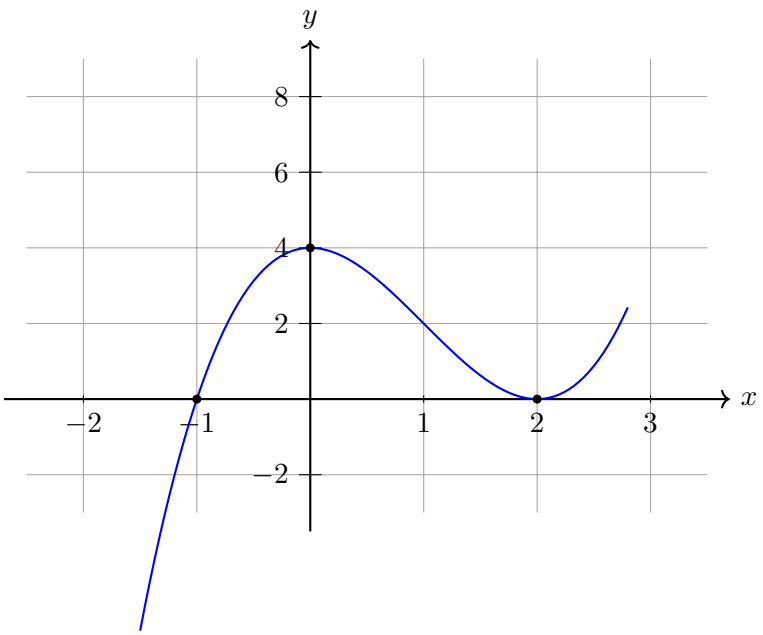
(c) The equation of Graph C is $y = \dfrac{q}{x}$.

The graph passes through the point $(0.5, 8)$.

Find the value of q.

$q =$ ..... **[2]**

240 The diagram shows a sketch of the curve $f(x) = x^3 + ax^2 + bx + c$.
The curve crosses the y-axis at $(0, 4)$.
It crosses the x-axis at $(-1, 0)$ and touches the x-axis at $(2, 0)$.



(a) Write down the value of c.

$c =$ ..... **[1]**

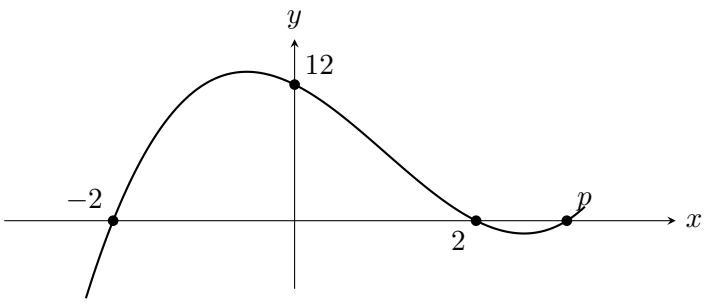
(b) By using the x-intercepts, find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[4]**

(c) Use the graph to state the number of solutions to the equation $f(x) = 3$.

..... **[1]**

241 The diagram shows the graph of $y = x^3 + bx^2 + cx + d$.
The graph crosses the axes at $(-2, 0)$, $(2, 0)$, $(p, 0)$ and $(0, 12)$.



(a) Write down the value of d.

$d =$ ..... **[1]**

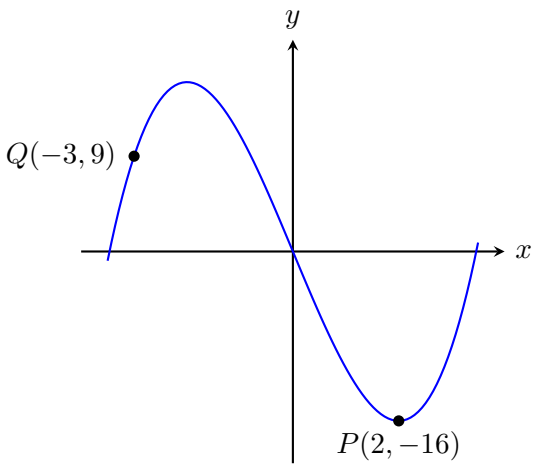
(b) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[4]**

(c) Find the value of p.

$p =$ ..... **[2]**

242 The diagram shows a sketch of the graph of $f(x) = ax^3 + bx$.
The curve passes through the points $P(2, -16)$ and $Q(-3, 9)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

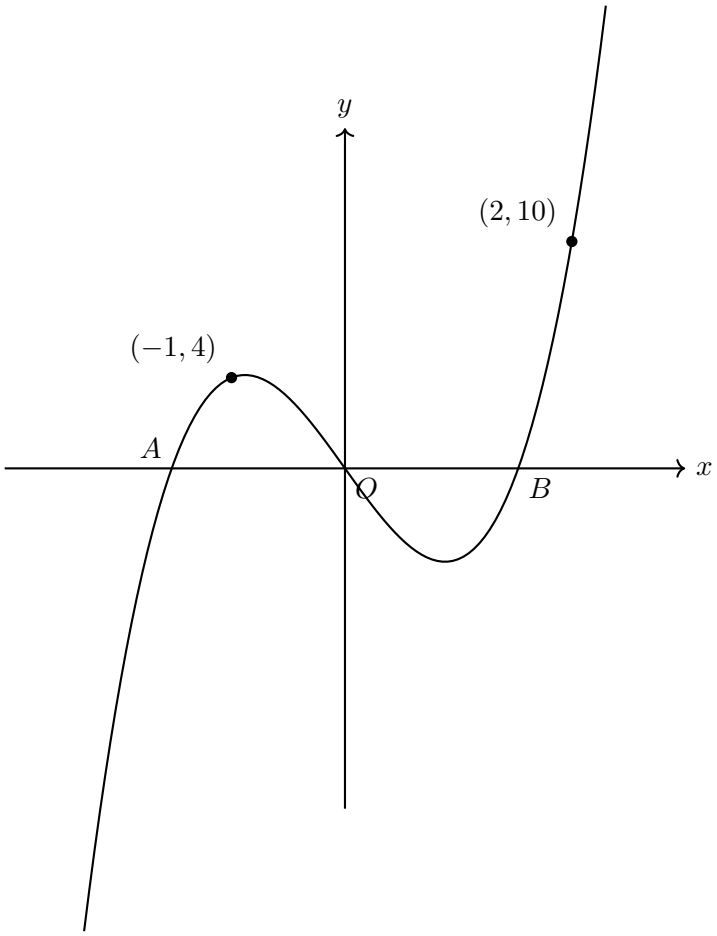
(b) Find the coordinates of the local maximum point of the graph.

(..... , ...) **[2]**

(c) The equation $f(x) = k$ has exactly one solution.
Write down the range of values for k.

..... **[2]**

243 The diagram shows a sketch of the curve $y = ax^3 + bx$.
The curve passes through the origin O, and the points $(2, 10)$ and $(-1, 4)$.
The curve crosses the x-axis at the points A, O, and B.



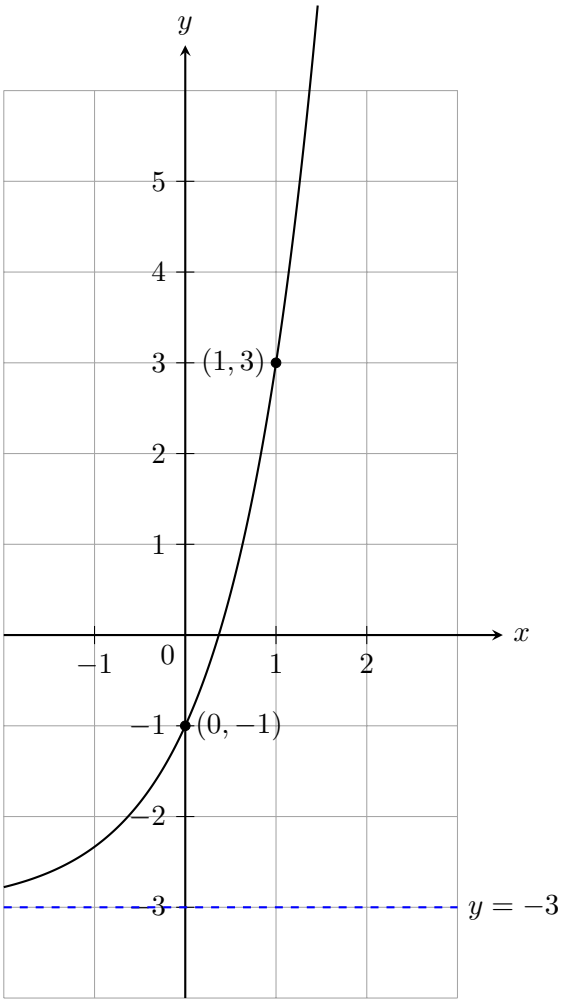
(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[4]**

(b) Calculate the exact length of the line segment AB.

..... **[3]**

244 The diagram shows the graph of the exponential function $f(x) = a \cdot b^x + c$.
The horizontal line $y = -3$ is an asymptote to the graph.
The graph passes through the points $(0, -1)$ and $(1, 3)$.



(a) State the value of c.

$c =$ ..... **[1]**

(b) Work out the value of a and the value of b.

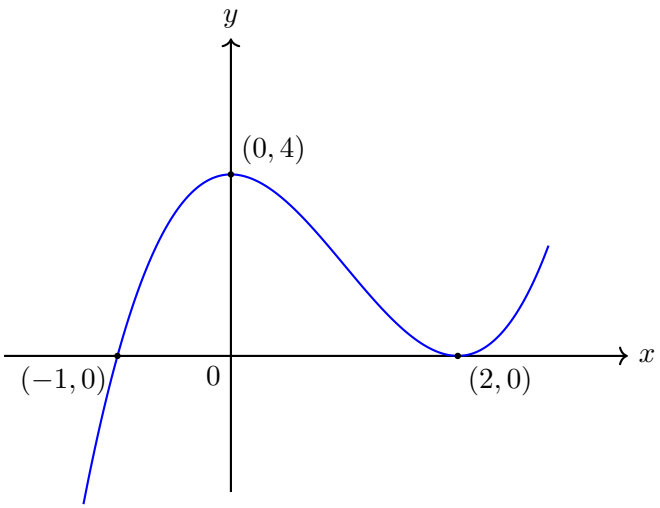
$a =$ ..... , $b =$ ..... **[4]**

(c) The graph of $g(x) = \dfrac{k}{x}$ intersects the graph of $f(x)$ at the point where $x = 1$.

Find the value of k.

$k =$ ..... **[2]**

245 The diagram shows a sketch of the graph of $f(x) = x^3 + bx^2 + cx + 4$.
The graph crosses the x-axis at $(-1, 0)$ and touches the x-axis at $(2, 0)$.
The graph crosses the y-axis at $(0, 4)$.



(a) **(i)** Use the point $(-1, 0)$ to show that $b - c = -3$.

[2]

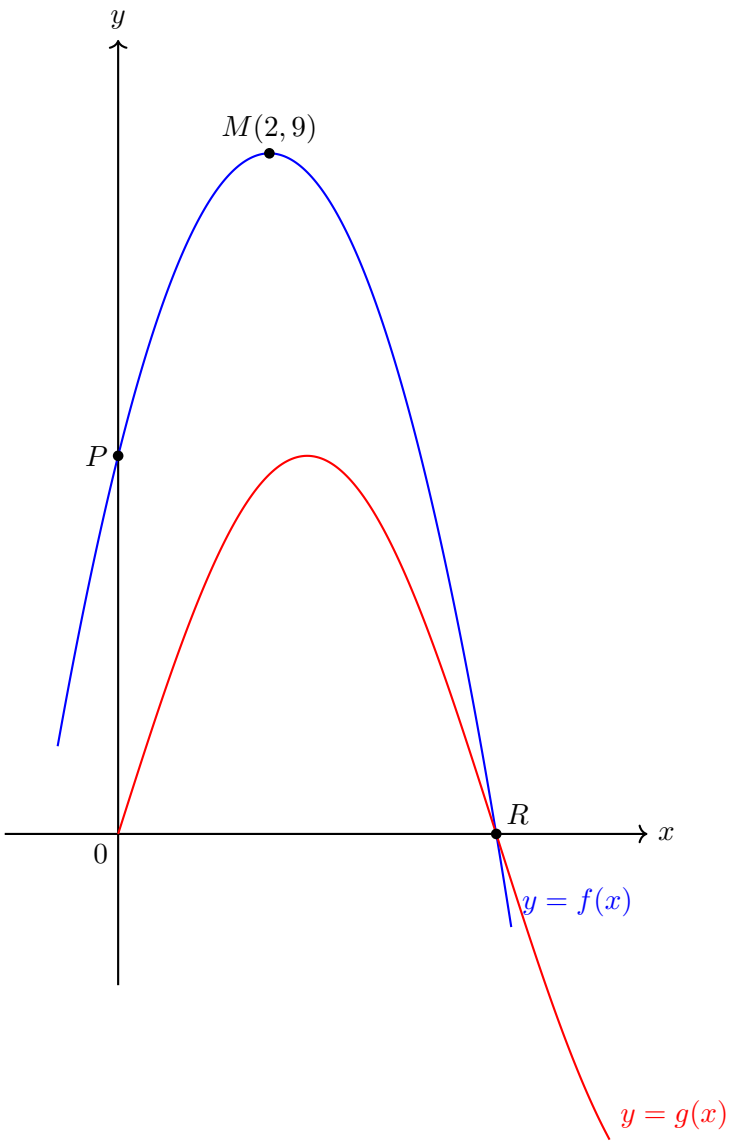
(ii) Find the values of b and c.

$b =$ ..... , $c =$ ..... **[4]**

(b) The graph of $g(x) = p \cdot q^x$ passes through the points $(0, 2)$ and $(3, 54)$.
Given that $q > 0$, work out the values of p and q.

$p =$ ..... , $q =$ ..... **[3]**

246 The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ for $x \geq 0$.
It is given that $f(x) = px^2 + qx + r$.
The graph of $y = f(x)$ has a maximum point at $M(2, 9)$ and intersects the y-axis at $P(0, 5)$.



(a) Find the values of p, q, and r.

$p =$ ..... , $q =$ ..... , $r =$ ..... **[4]**

(b) The graph of $y = f(x)$ intersects the positive x-axis at the point R.
Find the coordinates of R.

(..... , ...) **[2]**

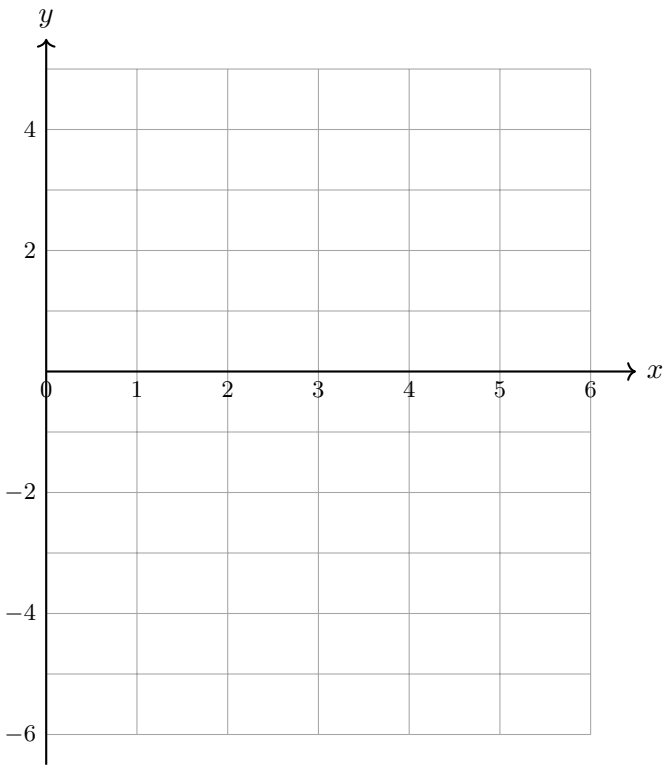
(c) It is given that $g(x) = a\sin(bx^\circ)$, where $a > 0$ and $b > 0$.
The graph of $y = g(x)$ has its first positive x-intercept at the point R.
The amplitude of $g(x)$ is equal to the y-coordinate of P.

Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

247 Let $f(x) = -x^2 + 6x - 5$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $0 \leq x \leq 6$.



[2]

(b) Find the zeros of $f(x)$.

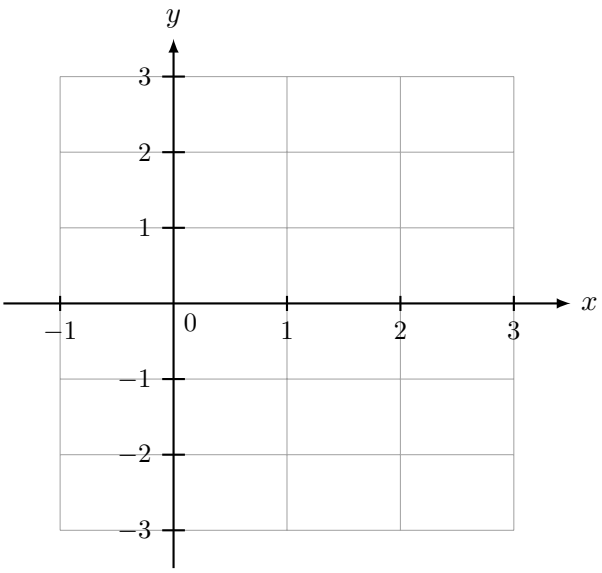
$x =$ ..... , $x =$ ..... **[2]**

(c) Find the coordinates of the vertex of the graph.

(..... , ...) **[2]**

248 Consider the function $f(x) = x^3 - 3x^2 + 2$ for $-1 \leq x \leq 3$.

(a) On the grid, sketch the graph of $y = f(x)$.



[2]

(b) Find the zeros of $f(x)$.

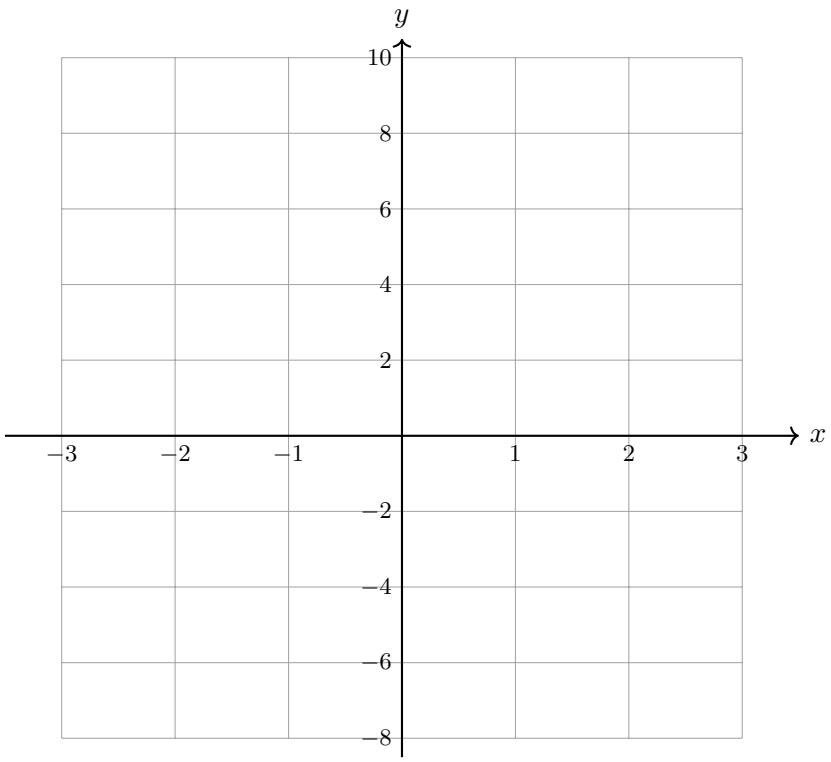
$x =$ ..... **[2]**

(c) Find the coordinates of the local minimum point of the graph.

(..... , ...) **[2]**

249 Let $f(x) = x^3 - 3x + 1$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.



[2]

(b) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

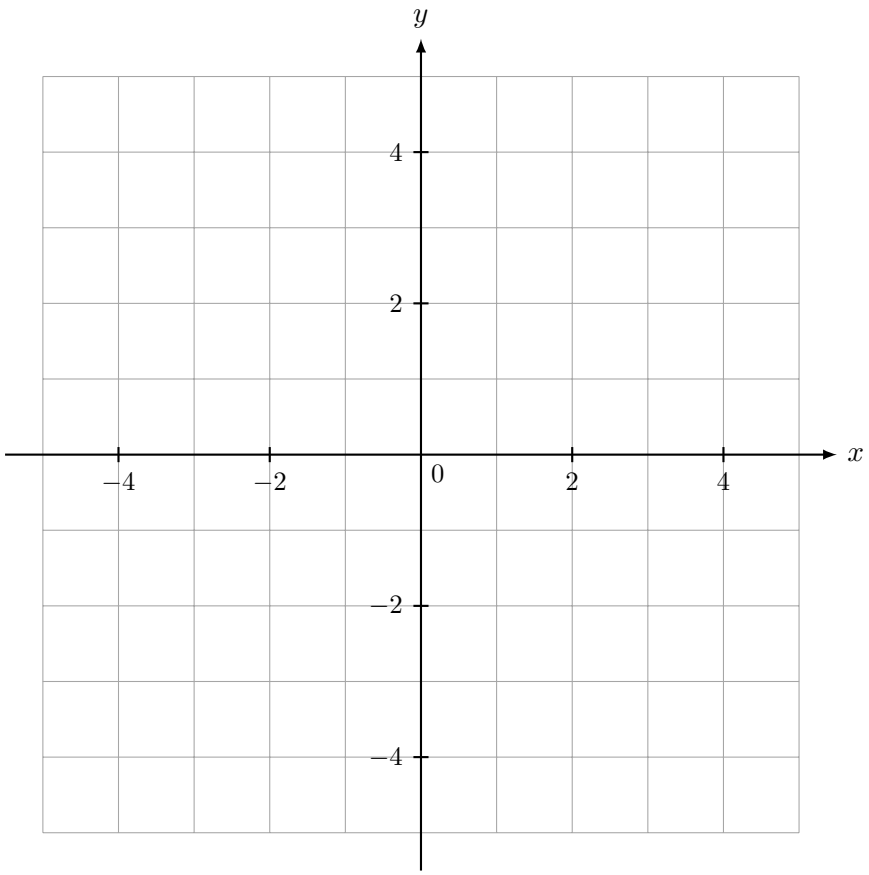
(c) On the same axes, sketch the graph of $y = 2x - 1$. **[1]**

(d) Find the x-coordinates of the points of intersection of the two graphs.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[2]**

250 Let $f(x) = \dfrac{2}{x+1}$ and $g(x) = x - 2$ for $-5 \leq x \leq 5$.

(a) On the grid, sketch the graphs of $y = f(x)$ and $y = g(x)$.



[3]

(b) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

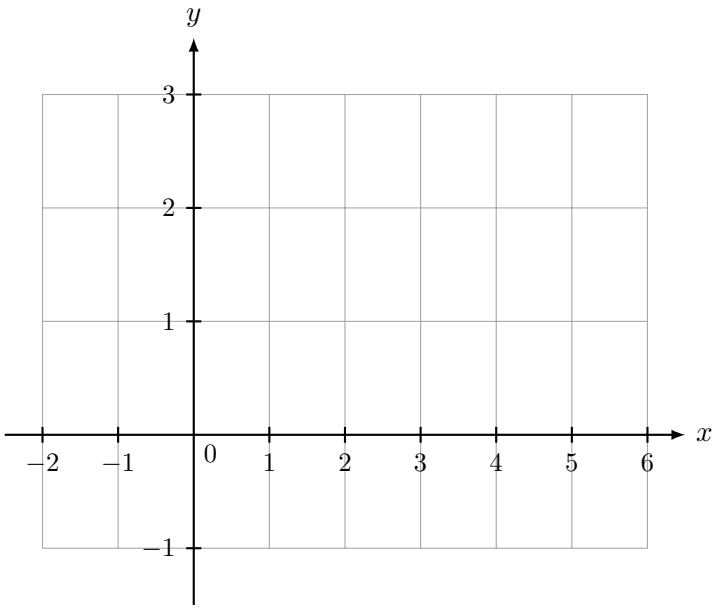
..... **[1]**

(c) Find the coordinates of the points of intersection of the two graphs.

Point 1 = ... , Point 2 = ... **[3]**

251 The function $f(x)$ is defined as $f(x) = \dfrac{x^2}{2^x}$ for the domain $-1 \leq x \leq 5$.

(a) On the grid, sketch the graph of $y = f(x)$.



[3]

(b) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

(c) Find the x-coordinates of the points where $f(x) = 0.5$.

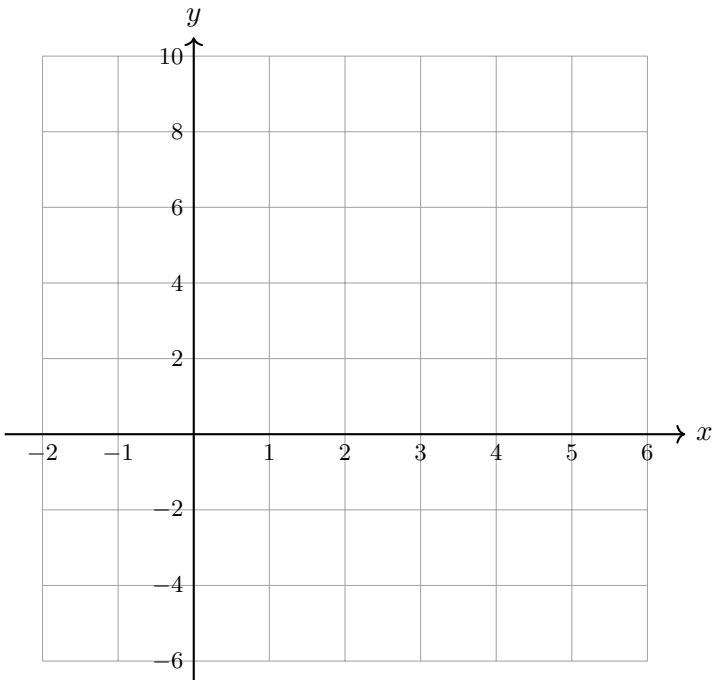
$x =$ ..... **[2]**

(d) Write down the range of values of k for which the equation $f(x) = k$ has exactly 3 different solutions.

..... **[2]**

252 Let $f(x) = \dfrac{x^2 - 2x + 4}{x - 2}$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 6, x \neq 2$.



[3]

(b) Write down the equation of the vertical asymptote.

..... **[1]**

(c) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

(d) Find the coordinates of the local minimum point.

(..... , ...) **[2]**

(e) The equation $f(x) = k$ has no real solutions.
Write down the range of values of k.

..... **[2]**

253 Functions f and g are defined as follows.

$f(x) = 2x - 5$, $g(x) = x^2 + 1$

(a) Find the value of $f(g(3))$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

254 It is given that $f(x) = 3x - 2$ and $g(x) = \dfrac{10}{x}$.

(a) Find $f(4)$.

..... **[1]**

(b) Find x when $g(x) = 2$.

$x =$ ..... **[1]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

255 The functions f and g are defined as $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

(a) Work out

(i)

hspace*0.8cm $f(-4)$

..... **[1]**

(ii)

hspace*0.8cm $gf(2)$

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

256 The functions f and g are defined as $f(x) = 5x - 2$, $g(x) = x^2 + 3$.

(a) Find the value of $f(4)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $fg(2)$.

..... **[2]**

257 Consider the functions $f(x) = 3x - 2$ and $g(x) = x^2 + 4$.

(a) Find $f(4)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $gf(2)$.

..... **[2]**

258 Let $f(x) = 5 - 2x$, $g(x) = x^2 - 3$.

(a) Work out $f(-4)$.

..... **[1]**

(b) Work out $gf(2)$.

..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

259 $f(x) = 4x - 3$, $g(x) = 2 - 5x$

(a) Work out $f(2)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Work out $gf(1)$.

..... **[2]**

260 Let $f(x) = 4x - 3$ and $g(x) = 2x^2 + 1$.

(a) Work out $fg(3)$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Solve the equation

$$f(x) = 17$$

$x =$ ..... **[2]**

261 Two functions, f and g, are defined as follows:

$$f(x) = 3x - 2 \qquad g(x) = x^2 + 1$$

(a) Work out $f(4)$.

..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

262 The functions f and g are defined as $f(x) = \dfrac{3x+2}{x-1}$, $x \neq 1$, $g(x) = x - 2$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[4]**

263 Consider the functions $h(x) = \dfrac{12}{x-1}$ and $k(x) = 2x + 3$.

(a) State the value of x which must be excluded from the domain of $h(x)$.

$x =$ ..... **[1]**

(b) Find $k^{-1}(x)$.

$k^{-1}(x) =$ ..... **[2]**

(c) Find $hk(x)$, giving your answer in its simplest form.

$hk(x) =$ ..... **[2]**

(d) Solve the equation.

$h(x) = k(2)$

$x =$ ..... **[3]**

264 Let $f(x) = \dfrac{4}{2x-3}$ for $x \neq 1.5$, and $g(x) = x + 5$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation $f(x) = g(-3)$.

$x =$ ..... **[3]**

(c) Express $gf(x)$ as a single fraction in its simplest form.

$gf(x) =$ ..... **[3]**

265 The functions f and g are defined as $f(x) = \dfrac{x+3}{x-1}$, where $x \neq 1$, and $g(x) = 2x - 5$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Show that $ff(x) = x$.

[3]

(c) Solve the equation $g(x^2) = f(3)$.

$x =$ ..... **[4]**

266 $f(x) = \dfrac{3x+2}{x-1}$, $g(x) = \dfrac{2}{x+1}$

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Find $fg(x)$, giving your answer as a single fraction in its simplest form.

..... **[3]**

(c) Solve the equation.

$f(x) = f^{-1}(x)$

Give your answers in the form $a \pm \sqrt{b}$, where a and b are integers.

$x =$ ..... **[4]**

267 The functions f and g are defined as follows:

$f(x) = \dfrac{3x+5}{x-1}, x \neq 1$, and $g(x) = x^2 + 1$

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(b) Solve the equation $f(x) = x$.

$x =$ ..... , $x =$ ..... **[4]**

(c) Find $gf(2)$.

$gf(2) =$ ..... **[2]**

268 The functions h and k are defined as $h(x) = \dfrac{2x+1}{3}$ and $k(x) = 5 - x$.

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

(b) Find an expression for $hk(x)$ in its simplest form.

$hk(x) =$ ..... **[2]**

(c) Solve the equation $h(x) = k(x)$.

$x =$ ..... **[3]**

269 Functions h and k are defined as follows.

$h(x) = \dfrac{4}{x-3}$ for $x \neq 3$

$k(x) = 2x + 1$

(a) Find $k(h(5))$.

..... **[2]**

(b) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[2]**

(c) Solve the equation $h(k(x)) = 2$.

$x =$ ..... **[3]**

270 The functions f, g and h are defined as follows:
$f(x) = 2x + 3$, $g(x) = x^2 - 4$, $h(x) = 5^x$

(a) Find $h(-1)$ as a fraction in its simplest form.

..... **[1]**

(b) Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[3]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(d) Find the value of x when $h(x) = 125$.

$x =$ ..... **[1]**

271 Two functions, f and g, are defined as follows:

$$f(x) = \frac{5}{x-2} \qquad g(x) = x + 2$$

(a) State the value of x that must be excluded from the domain of $f(x)$.

$x =$ ..... **[1]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation $f(x) = g(x)$.

$x =$ ..... , $x =$ ..... **[4]**

272 $f(x) = 2x - 3$, $g(x) = x^2 - x - 7$

(a) Work out $g(-2)$.

..... **[2]**

(b) Find $gf(x)$ in its simplest form.

..... **[3]**

(c) Solve the equation.

$f(x) = g(x)$

$x =$ ..... , $x =$ ..... **[3]**

273 Let $f(x) = \dfrac{2x+1}{3}$ and $g(x) = \dfrac{5}{x-2}$, where $x \neq 2$.

(a) Find $gf(x)$.
Give your answer as a single fraction in its simplest form.

..... **[3]**

(b) Find $g^{-1}(x)$.

$g^{-1}(x) =$ ..... **[3]**

(c) Solve the equation

$$f(x) = g(3)$$

$x =$ ..... **[3]**

274 The functions f, g and h are defined as $f(x) = 4 - 3x$, $g(x) = \dfrac{5}{x-2}$, $x \neq 2$, $h(x) = x^2 - x - 7$.

(a) **(i)** Find the value of $f(0.5)$.

..... **[1]**

(ii) Find $gf(x)$ in its simplest form.

$gf(x) =$ ..... **[2]**

(b) Find the range of $h(x)$.

..... **[2]**

(c) Solve the equation.

$h(x) = 2x - 1$ Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

275 Functions f, g and h are defined as follows.

$f(x) = 3x - 2$

$g(x) = x^2 - 4x$

$h(x) = \dfrac{2x+1}{x-1}$ for $x \neq 1$

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

(b) Find an expression for $g(f(x))$ in the form $ax^2 + bx + c$.

..... **[3]**

(c) Solve the equation $f(g(x)) = x$.
Give your answers correct to 2 decimal places.
You must show all your working.

$x =$ ..... , x =

..... **[4]**

276 Let $f(x) = 3x + 2$, $g(x) = x^2 - x$ and $h(x) = 5^x$.

(a) Find $fg(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[2]**

(b) Find $gf(x)$.
Give your answer in the form $ax^2 + bx + c$.

..... **[3]**

(c) Solve the equation

$$gf(x) = 0$$

$x =$ ..... , $x =$ ..... **[3]**

(d) Find the value of x when $h(x) = \dfrac{1}{125}$.

$x =$ ..... **[2]**

277 Let $f(x) = \dfrac{2x+5}{x-3}$ for $x \neq 3$, and $g(x) = 3x - 1$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Express $fg(x)$ as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

(c) Solve the equation $gf(x) = x$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[5]**

278 Consider the functions $f(x) = \dfrac{3}{x+2}$, $g(x) = (3x+5)^2$ and $h(x) = \dfrac{3x+5}{x-2}$.

(a) Find $h^{-1}(x)$.

$h^{-1}(x) =$ ..... **[3]**

(b) Find $fg(x)$.
Give your answer as a single fraction in its simplest form.

$fg(x) =$ ..... **[3]**

(c) Solve the equation.

$h(x) = x$

Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , or $x =$ ..... **[5]**

279 The functions f and g are defined as:

$$f(x) = \frac{2x+3}{x-1} \qquad g(x) = x^2 - 4x$$

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(b) Find $ff(x)$, giving your answer as a single fraction in its simplest form.

$ff(x) =$ ..... **[4]**

(c) Solve the equation $g(x) = f(2)$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[4]**

280 A quadratic function has x-intercepts at $x = 4$ and $x = -2$.
The coefficient of x^2 is 1.

Find the equation of the quadratic function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

281 A quadratic function $f(x)$ has x-intercepts at $x = -4$ and $x = 3$.
The coefficient of x^2 is 1.

Find the equation of the function in the form $f(x) = x^2 + bx + c$.

$f(x) =$ ..... **[2]**

282 A quadratic function has the equation $y = x^2 + bx + c$.
The vertex of the graph of this function is $(4, -3)$.

Find the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

283 A quadratic function is given by

$$y = x^2 + bx + c$$

The vertex of the graph of this function is $(3, -4)$.

Find the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[3]**

284 A quadratic function has a minimum point at $(3, -4)$ and passes through the point $(5, 0)$.
The coefficient of x^2 is 1.

Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[3]**

285 The graph of a quadratic function, $y = x^2 + bx + c$, crosses the x-axis at $(-4, 0)$ and $(2, 0)$.

(a) Work out the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Find the coordinates of the vertex of the graph.

(..... , ...) **[2]**

286 A quadratic function $y = f(x)$ has its vertex at $(3, -4)$.
The coefficient of x^2 is 1.

(a) Find the equation of the function in the form $y = x^2 + bx + c$.

$y =$ ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis.

..... **[2]**

287 A quadratic function is given by the equation $y = (x - h)^2 + k$.
The vertex of the graph of this function is $(3, -2)$.

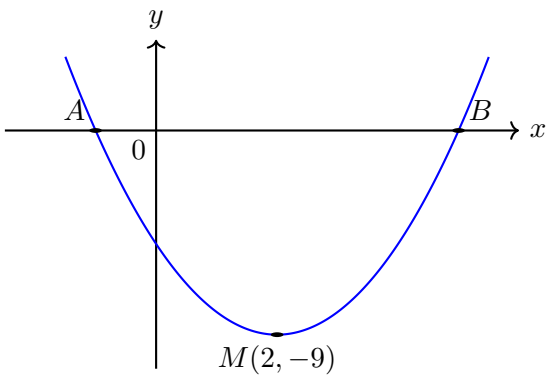
(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) The graph passes through the point $(5, p)$.
Work out the value of p.

$p =$ ..... **[2]**

288 The diagram shows the graph of a quadratic function $y = x^2 + px + q$.
The graph has a minimum point (vertex) at $M(2, -9)$.



(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(b) The graph crosses the x-axis at the points A and B.
Find the coordinates of A and the coordinates of B.

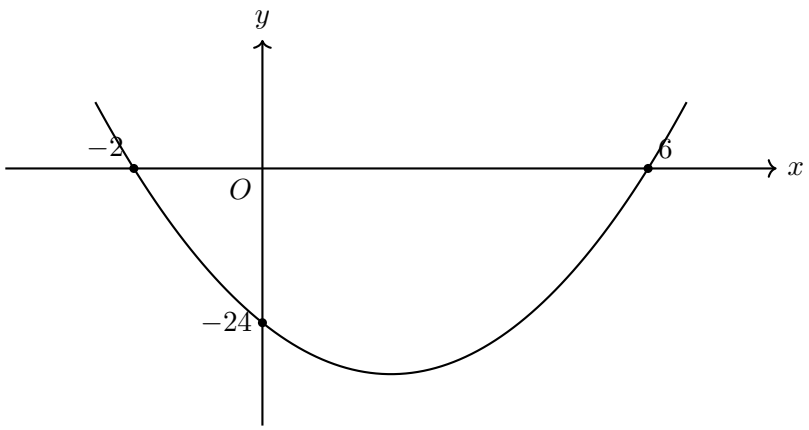
$A =$ ..... , $B =$ ..... **[3]**

289 The graph of a quadratic function $y = f(x)$ intersects the x-axis at $(-2, 0)$ and $(5, 0)$.
The graph intersects the y-axis at $(0, -20)$.

Work out the function $f(x)$, giving your answer in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

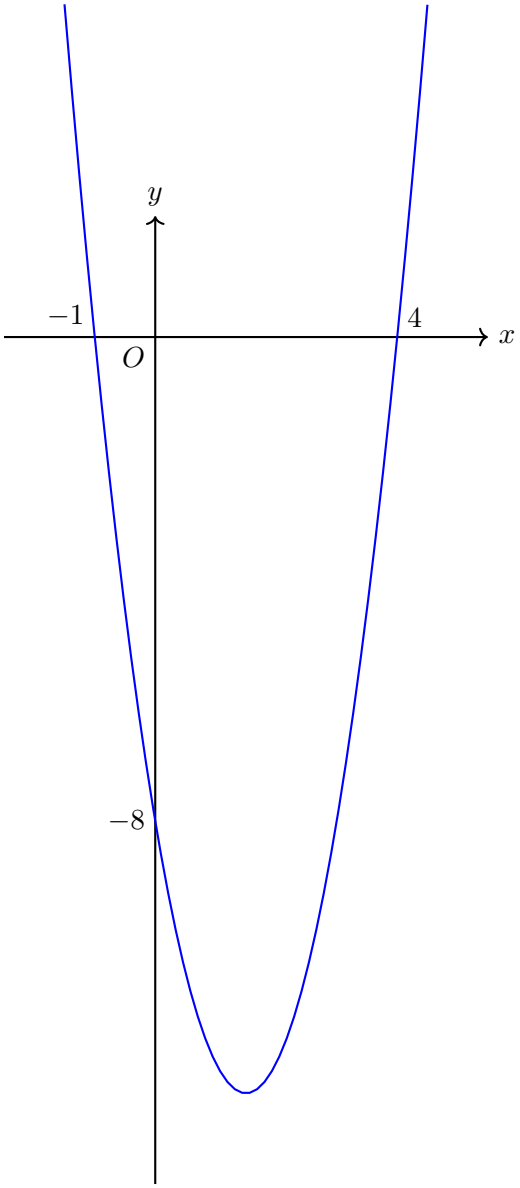
290 The diagram shows a sketch of the quadratic graph $y = f(x)$.
The graph crosses the x-axis at $(-2, 0)$ and $(6, 0)$.
It crosses the y-axis at $(0, -24)$.



Find the expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

291 The diagram shows a sketch of the quadratic function $y = f(x)$.
The graph crosses the x-axis at $(-1, 0)$ and $(4, 0)$.
The graph crosses the y-axis at $(0, -8)$.



(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

(b) Find the coordinates of the minimum point of the graph.

(..... , ...) **[3]**

292 A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The graph of $y = f(x)$ has a maximum point at $(-1, 8)$ and passes through the point $(2, -10)$.

(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

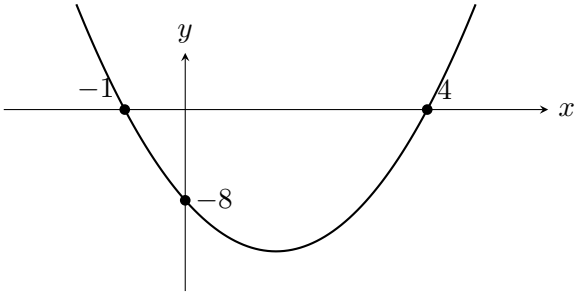
(b) Find the value of a.

$a =$ ..... **[3]**

(c) Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[3]**

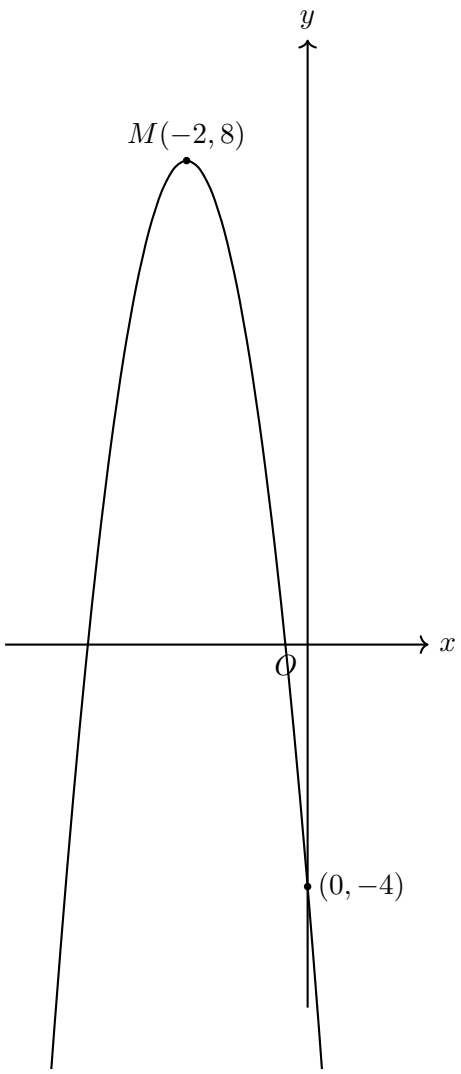
293 The diagram shows the graph of a quadratic function $y = f(x)$.
The graph crosses the x-axis at $x = -1$ and $x = 4$.
The graph crosses the y-axis at $(0, -8)$.



Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

294 The diagram shows a sketch of the curve $y = f(x)$ where $f(x)$ is a quadratic function.
The maximum point on the curve is $M(-2, 8)$ and the curve crosses the y-axis at $(0, -4)$.



(a) Find the equation of the curve.
Give your answer in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Solve $f(x) = 0$ to find the exact x-coordinates of the points where the curve crosses the x-axis.

Give your answers in the form $\frac{p \pm q\sqrt{6}}{r}$, where p, q and r are integers.

$x =$ ..... **[3]**

295 A quadratic curve has its vertex at $(3, -5)$ and passes through the point $(1, 7)$.
Find the equation of the curve in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

296 The graph of a quadratic function has its vertex at $(2, 7)$ and passes through the point $(4, -5)$.

(a) Find the equation of the function in the form $y = a(x - h)^2 + k$.

$y =$ ..... **[3]**

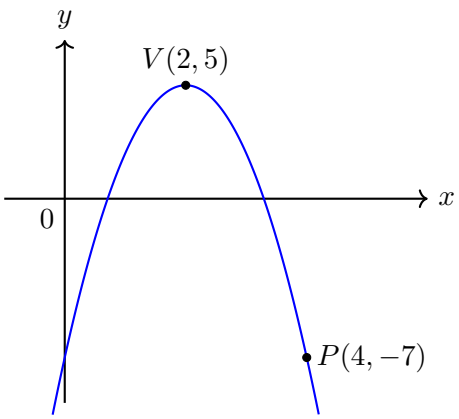
(b) Find the y-intercept of the graph.

..... **[1]**

297 A quadratic curve crosses the x-axis at $(-1, 0)$ and $(5, 0)$.
The curve also passes through the point $(2, -18)$.
Find the equation of the curve in the form $y = ax^2 + bx + c$.

$y =$ ..... **[4]**

298 The diagram shows a sketch of the graph of a quadratic function $y = a(x - h)^2 + k$.
The vertex of the graph is $V(2, 5)$.
The graph passes through the point $P(4, -7)$.



(a) Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

(c) Find the coordinates of the y-intercept of the graph.

(..... , ...) **[2]**

299 The graph of a quadratic function $f(x)$ intersects the x-axis at $x = -2$ and $x = 5$.
The graph passes through the point $(3, 10)$.

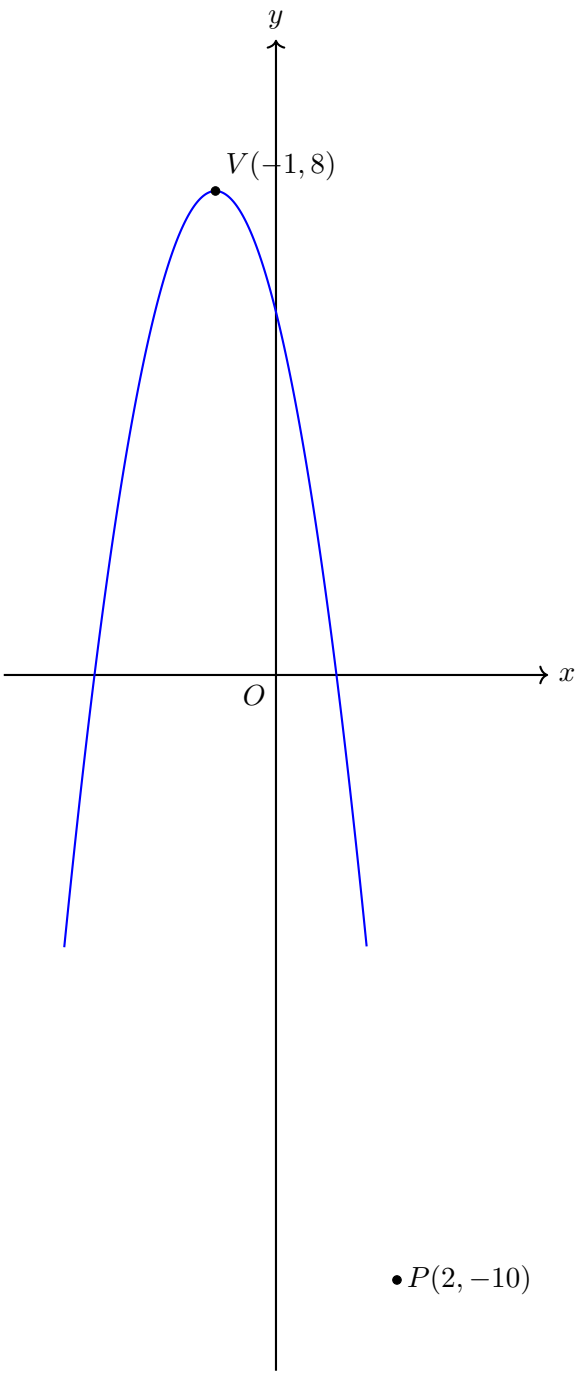
(a) Find an expression for $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Find the coordinates of the vertex of the graph of $y = f(x)$.

(..... , ...) **[2]**

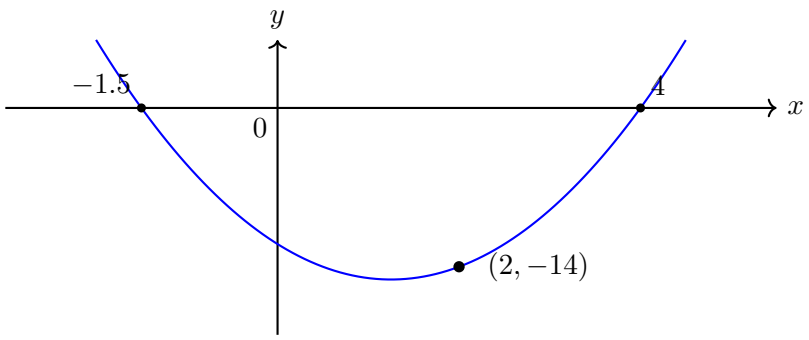
300 The diagram shows a sketch of the graph of $y = ax^2 + bx + c$.
The vertex of the graph is $V(-1, 8)$ and it passes through the point $P(2, -10)$.



Work out the values of a, b and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

301 The diagram shows a sketch of the quadratic function $y = f(x)$.
The curve crosses the x-axis at $x = -1.5$ and $x = 4$.
It passes through the point $(2, -14)$.



(a) Find the equation of the quadratic function in the form $f(x) = ax^2 + bx + c$.

$f(x) =$ ..... **[4]**

(b) Find the exact coordinates of the turning point of the graph.

(..... , ...) **[3]**

302 A quadratic function is given by $f(x) = a(x - h)^2 + k$.
The maximum point of the graph of $y = f(x)$ is $(-4, 20)$.
The graph intersects the x-axis at $x = -2$ and $x = p$.

(a) **(i)** Write down the value of h and the value of k.

$h =$ ..... , $k =$ ..... **[2]**

(ii) Find the value of a.

$a =$ ..... **[2]**

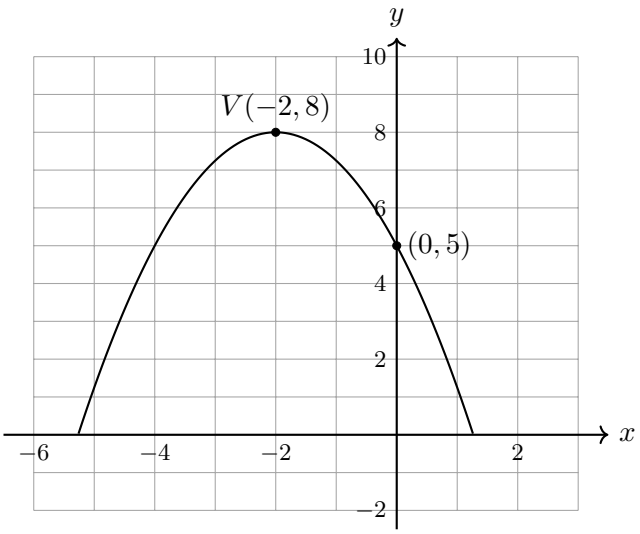
(iii) Find the value of p.

$p =$ ..... **[1]**

(iv) Express $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

303 The diagram shows the graph of a quadratic function $y = f(x)$.
The vertex of the graph is at $V(-2, 8)$ and it intersects the y-axis at $(0, 5)$.



(a) Find the equation of the function $f(x)$ in the form $f(x) = a(x - h)^2 + k$.

$f(x) =$ ..... **[3]**

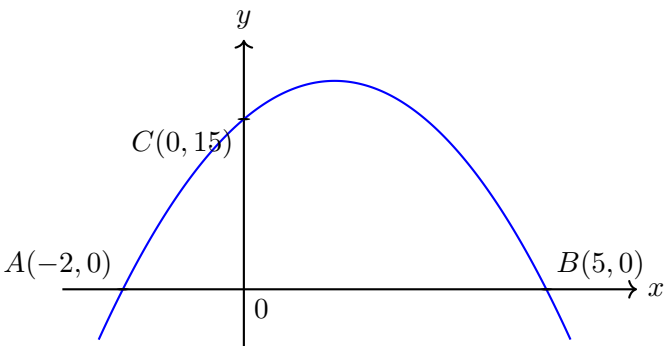
(b) Write $f(x)$ in the form $ax^2 + bx + c$.

$f(x) =$ ..... **[2]**

(c) Calculate the exact x-coordinates of the points where the graph intersects the x-axis.
Give your answers correct to 3 significant figures.

$x =$ ..... , $x =$ ..... **[3]**

304 The diagram shows a sketch of the quadratic curve $y = ax^2 + bx + c$.
The curve crosses the x-axis at $A(-2, 0)$ and $B(5, 0)$, and crosses the y-axis at $C(0, 15)$.



(a) Find the values of a, b, and c.

$a =$ ..... , $b =$ ..... , $c =$ ..... **[5]**

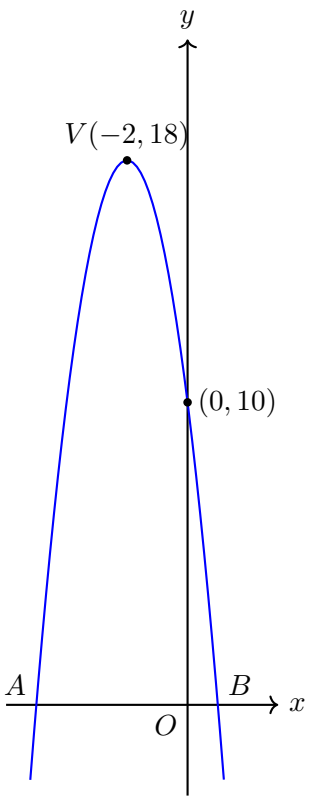
(b) Find the coordinates of the vertex of the curve.

(..... , ...) **[3]**

(c) The equation $ax^2 + bx + c = k$ has no real roots.
Write down the inequality for k.

..... **[1]**

305 The diagram shows a sketch of the graph of a quadratic function $y = f(x)$.
The maximum point of the graph is $V(-2, 18)$.
The graph crosses the y-axis at $(0, 10)$.



(a) Find the equation of the quadratic function in the form $y = ax^2 + bx + c$.

$y =$ ..... **[3]**

(b) The graph crosses the x-axis at A and B.
Find the length of the line segment AB.

$AB =$ ..... **[3]**

(c) The line $y = -2x + k$ is a tangent to the curve $y = f(x)$.

Find the value of k.

$k =$ ..... **[4]**

306 The quadratic function $f(x) = ax^2 + bx + c$ has a minimum turning point at $(1, -8)$.
The graph of $y = f(x)$ crosses the x-axis at $x = -1$ and $x = 3$.

(a) Show that $f(x) = 2x^2 - 4x - 6$.

[3]

(b) Solve the equation.

$f(x) = 10$

$x =$ ..... , $x =$ ..... **[3]**

(c) A straight line has equation $y = mx - 8$.
The line is a tangent to the graph of $y = f(x)$.
Find the possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

(d) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ k \end{pmatrix}$.
The new graph has no x-intercepts.
Write down the inequality for k.

..... **[2]**

307 Let $f(x) = \dfrac{7}{x-4} + 3$.

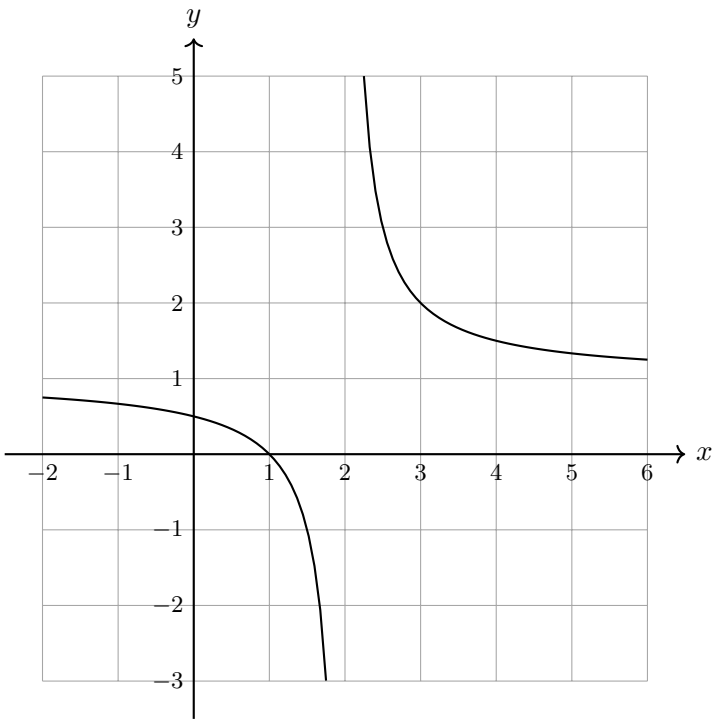
(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

308 The diagram shows the graph of a function $y = f(x)$.
The graph has two asymptotes, each parallel to one of the coordinate axes.
The equations of the asymptotes involve only integer values.



(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) The graph of $y = f(x)$ is reflected in the y-axis to form the graph of $y = g(x)$.
Write down the equation of the vertical asymptote of the graph of $y = g(x)$.

..... **[1]**

309 A function is defined as $f(x) = \dfrac{3}{x+2} + 1$.

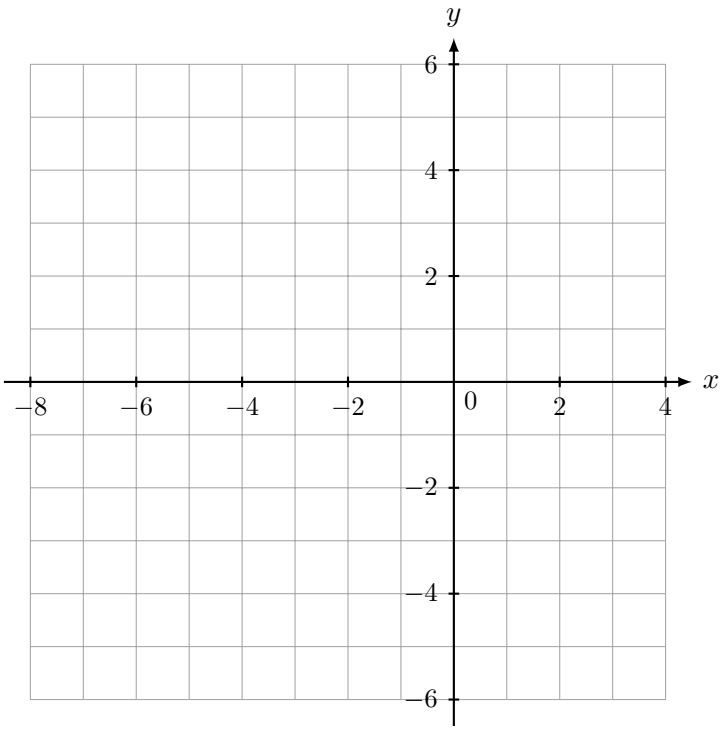
(a) Write down the equations of the two asymptotes of the graph of $y = f(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis and the y-axis.

x-axis at ..... , y-axis at ..... **[2]**

(c) On the grid, sketch the graph of $y = f(x)$.



[3]

310 Let $f(x) = \dfrac{3}{x+2} - 4$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) Find the coordinates of the points where the graph of $y = f(x)$ intersects the coordinate axes.

x-intercept = ... , y-intercept = ... **[5]**

(d) The graph of $y = \tan x$ has an infinite number of vertical asymptotes.
Write down the equations of the two asymptotes in the interval $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

311 A function is given by $f(x) = \dfrac{a}{x-b} + c$, where a, b and c are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = -1$.
The graph intersects the x-axis at the point $(5, 0)$.

(a) Write down the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Work out the value of a.

$a =$ ..... **[2]**

(c) The graph of $y = f(x)$ intersects the straight line $y = -x + 5$ at two points.
Find the x-coordinates of these two points.

$x =$ ..... , $x =$ ..... **[2]**

312 The function $f(x)$ is given by $f(x) = \dfrac{p}{x-q} + r$, where p, q, and r are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 5$ and a horizontal asymptote at $y = -3$.
The graph passes through the point $(7, -2)$.

(a) Write down the value of q and the value of r.

$q =$ ..... , $r =$ ..... **[2]**

(b) Find the value of p.

$p =$ ..... **[3]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(d) Write down the equations of the asymptotes of the graph of $y = f^{-1}(x)$.

$x =$ ..... , $y =$ ..... **[2]**

313 **(a)** The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ to map onto the graph of $y = g(x)$. Write down $g(x)$ in terms of x.

$g(x) =$ ..... **[1]**

(b) The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ to map onto the graph of $y = h(x)$. Write down $h(x)$ in terms of x.

$h(x) =$ ..... **[1]**

(c) Describe fully the single transformation that maps the graph of $y = x^2$ onto the graph of $y = (x + 4)^2 - 1$.

.....

.....

..... **[2]**

314 A function is defined as $f(x) = 2x - 5$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of x, giving your answer in its simplest form.

$g(x) =$ ..... **[2]**

(b) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

.....

..... **[2]**

315 The graph of $y = f(x)$ has a local maximum at the point $P(-2, 5)$ and a local minimum at the point $Q(3, -1)$.

(a) Write down the coordinates of the local maximum of the graph of $y = f(x) - 4$.

(..... , ...) **[1]**

(b) Write down the coordinates of the local minimum of the graph of $y = f(x - 2)$.

(..... , ...) **[1]**

(c) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ such that the image of the local maximum P is at the origin $(0, 0)$.

(i) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(ii) Write down the coordinates of the image of the local minimum Q under the same translation.

(..... , ...) **[1]**

316 Let $f(x) = 3x^2$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of x.
Give your answer in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$ to give the graph of $y = h(x)$.
Write down the coordinates of the minimum point of the graph of $y = h(x)$.

(..... , ...) **[2]**

(c) Find the y-intercept of the graph of $y = h(x)$.

..... **[1]**

317 The functions $f(x)$ and $g(x)$ are defined as $f(x) = x^2 - 6x + 14$ and $g(x) = \dfrac{3}{x+2} - 5$.

(a) By writing $f(x)$ in the form $(x-a)^2 + b$, find the column vector of the translation that maps the graph of $y = x^2$ onto the graph of $y = f(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ to map onto the graph of $y = h(x)$.
Find $h(x)$ in its simplest form.

$h(x) =$ ..... **[3]**

(c) Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

$x =$ ..... , $y =$ ..... **[2]**

318 A function is defined as $f(x) = \dfrac{20}{x}$.

(a) The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

(i) Write down an expression for $g(x)$.

$g(x) =$ ..... **[1]**

(ii) Solve the equation $g(x) = 0$.

$x =$ ..... **[2]**

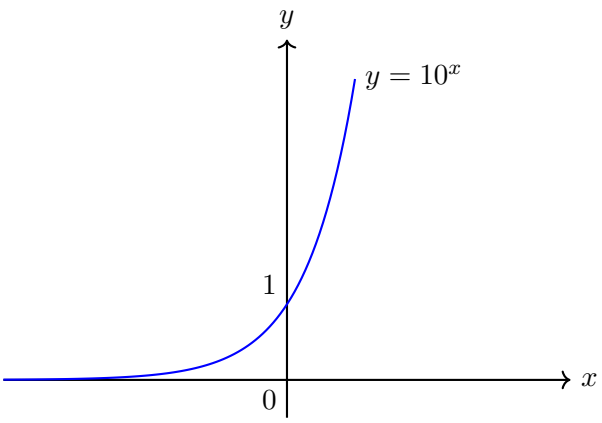
(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ to give the graph of $y = h(x)$.
The graph of $y = h(x)$ passes through the point $(8, 5)$.
Work out the value of k.

$k =$ ..... **[3]**

(c) Let $p(x) = f(x - 3) + 7$.
Write down the equations of the vertical and horizontal asymptotes of the graph of $y = p(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

319 The diagram shows a sketch of the function $f(x) = 10^x$.



(a) Write down the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

(b) Work out the exact value of $f^{-1}\left(\frac{1}{100}\right)$.

..... **[1]**

(c) Write down the exact solution to the equation $f(x) = 45$.

$x =$ ..... **[1]**

320 **(a)** Write $10^3 = 1000$ as a logarithmic equation.

..... **[1]**

(b) Find the exact value of x when $\log x = -2$.

$x =$ ..... **[1]**

(c) Solve $5^x = 12$, giving your answer in the form $\dfrac{\log a}{\log b}$.

$x =$ ..... **[1]**

321 **(a)** Find the value of x when $\log_2 x = 5$.

$x =$ ..... **[1]**

(b) Find the value of y when $10^y = 0.01$.

$y =$ ..... **[1]**

(c) Write the equation $P = k^t$ in logarithmic form.

..... **[1]**

322 **(a)** Write the following as a logarithmic equation.

$5^3 = 125$

..... **[1]**

(b) Find the value of $\log 1000$.

..... **[1]**

(c) Solve the equation.

$\log_2 x = 5$

$x =$ ..... **[2]**

323 **(a)** Write the equation $10^5 = 100000$ in the form $\log a = b$.

..... **[1]**

(b) Find the value of x when $\log x = 3$.

$x =$ ..... **[2]**

(c) The function $f(x)$ is defined as $f(x) = 10^x$.
Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

324 **(a)** Find the value of x in each of the following equations.

(i) $10^x = 100000$

$x =$ ..... **[1]**

(ii) $\log x = -2$

$x =$ ..... **[1]**

(iii) $\log_2 64 = x$

$x =$ ..... **[1]**

(b) Given that $f(x) = 10^x$, write down an expression for $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[1]**

325 **(a)** Write down the value of

$\log 1000$

..... **[1]**

(b) Solve the equation.

$\log x = -2$

$x =$ ..... **[2]**

(c) Write the equation $5^x = 125$ in the form $x = \log_a b$.

$x =$ ..... **[1]**

326 Answer the following independent questions on logarithmic and exponential equations.

(a) Write the exponential equation $10^4 = 10000$ in logarithmic form.

..... **[1]**

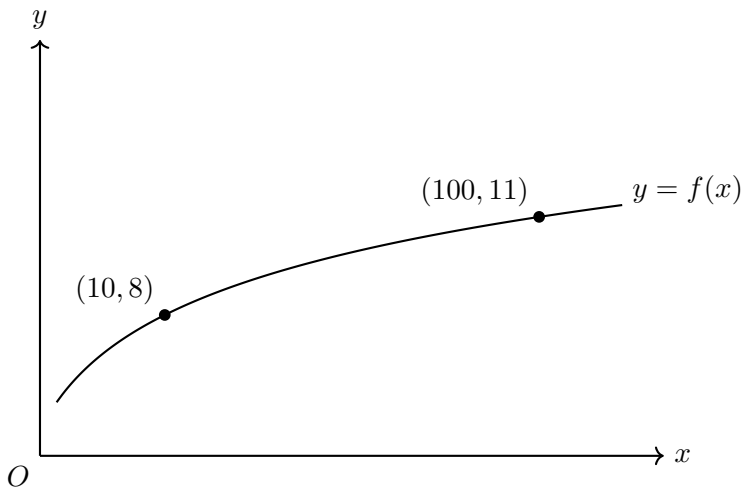
(b) Find the exact value of x when $\log_5 x = 3$.

$x =$ ..... **[2]**

(c) The solution to the equation $7^x = 25$ can be written in the form $x = \dfrac{\log b}{\log a}$.
Write down the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

327 The diagram shows a sketch of the graph of $y = f(x)$ for $x > 0$.
The function is defined as $f(x) = p \log x + q$, where $\log x$ denotes the logarithm to base 10.
The graph passes through the points $(10, 8)$ and $(100, 11)$.



(a) **(a)** Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[3]**

(b) Solve the equation $f(x) = 23$.

$x =$ ..... **[2]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

328 The logarithmic function is the inverse of the exponential function.

(a) Write the equation $y = 6^x$ in logarithmic form.

(a) Write the equation $y = 6^x$ in logarithmic form.

$x =$ ..... **[1]**

(b) Solve the equation.

$3^x = 20$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

329 A population of insects is growing exponentially.
The number of insects, N, after t days is given by

$$N = 250 \times 1.08^t$$

(a) Work out the number of insects after 10 days.
Give your answer to the nearest integer.

..... **[2]**

(b) Calculate the number of days it takes for the population to reach 1000.
Give your answer correct to 1 decimal place.

..... days **[3]**

330 A machine is purchased for \$25000. Its value depreciates exponentially by 12% each year.

(a) Calculate the value of the machine after 4 years.
Give your answer correct to the nearest dollar.

% ..... **[2]**

(b) Find the number of complete years it takes for the value of the machine to fall below \$10000.
You must show all your working.

..... years **[3]**

331 **(a)** Solve the equation.

$3^{2x} = 50$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$y = 10^{x-4}$

$x =$ ..... **[2]**

332 A population of bacteria is modelled by the formula

$P = 5000 \times 1.12^t$ where P is the population and t is the time in hours after the start of an experiment.

(a) **(a)** Write down the initial population of the bacteria.

..... **[1]**

(b) Find the time it takes for the population to reach 20 000.
Give your answer correct to 1 decimal place.

$t =$ ..... hours **[4]**

333 An investment of \$8000 earns compound interest at a rate of 3.5% per year. The value of the investment after n years is \$$V$.

(a) Write down an equation for V in terms of n.

$V =$ ..... **[1]**

(b) Calculate the minimum number of complete years it takes for the value of the investment to exceed \$12000.
Show all your working.

..... years **[3]**

(c) Solve the equation.

$2^x = 55$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

334 **(a)** Solve the equation $3^x = 20$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

(b) The mass, M grams, of a radioactive substance after t days is given by the formula
$M = 500 \times 0.85^t$ Calculate the value of t when $M = 100$.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

335 The value, V dollars, of a painting t years after it was bought is given by the formula

$$V = 4500 \times 1.08^t$$

(a) Calculate the value of the painting 5 years after it was bought.
Give your answer correct to the nearest dollar.

..... **[2]**

(b) Calculate the time it takes for the value of the painting to reach \$10 000.
Give your answer in years, correct to 2 decimal places.

..... years **[4]**

336 An amount of \$8000 is invested in an account that pays compound interest at a rate of 4.5% per year. The value of the investment, V dollars, after t years is given by the formula

$$V = 8000 \times 1.045^t$$

(a) Calculate the value of the investment after 5 years.

\$ ..... **[2]**

(b) Use logarithms to calculate the number of complete years it takes for the value of the investment to first exceed \$15000.

..... years **[3]**

(c) Solve the equation.

$7^{x-1} = 250$ Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

337 A population of insects grows exponentially.
The number of insects, N, after t weeks is given by the formula

$$N = 800 \times 2^t$$

(a) Work out the number of insects after 4 weeks.

..... **[1]**

(b) The number of insects reaches 102 400.
Work out the exact value of t.

$t =$ ..... **[2]**

(c) Another population of insects can be modelled by the formula $P = 1500 \times 3^t$.
Find the value of t when this population reaches 20 000.
Give your answer correct to 3 significant figures.

$t =$ ..... **[3]**

(d) A sum of \$2000 is invested at an annual compound interest rate of 5%. The total value, V, of the investment after n years is given by

$$V = 2000 \times 1.05^n$$

Calculate the minimum number of complete years required for the investment to double in value.

..... years **[3]**

338 A function f is defined as $f(x) = 4 \times 10^{2x} - 7$.

(a) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(b) Solve the equation $f(x) = 53$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

339 **(a)** The mass, M grams, of a radioactive substance after t days is given by $M = 250 \times 0.92^t$.

(i) Write down the initial mass of the substance.

..... grams **[1]**

(ii) Find the number of days it takes for the mass to reduce to 50 grams.
Give your answer correct to 3 significant figures.

..... days **[3]**

(b) An amount of \$800 is invested at a rate of 6% per year compound interest. Calculate the number of complete years it takes for the investment to first exceed \$2000.

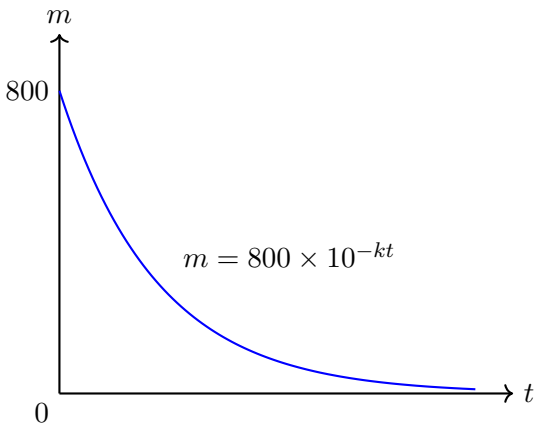
..... years **[3]**

340 The mass, m grams, of a radioactive isotope decays according to the formula

$m = 800 \times 10^{-kt}$ where t is the time in years and k is a constant.

The diagram shows a sketch of the decay curve.

After 15 years, the mass is 250 grams.



(a) Show that $k = 0.0337$ correct to 3 significant figures.

[3]

(b) Find the number of years it takes for the mass to become 50 grams.

..... years **[3]**

(c) Make t the subject of the formula $m = 800 \times 10^{-kt}$.

$t =$ ..... **[2]**

341 Let $f(x) = 5 \times 10^x$ and $g(x) = \log x$.

(a) Find the value of $f(g(100))$.

..... **[2]**

(b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[2]**

(c) Solve the equation.

$f(x) = 200$ Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

(d) Find $g(f(x))$ in the form $x + k$, where k is a constant.

..... **[2]**

342 A function is defined as $f(x) = 5 \times 3^x - 2$.

(a) Find the value of $f(-2)$, giving your answer as a fraction in its simplest form.

..... **[2]**

(b) Find $f^{-1}(x)$.

Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

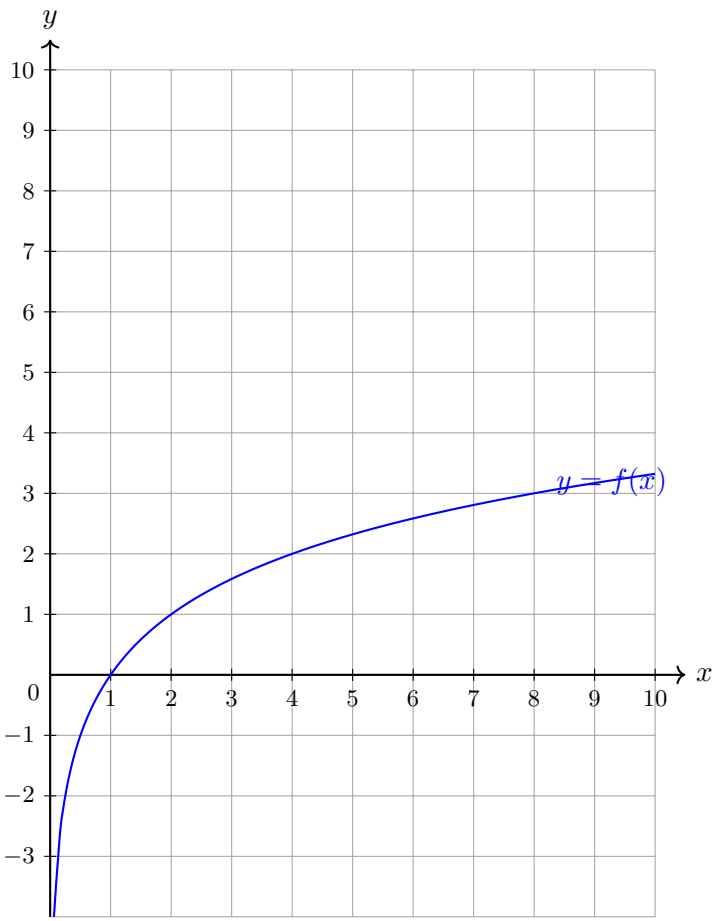
(c) Solve the equation.

$f(x) = 500$

Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

343 The diagram shows the graph of $y = f(x)$.



(a) The graph of $y = f(x)$ passes through the point $(8, 3)$.
Given that $f(x) = \log_a x$, show that $a = 2$.

[2]

(b) **(i)** On the grid, sketch the graph of $y = f^{-1}(x)$. **[2]**

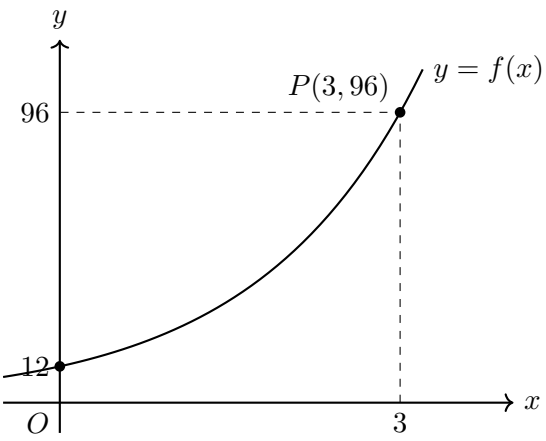
(ii) Write down the equation of the asymptote of the graph of $y = f^{-1}(x)$.

..... **[1]**

(c) By drawing a suitable straight line on the grid, find an approximate solution to the equation $\log_2 x + x = 8$.

$x =$ ..... **[3]**

344 The diagram shows the graph of $y = f(x)$ where $f(x) = a \times b^x$.
The graph intersects the y-axis at the point $(0, 12)$ and passes through the point $P(3, 96)$.



(a) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[3]**

(b) Find $f^{-1}(x)$.
Give your answer in terms of base 10 logarithms.

$f^{-1}(x) =$ ..... **[3]**

(c) Solve the equation $f(x) = 500$.
Give your answer correct to 3 significant figures.

$x =$ ..... **[3]**

345 The function $f(x) = A \times b^x$, where $b > 0$, passes through the points $(0, 7)$ and $(3, 56)$.

(a) Find the value of A and the value of b.

$A =$ ..... , $b =$ ..... **[3]**

(b) Find the inverse function, $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[3]**

(c) The equation $f(x) = 600$ is equivalent to $2^x = k$.

(i) Work out the exact value of k as a fraction.

$k =$ ..... **[1]**

(ii) Solve the equation $f(x) = 600$.
Give your answer correct to 3 decimal places.

$x =$ ..... **[2]**

(d) A different function $g(x)$ is defined as $g(x) = \log_c(x)$.
The graph of $y = g(x)$ passes through the point $(81, 4)$.

(i) Find the value of c.

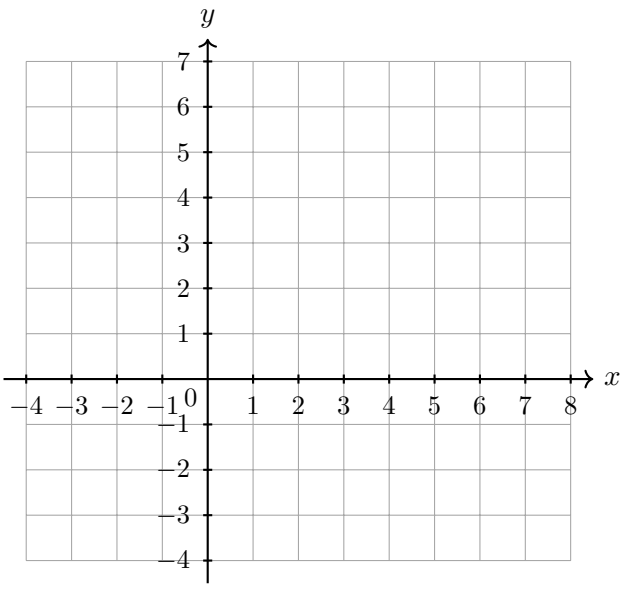
$c =$ ..... **[2]**

(ii) Solve the equation $g(x) = -2$.

$x =$ ..... **[1]**

346 The points $A(-1,3)$, $B(2,-1)$ and $C(6,1)$ are three vertices of a parallelogram $ABCD$.

(a) On the grid, plot the points A, B and C.



[1]

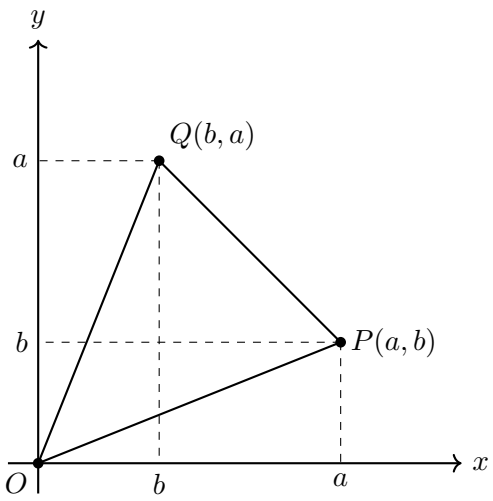
(b) Find the coordinates of the fourth vertex, D.

(..... , ...) **[2]**

(c) Write down the coordinates of the centre of the parallelogram.

(..... , ...) **[2]**

347 The points $P(a, b)$ and $Q(b, a)$ are in the first quadrant, with $a > b$.
The origin is denoted by $O(0, 0)$.



(a) By drawing a suitable rectangle or otherwise, show that the area of triangle OPQ is $\frac{1}{2}(a^2 - b^2)$.

[3]

(b) The area of triangle OPQ is 24 and the length of PQ is $4\sqrt{2}$.
Find the values of a and b.

$a =$ ..... , $b =$ ..... **[5]**

348 A straight line L has the equation $3x + 2y = 18$.

(a) The line L crosses the x-axis at point A and the y-axis at point B.
Find the coordinates of A and B.

$A =$

$B =$ **[2]**

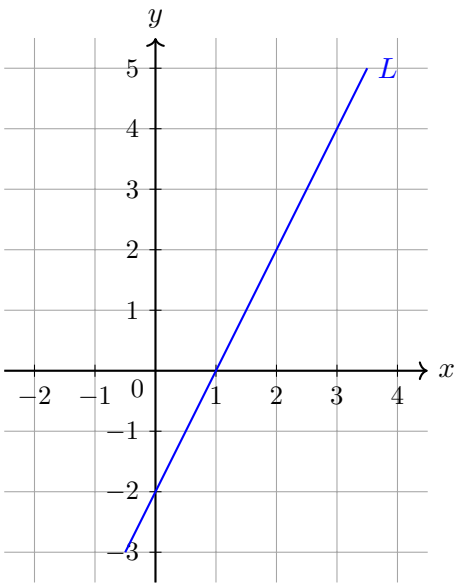
(b) Point P lies on the line segment AB such that the ratio $AP : PB = 2 : 1$.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Does the point $(4, 3)$ lie on the line L?
Show how you decide.

[2]

349 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the gradient of line L.

..... **[2]**

(b) Write down the coordinates of the y-intercept of line L.

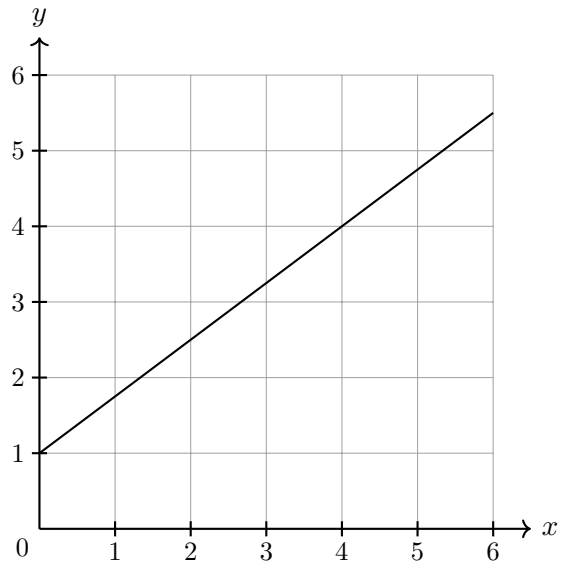
(..... , ...) **[1]**

350 The points A and B have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

(a) Work out the gradient of the line AB.

..... **[2]**

(b) A straight line L is drawn on the grid.
Find the gradient of line L.



..... **[2]**

351 The line L_1 passes through the points $P(-2, 3a)$ and $Q(4, a + 6)$.

(a) Find an expression for the gradient of L_1 in terms of a.
Give your answer in its simplest form.

..... **[3]**

(b) The line L_2 has equation $y = 3x - 5$.
Given that L_1 is parallel to L_2, find the value of a.

$a =$ ..... **[2]**

(c) Using your value of a, find the coordinates of the midpoint of PQ.

(..... , ...) **[3]**

352 The points A, B and C have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

(a) The points A, B and C lie on the same straight line.
Work out the value of y.

$y =$ ..... **[2]**

(b) A different point D has coordinates $(p, p^2 - p - 3)$.
The gradient of the line AD is 2.
Work out the possible values of p.

$p =$ ..... , $p =$ ..... **[4]**

(c) A line L has gradient m.
The line L passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

353 Point A has coordinates $(-3, 5)$ and point B has coordinates $(4, -9)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(k, 1)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of k.

$k =$ ..... **[3]**

354 Point P has coordinates $(k-3, 2k)$ and point Q has coordinates $(2, 5)$.

(a) The gradient of the line PQ is -3.
Work out the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), find the coordinates of the midpoint of PQ.

(..... , ...) **[2]**

355 Point A has coordinates $(1, 4)$ and point B has coordinates $(7, -4)$.

(a) Calculate the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the length of the line segment AB.

..... **[3]**

356 A circle has its centre at the origin, $O(0, 0)$.
The point $P(-3, 4)$ lies on the circumference of this circle.

(a) Work out the radius of the circle.

..... **[2]**

(b) The line segment PQ is a diameter of the circle.
Find the coordinates of Q.

(..... , ...) **[2]**

(c) The point R has coordinates $(k, \sqrt{11})$.
R also lies on the circumference of the circle, and k is a negative number.
Work out the exact value of k.

$k =$ ..... **[3]**

357 A circle has diameter AB, where A is the point $(-1, 2)$ and B is the point $(5, 8)$.

(a) Find the coordinates of the centre of the circle, C.

(..... , ...) **[2]**

(b) Find the exact radius of the circle.
Give your answer in the form $a\sqrt{b}$, where a and b are positive integers and b is as small as possible.

..... **[3]**

(c) Point D is such that the midpoint of CD is $M\left(\frac{5}{2}, \frac{3}{2}\right)$.

Find the exact length of AD, giving your answer in the form $c\sqrt{d}$, where c and d are integers and d is as small as possible.

..... **[5]**

358 The points A and B have coordinates $(-2, 5)$ and $(4, -3)$ respectively.

(a) Calculate the exact length of the line segment AB.

..... **[3]**

(b) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

359 Point P has coordinates $(-3, 5)$ and point Q has coordinates $(k, -1)$.
The length of the line segment PQ is 10 units.

(a) Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

(b) Given that $k > 0$, point M is the midpoint of PQ.
Find the coordinates of M.

(..... , ...) **[2]**

360 The points $A(2p, p)$ and $B(3p, 4p)$ are joined by a straight line, where $p > 0$.

(a) The length of the line segment AB is $5\sqrt{10}$.
Find the value of p.

$p =$ ..... **[4]**

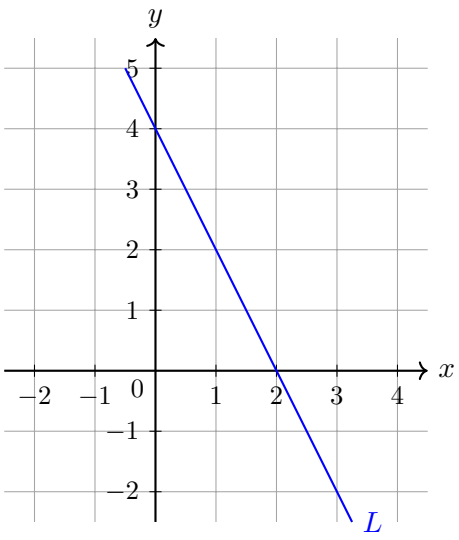
(b) Point C lies on the line segment AB such that the ratio $AC : CB = 2 : 3$.
Using your value of p from part (a), find the coordinates of C.

(..... , ...) **[3]**

(c) The point $D(x, y)$ is such that $ABCD$ forms a rhombus.
Work out the coordinates of D.

(..... , ...) **[2]**

361 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Write the equation of line L in the form $ax + by = c$, where a, b and c are integers.

..... **[1]**

362 **(a)** Find the gradient and the y-intercept of the line $3x + 5y = 15$.

Gradient = ... , y-intercept = ... **[3]**

(b) Find the equation of the line parallel to $3x + 5y = 15$ that passes through the point $(0, -4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

363 The line L passes through the points $A(2a, a+4)$ and $B(a-1, 3a+2)$.
The gradient of line L is -4.

(a) Find the value of a.

$a =$ ..... **[3]**

(b) Find the equation of line L.
Give your answer in the form $px + qy = r$, where p, q and r are integers and $p > 0$.

..... **[3]**

(c) Line L intersects the line $y = 3x - 2$ at the point M.
Find the coordinates of M.

(..... , ...) **[3]**

364 A is the point $(-3, 1)$ and B is the point $(5, 5)$.

(a) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) The perpendicular bisector of AB intersects the line $x - 2y = 5$ at the point P.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Find the exact length of AP.

..... **[3]**

365 Point A has coordinates $(-4, 3)$ and point B has coordinates $(6, -2)$.

(a) Find the equation of the line passing through A and B.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

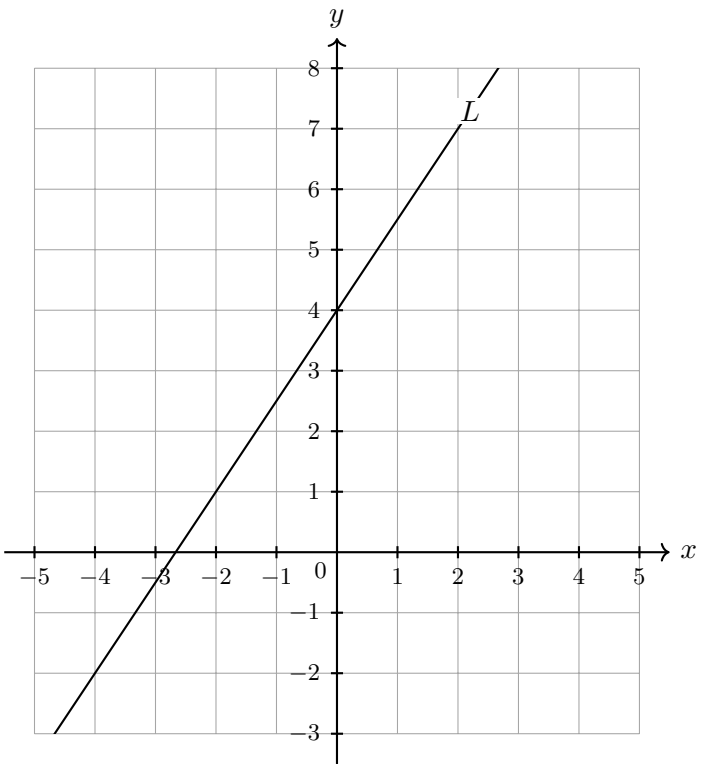
(b) The line $3x - 5y = 15$ intersects the x-axis at point C and the y-axis at point D.
Find the coordinates of C and D.

$C =$ ..... , $D =$ ..... **[2]**

(c) Write down the equation of the line that is parallel to the y-axis and passes through point A.

..... **[1]**

366 The diagram shows a straight line L.



(a) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(b) Line M is perpendicular to line L and passes through the point $(9, -1)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

367 Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

368 Line L has the equation $y = 6 - 3x$.

Find the equation of the line parallel to L that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

369 The points A and B have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line L is parallel to the line segment AB and passes through the point $(4, -2)$.

(a) Find the gradient of the line segment AB.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

370 Line M has the equation $5x - 2y = 15$.
Point P has coordinates $(-4, 1)$.
Find the equation of the line parallel to M that passes through P.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

371 Points $A(-1, 5)$ and $B(3, -3)$ lie on line L_1.
Line L_2 is parallel to L_1 and passes through the point $C(2, 7)$.
Line L_2 crosses the x-axis at D and the y-axis at E.

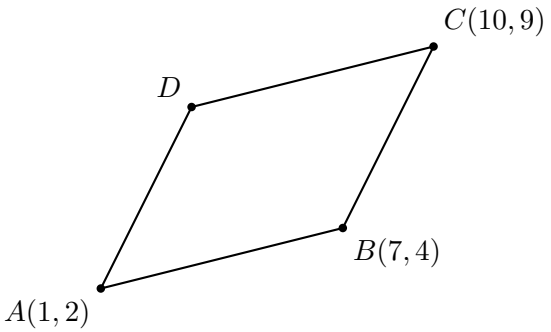
(a) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) Calculate the exact area of triangle ODE, where O is the origin $(0, 0)$.

..... $units^2$ **[3]**

372 The points $A(1, 2)$, $B(7, 4)$, and $C(10, 9)$ are three vertices of a parallelogram $ABCD$.
Line CD is parallel to line AB and line AD is parallel to line BC.



Not to scale

(a) Find the equation of the line CD.

..... **[3]**

(b) Find the equation of the line AD.

..... **[3]**

(c) Work out the coordinates of the vertex D.

(..... , ...) **[3]**

373 Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

374 Line L_1 has the equation $y = 5 - 2x$.
Line L_2 is perpendicular to L_1 and passes through the point $(4, 1)$.

(a) Find the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

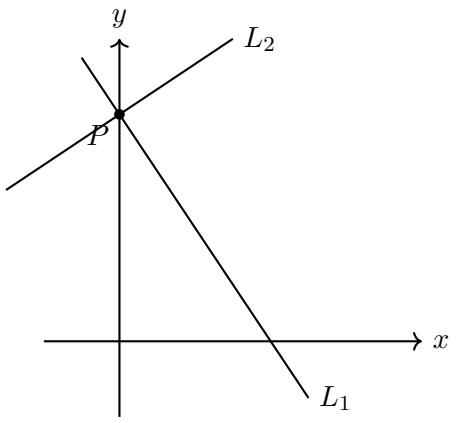
$y =$ ..... **[3]**

375 A is the point $(-4, 7)$ and B is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[5]**

376 Line L_1 has the equation $3x + 2y = 12$.
Line L_2 is perpendicular to L_1 and intersects L_1 at a point P on the y-axis.



Not to scale

(a) Find the coordinates of point P.

(..... , ...) **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Point Q lies on L_2 and has an x-coordinate of 9.
Point R lies on L_1 and has an x-coordinate of 4.
Work out the area of triangle PQR.

..... **[5]**

377 The points A and B have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

378 The line L_1 has equation $2y + 3x = 12$.
The line L_2 is perpendicular to L_1 and intersects L_1 at the point P on the y-axis.

(a) Find the coordinates of P.

(..... , ...) **[2]**

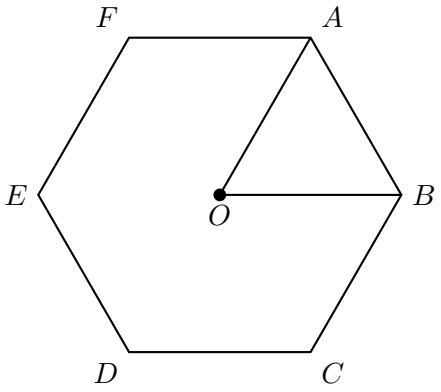
(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line L_2 crosses the x-axis at the point Q.
Calculate the exact length of PQ.

..... **[4]**

379 The diagram shows a regular polygon $ABCDEF$ with centre O.



(a) Write down the mathematical name of the polygon $ABCDEF$.

..... **[1]**

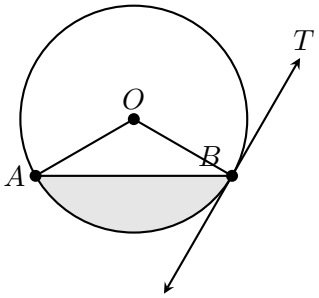
(b) Write down the mathematical name for the angle AFE.

..... **[1]**

(c) Write down the mathematical name for the triangle AOB.

..... **[1]**

380 The diagram shows a circle, centre O.
A and B are points on the circumference of the circle.
The line passing through T touches the circle at B.



(a) Write down the geometrical name of the line AB.

..... **[1]**

(b) Write down the geometrical name of the line BT.

..... **[1]**

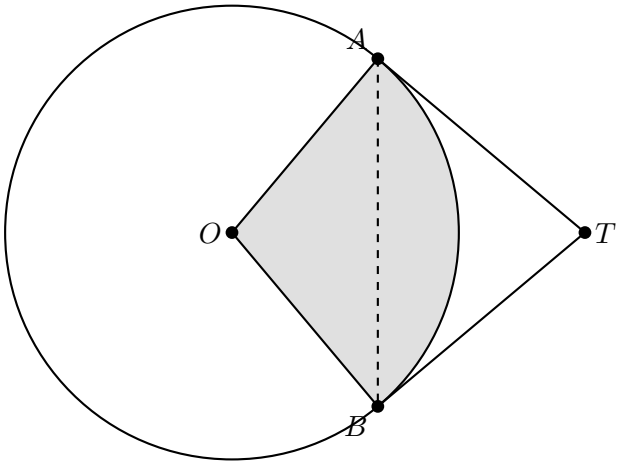
(c) Write down the geometrical name of the shaded region.

..... **[1]**

(d) Write down the mathematical name of triangle OAB.

..... **[1]**

381 Points A and B lie on a circle, centre O.
The lines TA and TB are tangents to the circle.
The region enclosed by the arc AB and the radii OA and OB is shaded.



(a) Write down the mathematical name for the quadrilateral $OATB$.

..... **[1]**

(b) Write down the mathematical name for the shaded region.

..... **[1]**

(c) Write down the mathematical name for the straight line AB.

..... **[1]**

(d) Triangle OAT and triangle OBT are exactly the same size and shape.
Write down the mathematical term for two identical shapes.

..... **[1]**

382 **(a)** The interior angle of a regular polygon is 144°.

(i) Work out the number of sides of this polygon.

..... **[2]**

(ii) Write down the geometrical name of this polygon.

..... **[1]**

(b) A hexagon has interior angles of $x°$, $2x°$, $(x + 30)°$, $(3x + 60)°$, 110° and 100°.

(i) Calculate the value of x.

$x =$ ..... **[3]**

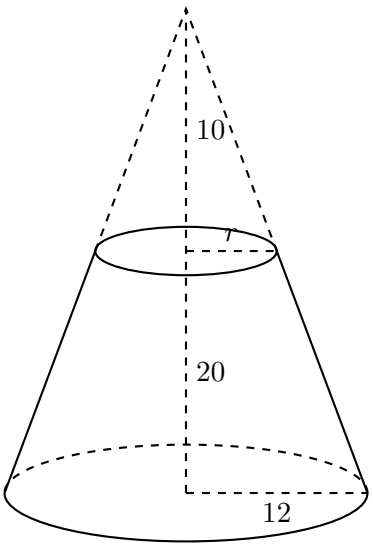
(ii) Write down the geometrical name for an angle of $(x + 30)°$.

..... **[1]**

(iii) Write down the geometrical name for an angle of $(3x + 60)°$.

..... **[1]**

383 The diagram shows a large solid cone with a smaller solid cone removed from the top.
The cut is made parallel to the base of the large cone.
All dimensions are in centimetres.



(a) Write down the mathematical name of the solid with height 20.

..... **[1]**

(b) The small cone is geometrically related to the original large cone.

(i) Write down the mathematical word that describes the relationship between the small cone and the large cone.

..... **[1]**

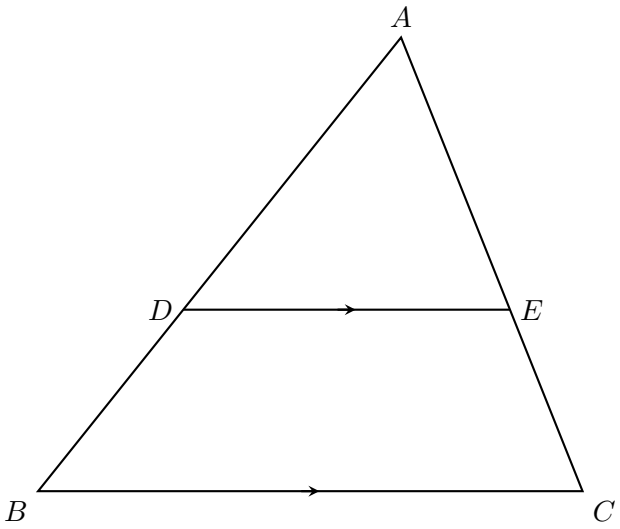
(ii) Work out the value of the radius, r.

$r =$ ..... **[2]**

(iii) Calculate the slant edge of the solid with height 20.

..... *cm* **[3]**

384 The diagram shows triangle ABC.
D is a point on AB and E is a point on AC such that DE is parallel to BC.
$AD = 6$ cm, $DB = 4$ cm, $AE = 7.5$ cm and $BC = 15$ cm.



(a) Explain why triangle ADE is similar to triangle ABC.

[2]

(b) Calculate the length of DE.

..... cm **[2]**

(c) Calculate the length of EC.

..... cm **[2]**

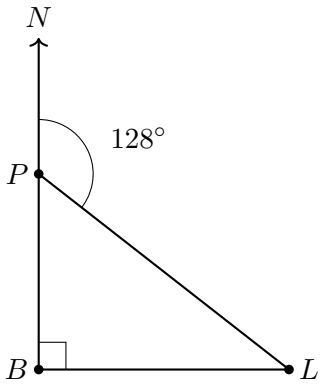
(d) State the geometrical name of the quadrilateral $DBCE$.

..... **[1]**

(e) The area of triangle ADE is 18 cm^2.
Calculate the area of the quadrilateral $DBCE$.

..... cm^2 **[3]**

385 The bearing of a lighthouse L from a port P is 128°.
A boat B is positioned such that it is due South of P and due West of L.



(a) Find the bearing of P from L.

..... ° **[2]**

(b) Work out the size of angle BPL.

..... ° **[1]**

(c) Work out the size of angle PLB.

..... ° **[2]**

386 The bearing of a lighthouse, L, from a boat, B, is 148°.

(a) Work out the bearing of B from L.

..... ° **[2]**

(b) A port, P, is due north of L and due east of B.

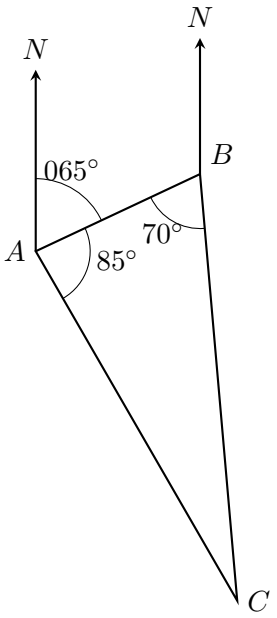
(i) Write down the bearing of B from P.

..... ° **[1]**

(ii) Work out the size of angle LBP.

..... ° **[2]**

387 The diagram shows the positions of three towns, A, B and C.
The bearing of B from A is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.



(a) Calculate the bearing of C from A.

..... ° **[1]**

(b) Calculate the bearing of A from B.

..... ° **[2]**

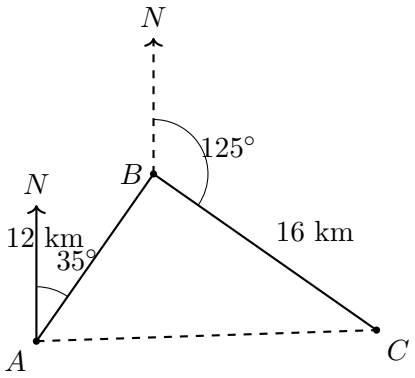
(c) Calculate the bearing of C from B.

..... ° **[2]**

(d) Calculate angle ACB.

..... ° **[1]**

388 A ship leaves a port A and travels 12 km on a bearing of 035° to point B.
From B, the ship travels 16 km on a bearing of 125° to point C.



(a) Show that angle $ABC = 90°$.

[2]

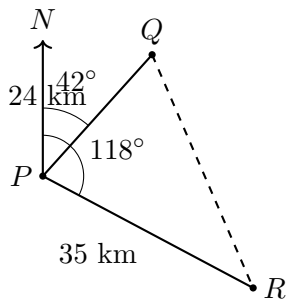
(b) Calculate the direct distance AC.

$AC =$ ..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[3]**

389 Three towns are located at points P, Q, and R.
Town Q is 24 km from P on a bearing of 042°.
Town R is 35 km from P on a bearing of 118°.



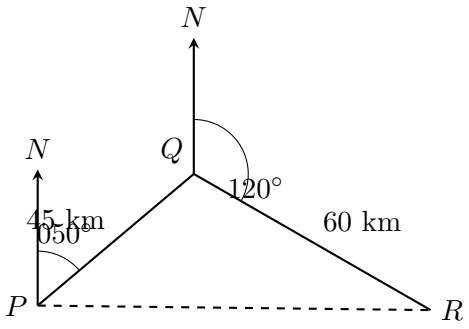
(a) Calculate the distance QR.

$QR =$ ..... km **[3]**

(b) Calculate the bearing of R from Q.

..... ° **[4]**

390 A ship sails 45 km from Port P on a bearing of 050° to Point Q.
It then sails 60 km on a bearing of 120° to Point R.



(a) Calculate the distance PR.

..... km **[4]**

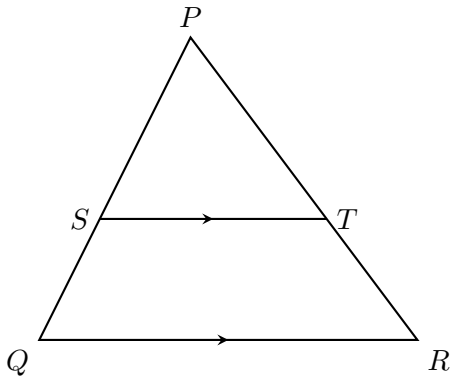
(b) Calculate the bearing of R from P.

..... ° **[3]**

(c) The ship sails directly back from R to P.
Calculate the bearing of P from R.

..... ° **[2]**

391 The diagram shows triangle PQR.
Point S lies on PQ and point T lies on PR.
The line ST is parallel to the line QR.

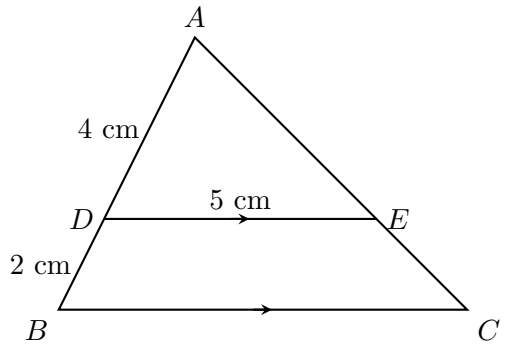


NOT TO SCALE

Given that $PS = 6$ cm, $SQ = 3$ cm and $ST = 8$ cm, calculate the length of QR.

$QR =$ ..... cm **[2]**

392 The diagram shows triangle ABC.
D is a point on AB and E is a point on AC such that DE is parallel to BC.
$AD = 4$ cm, $DB = 2$ cm and $DE = 5$ cm.



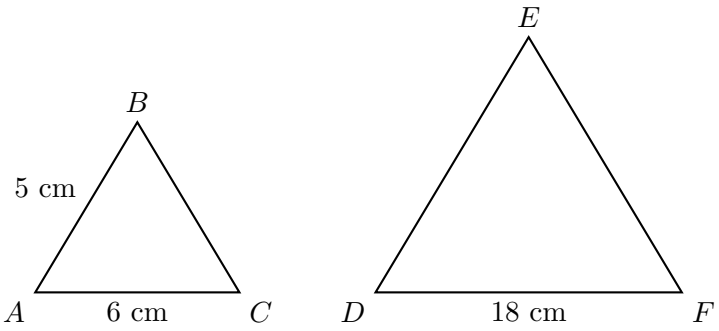
(a) Calculate the length of BC.

..... *cm* **[2]**

(b) The area of triangle ADE is 8 cm^2.
Work out the area of the trapezium $DBCE$.

..... *cm*^{2} **[2]**

393 Triangle ABC and triangle DEF are mathematically similar.



NOT TO SCALE

(a) Calculate the length DE.

$DE =$ ..... cm **[2]**

(b) The area of triangle DEF is 72 cm^2.
Calculate the area of triangle ABC.

..... cm^2 **[2]**

394 Two mathematically similar solid cylinders are shown as Cylinder A and Cylinder B.
Cylinder A has a height of 6 cm.
Cylinder B has a height of 15 cm.

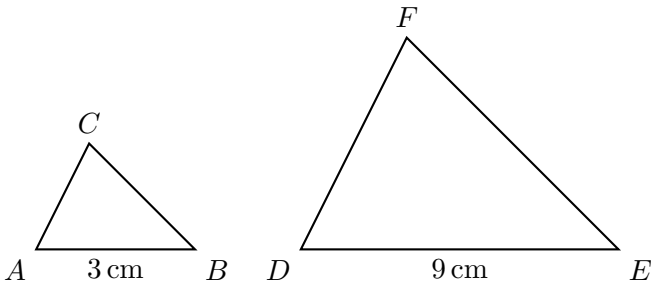
(a) The volume of Cylinder A is 48 cm^3.
Calculate the volume of Cylinder B.

..... cm^3 **[2]**

(b) The curved surface area of Cylinder B is 200 cm^2.
Calculate the curved surface area of Cylinder A.

..... cm^2 **[2]**

395 Triangle ABC and triangle DEF are mathematically similar.
$AB = 3$ cm and $DE = 9$ cm.



NOT TO SCALE

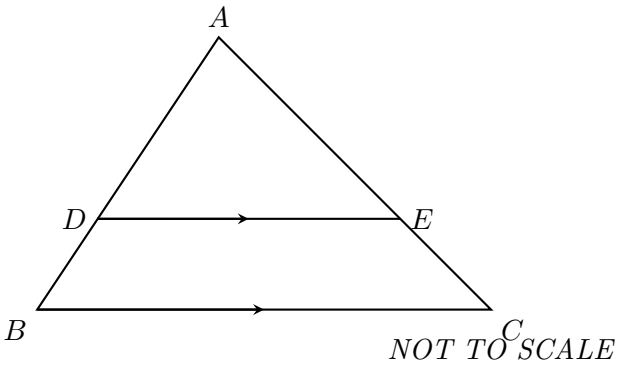
(a) The length of BC is 5 cm.
Calculate the length of EF.

..... cm **[2]**

(b) The area of triangle DEF is 54 cm^2.
Calculate the area of triangle ABC.

..... cm^2 **[2]**

396 In the diagram, ADE is a triangle and DE is parallel to BC.
$AD = 4$ cm, $DB = 2$ cm and $DE = 5$ cm.



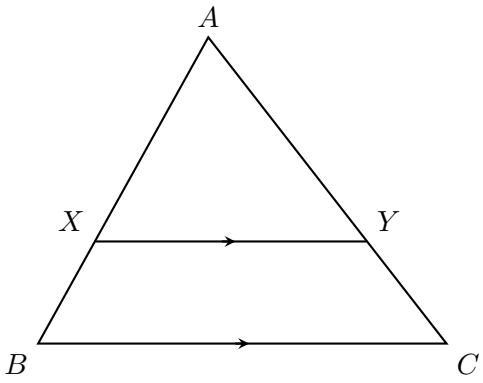
(a) Calculate the length of BC.

$BC =$ ..... cm **[2]**

(b) The area of triangle ADE is 8 cm^2.
Calculate the area of the trapezium $DBCE$.

Area = ... cm^2 **[3]**

397 In the diagram, ABC is a triangle and the line XY is parallel to BC.
$AX = 4$ cm, $XB = 2$ cm and $XY = 6$ cm.



(a) Calculate the length of BC.

$BC =$ ..... cm **[2]**

(b) The area of triangle AXY is 32 cm^2.
Calculate the area of the trapezium $XBCY$.

Area = ... cm^2 **[3]**

398 Two mathematically similar bottles have heights of 10 cm and 15 cm.

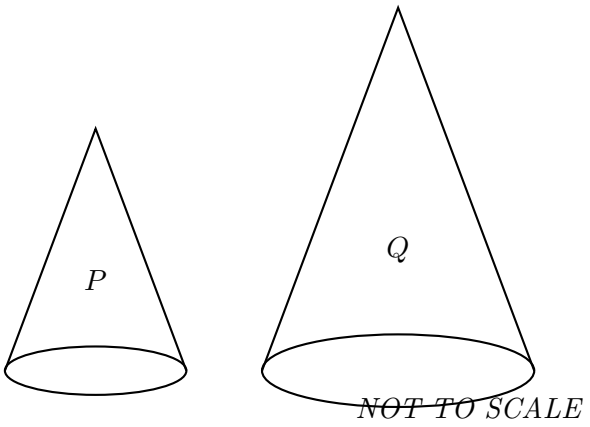
(a) The surface area of the smaller bottle is 120 cm^2.
Work out the surface area of the larger bottle.

..... cm^2 **[3]**

(b) The volume of the larger bottle is 540 cm^3.
Work out the volume of the smaller bottle.

..... cm^3 **[3]**

399 The diagram shows two similar cones, P and Q.
Cone P has a volume of 400 cm^3 and a height of 12 cm.
Cone Q has a volume of 1350 cm^3.



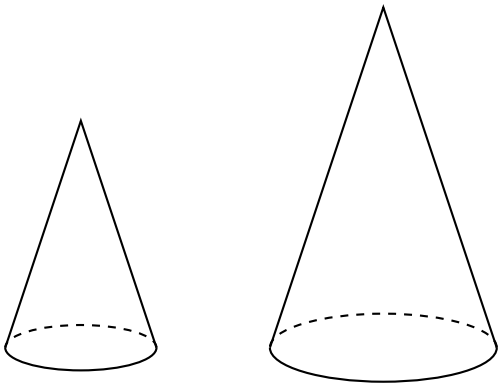
(a) Calculate the height of cone Q.

Height = ... cm **[3]**

(b) The curved surface area of cone Q is 360 cm^2.
Calculate the curved surface area of cone P.

Surface area = ... cm^2 **[2]**

400 Two solid cones, A and B, are mathematically similar.



Cone A Cone B

(a) Cone A has a volume of 150 cm^3 and a height of 12 cm.
Cone B has a volume of 1200 cm^3.
Calculate the height of cone B.

..... cm **[3]**

(b) The surface area of cone B is 440 cm^2.
Calculate the surface area of cone A.

..... cm^2 **[2]**

401 Two mathematically similar containers, A and B, have volumes of 320 cm^3 and 1080 cm^3 respectively.

(a) Container A has a height of 18 cm.

Calculate the height of container B.

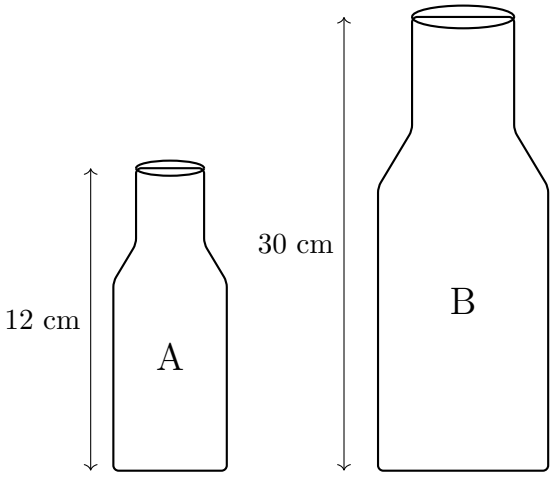
Height = ... cm **[3]**

(b) Container B has a surface area of 630 cm^2.

Calculate the surface area of container A.

Surface area = ... cm^2 **[3]**

402 Two mathematically similar containers, A and B, are shown in the diagram.
Container A has a height of 12 cm and container B has a height of 30 cm.



NOT TO SCALE

(a) Container A has a volume of 320 cm^3.
Calculate the volume of container B.

Volume = ... cm^3 **[2]**

(b) Container B has a surface area of 1125 cm^2.
Calculate the surface area of container A.

Surface area = ... cm^2 **[2]**

(c) The containers are completely filled with liquid.
The cost of filling container A is \$4.80.
Assuming the cost is directly proportional to the volume, calculate the cost of filling container B.

..... **[2]**

403 Two mathematically similar solids, P and Q, have volumes of 320 cm^3 and 8640 cm^3 respectively.

(a) The height of solid P is 12 cm.
Calculate the height of solid Q.

..... cm **[3]**

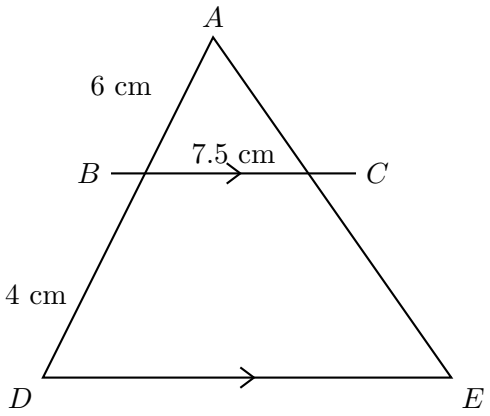
(b) The surface area of solid Q is 2250 cm^2.
Calculate the surface area of solid P.

..... cm^2 **[2]**

(c) Solid P is made of a metal with a density of 8.5 g/cm^3.
Calculate the mass of solid P in kilograms.

..... kg **[2]**

404 The diagram shows triangle ADE with a line BC drawn parallel to DE.
$AB = 6$ cm, $BD = 4$ cm and $BC = 7.5$ cm.



NOT TO SCALE

(a) Explain why triangle ABC is mathematically similar to triangle ADE.

[2]

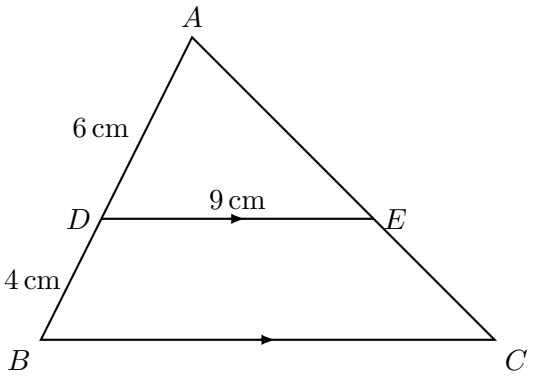
(b) Calculate the length of DE.

$DE =$ ..... cm **[2]**

(c) The area of triangle ABC is 18 cm^2.
Calculate the area of the quadrilateral $BCED$.

..... cm^2 **[3]**

405 In the diagram, DE is parallel to BC.
$AD = 6$ cm, $DB = 4$ cm and $DE = 9$ cm.



NOT TO SCALE

(a) Explain why triangle ADE is similar to triangle ABC.

.....

.....

..... **[2]**

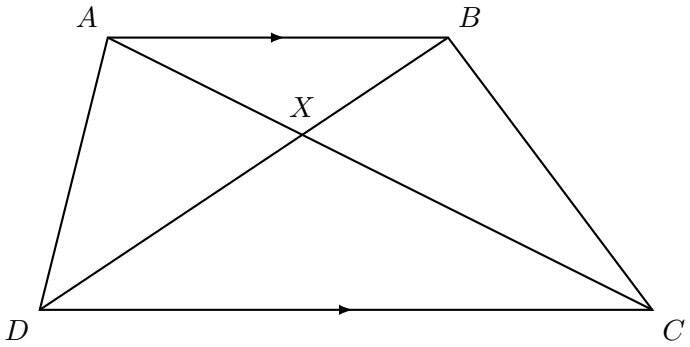
(b) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(c) The area of triangle ADE is 27 cm^2.
Calculate the area of the trapezium $DBCE$.

..... cm^2 **[3]**

406 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The diagonals AC and BD intersect at point X.
$AB = 9$ cm, $DC = 15$ cm and $BX = 6$ cm.
The area of triangle ABX is 27 cm^2.



NOT TO SCALE

(a) Explain why triangle ABX is similar to triangle CDX.

[2]

(b) Calculate the length of BD.

..... cm **[3]**

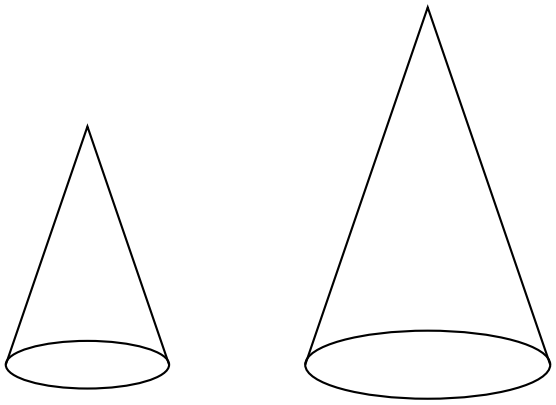
(c) Calculate the area of triangle CDX.

..... cm^2 **[2]**

(d) Calculate the area of triangle BCX.

..... cm^2 **[2]**

407 Cone A and Cone B are mathematically similar solids.



NOT TO SCALE

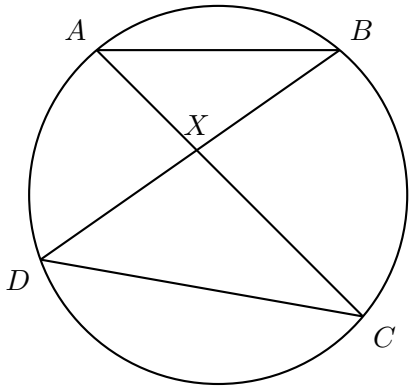
(a) The volume of Cone A is 108 cm^3 and the volume of Cone B is 500 cm^3.
The sum of the surface areas of Cone A and Cone B is 340 cm^2.
Calculate the surface area of Cone B.

..... cm^2 **[4]**

(b) Cone B has a total vertical height of 20 cm.
A smaller cone, mathematically similar to Cone B, is removed from the top of Cone B to leave a frustum.
The volume of the frustum is $\frac{7}{8}$ of the volume of Cone B.
Calculate the vertical height of the removed small cone.

..... cm **[3]**

408 The diagram shows a circle with points A, B, C and D on its circumference.
The chords AC and BD intersect at X.



(a) Show that triangle AXB is similar to triangle DXC.
Give a geometric reason for each statement you make.

[3]

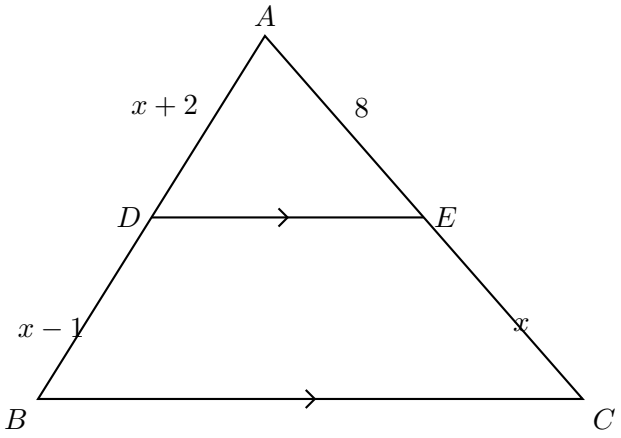
(b) $AX = 4$ cm, $BX = 5$ cm, $CX = 7.5$ cm and $AB = 6$ cm.
Calculate the length of DX and the length of DC.

$DX =$ ..... , $DC =$ ..... **[3]**

(c) The area of triangle AXB is 11.8 cm^2.
Calculate the area of triangle DXC.

..... cm^2 **[2]**

409 In the diagram, DE is parallel to BC.
$AD = (x + 2)$ cm, $DB = (x - 1)$ cm, $AE = 8$ cm and $EC = x$ cm.



NOT TO SCALE

(a) Show that $x^2 - 6x + 8 = 0$.

[3]

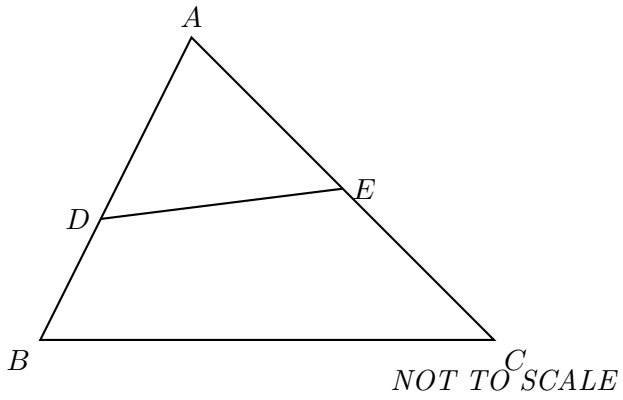
(b) Solve the equation $x^2 - 6x + 8 = 0$.

$x =$ ..... , $x =$ ..... **[2]**

(c) Given that the area of triangle ADE is 24 cm^2 and $x = 4$, calculate the area of the trapezium $DBCE$.

..... cm^2 **[3]**

410 In the diagram, D is a point on AB and E is a point on AC.
Angle AED = Angle ABC.
$AD = 6$ cm, $AE = 5$ cm and $DB = 9$ cm.



(a) Show that triangle ADE is similar to triangle ACB.
Give reasons for your statements.

[3]

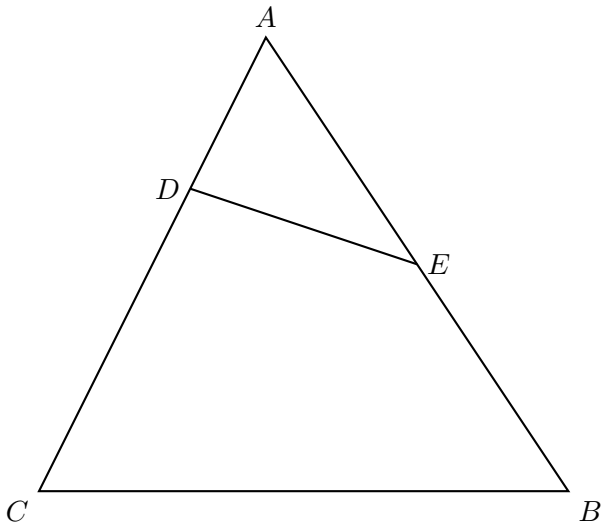
(b) Calculate the length of EC.

$EC =$ ..... cm **[3]**

(c) Find the ratio of the area of triangle ADE to the area of quadrilateral $BCED$ in its simplest form.

..... **[2]**

411 The diagram shows triangle ABC.
D is a point on AC and E is a point on AB.
Angle ABC = Angle AED.



NOT TO SCALE

(a) Explain why triangle ABC is similar to triangle AED.

.....

.....

..... **[2]**

(b) Given that $AC = 15$ cm and $AD = 6$ cm, write down the scale factor of the enlargement that maps triangle AED onto triangle ABC.

..... **[1]**

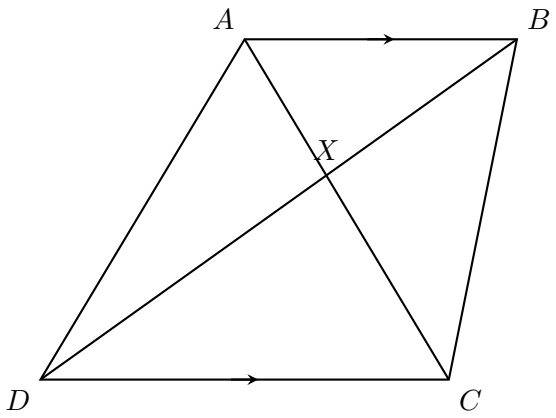
(c) $AE = x$ cm and $AB = (2x + 2.5)$ cm.
Use similarity to set up an equation and find the value of x.

$x =$ ..... **[3]**

(d) The area of the quadrilateral $BCDE$ is 42 cm^2.
Calculate the area of triangle ADE.

..... cm^2 **[3]**

412 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The diagonals AC and BD intersect at X.



(a) Explain, giving geometric reasons, why triangle ABX is similar to triangle CDX.

[3]

(b) $AX = 4$ cm and $CX = 6$ cm.
$AB = x$ cm and $DC = (2x - 3)$ cm.

Work out the value of x.

$x =$ ..... **[3]**

(c) The area of triangle ABX is 24 cm^2.
Calculate the area of the trapezium $ABCD$.

Area = ... cm^2 **[4]**

413 Two solid statues, A and B, are mathematically similar.

(a) Statue A has a height of 12 cm and a volume of 800 cm^3.
Statue B has a volume of 2700 cm^3.
Calculate the height of Statue B.

..... cm **[3]**

(b) The cost to paint a statue is directly proportional to its surface area.
It costs \$18 to paint Statue A.
Calculate the cost to paint Statue B.

\$ ..... **[3]**

(c) A third mathematically similar statue, C, has a surface area that is 44% greater than the surface area of Statue A.
Calculate the volume of Statue C.

..... cm^3 **[4]**

414 Solid Cone A has a base radius of 10 cm and a perpendicular height of 24 cm.

(a) Calculate the total surface area of Cone A.
Give your answer in terms of π.

..... cm^2 **[3]**

(b) Cone B is mathematically similar to Cone A.
The volume of Cone B is 100π cm^3.
Calculate the total surface area of Cone B.
Give your answer in terms of π.

..... cm^2 **[4]**

(c) A frustum is made by cutting a smaller cone, which is geometrically identical to Cone B, from the top of Cone A.
This frustum is melted down and recast completely into a solid sphere.
Calculate the radius of this sphere.

..... cm **[4]**

415 Answer the following questions about the symmetry of two-dimensional shapes.

(a) Write down the number of lines of symmetry of a regular nonagon (a polygon with 9 sides).

..... **[1]**

(b) Write down the order of rotational symmetry of a parallelogram.

..... **[1]**

(c) Write down the mathematical name of a triangle that has exactly 1 line of symmetry and rotational symmetry of order 1.

..... **[1]**

416 Answer the following questions about 2D symmetry.

(a) Write down the order of rotational symmetry of a regular decagon.

..... **[1]**

(b) A quadrilateral has exactly two lines of symmetry and rotational symmetry of order 2.
State the mathematical name of this quadrilateral.

..... **[1]**

(c) A shape has rotational symmetry of order 5.
Calculate the smallest angle of rotation that maps the shape onto itself.

..... $^\circ$ **[1]**

417 Consider the symmetry properties of the following 3D solid shapes.

(a) A right square-based pyramid has a base of side length 8 cm and perpendicular height 10 cm.
Write down the number of planes of symmetry for this pyramid.

..... **[1]**

(b) Write down the total number of planes of symmetry of a solid regular pentagonal prism.

..... **[2]**

(c) A right circular cone has base radius r and slant height l.
Describe fully its axis of symmetry.

..... **[2]**

418 **(a)** A regular polygon has an exterior angle of 15°.

(i) Calculate the number of sides of this polygon.

..... **[2]**

(ii) Write down the number of lines of symmetry of this polygon.

..... **[1]**

(b) Write down the mathematical name of a quadrilateral that has exactly 1 line of symmetry and rotational symmetry of order 1.

..... **[1]**

(c) A solid cone has a base radius of 5 cm and a perpendicular height of 12 cm.

State the number of planes of symmetry of this cone.

..... **[1]**

419 **(a)** A solid wooden block is in the shape of a right prism.
The cross-section of the prism is a regular octagon.

(i) Write down the order of rotational symmetry of the cross-section.

..... **[1]**

(ii) State the total number of planes of symmetry of this right prism.

..... **[2]**

(b) A different solid is a right pyramid with a regular polygon of n sides as its base.
The pyramid has exactly 12 planes of symmetry.

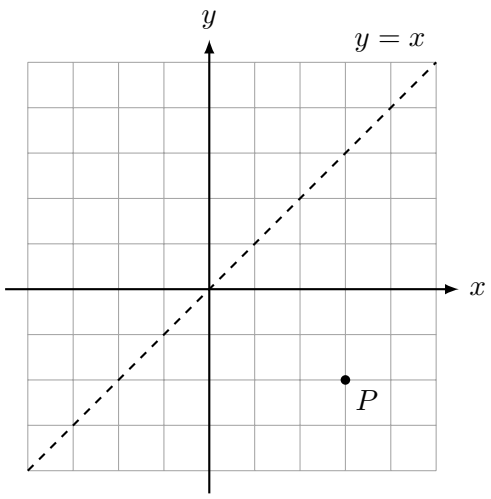
(i) Write down the value of n.

$n =$ ..... **[1]**

(ii) Calculate the sum of the interior angles of the base of this pyramid.

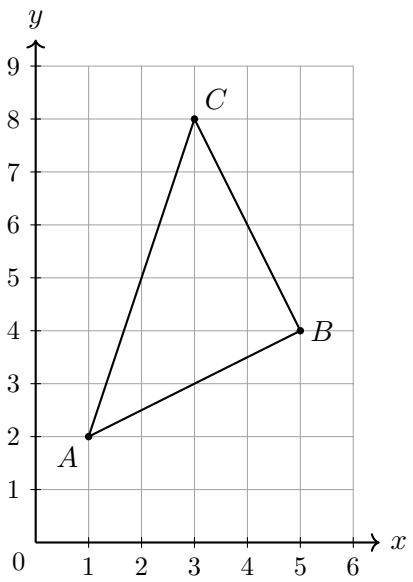
..... ° **[2]**

(c) A shape S is drawn on a coordinate grid.
The shape S has a line of symmetry with equation $y = x$.
One vertex of the shape is at the point $P(3, -2)$.
Write down the coordinates of the image of P after a reflection in this line of symmetry.



(..... , ...) **[2]**

420 Triangle ABC has vertices at $A(1, 2)$, $B(5, 4)$ and $C(3, 8)$.



(a) Show that triangle ABC is an isosceles triangle.

[3]

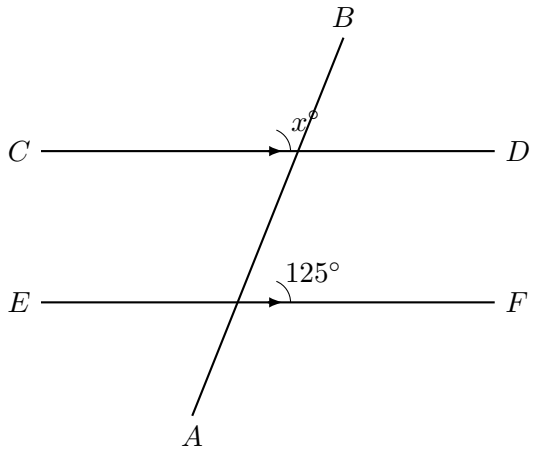
(b) Triangle ABC has exactly one line of symmetry.
Find the equation of this line of symmetry.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(c) A different shape is a regular polygon with n sides.
It has exactly 15 lines of symmetry.
Calculate the size of one interior angle of this polygon.

..... ° **[2]**

421 The diagram shows a straight line AB intersecting two parallel lines, CD and EF.



(a) Find the value of x.

$x =$ ..... **[1]**

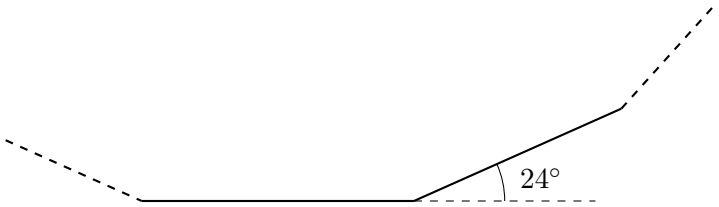
(b) Give a geometric reason for your answer to part (a).

..... **[1]**

(c) Calculate the interior angle of a regular decagon (10-sided polygon).

..... ° **[1]**

422 The diagram shows part of a regular polygon with n sides.
The exterior angle of this polygon is 24°.



(a) Calculate the value of n.

$n =$ ..... **[2]**

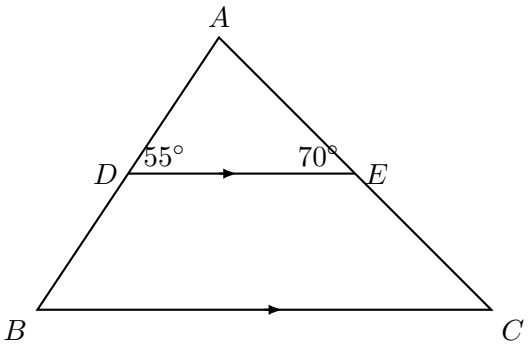
(b) Calculate the sum of the interior angles of this polygon.

..... ° **[2]**

(c) A different regular polygon has an interior angle of 165°.
Work out the number of sides of this polygon.

..... **[2]**

423 The diagram shows a triangle ABC with a line DE drawn parallel to BC.
D lies on AB and E lies on AC.
Angle $ADE = 55°$ and angle $AED = 70°$.



(a) Calculate angle DAE.

Angle $DAE =$ ..... ° **[2]**

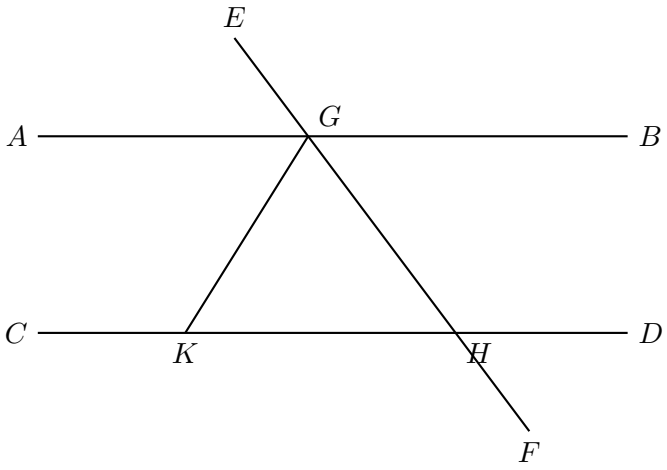
(b) Calculate angle ABC.

Angle $ABC =$ ..... ° **[1]**

(c) Explain why triangle ADE is an isosceles triangle.

[2]

424 In the diagram, the line AB is parallel to the line CD.
The transversal EF intersects AB at G and CD at H.
The point K lies on CD.



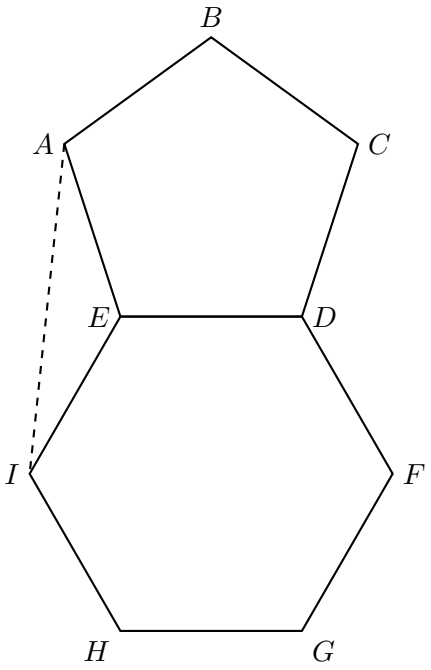
(a) Angle $EGB = (5x + 10)°$ and angle $GHD = (7x - 30)°$.
Find the value of x.
Give a geometrical reason for your working.

$x =$ ..... , Reason: ..... **[3]**

(b) Angle $HGK = 35°$.
Calculate angle GKH.

..... ° **[3]**

425 The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $EDFGHI$ sharing a common edge ED.



(a) Calculate angle CDE.

Angle $CDE =$ ..... ° **[2]**

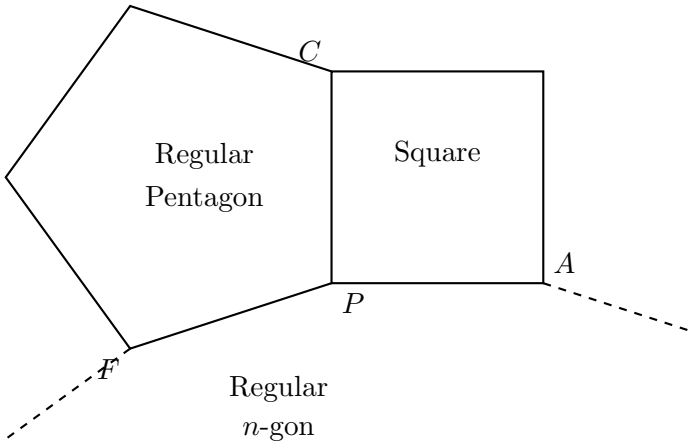
(b) Calculate angle AEI.

Angle $AEI =$ ..... ° **[2]**

(c) Calculate angle EAI.

Angle $EAI =$ ..... ° **[3]**

426 The diagram shows three regular polygons meeting at the point P.
The polygons are a square, a regular pentagon, and a regular polygon with n sides.
The square shares the side PC with the pentagon and the side PA with the n-gon.
The pentagon shares the side PF with the n-gon.
They meet with no gaps and do not overlap.



(a) Calculate the interior angle of a regular pentagon.

..... ° **[2]**

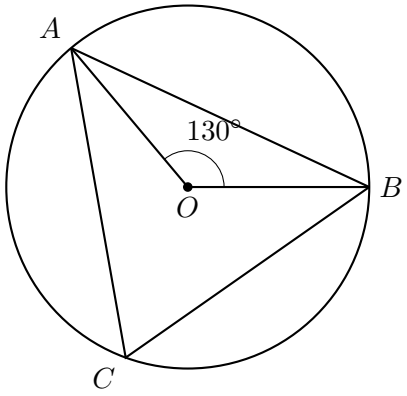
(b) Find the value of n.

$n =$ ..... **[3]**

(c) The vertices A and F of the n-gon are joined by a straight line.
Calculate angle PFA.

..... ° **[2]**

427 Points A, B and C lie on a circle, centre O.
Angle $AOB = 130°$.



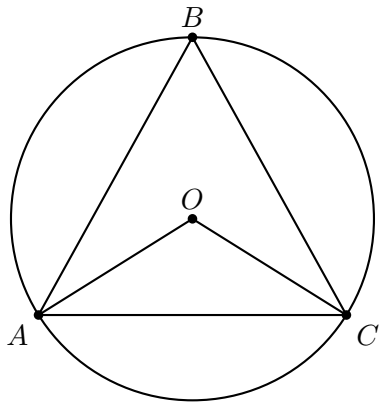
(a) Find the size of angle ACB.

..... ° **[1]**

(b) Find the size of angle OAB.

..... ° **[2]**

428 A, B and C are points on a circle, center O.
Angle $ABC = 58°$.



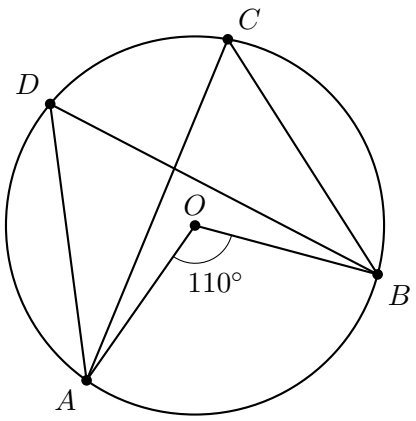
(a) Work out the obtuse angle AOC.

Angle AOC = ... ° **[1]**

(b) Work out angle OAC.

Angle OAC = ... ° **[2]**

429 A, B, C and D are points on a circle, centre O.
Angle $AOB = 110°$.



NOT TO SCALE

(a) Find the size of angle ACB.

Angle $ACB =$ ..... ° **[1]**

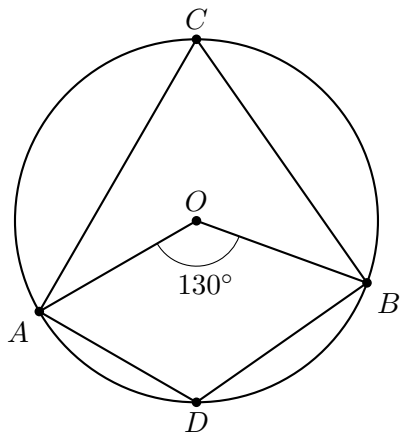
(b) Find the size of angle ADB.

Angle $ADB =$ ..... ° **[1]**

(c) State the geometrical reason for your answer to part (b).

..... **[1]**

430 Points A, B, C and D lie on the circumference of a circle, center O.
Angle $AOB = 130°$.



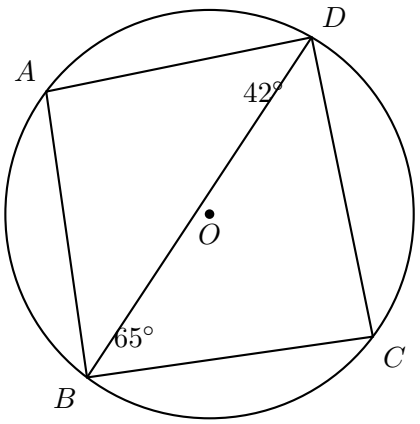
(a) Calculate angle ACB.

..... ° **[1]**

(b) Calculate angle ADB.

..... ° **[2]**

431 A, B, C and D are points on the circumference of a circle, centre O.
Angle $BDC = 42°$ and angle $CBD = 65°$.



(a) Find angle BAC.

Angle $BAC =$ ..... ° **[1]**

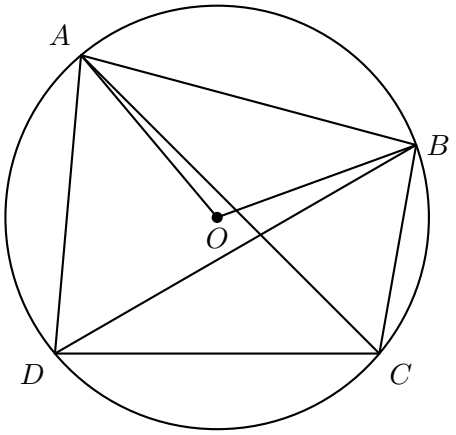
(b) Find angle BCD.

Angle $BCD =$ ..... ° **[1]**

(c) Find angle BAD.

Angle $BAD =$ ..... ° **[2]**

432 The diagram shows a circle with centre O.
The points A, B, C and D lie on the circumference of the circle.
Angle $AOB = 110°$ and angle $BDC = 30°$.



(a) Calculate angle ACB.

..... ° **[1]**

(b) State the geometrical reason for your answer to part (a).

..... **[1]**

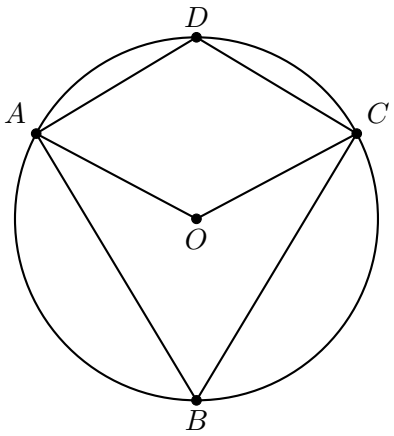
(c) Calculate angle BAC.

..... ° **[1]**

(d) Calculate angle OAB.

..... ° **[2]**

433 A, B, C and D are points on the circumference of a circle, centre O.
Angle $AOC = 124°$.



(a) Calculate angle ABC.

..... ° **[2]**

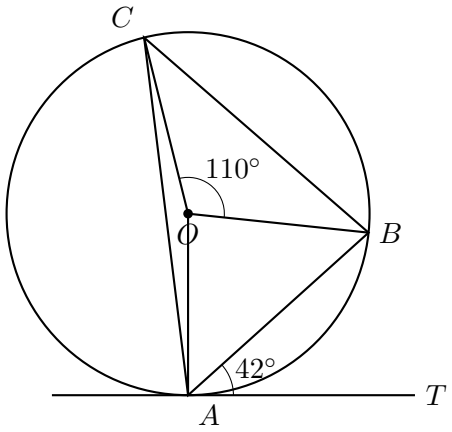
(b) Calculate angle ADC.

..... ° **[2]**

(c) Calculate angle OAC.

..... ° **[2]**

434 A, B and C are points on a circle, centre O.
TA is a tangent to the circle at A.
C is a point on the major arc AB.
Angle $TAB = 42°$ and angle $BOC = 110°$.



(a) Write down the value of

(i) angle BCA,

Angle $BCA =$ ..... ° **[1]**

(ii) angle BAC.

Angle $BAC =$ ..... ° **[1]**

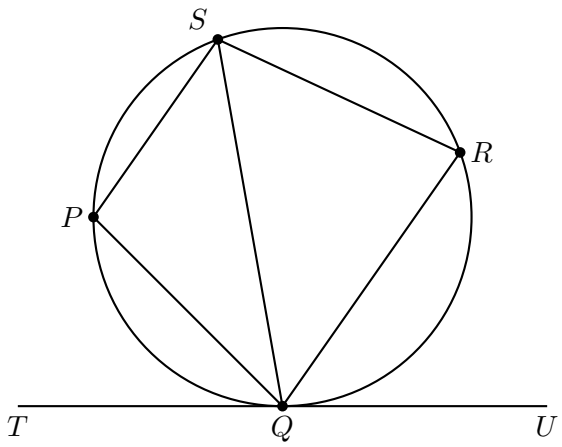
(b) Calculate angle OAB.

Angle $OAB =$ ..... ° **[2]**

(c) Calculate angle ABC.

Angle $ABC =$ ..... ° **[2]**

435 P, Q, R and S are points on a circle.
TQU is a tangent to the circle at Q.
Angle $RQU = 55°$ and Angle $QPS = 100°$.



NOT TO SCALE

(a) Find the size of angle QSR.

Angle $QSR =$ ..... ° **[1]**

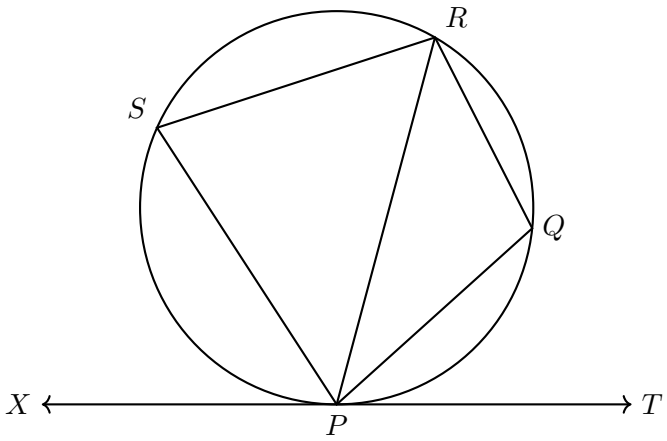
(b) Find the size of angle QRS.

Angle $QRS =$ ..... ° **[2]**

(c) Calculate the size of angle SQR.

Angle $SQR =$ ..... ° **[2]**

436 Points P, Q, R and S lie on a circle.
The straight line XPT is a tangent to the circle at P.
Angle $QPT = 42°$, angle $PQR = 105°$ and angle $SPR = 48°$.



(a) Find the size of angle PRQ.

..... ° **[1]**

(b) Find the size of angle PSR.

..... ° **[1]**

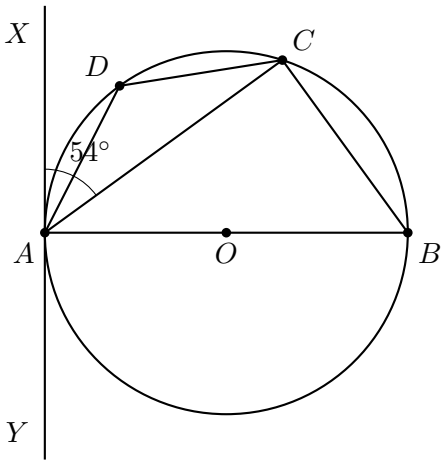
(c) Find the size of angle PRS.

..... ° **[2]**

(d) Find the size of angle XPS.

..... ° **[1]**

437 In the diagram, AB is a diameter of the circle, center O.
The line XAY is a tangent to the circle at A.
C and D are points on the circle.
Angle $XAC = 54°$.



(a) **(i)** Write down the size of angle ACB.

..... ° **[1]**

(ii) Give a geometric reason for your answer.

..... **[1]**

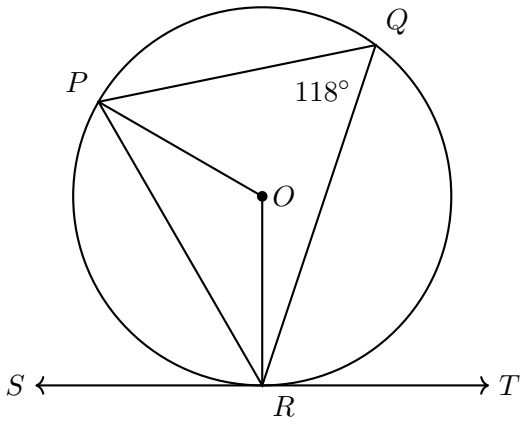
(b) Find angle ABC.

..... ° **[1]**

(c) Find angle ADC.

..... ° **[2]**

438 P, Q and R are points on a circle, centre O.
ST is a tangent to the circle at R.
Angle $PQR = 118°$.



(a) Find reflex angle POR.

Reflex angle $POR =$ ..... ° **[2]**

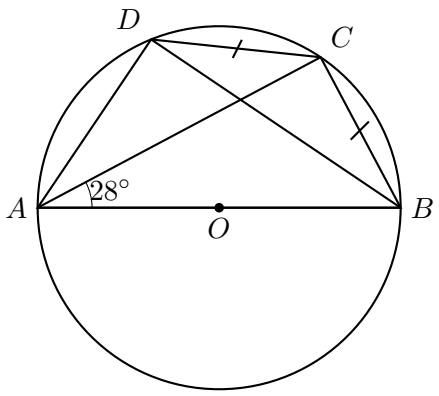
(b) Find obtuse angle POR.

Obtuse angle $POR =$ ..... ° **[1]**

(c) Find angle PRS.

Angle $PRS =$ ..... ° **[2]**

439 A, B, C and D are points on a circle.
AB is a diameter of the circle.
$CD = CB$ and angle $CAB = 28°$.



(a) **(i)** Find angle CBA.

Angle $CBA =$ ..... ° **[1]**

(ii) Give a geometrical reason for your answer.

..... **[1]**

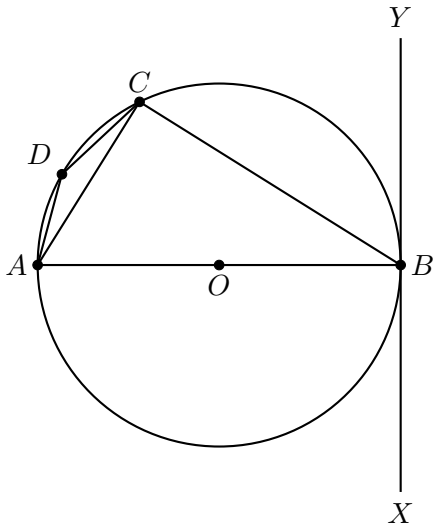
(b) Find angle CAD.

Angle $CAD =$ ..... ° **[1]**

(c) Find angle ADC.

Angle $ADC =$ ..... ° **[2]**

440 A, B, C and D lie on a circle, centre O.
AB is a diameter of the circle.
The line XY is a tangent to the circle at B.
Angle $CBY = 58°$.



(a) **(i)** Find angle BAC.

..... ° **[1]**

(ii) State the circle theorem that gives this angle directly from angle CBY.

..... **[1]**

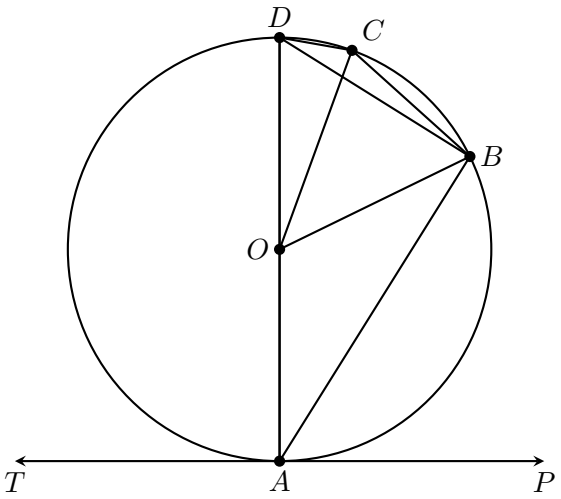
(b) Calculate angle ABC.

..... ° **[3]**

(c) Calculate angle ADC.

..... ° **[2]**

441 A, B, C and D are points on a circle, center O.
AD is a diameter of the circle.
PAT is a tangent to the circle at A.
Angle $PAB = 58°$ and angle $BDC = 22°$.



(a) Calculate angle ADB.

Angle ADB = ... ° **[2]**

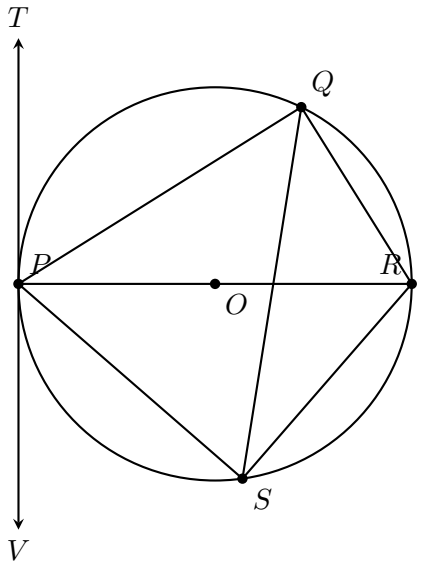
(b) Calculate angle ABC.

Angle ABC = ... ° **[2]**

(c) Calculate angle OBC.

Angle OBC = ... ° **[3]**

442 P, Q, R and S are points on a circle, center O.
PR is a diameter of the circle.
TPV is a tangent to the circle at P.
Angle $QPR = 32°$ and angle $RPS = 41°$.



(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[2]**

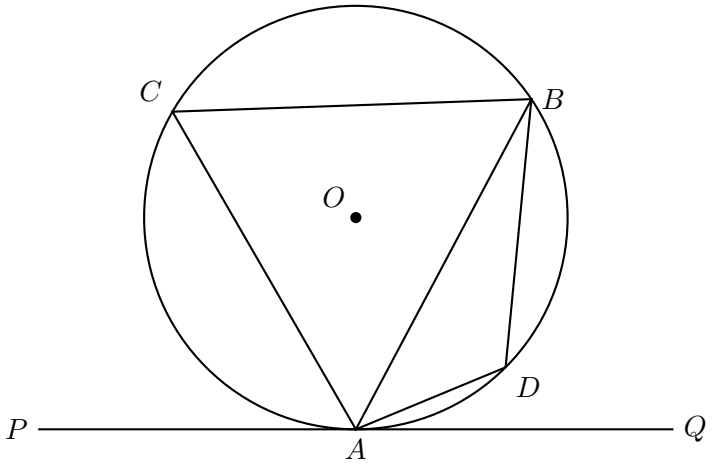
(b) Calculate angle TPQ.

Angle $TPQ =$ ..... ° **[2]**

(c) Calculate angle PQS.

Angle $PQS =$ ..... ° **[2]**

443 The diagram shows a circle with centre O.
The line PAQ is a tangent to the circle at the point A.
The points B, C and D lie on the circle.
Angle $QAB = 62°$.



(a) Calculate angle ACB and give a geometrical reason for your answer.

$Angle ACB =$ ..... ° , Reason: ..... **[2]**

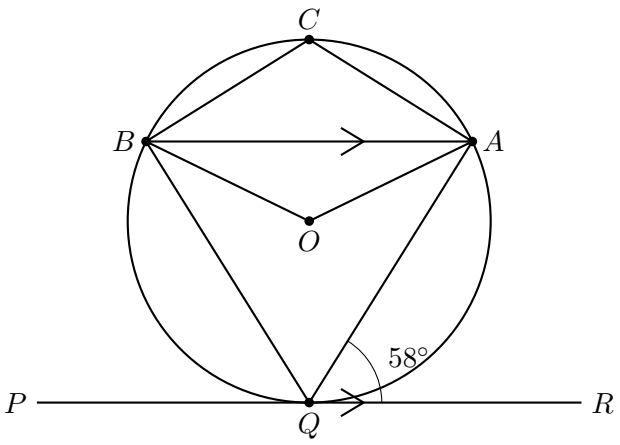
(b) Calculate angle ADB.

..... ° **[2]**

(c) Calculate angle OBA.

..... ° **[2]**

444 A, B, C and Q are points on a circle, center O.
The line PQR is a tangent to the circle at Q.
The chord AB is parallel to PR.
Angle $AQR = 58°$.



(a) **(i)** Find the size of angle ABQ.

..... ° **[1]**

(ii) Give a geometric reason for your answer.

..... **[1]**

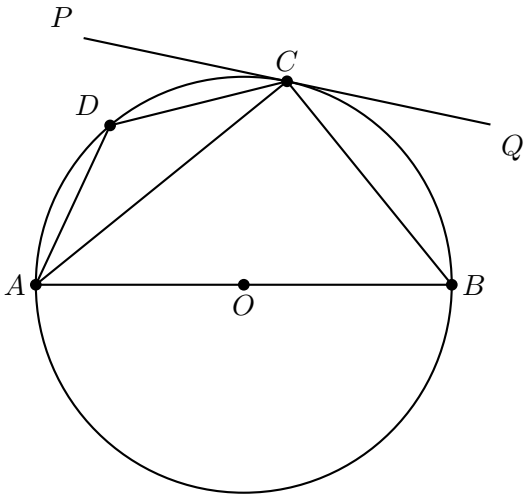
(b) Find angle AOB.

..... ° **[2]**

(c) Find angle ACB.

..... ° **[2]**

445 A, B, C and D are points on a circle, centre O.
AB is a diameter of the circle.
PQ is a tangent to the circle at C.
Angle $CAB = (2x - 5)^\circ$ and Angle $BCQ = (x + 17)^\circ$.



NOT TO SCALE

(a) Set up an equation in terms of x and solve it to show that $x = 22$.
Give a geometrical reason to justify your equation.

[3]

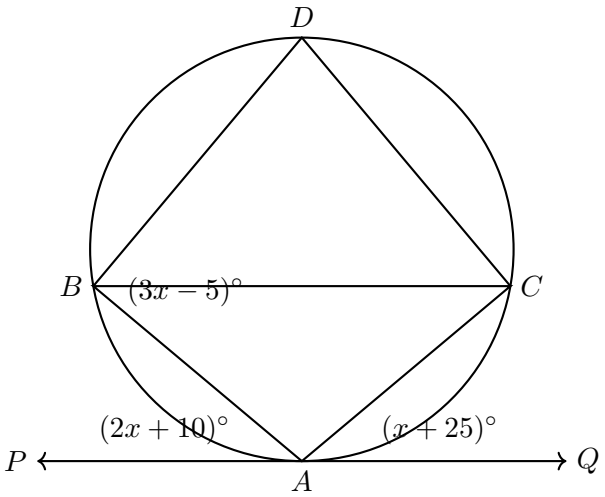
(b) Work out the size of angle CBA.

Angle $CBA =$ ..... ° **[2]**

(c) Find the size of angle ADC.

Angle $ADC =$ ..... ° **[2]**

446 Points A, B, C and D lie on a circle.
The straight line PAQ is a tangent to the circle at A.
Angle $PAB = (2x + 10)°$, angle $ABC = (3x - 5)°$ and angle $CAQ = (x + 25)°$.
D is a point on the major arc BC.



(a) Find the value of x.
Show all your working.

$x =$ ..... **[3]**

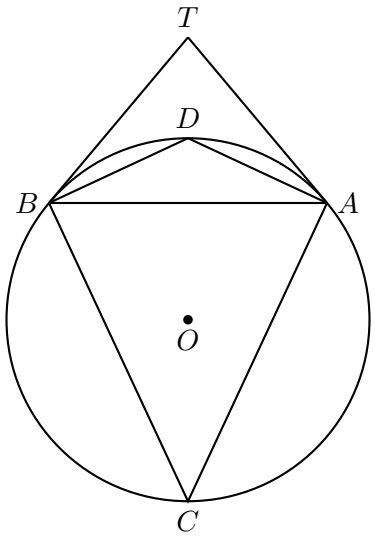
(b) Find the size of angle BAC.

..... ° **[2]**

(c) Find the size of angle BDC.

..... ° **[2]**

447 TA and TB are tangents to a circle, centre O, at the points A and B.
C is a point on the major arc AB and D is a point on the minor arc AB.
Angle $ATB = (x + 10)^\circ$
Angle $TAB = (2y - 10)^\circ$
Angle $ACB = (3y - 40)^\circ$



(a) Find the value of y.

$y =$ ..... **[3]**

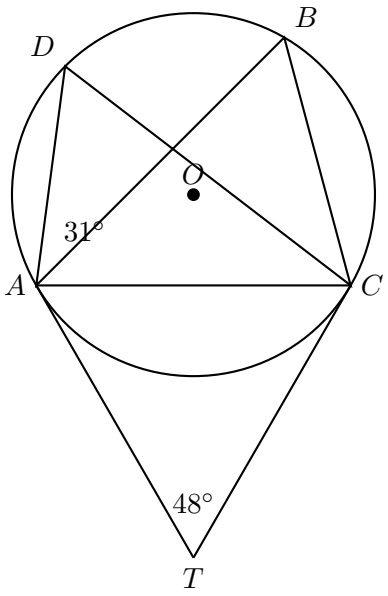
(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find angle ADB.

Angle $ADB =$ ..... $^\circ$ **[2]**

448 A, B, C and D are points on a circle, centre O.
TA and TC are tangents to the circle at A and C.
Angle $ATC = 48°$ and angle $CAD = 31°$.



(a) **(i)** Find angle ABC.

Angle $ABC =$ ..... ° **[2]**

(ii) Find angle ADC.

Angle $ADC =$ ..... ° **[2]**

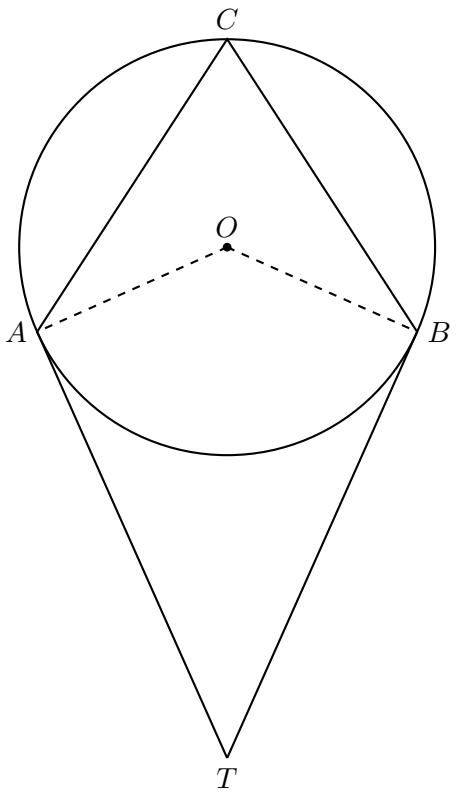
(iii) Find angle ACD.

Angle $ACD =$ ..... ° **[1]**

(iv) Calculate angle AOC.

Angle $AOC =$ ..... ° **[2]**

449 The diagram shows a circle with centre O.
The lines TA and TB are tangents to the circle at the points A and B.
C is a point on the major arc AB.
Angle $ATB = x°$ and angle $ACB = (2x - 30)°$.



(a) (i) Find the value of x.

$x =$ ..... **[4]**

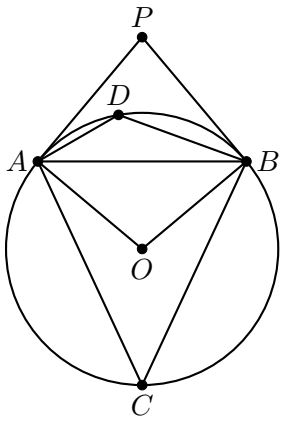
(ii) Calculate angle OAB.

..... ° **[2]**

(b) D is a point on the minor arc AB.
Calculate angle ADB.

..... ° **[2]**

450 PA and PB are tangents to a circle, centre O, from a point P.
C is a point on the major arc AB.
Angle $APB = 4x°$.
Angle $ACB = (3x - 10)°$.



(a) Find an expression for angle PAB in terms of x.

..... ° **[3]**

(b) Use the alternate segment theorem to form an equation in x, and show that $x = 20$.

[3]

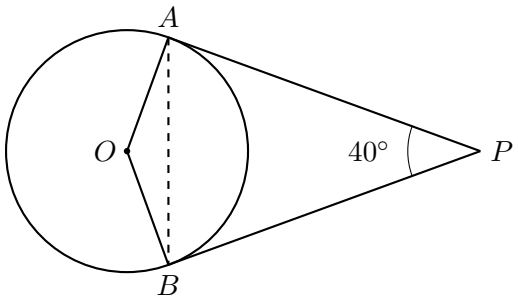
(c) Calculate angle AOB.

..... ° **[3]**

(d) D is a point on the minor arc AB.
Calculate angle ADB.

..... ° **[3]**

451 P is a point outside a circle, centre O.
PA and PB are tangents to the circle at A and B.
Angle $APB = 40°$.



(a) Write down the geometrical reason why triangle PAB is isosceles.

..... **[1]**

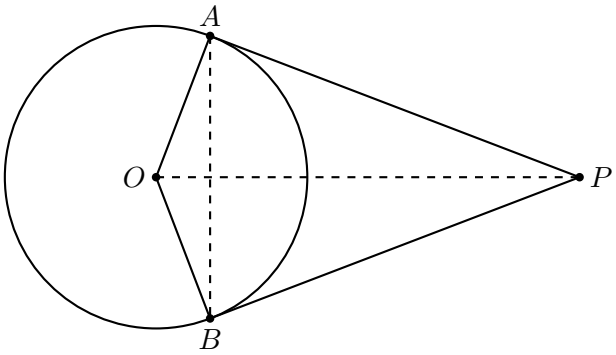
(b) Calculate angle PAB.

Angle $PAB =$ ..... ° **[1]**

(c) Calculate angle AOB.

Angle $AOB =$ ..... ° **[2]**

452 P is an external point.
PA and PB are tangents to a circle, centre O.
The line OP intersects the chord AB at right angles.
$OA = 7$ cm and $PA = 24$ cm.



(a) Write down the length of PB.

..... cm **[1]**

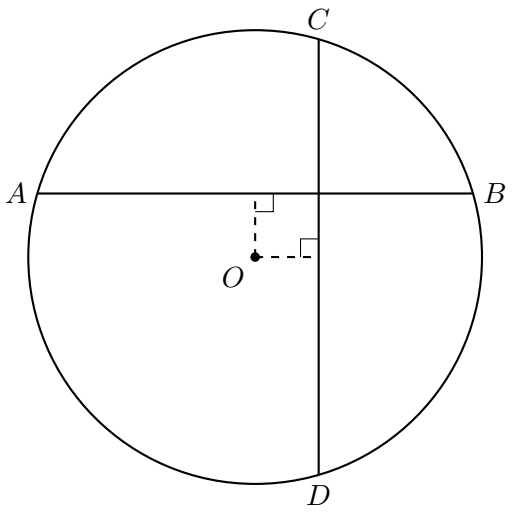
(b) Calculate the length of OP.

..... cm **[2]**

(c) Calculate the shortest distance from O to the chord AB.

..... cm **[3]**

453 The diagram shows a circle, centre O, with radius 25 cm.
AB and CD are two chords of the circle such that $AB = CD$.
The perpendicular distance from O to AB is $(x + 3)$ cm.
The perpendicular distance from O to CD is $(3x - 5)$ cm.



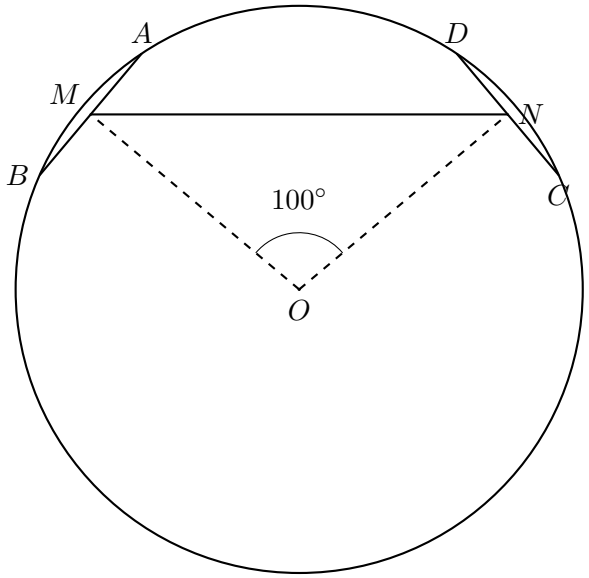
(a) Find the value of x.

$x =$ ..... **[2]**

(b) Calculate the length of the chord AB.

$AB =$ ..... cm **[3]**

454 A, B, C and D are points on a circle, centre O and radius 25 cm.
The chords AB and CD each have a length of 14 cm.
M is the midpoint of AB and N is the midpoint of CD.
Angle $MON = 100°$.



(a) Show that $OM = 24$ cm.

[2]

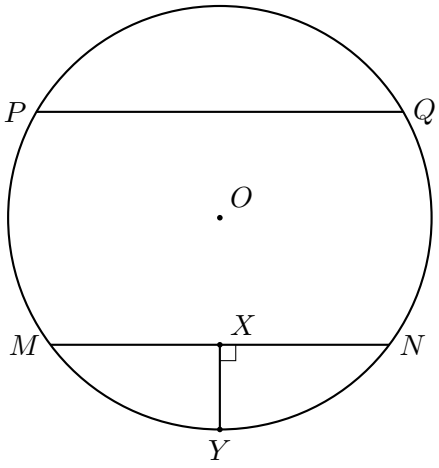
(b) Calculate the length of MN.

..... cm **[3]**

(c) Calculate the area of triangle MON.

..... cm^2 **[2]**

455 A circle has centre O and radius 10 cm.
MN is a chord of length 16 cm.
The perpendicular bisector of MN intersects MN at X and the minor arc MN at Y.
PQ is a second chord which is parallel to MN.
The perpendicular distance between MN and PQ is 11 cm.



(a) Explain why the line containing XY passes through the centre O.

..... **[1]**

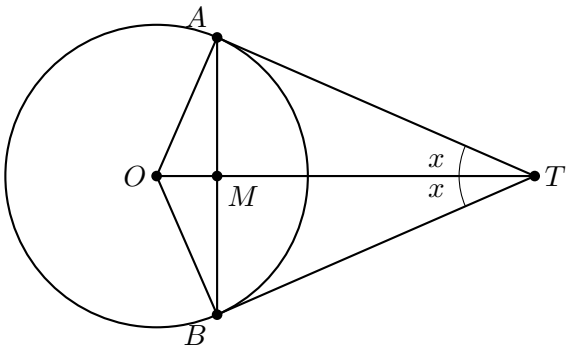
(b) Calculate the length of XY.

$XY =$ ..... cm **[3]**

(c) Calculate the length of the chord PQ.

$PQ =$ ..... cm **[4]**

456 TA and TB are tangents to a circle, centre O and radius r, at the points A and B.
The line TO intersects the chord AB at M.
Angle ATO = Angle $BTO = x$.



(a) Show that the length of the chord AB is $2r\cos x$.

[2]

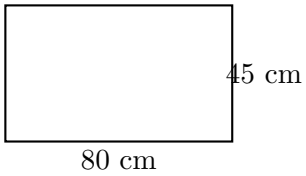
(b) A second chord BC is drawn such that $BC = AB$.
Write down the shortest distance from O to the chord BC, in terms of r and x.

..... **[1]**

(c) The area of triangle OAB is equal to the area of triangle ATB.
Find the exact value of x.

$x =$ ..... $^\circ$ **[5]**

457 A rectangular piece of card and a water jug are being measured.



(a) Calculate the area of the rectangular piece of card in square metres (m^2).

..... m^2 **[2]**

(b) The jug holds 3.2 litres of water.
Write down this volume in cm^3.

..... cm^3 **[1]**

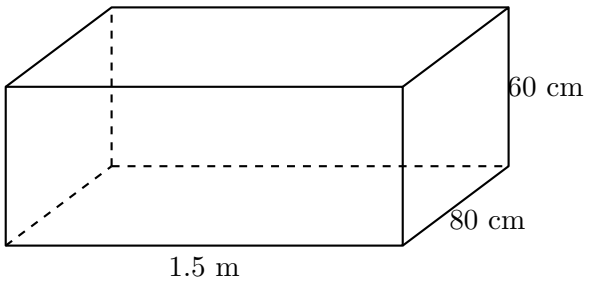
458 **(a)** Change 3.5 m^2 into cm^2.

..... cm^2 **[1]**

(b) A rectangular tank measures 60 cm by 45 cm by 30 cm.
Calculate the volume of the tank in litres.

..... litres **[3]**

459 A metal water tank is made in the shape of a cuboid.



(a) Calculate the capacity of the water tank in litres.

..... litres **[3]**

(b) A solid metal sphere is melted down to make the tank.
The metal sphere has a volume of 0.024 m^3.
The mass of 1 cm^3 of this metal is 7.8 g.

Calculate the total mass of the sphere, giving your answer in kilograms.

..... kg **[3]**

460 During a storm, 15 mm of rain falls on a flat rectangular roof measuring 15 m by 12 m.

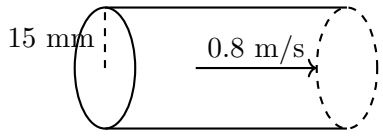
(a) Calculate the volume of rain that falls on the roof.
Give your answer in litres.

..... litres **[3]**

(b) The water from the roof flows into a cylindrical tank with a base radius of 60 cm.
Calculate the increase in the depth of the water in the tank.
Give your answer in centimetres, correct to 1 decimal place.

..... cm **[4]**

461 Water flows through a cylindrical pipe into an empty rectangular pool.
The pool measures 4 m by 2.5 m.



(a) The cylindrical pipe has a radius of 15 mm.
Water flows through the pipe at a speed of 0.8 m/s.

Show that the volume of water flowing out of the pipe in one second is 0.565 litres, correct to 3 significant figures.

[3]

(b) Calculate the time taken for the depth of water in the pool to reach 1.2 m.
Give your answer in hours and minutes, correct to the nearest minute.

..... hours ..... minutes **[4]**

462 Water flows through a cylindrical pipe of radius 2.5 cm at a speed of 18 km/h.



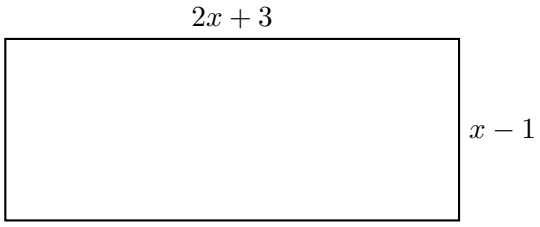
(a) Calculate the volume of water, in litres, that flows through the pipe in 1 minute.

..... litres **[5]**

(b) The water from the pipe is used to fill an empty rectangular reservoir measuring 50 m by 30 m by 4 m.
Calculate the time it takes to fill the reservoir completely.
Give your answer in days and hours, correct to the nearest hour.

days ..... , hours ..... **[7]**

463 The diagram shows a rectangle.
All measurements are in centimetres.



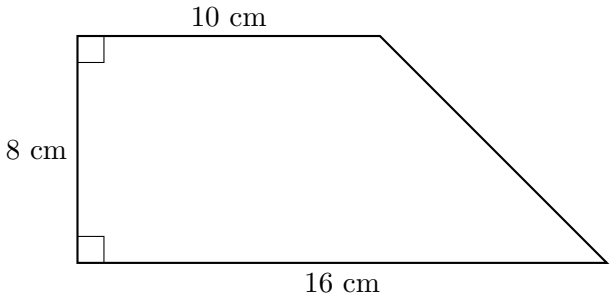
(a) The perimeter of the rectangle is 34 cm.
Find the value of x.

$x =$ ..... **[3]**

(b) Calculate the area of the rectangle.

..... cm^2 **[2]**

464 The diagram shows a right-angled trapezium.



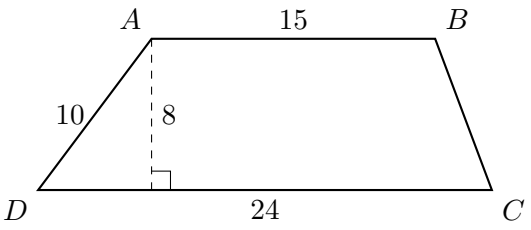
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[3]**

465 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



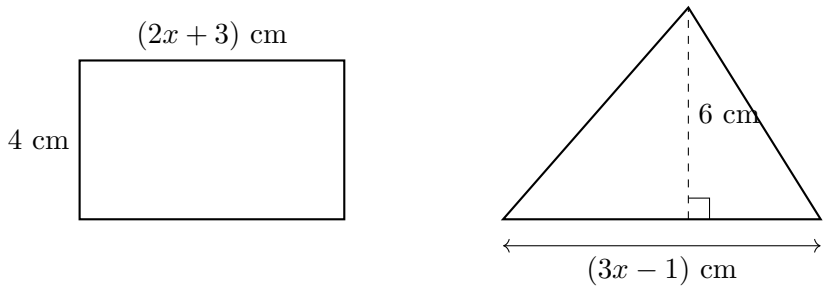
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[4]**

466 The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



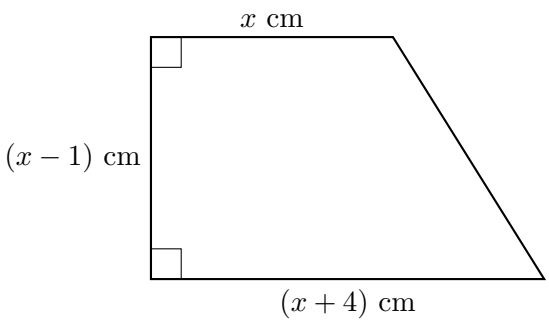
(a) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[4]**

(b) Work out the perimeter of the rectangle.

..... cm **[2]**

467 The diagram shows a right-angled trapezium with parallel sides of length x cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm^2.



(a) Show that $x^2 + x - 42 = 0$.

[3]

(b) Solve the equation $x^2 + x - 42 = 0$ to find the value of x.

$x =$ ..... **[2]**

(c) Calculate the perimeter of the trapezium.

..... cm **[4]**

468 A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be y cm.

(a) Write down an expression, in terms of y, for the side length of the square.

..... cm **[2]**

(b) The total area enclosed by the square and the rectangle is 113 cm^2.
Show that $17y^2 - 180y + 448 = 0$.

[3]

(c) Solve the equation.

$$17y^2 - 180y + 448 = 0$$

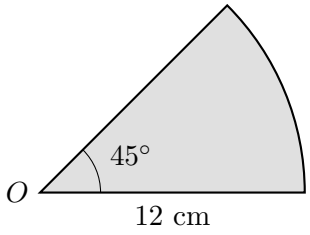
Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

(d) Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... cm^2 **[2]**

469 The diagram shows a sector of a circle, center O, with radius 12 cm and sector angle 45°.



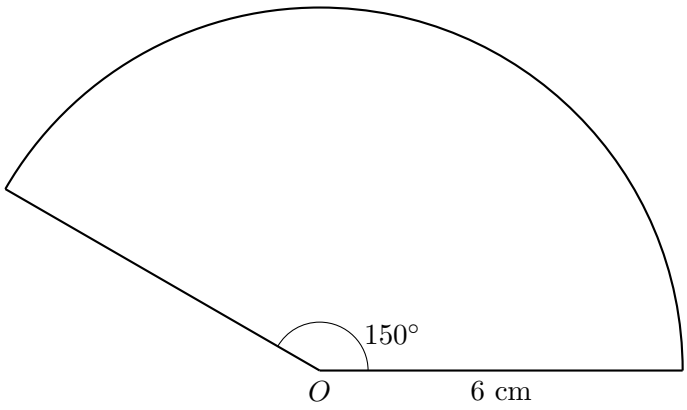
(a) Work out the exact length of the arc of the sector.
Give your answer in terms of π.

..... cm **[2]**

(b) Work out the exact area of the sector.
Give your answer in terms of π.

..... cm^2 **[2]**

470 The diagram shows a sector of a circle with centre O and radius 6 cm.
The sector angle is 150°.



(a) Calculate the area of the sector.
Give your answer in terms of π.

..... cm^2 **[2]**

(b) Calculate the perimeter of the sector.
Give your answer in terms of π.

..... cm **[3]**

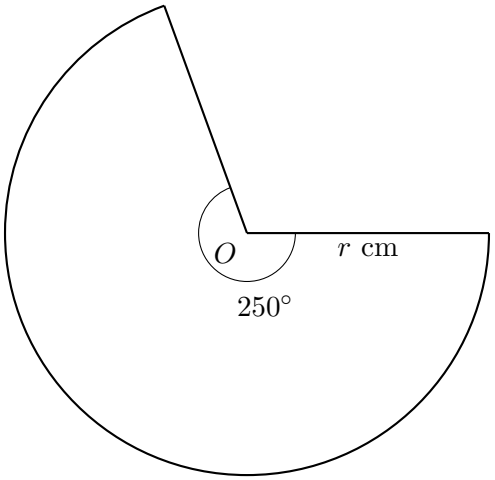
471 **(a)** A sector of a circle, center O, has radius r cm and sector angle $\theta°$.
The area of the sector is 10π cm^2 and the length of its arc is 2π cm.
Find the value of r and the value of θ.

$r =$ ..... , $\theta =$ ..... **[5]**

(b) Another sector of a circle has radius x cm and sector angle $(x + 20)°$.
The length of its arc is $\frac{16\pi}{5}$ cm.
Find the value of x.
You must show all your working.

$x =$ ..... **[4]**

472 The diagram shows a major sector of a circle with centre O and radius r cm.
The reflex angle is 250°.
The length of the major arc is 35 cm.



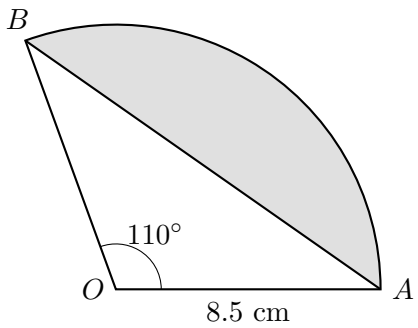
(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Calculate the area of the major sector.

..... cm^2 **[3]**

473 The diagram shows a sector OAB of a circle with center O and radius 8.5 cm.
The sector angle is 110°.
A straight line connects A and B, forming a triangle OAB and a shaded segment.



(a) Calculate the area of the sector OAB.

..... cm^2 **[2]**

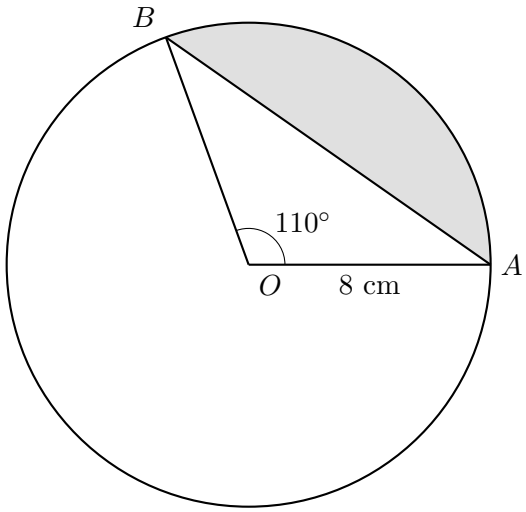
(b) Calculate the area of the shaded segment.

..... cm^2 **[3]**

(c) Calculate the perimeter of the shaded segment.

..... cm **[4]**

474 The diagram shows a circle with centre O and radius 8 cm.
A and B are points on the circumference such that angle $AOB = 110°$.



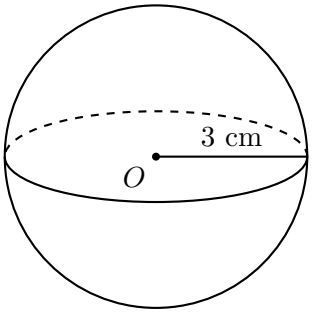
(a) Calculate the area of the shaded segment.

..... cm^2 **[4]**

(b) Calculate the perimeter of the shaded segment.

..... cm **[3]**

475 The diagram shows a solid sphere with center O and radius 3 cm.



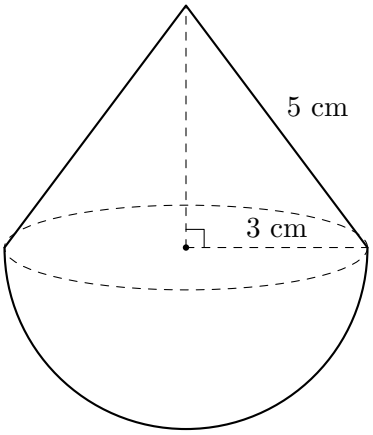
(a) Calculate the exact volume of the sphere.
Give your answer in terms of π.

..... cm^3 **[2]**

(b) Calculate the exact surface area of the sphere.
Give your answer in terms of π.

..... cm^2 **[2]**

476 The diagram shows a solid formed by joining a cone to a hemisphere with the same radius.
The radius of the hemisphere is 3 cm and the slant height of the cone is 5 cm.



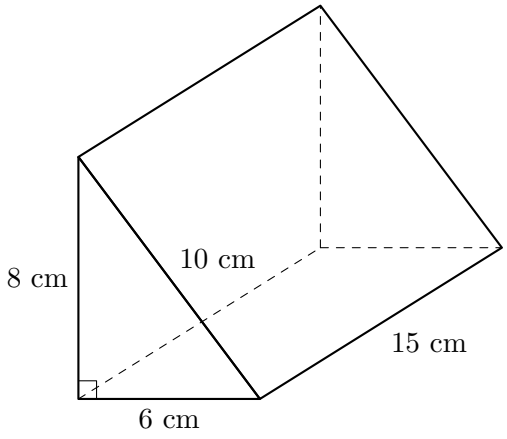
(a) Calculate the exact volume of the solid.
Give your answer in terms of π.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

477 The diagram shows a solid triangular prism.
The cross-section of the prism is a right-angled triangle with sides 6 cm, 8 cm and 10 cm.
The length of the prism is 15 cm.



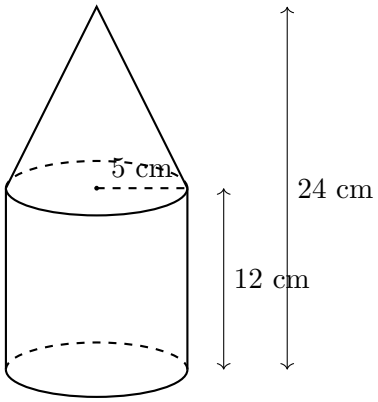
(a) Calculate the volume of the prism.

..... cm^3 **[2]**

(b) Calculate the total surface area of the prism.

..... cm^2 **[3]**

478 A solid is formed by attaching a right circular cone to the top of a solid cylinder.
The cylinder and the base of the cone both have a radius of 5 cm.
The height of the cylinder is 12 cm and the total height of the solid is 24 cm.



(a) Calculate the volume of the solid.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[5]**

479 A solid metal cylinder has radius x cm and height $3x$ cm.
The cylinder is melted down and completely recast into solid spheres.
Each sphere has a radius of $\frac{x}{2}$ cm.

(a) Find an expression, in terms of x and π, for the volume of the cylinder.
Give your answer in its simplest form.

..... cm^3 **[2]**

(b) Find an expression, in terms of x and π, for the volume of one sphere.
Give your answer in its simplest form.

..... cm^3 **[2]**

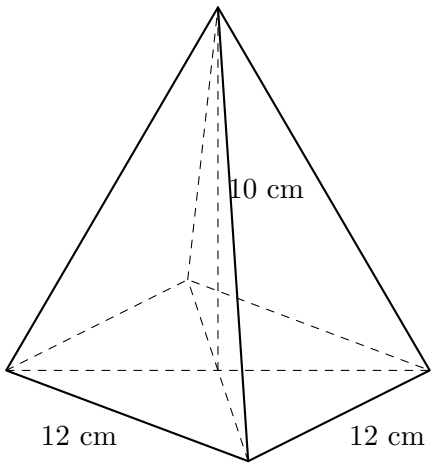
(c) Work out the number of complete spheres that can be made.

..... **[1]**

(d) The total surface area of the original cylinder is 140 cm^2.
Work out the value of x.

$x =$ ..... **[3]**

480 A solid metal pyramid has a square base of side 12 cm and a perpendicular height of 10 cm.



(a) Calculate the volume of the pyramid.

..... cm^3 **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

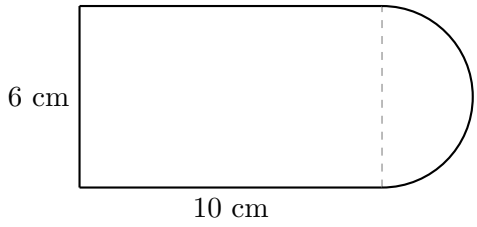
(c) The metal pyramid is melted down and recast to make solid spheres of radius 2.5 cm.
Calculate the number of complete spheres that can be made.

..... **[3]**

(d) A mathematically similar pyramid has a volume of 1620 cm^3.
Calculate the perpendicular height of this similar pyramid.

..... cm **[3]**

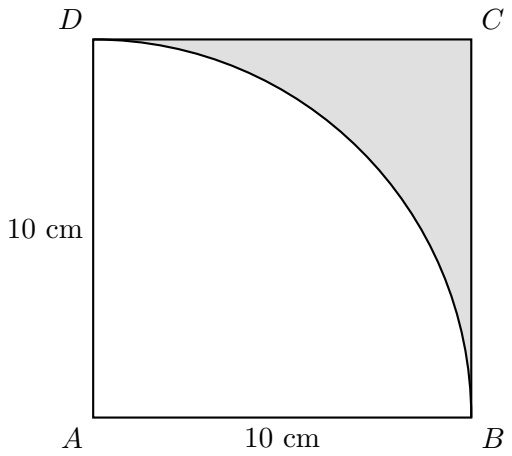
481 A shape is made from a rectangle and a semicircle.
The rectangle has a length of 10 cm and a width of 6 cm.
The semicircle is attached to one of the 6 cm sides.



Find the exact perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

482 The diagram shows a square $ABCD$ with side length 10 cm.
A quarter circle with center A and radius 10 cm is drawn inside the square.
The region inside the square but outside the quarter circle is shaded.



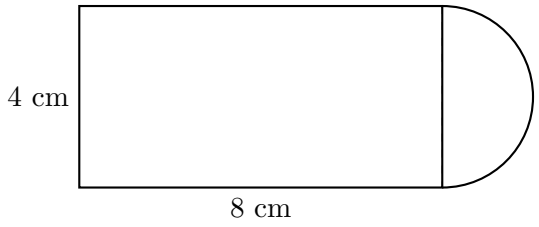
(a) Find the exact perimeter of the shaded region.
Give your answer in terms of π.

..... cm **[2]**

(b) Find the exact area of the shaded region.
Give your answer in terms of π.

..... cm^2 **[2]**

483 The diagram shows a compound shape formed by joining a semi-circle to a rectangle.
The rectangle has a length of 8 cm and a width of 4 cm.
The diameter of the semi-circle is 4 cm.



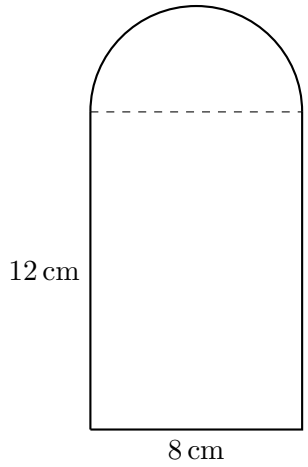
(a) Calculate the perimeter of the shape.
Give your answer in its simplest form, in terms of π.

..... *cm* **[2]**

(b) Calculate the area of the shape.
Give your answer in its simplest form, in terms of π.

..... cm^2 **[2]**

484 A compound shape is made from a rectangle and a semicircle.
The rectangle has width 8 cm and length 12 cm. The semicircle is attached to one of the sides of width 8 cm.



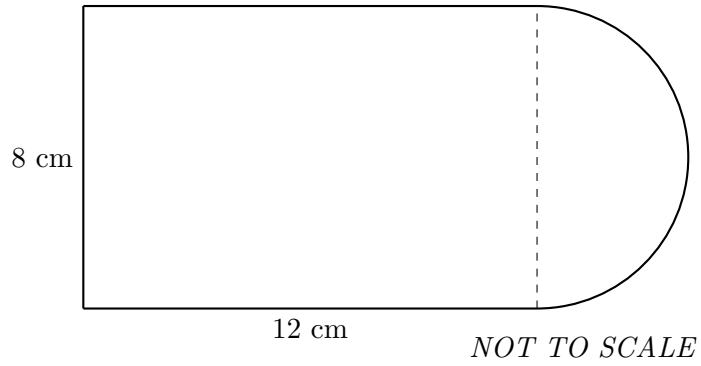
(a) Calculate the exact perimeter of the compound shape.
Give your answer in its simplest form, in terms of π.

..... cm **[2]**

(b) Calculate the exact area of the compound shape.
Give your answer in its simplest form, in terms of π.

..... cm^2 **[2]**

485 The diagram shows a shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 8 cm.
The semicircle has a diameter of 8 cm.



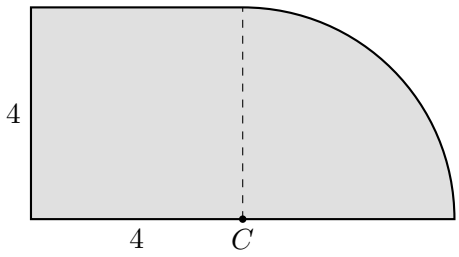
(a) Find the perimeter of the shape.
Give your answer in its simplest form in terms of π.

Perimeter = ... cm **[2]**

(b) Find the area of the shape.
Give your answer in its simplest form in terms of π.

Area = ... cm^2 **[3]**

486 The diagram shows a shape formed by a square of side 4 cm and a quarter of a circle with center C.



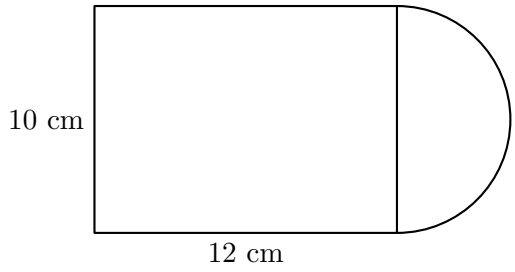
(a) Find the area of the shape.
Give your answer in terms of π.

Area = ... cm^2 **[3]**

(b) Find the perimeter of the shape.
Give your answer in terms of π.

Perimeter = ... cm **[3]**

487 The diagram shows a compound shape made from a rectangle and a semicircle.
The rectangle has length 12 cm and width 10 cm.
The semicircle has a diameter of 10 cm.



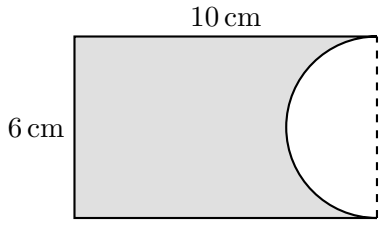
(a) Calculate the perimeter of the shape.
Give your answer in the form $a + b\pi$.

..... cm **[3]**

(b) Calculate the area of the shape.
Give your answer in the form $c + d\pi$.

..... cm^2 **[3]**

488 The diagram shows a shape made from a rectangle with dimensions 10 cm by 6 cm. A semicircle of diameter 6 cm has been removed from one end.



NOT TO SCALE

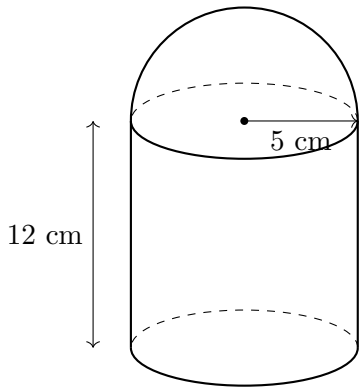
(a) Calculate the area of the shape.
Give your answer in terms of π.

..... cm^2 **[3]**

(b) Calculate the perimeter of the shape.
Give your answer in terms of π.

..... cm **[3]**

489 A solid is made by attaching a hemisphere of radius 5 cm to one end of a solid cylinder.
The cylinder has a radius of 5 cm and a height of 12 cm.



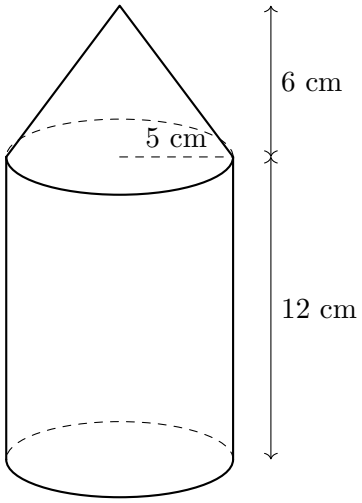
(a) Calculate the total volume of the solid.
Give your answer correct to 3 significant figures.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... cm^2 **[3]**

490 The diagram shows a solid compound shape made from a cylinder and a cone.
The cylinder and the cone both have a base radius of 5 cm.
The height of the cylinder is 12 cm and the height of the cone is 6 cm.



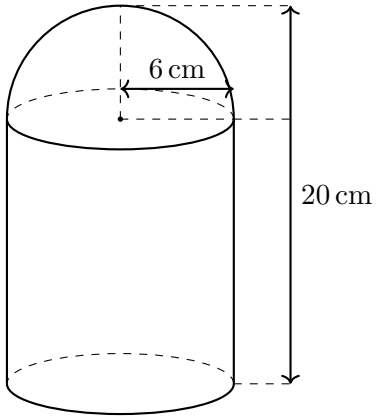
(a) Calculate the volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

491 A solid is made by placing a solid hemisphere on top of a solid cylinder.
The cylinder and the hemisphere both have radius 6 cm.
The total height of the solid is 20 cm.



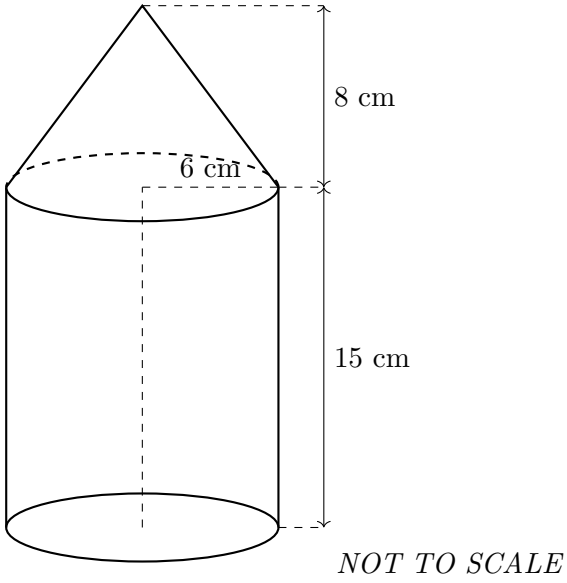
(a) Calculate the volume of the solid.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[4]**

492 A solid is made from a cylinder and a cone.
The cylinder has radius 6 cm and height 15 cm.
The cone has radius 6 cm and height 8 cm.



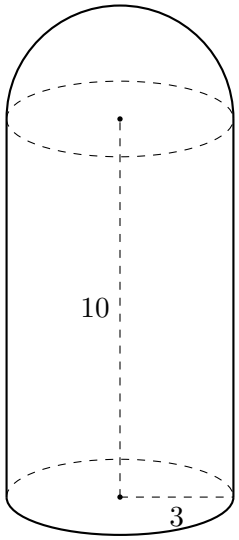
(a) Calculate the total volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[4]**

493 A solid is made by attaching a hemisphere of radius 3 cm to one end of a solid cylinder of radius 3 cm and height 10 cm.



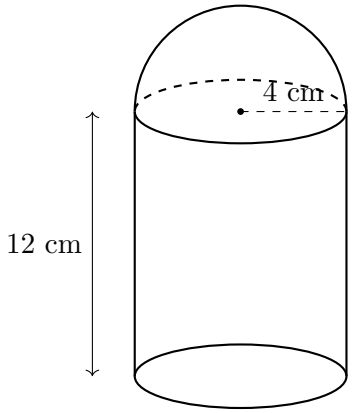
(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Total surface area = ... cm^2 **[4]**

494 A solid object is formed by joining a hemisphere to one end of a solid cylinder.
The cylinder has a radius of 4 cm and a length of 12 cm.
The hemisphere has the same radius as the cylinder.



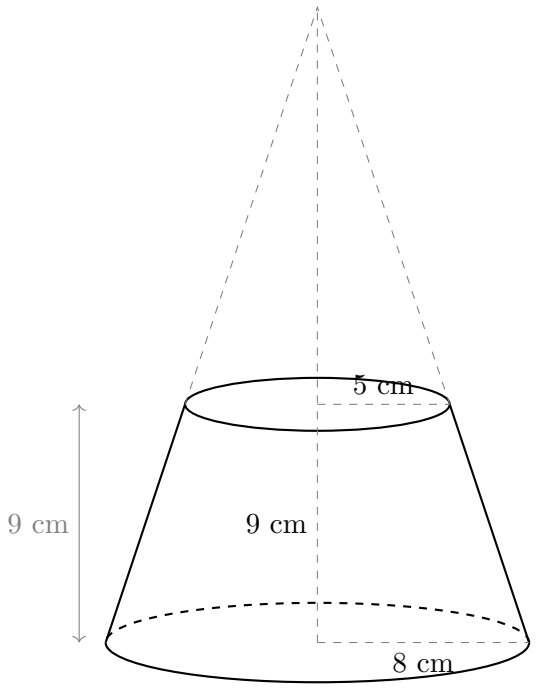
(a) Calculate the volume of the solid.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the solid.

Surface area = ... cm^2 **[4]**

495 The diagram shows a solid frustum.
The frustum is made by removing a smaller cone from a larger cone.
The larger cone has a base radius of 8 cm and a vertical height of 24 cm.
The smaller cone has a base radius of 5 cm and a vertical height of 15 cm.



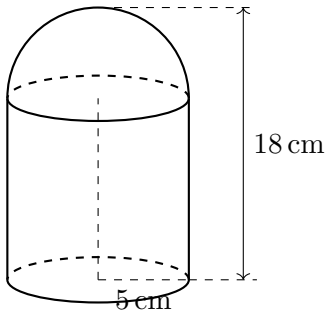
(a) Calculate the volume of the solid frustum.

..... cm^3 **[3]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[5]**

496 A solid is formed by placing a hemisphere of radius 5 cm on top of a cylinder of radius 5 cm.
The total height of the solid is 18 cm.



NOT TO SCALE

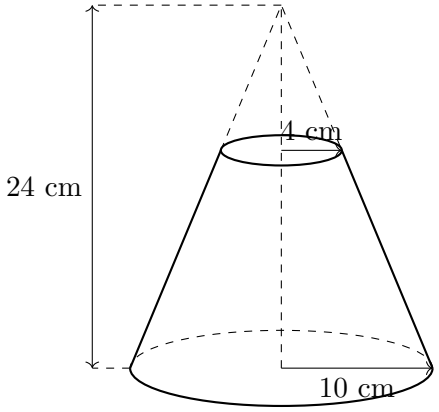
(a) Calculate the volume of the solid.
Give your answer correct to 3 significant figures.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer correct to 3 significant figures.

..... cm^2 **[4]**

497 A solid cone of height 24 cm and base radius 10 cm is cut parallel to its base to form a frustum and a smaller cone.
The radius of the top circular face of the frustum is 4 cm.



(a) Show that the height of the removed small cone is 9.6 cm.

[2]

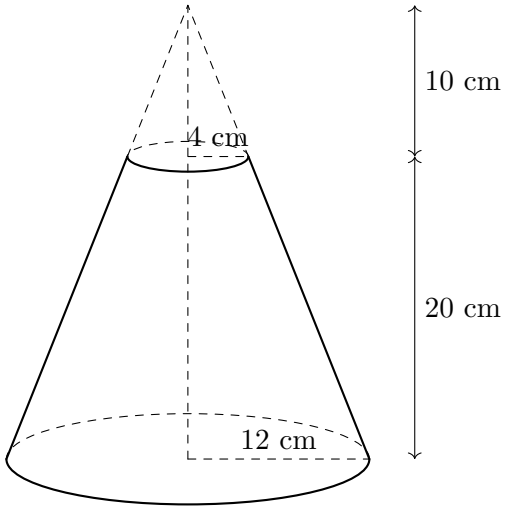
(b) Calculate the volume of the frustum.

..... cm^3 **[3]**

(c) The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of this sphere.

..... cm **[3]**

498 The diagram shows a solid metal frustum.
The frustum is formed by removing a small cone of height 10 cm from a larger cone.
The radius of the circular base of the frustum is 12 cm and the radius of the circular top is 4 cm.
The height of the frustum is 20 cm.



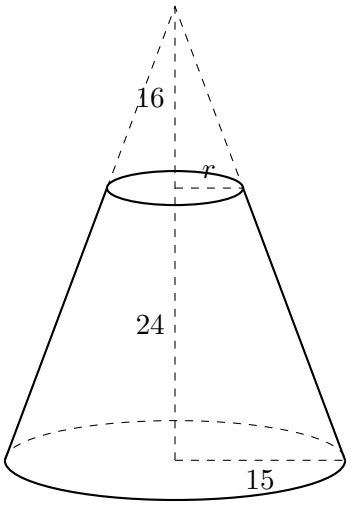
(a) Calculate the volume of the solid frustum.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[4]**

499 A container is in the shape of a frustum of a cone.
The original cone has a base radius of 15 cm and a vertical height of 40 cm.
A smaller cone of vertical height 16 cm is removed from the top to form the frustum.



(a) Show that the radius r of the top circular face of the frustum is 6 cm.

[2]

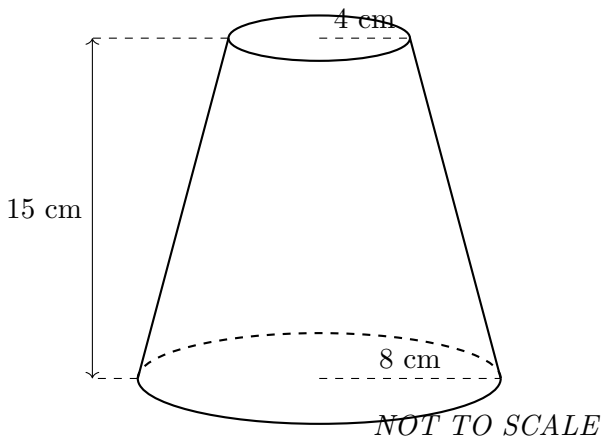
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) The solid frustum is melted down and recast entirely into a solid sphere.
Calculate the radius of the sphere.

Radius = ... cm **[3]**

500 A solid frustum of a cone has top radius 4 cm, bottom radius 8 cm and vertical height 15 cm.
It is formed by removing a smaller cone from the top of a larger cone.



(a) Show that the original height of the cone before the top was removed was 30 cm.

[2]

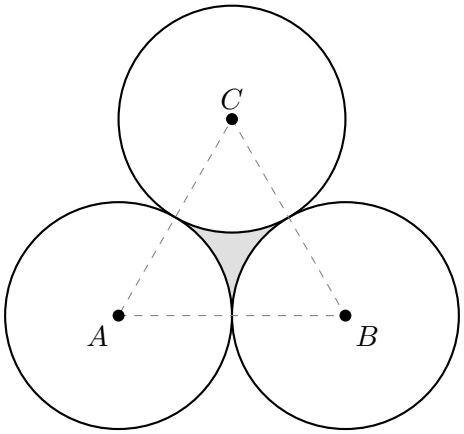
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the solid frustum.

Surface area = ... cm^2 **[4]**

501 The diagram shows three identical circles, each of radius r cm.
The circles touch each other externally.
The region enclosed between the three circles is shaded.



(a) Show that the area of the shaded region, A, is given by the formula

$$A = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$$

[4]

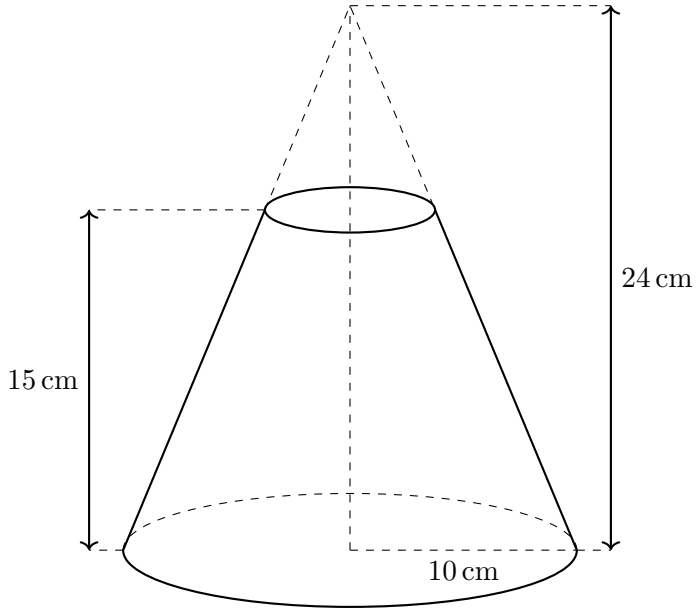
(b) The area of the shaded region is 15 cm^2.
Work out the value of r.

$r =$ ..... **[3]**

(c) Calculate the total perimeter of the shaded region.

..... cm **[3]**

502 A frustum is made by removing a smaller cone from the top of a larger cone.
The larger cone has base radius 10 cm and vertical height 24 cm.
The frustum has vertical height 15 cm.



(a) Find the radius of the top of the frustum.

..... cm **[2]**

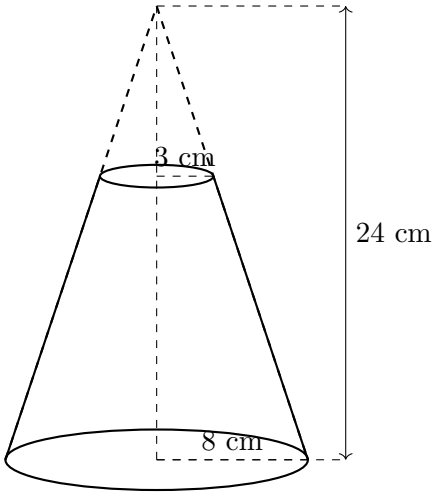
(b) Calculate the volume of the frustum.

..... cm^3 **[3]**

(c) Calculate the total surface area of the frustum.

..... cm^2 **[5]**

503 A frustum is made by removing a small cone from the top of a large cone.
The large cone has a base radius of 8 cm and a total height of 24 cm.
The removed small cone has a base radius of 3 cm.



(a) Show that the height of the small cone removed is 9 cm.

[2]

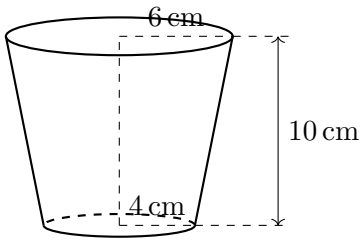
(b) Calculate the volume of the frustum.

Volume = ... cm^3 **[3]**

(c) Calculate the total surface area of the frustum.

Surface area = ... cm^2 **[5]**

504 The diagram shows a closed container in the shape of a frustum of a cone.
The radius of the top circular face is 6 cm and the radius of the base is 4 cm.
The vertical height of the container is 10 cm.



NOT TO SCALE

(a) The frustum is made by removing a smaller cone from a larger cone.
Calculate the vertical height of the larger cone.

..... cm **[3]**

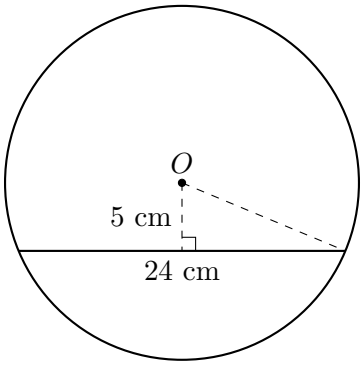
(b) Calculate the volume of the container.
Give your answer correct to 3 significant figures.

..... cm^3 **[3]**

(c) Calculate the total surface area of the closed container.
Give your answer correct to 3 significant figures.

..... cm^2 **[5]**

505 A chord of a circle, centre O, has length 24 cm. The perpendicular distance from the centre of the circle to the chord is 5 cm.



Calculate the radius of the circle.

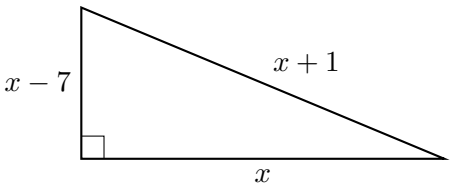
..... cm **[3]**

506 The point A has coordinates $(1, 4)$ and the point B has coordinates $(k, 19)$.
The exact distance between point A and point B is 17.

Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

507 The diagram shows a right-angled triangle.
All lengths are in centimetres.



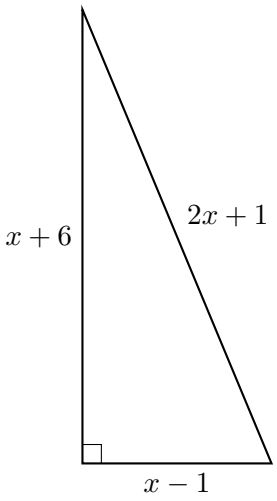
(a) Show that $x^2 - 16x + 48 = 0$.

[3]

(b) Solve the equation to find the value of x, and hence calculate the perimeter of the triangle.

Perimeter = ... cm **[4]**

508 The diagram shows a right-angled triangle.
The lengths of the sides are $x - 1$, $x + 6$, and $2x + 1$, where x is an integer.
All lengths are in centimetres.



(a) Show that $x^2 - 3x - 18 = 0$.

[4]

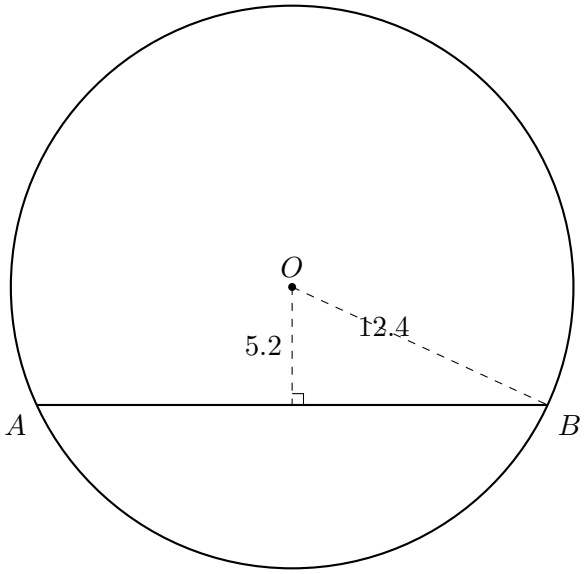
(b) Solve the equation by factorising to find the value of x.
You must show your working.

$x =$ ..... **[3]**

(c) Work out the area of the triangle.

..... cm^2 **[3]**

509 The diagram shows a circle, centre O, with radius 12.4 cm.
AB is a chord of the circle.
The perpendicular distance from O to the chord AB is 5.2 cm.



(a) Calculate the length of the chord AB.

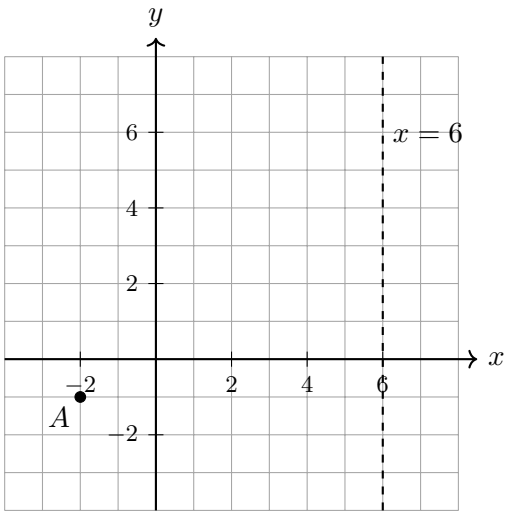
..... cm **[3]**

(b) Another chord CD is drawn parallel to AB and on the opposite side of the centre O.
The length of the chord CD is 18 cm.

Calculate the perpendicular distance between the two chords AB and CD.

..... cm **[4]**

510 The diagram shows a coordinate grid.
Point A is $(-2, -1)$ and point B lies on the line $x = 6$.
The y-coordinate of B is positive.
The distance AB is 10 units.



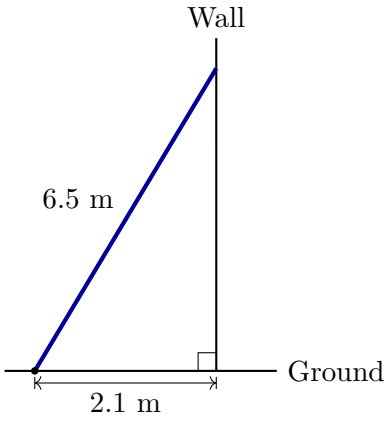
(a) Find the y-coordinate of point B.

$y =$ ..... **[3]**

(b) Point C lies on the x-axis.
The triangle ABC is isosceles with $AC = BC$.
Calculate the x-coordinate of C.

$x =$ ..... **[4]**

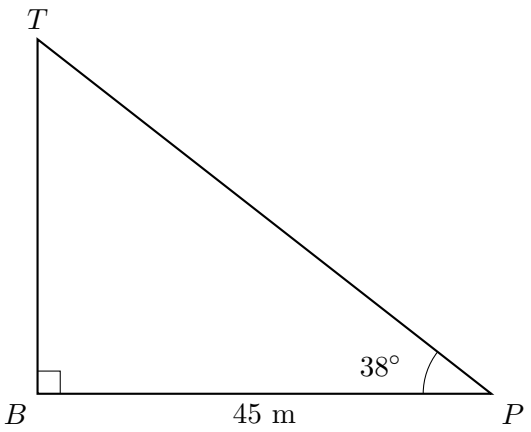
511 A straight ladder of length 6.5 m leans against a vertical wall.
The base of the ladder is 2.1 m from the bottom of the wall on horizontal ground.



Calculate the angle the ladder makes with the horizontal ground.

..... ° **[3]**

512 A surveyor stands at a point P, which is 45 m away from the base, B, of a vertical tower TB. The angle of elevation of the top of the tower, T, from P is 38°.



(a) Calculate the height of the tower, TB.

..... m **[2]**

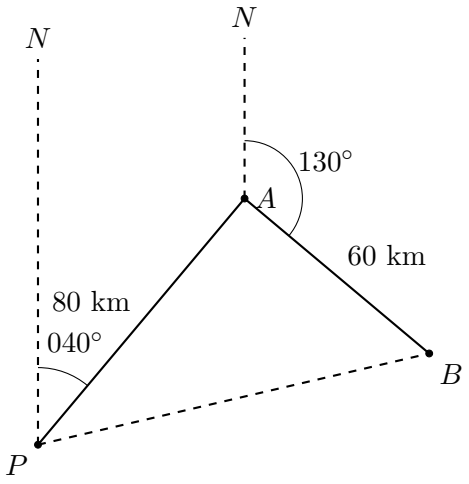
(b) Calculate the straight-line distance PT.

..... m **[2]**

(c) A bird sits at a point X on the tower, where $BX = 20$ m.
Calculate the angle of elevation of the bird from P.

..... ° **[3]**

513 A ship sails from port P on a bearing of 040° for 80 km to point A.
It then changes course and sails on a bearing of 130° for 60 km to point B.



(a) Calculate the direct distance from P to B.

$PB =$ ..... km **[3]**

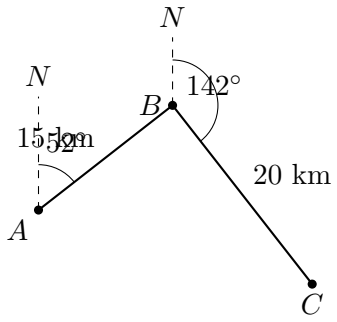
(b) Calculate the bearing of B from P.

Bearing = ... ° **[3]**

(c) Calculate the shortest distance from point A to the path PB.

..... km **[3]**

514 A ship sails from port A to point B on a bearing of 052° for 15 km.
It then sails from point B to point C on a bearing of 142° for 20 km.



(a) Show that angle $ABC = 90°$.

[2]

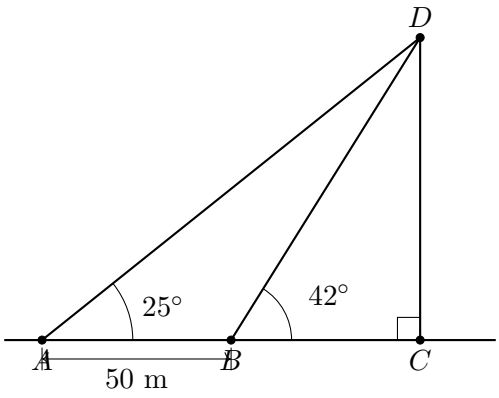
(b) Calculate the distance AC.

..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[4]**

515 Point A and point B are on horizontal ground, in a straight line with the base C of a vertical building CD.
The distance between A and B is 50 m.
The angle of elevation of the top of the building, D, from A is 25° and from B is 42°.



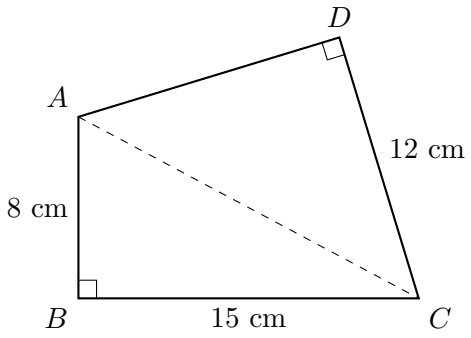
(a) Calculate the height of the building CD.

$CD =$ ..... m **[5]**

(b) A bird flies directly along the straight path from A to D.
Calculate the shortest distance from point B to the flight path AD.

..... m **[3]**

516 The diagram shows a quadrilateral $ABCD$.
Angle $ABC = 90°$ and angle $ADC = 90°$.
$AB = 8$ cm, $BC = 15$ cm, and $CD = 12$ cm.



(a) Calculate the length of AC.

..... cm **[2]**

(b) Calculate the length of AD.

..... cm **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

(d) Calculate the size of angle DAB.

..... ° **[4]**

517 Without using a calculator, answer the following questions.

(a) Write down the exact value of $\cos 30°$.

..... **[1]**

(b) Show that

$$\frac{1 - \sin 30°}{\tan^2 60°} = \frac{1}{6}$$

[2]

518 **(a)** Work out the exact value of

$$2\sin 60^\circ \times \tan 30^\circ$$

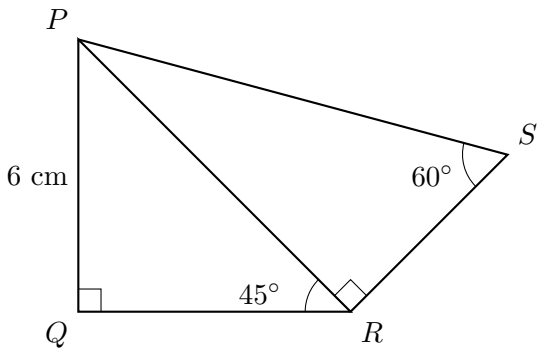
..... **[2]**

(b) Find the value of x, where $0^\circ \leq x \leq 90^\circ$, such that

$$\sqrt{2}\cos x = \tan 45^\circ$$

$x =$ ..... $^\circ$ **[2]**

519 The diagram shows two right-angled triangles, PQR and PRS.
$PQ = 6$ cm, angle $PQR = 90°$ and angle $PRQ = 45°$.
Angle $PRS = 90°$ and angle $PSR = 60°$.



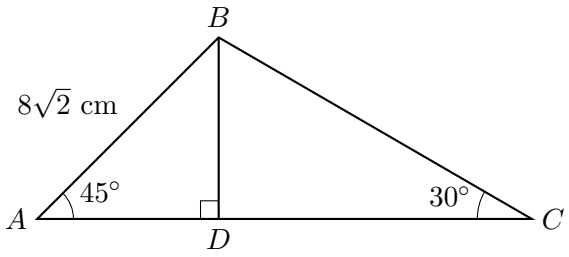
(a) Calculate the exact length of PR.

..... cm **[2]**

(b) Calculate the exact length of PS.
Give your answer in the form $a\sqrt{b}$ where a and b are integers.

..... cm **[3]**

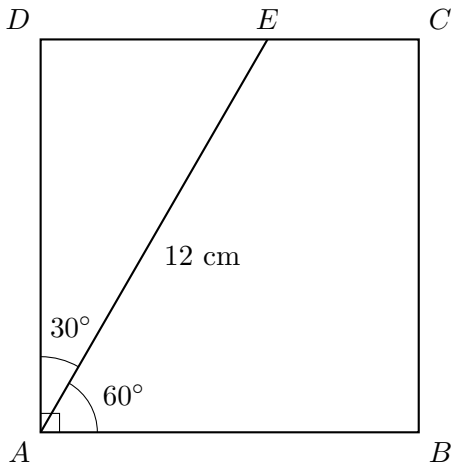
520 The diagram shows a triangle ABC with a point D on the side AC.
BD is perpendicular to AC.
$AB = 8\sqrt{2}$ cm, $\angle BAC = 45°$ and $\angle BCA = 30°$.



Calculate the exact perimeter of triangle ABC.
Give your answer in the form $a + b\sqrt{2} + c\sqrt{3}$ where a, b, and c are integers.

Perimeter = ... cm **[6]**

521 The diagram shows a rectangle $ABCD$.
E is a point on the side CD.
$AE = 12$ cm, angle $DAE = 30°$ and angle $EAB = 60°$.



(a) Show that the exact length of AD is $6\sqrt{3}$ cm.

[2]

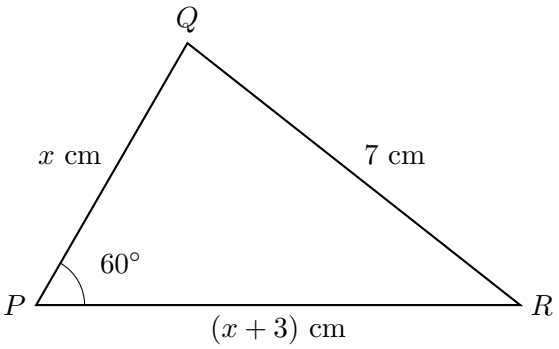
(b) The exact area of the rectangle $ABCD$ is $60\sqrt{3}$ cm^2.
Work out the exact length of CE.

..... cm **[3]**

(c) Find the exact value of $\tan(\angle EBC)$.
Give your answer in the form $\dfrac{a\sqrt{b}}{c}$ where a, b, and c are integers.

..... **[3]**

522 The diagram shows a triangle PQR.
$PQ = x$ cm, $PR = (x + 3)$ cm and $QR = 7$ cm.
Angle $QPR = 60°$.



(a) Show that $x^2 + 3x - 40 = 0$.

[4]

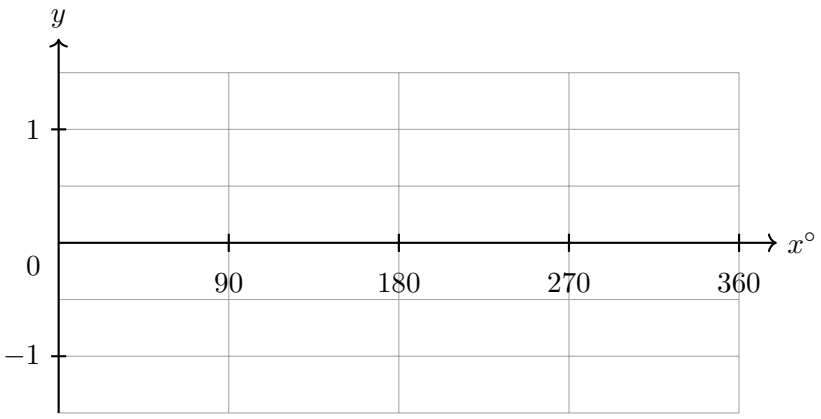
(b) Solve the equation to find the value of x, and hence find the perimeter of triangle PQR.

Perimeter = ... cm **[3]**

(c) Find the exact area of triangle PQR.

Area = ... cm^2 **[3]**

523 **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the exact solutions to the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

524 Answer the following questions for $0° \leq x \leq 360°$.

(a) Solve the equation

$$2\cos x - 1 = 0$$

$x =$ ..... **[3]**

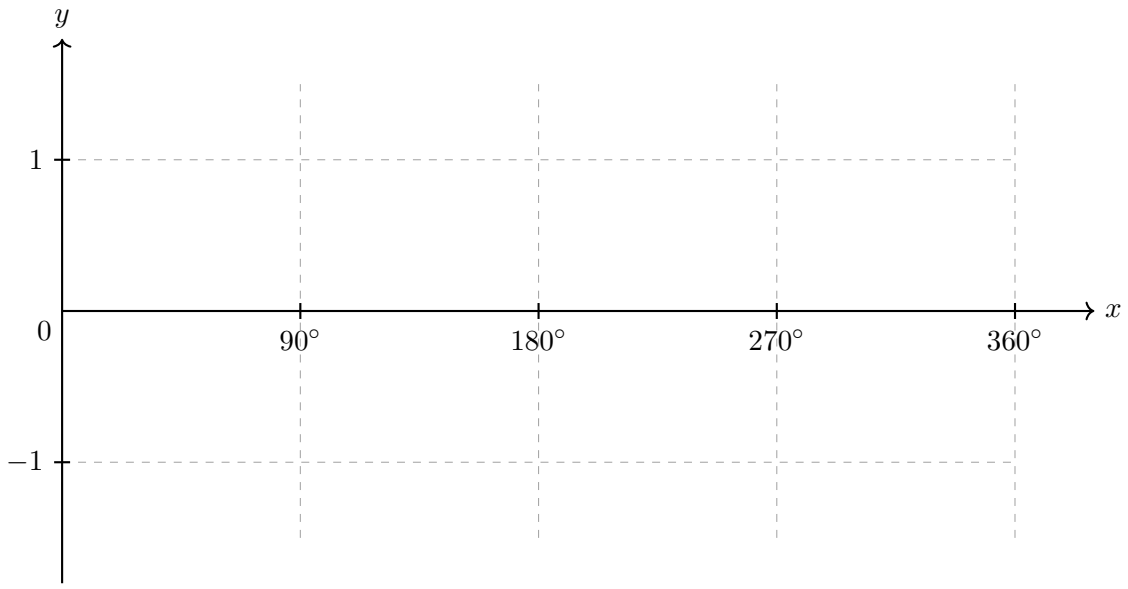
(b) Write down the number of solutions to the equation

$$3\sin x = 5$$

..... **[1]**

525 The functions $g(x) = \sin x$ and $h(x) = \cos x$ are defined for $0° \leq x \leq 360°$.

(a) On the grid provided, sketch the graphs of $y = g(x)$ and $y = h(x)$.



[3]

(b) Solve the equation $g(x) = h(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

(c) Use your graphs to write down the range of values of x for which $g(x) > h(x)$ for $0° \leq x \leq 360°$.

..... **[2]**

526 Consider the trigonometric function $f(x) = 4\sin x + 3$ for $0° \leq x \leq 360°$.

(a) Solve the equation $f(x) = 0$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Write down the exact coordinates of the minimum point on the graph of $y = f(x)$.

(..... , ...) **[2]**

527 **(a)** Solve the equation for $0° \leq x \leq 360°$.

$$4\cos x + 1 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$$2\tan x - 5 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

528 **(a)** The equation $a \sin x - b = 0$ has a solution $x = 30°$.
Given that a and b are positive integers with no common factors, find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

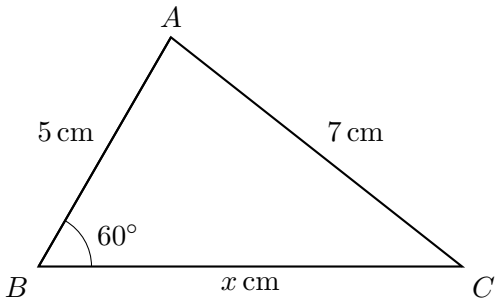
(b) Write down the second solution to the equation $a \sin x - b = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

(c) A horizontal line $y = k$ intersects the graph of $y = \cos x$ at exactly two points in the interval $0° \leq x \leq 360°$.
The x-coordinate of one of these points is $242°$.
Find the value of k correct to 3 significant figures and find the x-coordinate of the second point of intersection.

$k =$ ..... , $x =$ ..... **[3]**

529 In triangle ABC, $AB = 5\,\text{cm}$, $BC = x\,\text{cm}$ and $AC = 7\,\text{cm}$.
Angle $ABC = 60°$.



NOT TO SCALE

(a) Use the cosine rule to show that $x^2 - 5x - 24 = 0$.

[3]

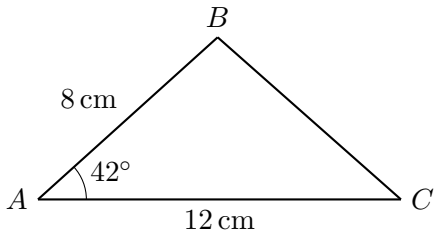
(b) Solve the equation $x^2 - 5x - 24 = 0$ to find the length of BC.

$BC =$ ..... cm **[2]**

(c) Calculate the exact area of triangle ABC.
Give your answer in the form $a\sqrt{b}$, where a and b are integers.

Area = ... cm^2 **[2]**

530 The diagram shows triangle ABC.
$AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 42°$.



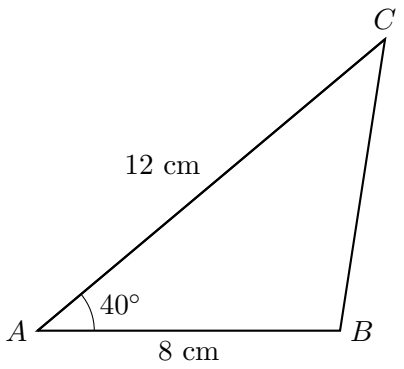
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

$BC =$ ..... cm **[3]**

531 The diagram shows a triangle ABC.



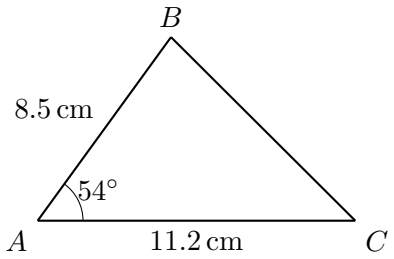
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

532 The diagram shows triangle ABC.



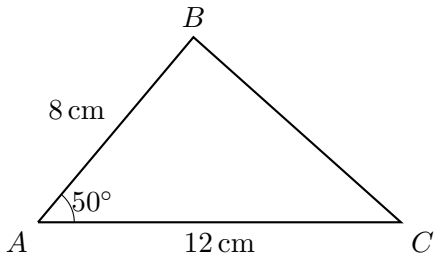
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

533 A triangle ABC is shown below.



NOT TO SCALE

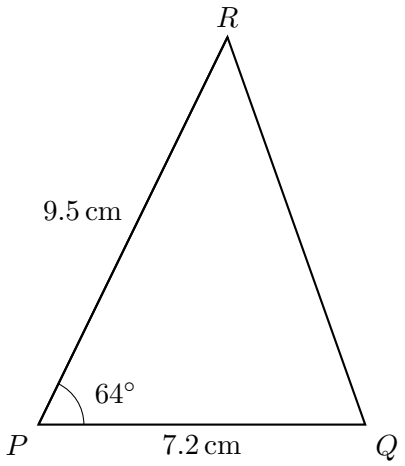
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

534 In triangle PQR, $PQ = 7.2\,\text{cm}$, $PR = 9.5\,\text{cm}$ and angle $QPR = 64°$.



NOT TO SCALE

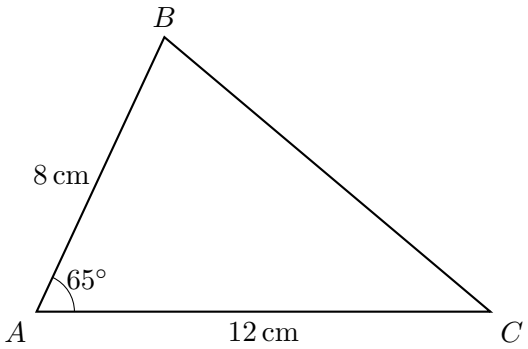
(a) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

(b) Calculate the length of QR.

QR = ... cm **[3]**

535 The diagram shows a triangle ABC with $AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 65°$.



NOT TO SCALE

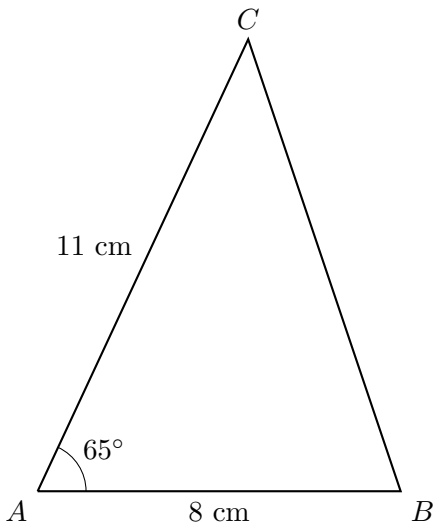
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

536 The diagram shows a triangle ABC.
$AB = 8$ cm, $AC = 11$ cm and angle $BAC = 65°$.



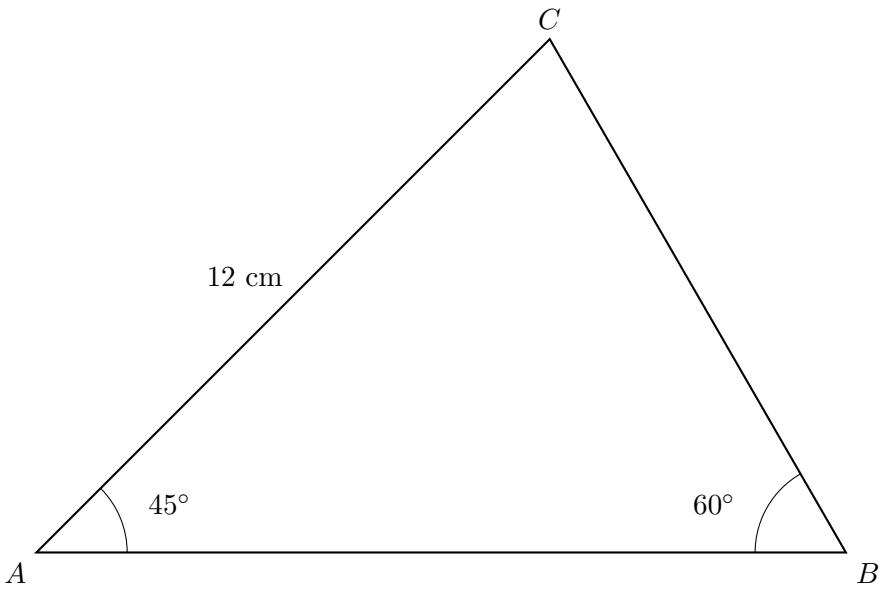
(a) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

(b) Calculate the length of BC.

BC = ... cm **[3]**

537 In triangle ABC, $AC = 12$ cm, angle $BAC = 45°$ and angle $ABC = 60°$.



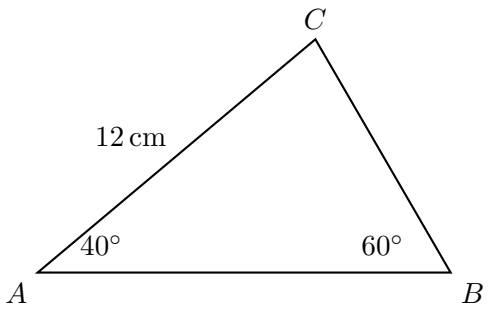
(a) Calculate the length of BC.

..... cm **[3]**

(b) Calculate the area of triangle ABC.

..... cm^2 **[3]**

538 The diagram shows triangle ABC.
$AC = 12$ cm, angle $BAC = 40°$ and angle $ABC = 60°$.



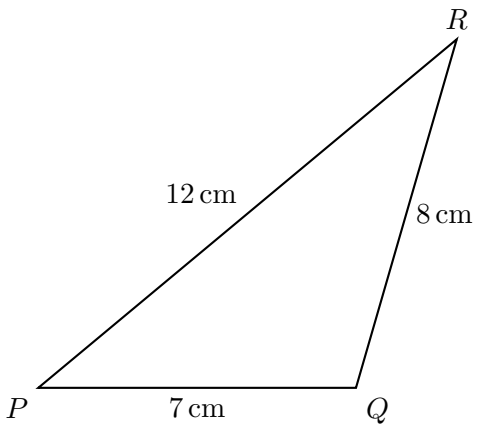
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

539 The diagram shows triangle PQR.
$PQ = 7$ cm, $QR = 8$ cm and $PR = 12$ cm.



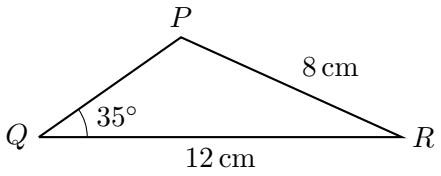
(a) Calculate the largest angle in triangle PQR.

..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

540 In triangle PQR, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle QPR is an obtuse angle.



NOT TO SCALE

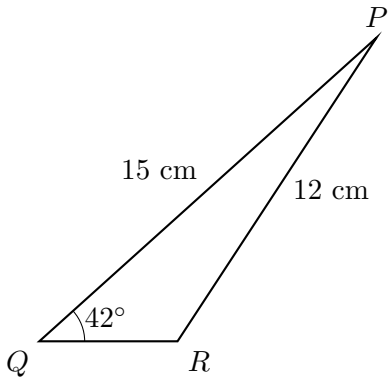
(a) Calculate the size of angle QPR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

541 The diagram shows a triangle PQR.
The length of PQ is 15 cm and the length of PR is 12 cm.
Angle $PQR = 42°$.
Angle PRQ is obtuse.



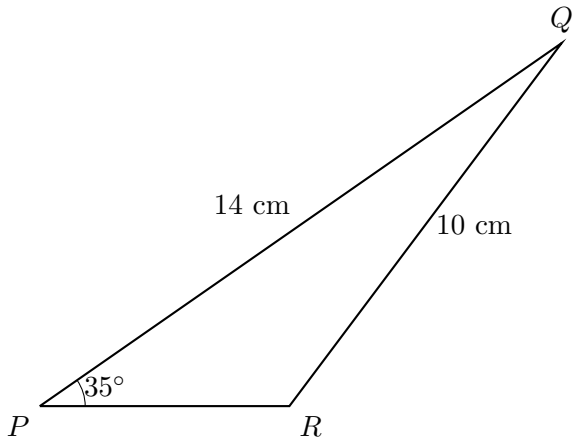
(a) Calculate angle PRQ.

..... ° **[3]**

(b) Calculate the length of QR.

..... cm **[3]**

542 The diagram shows triangle PQR.
$PQ = 14$ cm, $QR = 10$ cm and angle $QPR = 35°$.
Angle PRQ is obtuse.



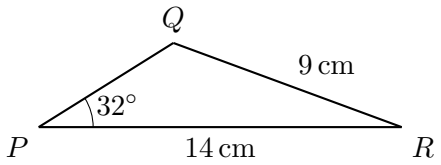
(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[3]**

543 The diagram shows triangle PQR.
The angle PQR is obtuse.



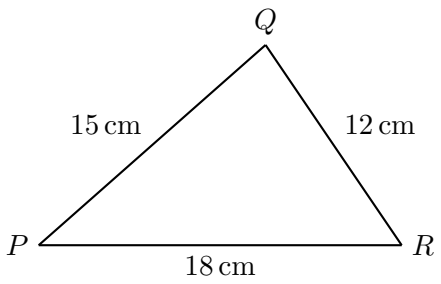
(a) Calculate the obtuse angle PQR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[3]**

544 The diagram shows triangle PQR.
$PQ = 15\,\text{cm}$, $QR = 12\,\text{cm}$ and $PR = 18\,\text{cm}$.



(a) Use the cosine rule to show that angle $PQR = 82.8°$, correct to 1 decimal place.

[3]

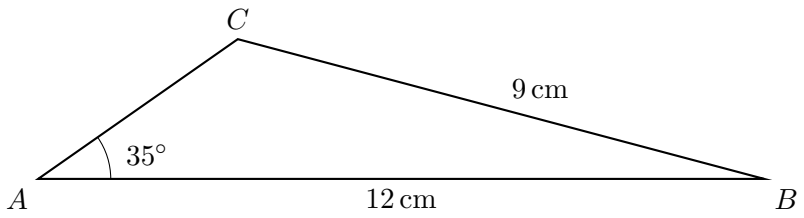
(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

(c) Calculate the shortest distance from R to the line PQ.

..... cm **[2]**

545 In triangle ABC, $AB = 12\,\text{cm}$, $BC = 9\,\text{cm}$ and angle $BAC = 35°$.



NOT TO SCALE

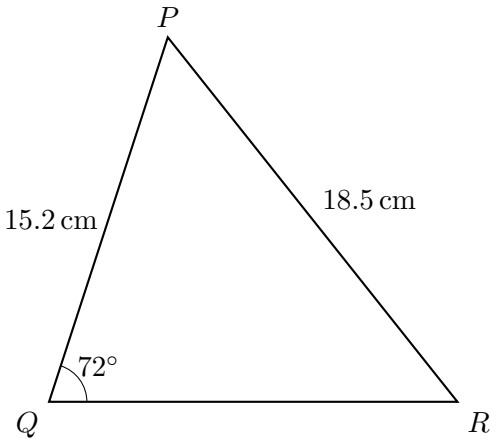
(a) Calculate the two possible values of angle ACB.
Give your answers correct to 1 decimal place.

..... ° **[4]**

(b) Given that angle ACB is obtuse, calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

546 The diagram shows a triangle PQR.
$PQ = 15.2\,\text{cm}$, $PR = 18.5\,\text{cm}$ and angle $PQR = 72°$.



NOT TO SCALE

(a) Calculate the size of angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

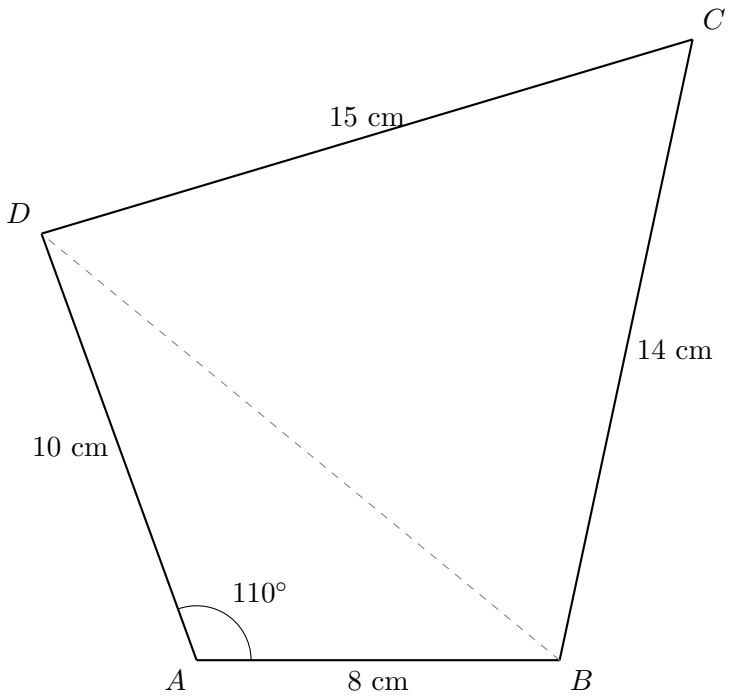
(b) Calculate the length of QR.

$QR =$ ..... cm **[3]**

(c) Calculate the shortest distance from P to the line QR.

Distance = ... cm **[2]**

547 The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $AD = 10$ cm, $BC = 14$ cm and $CD = 15$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of the diagonal BD.

..... cm **[3]**

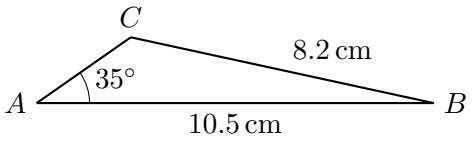
(b) Calculate the angle BCD.

..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

548 In triangle ABC, $AB = 10.5\,\text{cm}$, $BC = 8.2\,\text{cm}$ and angle $BAC = 35°$.
Angle BCA is obtuse.



(a) Calculate the size of angle BCA.

..... ° **[3]**

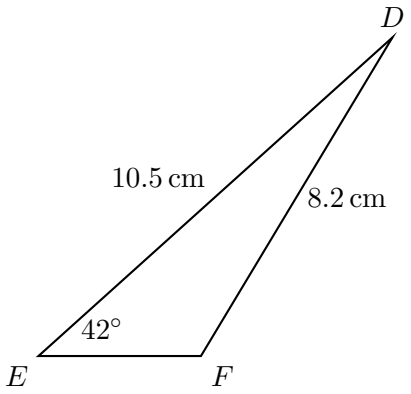
(b) Calculate the length of AC.

$AC =$ ..... cm **[3]**

(c) A point D lies outside the triangle ABC such that $CD = 6.4\,\text{cm}$ and angle ACD is acute.
The area of triangle ACD is $8.5\,\text{cm}^2$.
Calculate the size of angle ACD.

..... ° **[2]**

549 The diagram shows triangle DEF.
$DE = 10.5$ cm, $DF = 8.2$ cm and angle $DEF = 42°$.
Angle DFE is an obtuse angle.



(a) Calculate angle DFE.

Angle $DFE =$ ..... ° **[3]**

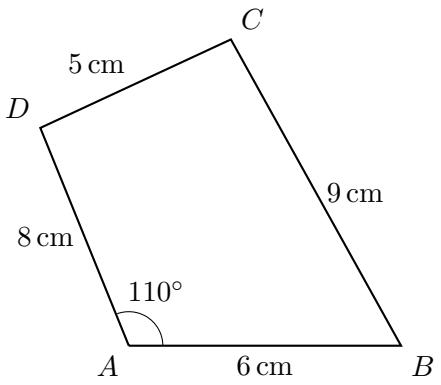
(b) Calculate the length of EF.

$EF =$ ..... cm **[3]**

(c) Calculate the shortest distance from D to the line EF.

..... cm **[2]**

550 The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.



NOT TO SCALE

(a) Calculate the length of the diagonal BD.

..... cm **[3]**

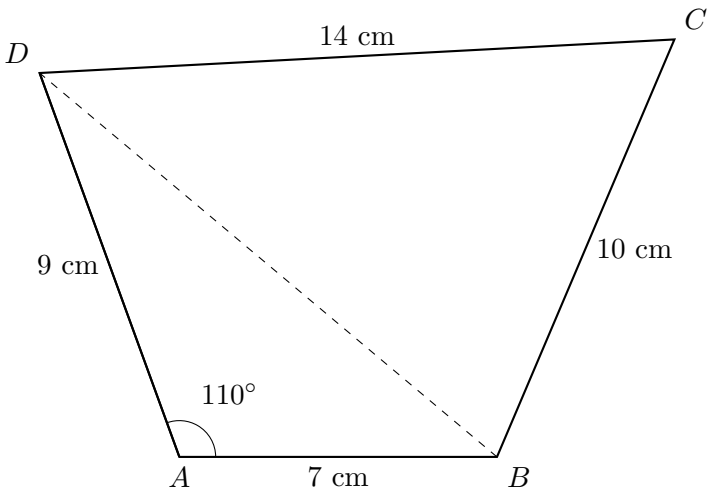
(b) Calculate the size of angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

551 The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of BD.

..... cm **[3]**

(b) Calculate angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

552 The diagram shows a quadrilateral $ABCD$.
$AB = 7.4$ cm, $BC = 9.1$ cm, and $CD = 10.5$ cm.
Angle $ABC = 112°$ and angle $CAD = 43°$.

(a) Calculate the length of AC.

..... cm **[3]**

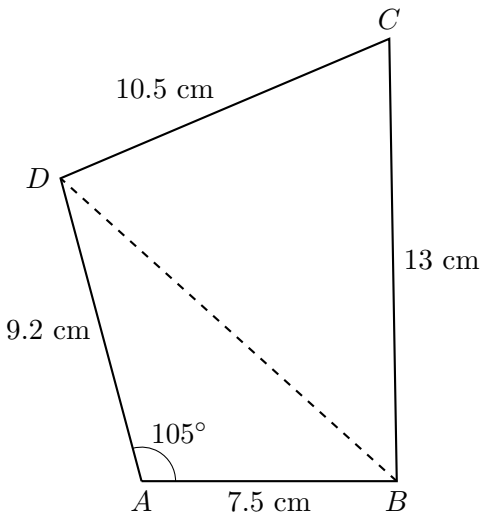
(b) Calculate the acute angle ADC.

..... ° **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

553 The diagram shows a quadrilateral $ABCD$.
$AB = 7.5$ cm, $AD = 9.2$ cm and angle $BAD = 105°$.
$BC = 13$ cm and $CD = 10.5$ cm.



(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

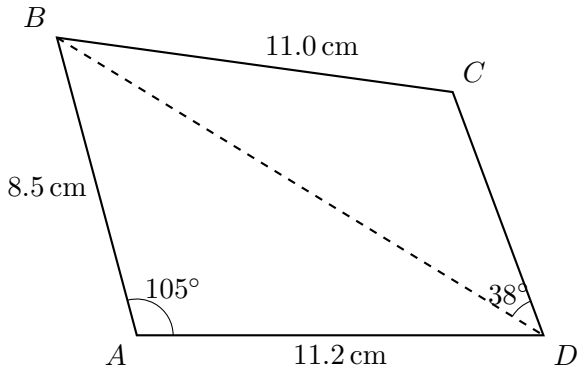
(b) Calculate angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[3]**

554 The diagram shows a quadrilateral $ABCD$.
$AB = 8.5\,\text{cm}$, $AD = 11.2\,\text{cm}$ and angle $BAD = 105°$.
$BC = 11.0\,\text{cm}$ and angle $BDC = 38°$.
Angle BCD is obtuse.



NOT TO SCALE

(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

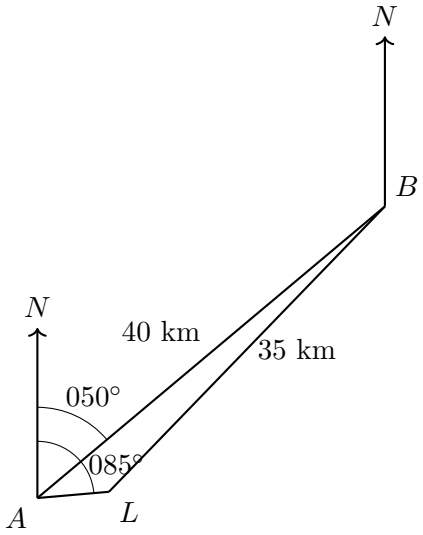
(b) Calculate the size of angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[4]**

555 Ship A and Ship B are at sea.
Ship B is 40 km from Ship A on a bearing of 050°.
A lighthouse L is 35 km from Ship B.
The bearing of L from A is 085°.
Given that the angle ALB is obtuse:



(a) Calculate the obtuse angle ALB.

..... ° **[4]**

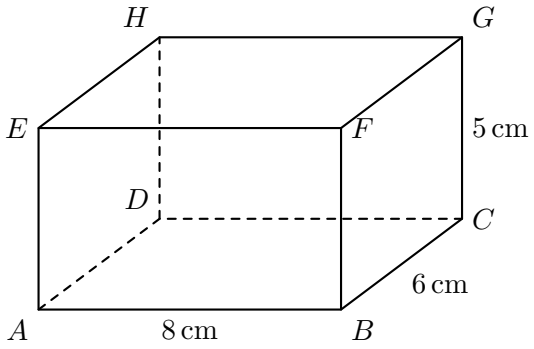
(b) Calculate the distance AL.

..... km **[3]**

(c) Calculate the bearing of L from B.

..... ° **[3]**

556 The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.



(a) Calculate the length of the diagonal AC.

..... cm **[2]**

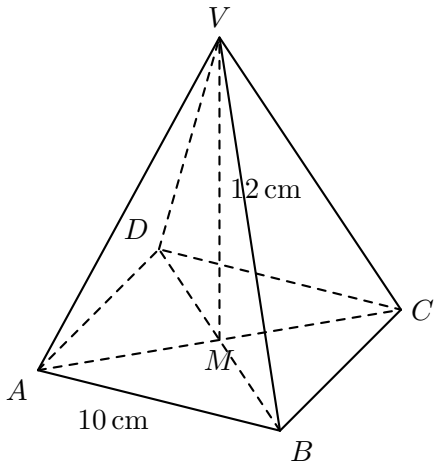
(b) Calculate the length of the space diagonal AG.

..... cm **[2]**

(c) Calculate the angle between the line AG and the base $ABCD$.

..... ° **[2]**

557 The diagram shows a pyramid $VABCD$ with a square base $ABCD$ of side 10 cm.
The point M is the centre of the base and V is vertically above M.
The vertical height VM of the pyramid is 12 cm.



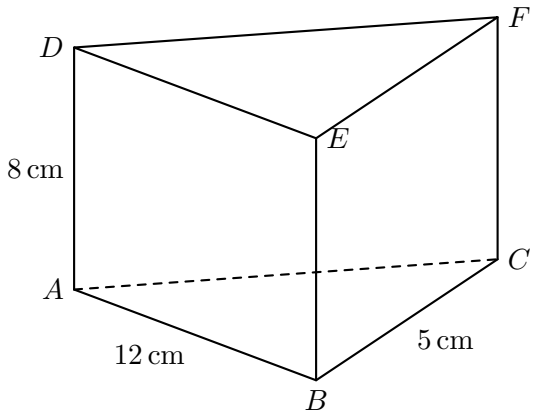
(a) Calculate the length of the edge VA.

..... cm **[3]**

(b) Calculate the angle between the edge VA and the base $ABCD$.

..... ° **[2]**

558 The diagram shows a right triangular prism $ABCDEF$.
The base ABC is a horizontal right-angled triangle with angle $ABC = 90°$.
$AB = 12$ cm, $BC = 5$ cm and the vertical height $AD = 8$ cm.



(a) Calculate the length of AF.

..... cm **[3]**

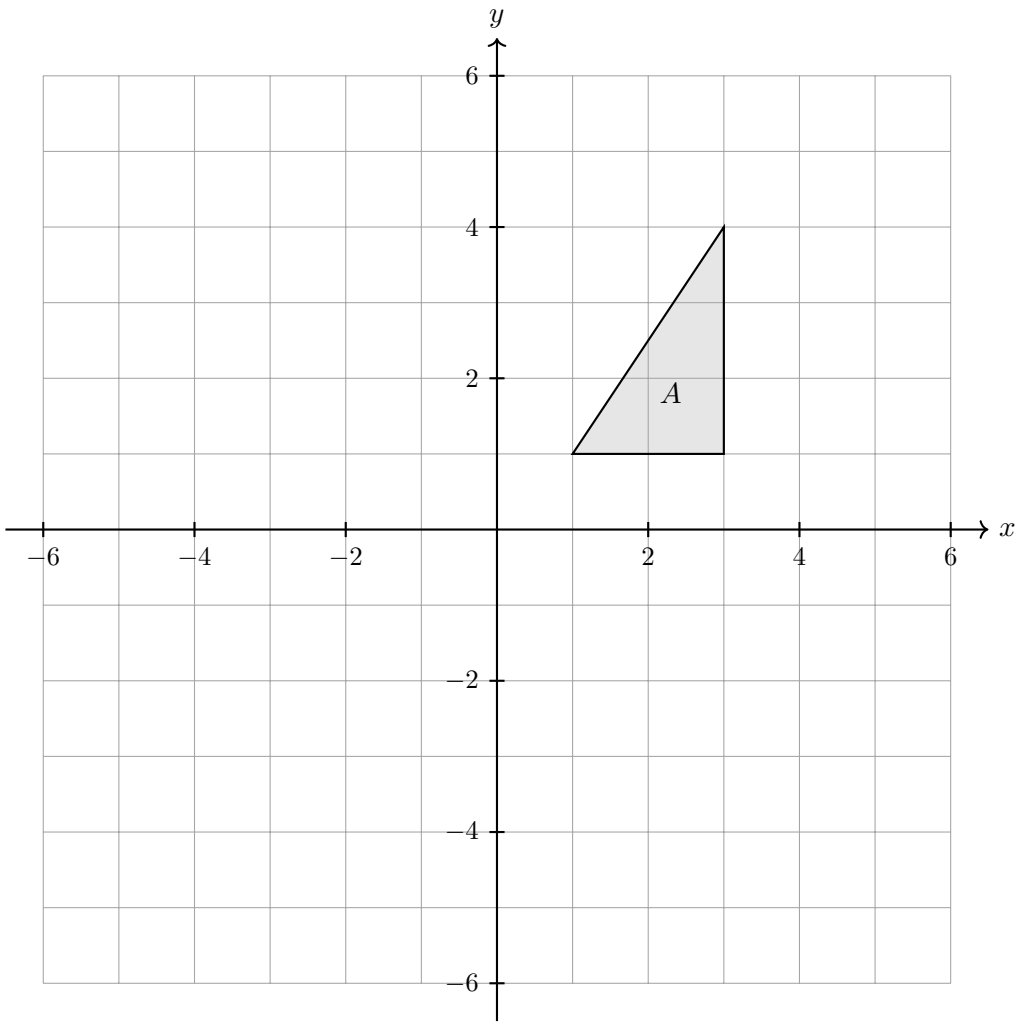
(b) Calculate the angle between the line AF and the base ABC.

..... ° **[2]**

(c) Calculate the angle between the line AF and the vertical plane $ABED$.

..... ° **[3]**

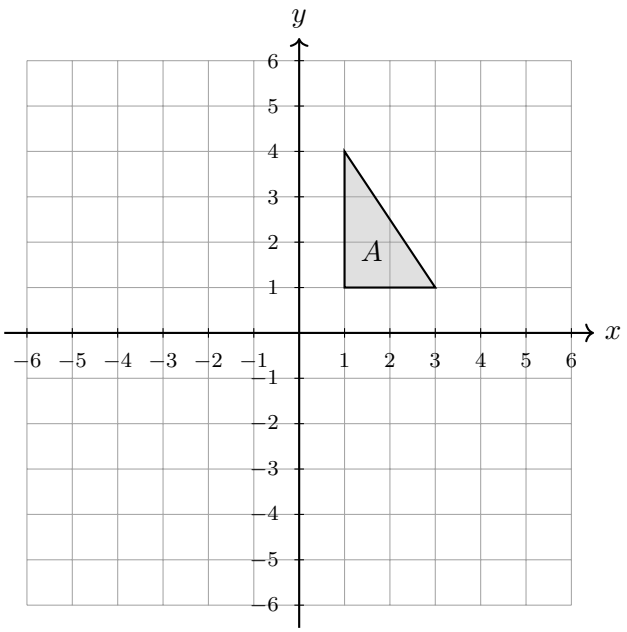
559 The grid shows triangle A.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $x = -1$.
Label the image C. **[2]**

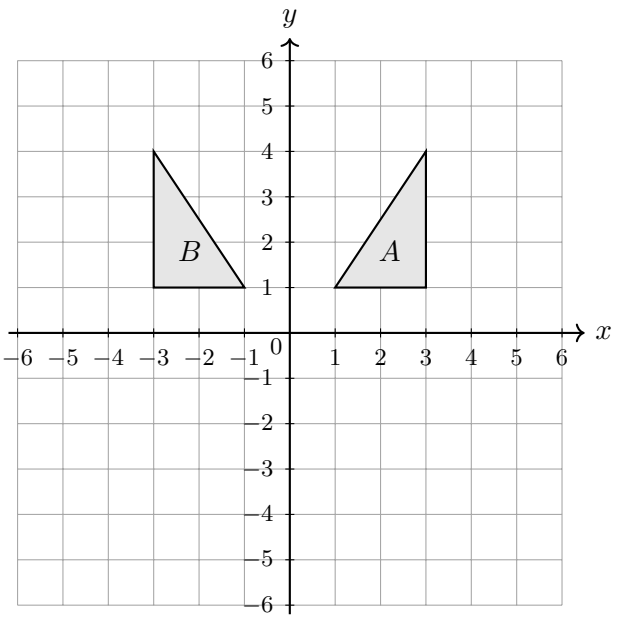
560 The diagram shows a triangle, Shape A, drawn on a coordinate grid.



(a) On the grid, draw the image of Shape A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label this image B. **[2]**

(b) On the grid, draw the image of Shape A after a reflection in the line $y = -1$.
Label this image C. **[2]**

561 The diagram shows two triangles, A and B, plotted on a coordinate grid.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$. **[2]**

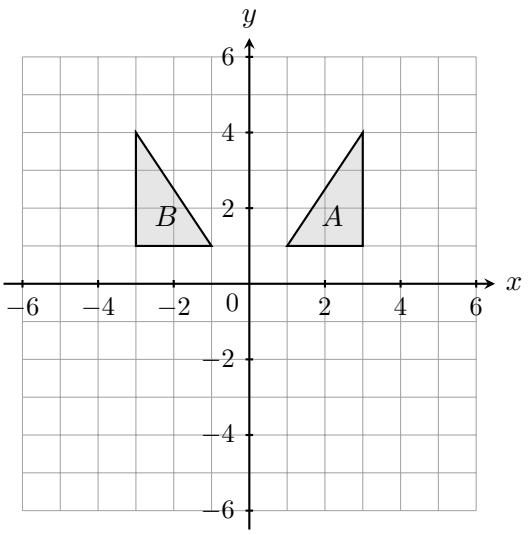
(b) Describe fully the single transformation that maps triangle A onto triangle B.

.....

.....

..... **[2]**

562 The grid shows triangle A and triangle B.



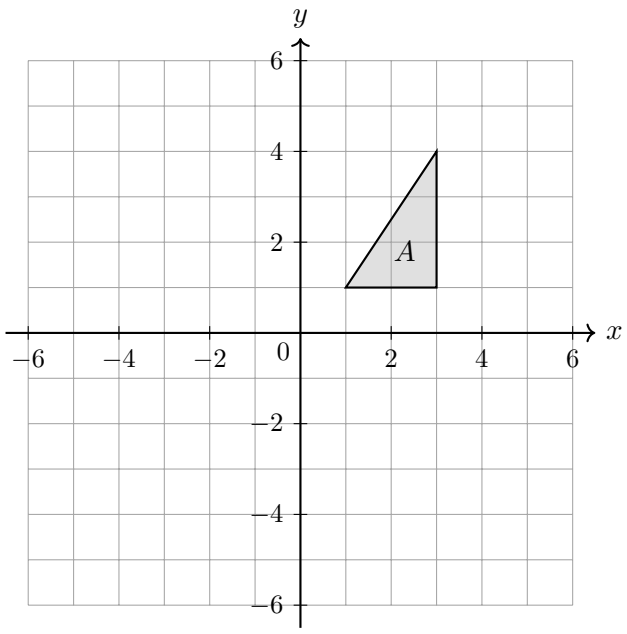
(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$. **[2]**

(b) Describe fully the single transformation that maps triangle A onto triangle B.

.....
.....
.....

[2]

563 The diagram shows triangle A on a coordinate grid.

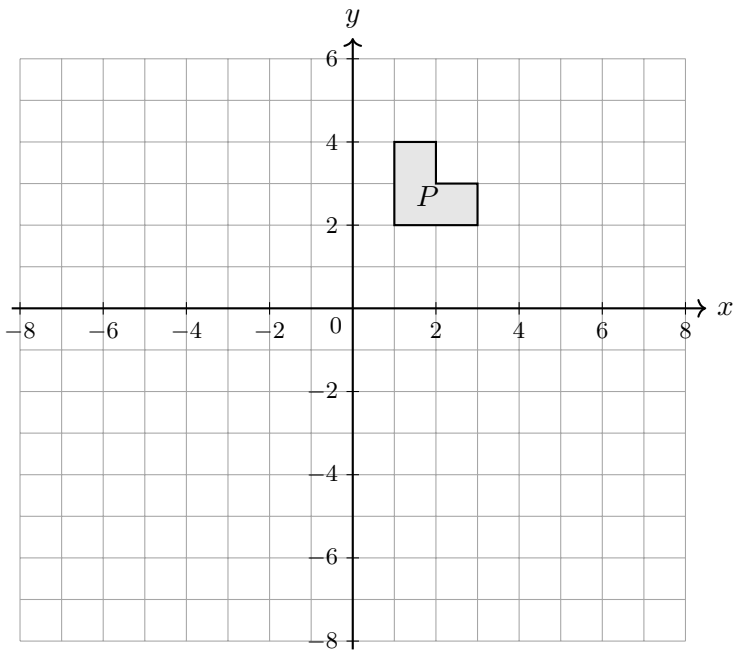


(a) On the grid, translate shape A by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, reflect shape A in the x-axis.
Label the image C. **[2]**

(c) On the grid, rotate shape A by 180° about the origin $(0, 0)$.
Label the image D. **[2]**

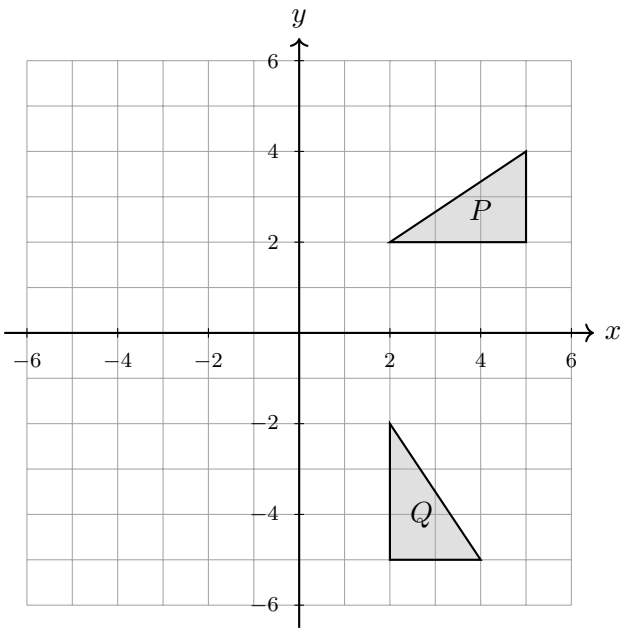
564 The diagram shows a shape P plotted on a coordinate grid.



(a) On the grid, draw the image of shape P after an enlargement, scale factor -2, centre $(0, 0)$. **[2]**

(b) On the grid, draw the image of shape P after a rotation of $90°$ clockwise about the point $(1, -1)$. **[2]**

565 The diagram shows two triangles, Shape P and Shape Q, drawn on a grid.



(a) Describe fully the single transformation that maps Shape P onto Shape Q.

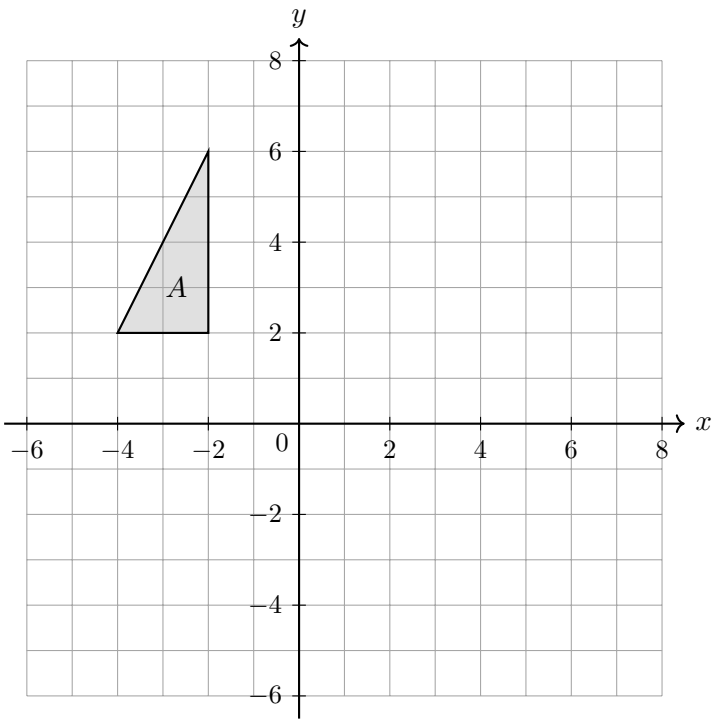
.....

.....

..... **[3]**

(b) Shape P is enlarged by a scale factor of $\frac{1}{2}$ with centre of enlargement $(-2, 2)$.
On the grid, draw the image of Shape P and label it R. **[2]**

566 The diagram shows shape A on a coordinate grid.



(a) Shape A is reflected in the line $y = x$ to give shape B.
On the grid, draw and label shape B. **[2]**

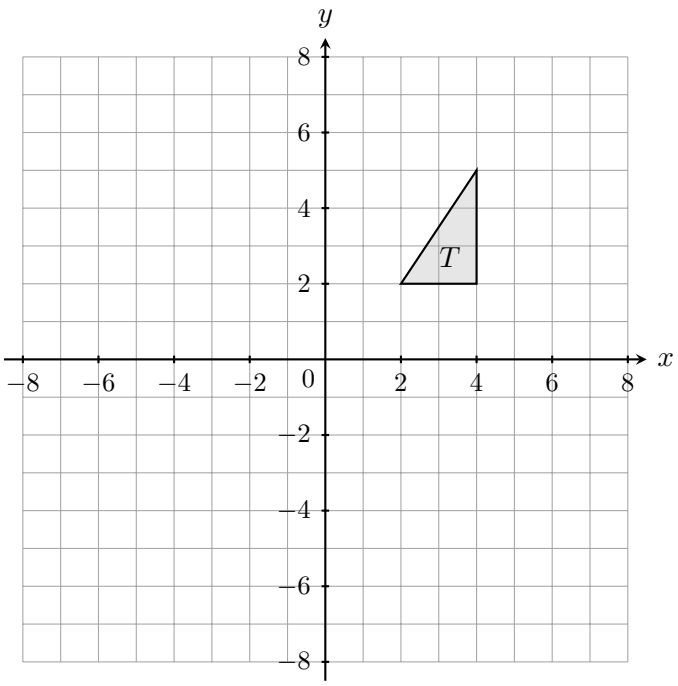
(b) Shape B is rotated by 90° clockwise about the point $(2, 2)$ to give shape C.
On the grid, draw and label shape C. **[2]**

(c) Describe fully the single transformation that maps shape A onto shape C.

.....

..... **[2]**

567 The grid shows triangle T.



(a) On the grid, draw the image of triangle T after an enlargement with scale factor $-\frac{1}{2}$ and centre $(0, 2)$. **[2]**

(b) Triangle T is reflected in the line $x = 1$ to give triangle U.
Triangle U is then reflected in the line $y = 2$ to give triangle V.
Describe fully the single transformation that maps triangle T onto triangle V.

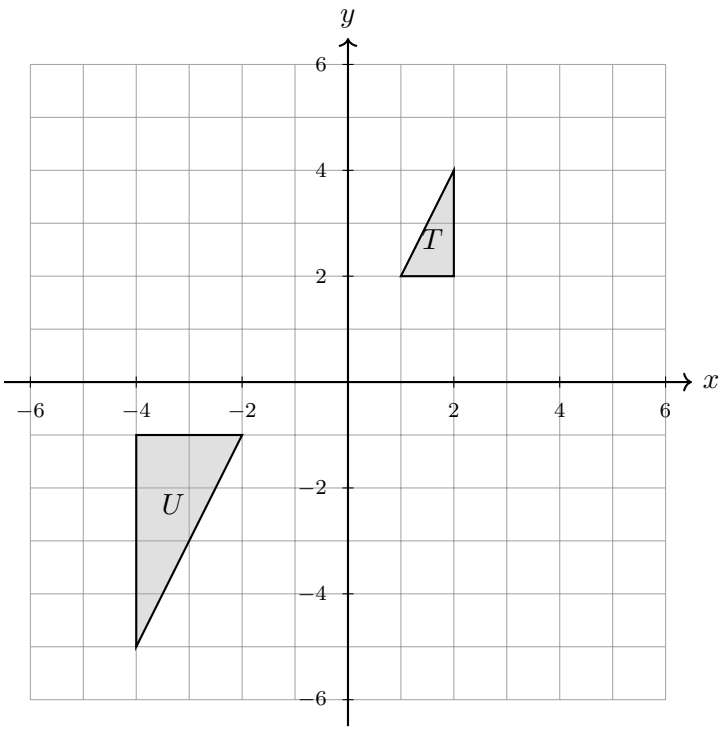
.....
.....
.....

[3]

(c) Triangle T is enlarged by a scale factor of k to give triangle W.
The area of triangle W is 48 square units.
Find the value of k, given that $k > 0$.

$k =$ ..... **[2]**

568 The diagram shows two triangles, Shape T and Shape U.



(a) Describe fully the single transformation that maps Shape T onto Shape U.

.....

.....

..... **[3]**

(b) Describe fully the single transformation that maps Shape U onto Shape T.

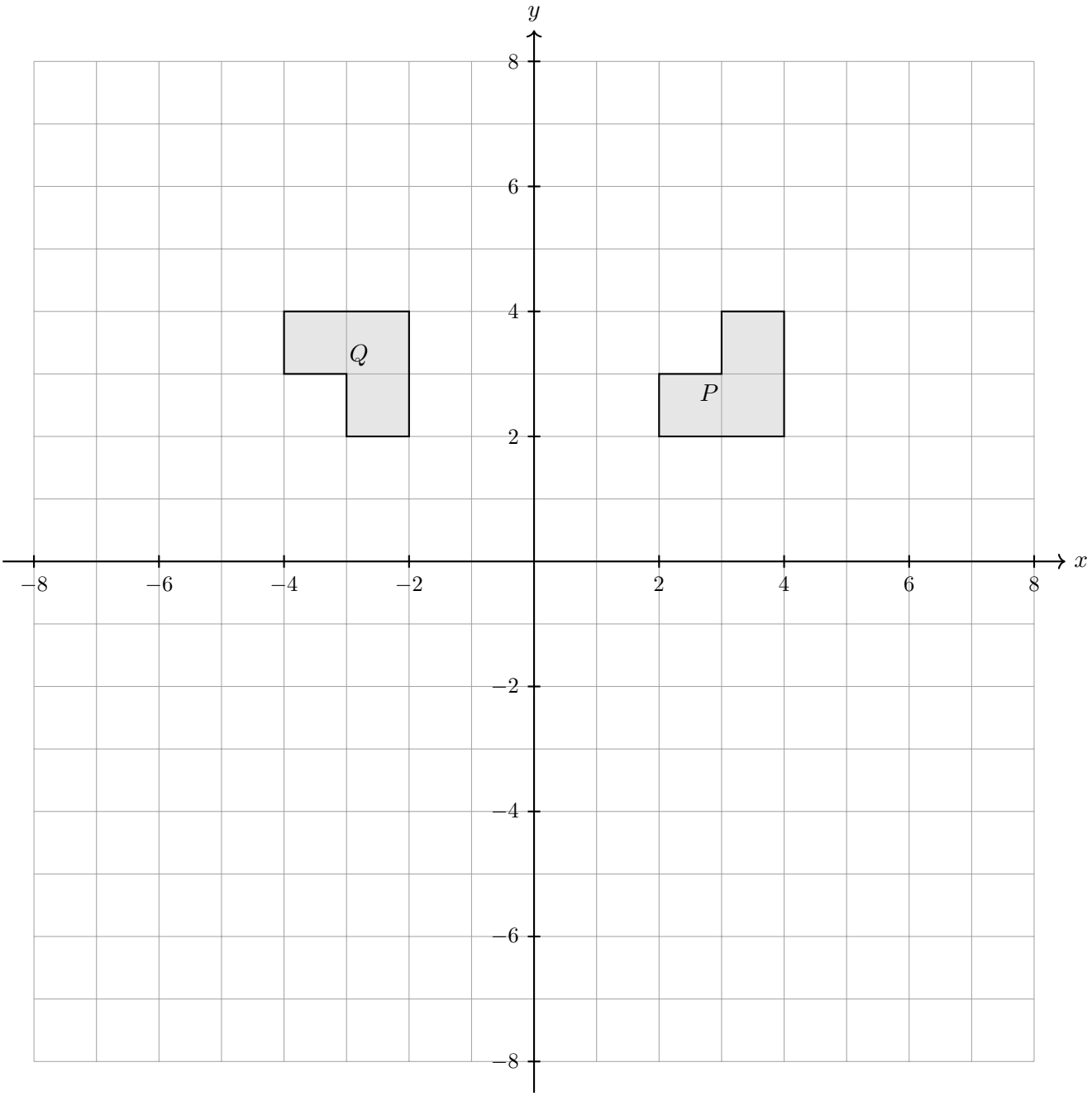
.....

..... **[2]**

(c) Shape T is translated by the vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ to give Shape V.
Shape V is then reflected in the line $y = x$ to give Shape W.
Find the coordinates of the three vertices of Shape W.

..... **[3]**

569 The grid shows shapes P and Q.



(a) Describe fully the single transformation that maps shape P onto shape Q.

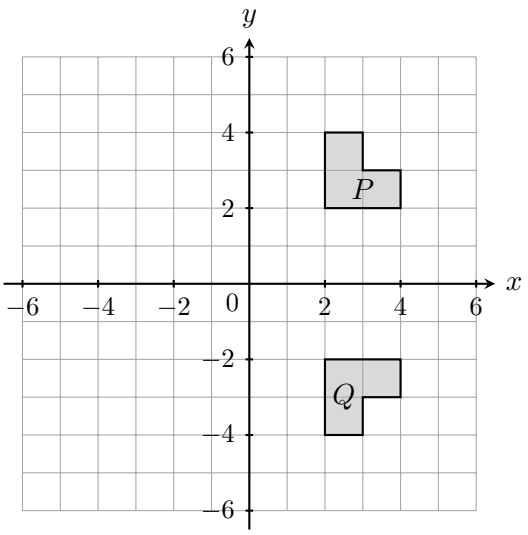
.....

.....

..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$, centre $(6, 6)$.
Label the image R. **[2]**

570 The grid shows shapes P and Q.



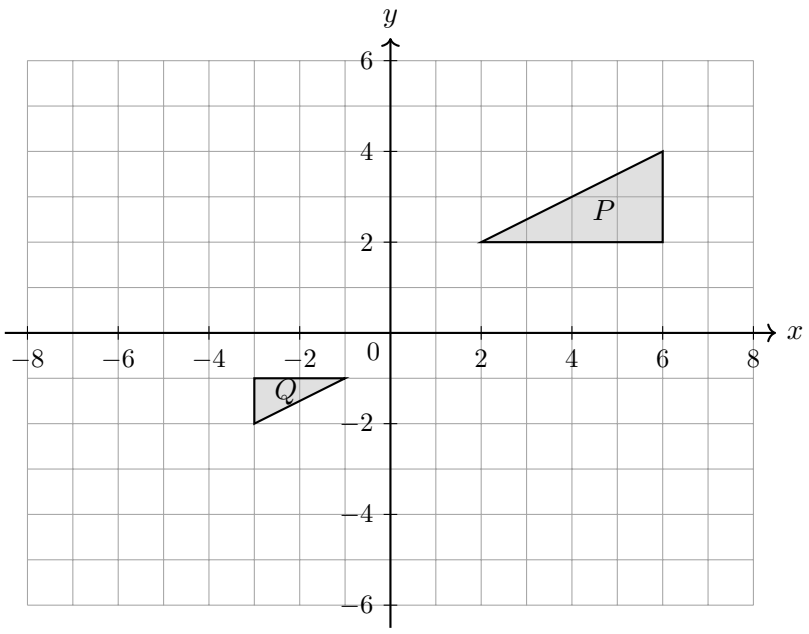
(a) On the grid, draw the image of shape P after an enlargement with scale factor 0.5 and centre $(0, 0)$. **[2]**

(b) Describe fully the single transformation that maps shape P onto shape Q.

.....
.....
.....

[3]

571 The diagram shows triangles P and Q on a coordinate grid.



(a) Describe fully the single transformation that maps shape P onto shape Q.

.....

.....

..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$ with centre $(0, 4)$.
Label the image R. **[3]**

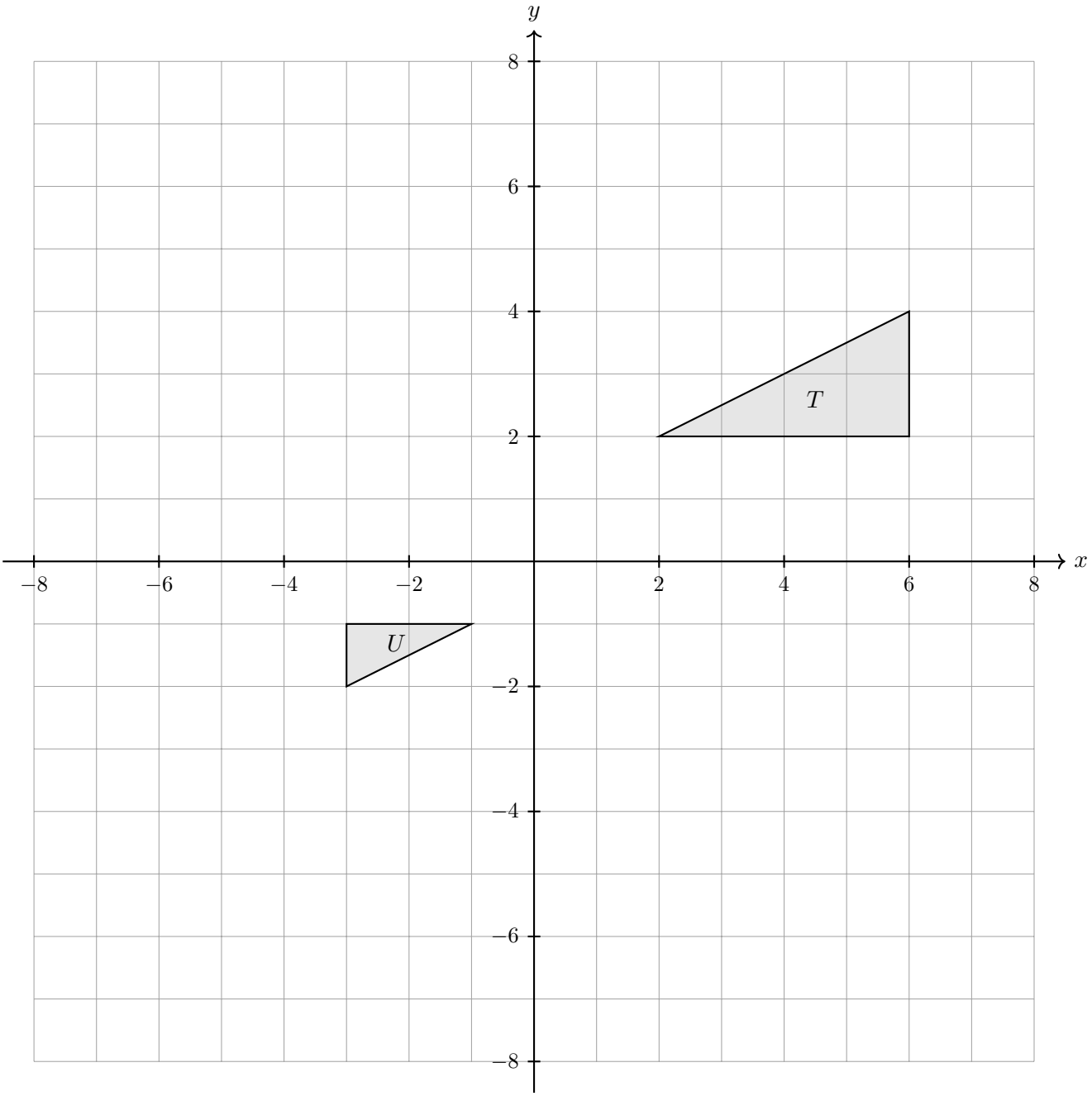
(c) Describe fully the single transformation that maps shape R onto shape Q.

.....

.....

..... **[3]**

572 The grid shows triangles T and U.



(a) Describe fully the single transformation that maps triangle T onto triangle U.

.....

.....

..... **[3]**

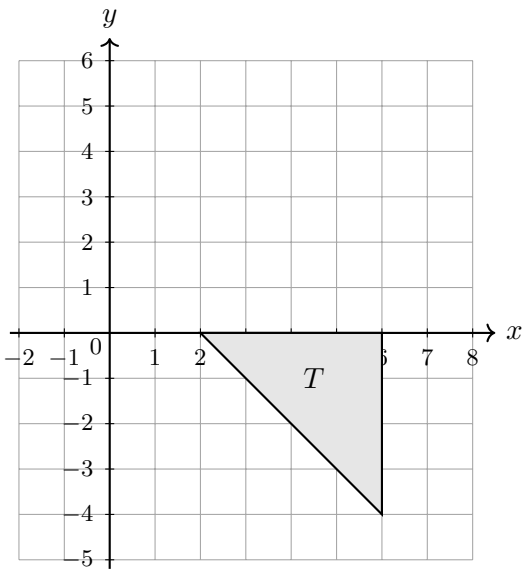
(b) Triangle T is reflected in the line $y = x$ to give triangle V.
Triangle V is then rotated 90° clockwise about the origin to give triangle W.
Describe fully the single transformation that maps triangle T onto triangle W.

.....

. . . . . .

. . . . . **[2]**

573 The diagram shows triangle T plotted on a coordinate grid.



(a) Triangle T is mapped onto triangle U by an enlargement, scale factor $\frac{1}{2}$, centre $(0, 4)$.
On the grid, draw triangle U. **[2]**

(b) Triangle U is mapped onto triangle V by a reflection in the line $y = x$.
On the grid, draw triangle V. **[2]**

(c) Describe fully the single transformation that maps triangle U back onto triangle T.

.....

.....

..... **[3]**

574 Vectors $\mathbf{p}$ and $\mathbf{q}$ are given as

$$\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \qquad \mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

(a) Work out

$$2\mathbf{p} + \mathbf{q}$$

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(1, 4)$.
The translation from A to B is given by the vector $\mathbf{p}$.
Work out the coordinates of B.

(..... , ...) **[1]**

575 Given the vectors $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.

(a) Work out $3\mathbf{p}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

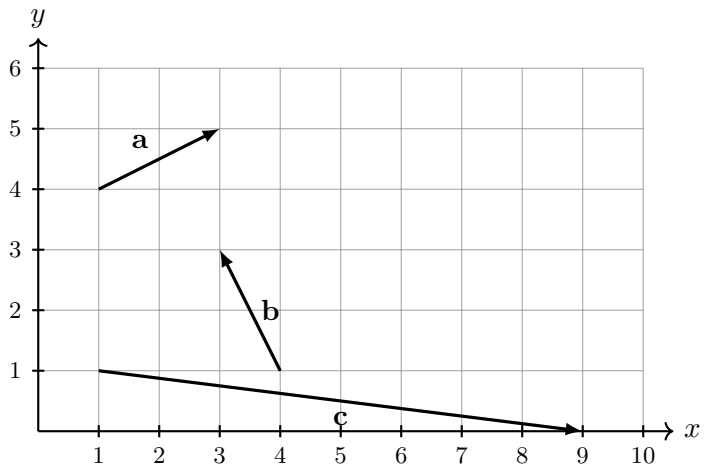
(b) Work out $\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(c) Point M is translated to point N by the vector $\mathbf{q}$.
Given that the coordinates of N are $(6, -1)$, find the coordinates of M.

(..... , ...) **[2]**

576 The diagram shows vectors **a**, **b** and **c** plotted on a Cartesian coordinate grid.



(a) Write **c** as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Vector **c** can be written in the form $p\mathbf{a} + q\mathbf{b}$, where p and q are integers.
Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[4]**

577 $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ $\mathbf{q} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$

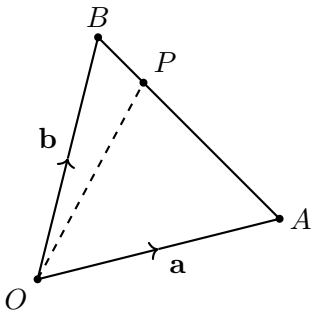
(a) Work out $3\mathbf{p} - 2\mathbf{q}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Point A has coordinates $(3, -1)$.
$\vec{AB} = \mathbf{p} + \mathbf{q}$.
Find the coordinates of B.

(..... , ...) **[2]**

578 In triangle OAB, $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$.
P is the point on AB such that $AP : PB = 3 : 1$.



(a) Find $\vec{OP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{OP} =$ ..... **[3]**

(b) M is the midpoint of OA.
Find $\vec{MP}$ in terms of **a** and **b**.
Give your answer in its simplest form.

$\vec{MP} =$ ..... **[2]**

579 **(a)** $\mathbf{u} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ y \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$.

Given that $3\mathbf{u} + k\mathbf{v} = \mathbf{w}$, work out the value of k and the value of y.

$k =$ ..... , $y =$ ..... **[3]**

(b) Points P, Q and R have coordinates $(2, -1)$, $(5, 5)$ and $(11, 17)$ respectively.

(i) Write $\vec{PQ}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(ii) Show that P, Q and R lie on a straight line.

[2]

580 Points A, B and C have coordinates $(1, -2)$, $(7, 10)$ and $(4, 4)$ respectively.

(a) Write $\vec{AB}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Show that $\vec{AC} = \frac{1}{2}\vec{AB}$.

[2]

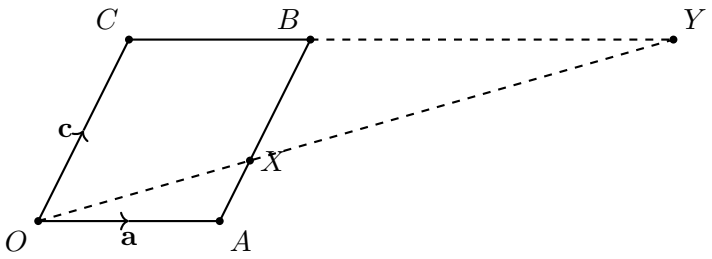
(c) Point C is translated by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$ to point D.
Find the coordinates of D.

(..... , ...) **[2]**

(d) Work out the vector $\vec{BD}$ as a column vector.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

581 $OABC$ is a parallelogram.
$\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.
X is a point on AB such that $AX : XB = 1 : 2$.
Y is a point on the line CB extended such that $CB : BY = 1 : 2$.



(a) Find $\vec{OX}$ in terms of **a** and **c**.

$\vec{OX} =$ ..... **[2]**

(b) Find $\vec{OY}$ in terms of **a** and **c**.

$\vec{OY} =$ ..... **[2]**

(c) Show that the points O, X and Y lie on a straight line.

[2]

(d) Find the ratio $OX : XY$.

$OX : XY =$ ..... **[1]**

582 Given the vectors $\mathbf{u} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$.

(a) Point P has coordinates $(5, -1)$.
The vector $\vec{PQ} = 2\mathbf{u} - \mathbf{v}$.
Find the coordinates of Q.

(..... , ...) **[3]**

(b) Point R has coordinates $(14, 3)$.
The vector $\vec{PR}$ can be written in the form $h\mathbf{u} + k\mathbf{v}$.
Work out the values of the constants h and k.

$h =$ ..... , $k =$ ..... **[5]**

583 Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

584 Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

(a) Calculate $|\mathbf{p}|$.

..... **[2]**

(b) Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

585 Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

(a) The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

[3]

(b) Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

(c) Hence write down the two possible column vectors for **u**.

[2]

586 Points A, B, and C have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

(a) Calculate $|\vec{AC}|$.

..... **[2]**

(b) The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of y.

$y =$ ..... , $y =$ ..... **[5]**

587 **(a)** The points A and B have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

(c) Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

588 Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

(a) Given that $|\mathbf{u} + \mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

[4]

(b) Solve the equation

$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

(c) Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**

589 Event A is that a train arrives on time.

$P(A) = 0.83$

(a) Work out $P(A')$.

..... **[1]**

(b) State in words what the event A' represents.

..... **[1]**

590 A biased spinner can land on 1, 2, 3, 4, or 5.
The table shows the probabilities of the spinner landing on each of the numbers.

Outcome	1	2	3	4	5
Probability	0.15	$2x$	0.31	x	0.24

(a) Find the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun once.
Find the probability that it does not land on 2.

..... **[2]**

591 A biased spinner can land on 1, 2, 3 or 4.
The table shows the probabilities of the spinner landing on each number.

Number	1	2	3	4
Probability	$\frac{1}{5}$	0.35	x	$2x$

(a) Work out the value of x.

$x =$ ..... **[2]**

(b) Find the probability that the spinner does not land on 4.

..... **[2]**

592 A bag contains only red counters, blue counters, and green counters.
The number of red counters is x^2.
The number of blue counters is $3x$.
The number of green counters is 2.

(a) Write down an expression, in its simplest form, for the total number of counters in the bag.

..... **[1]**

(b) A counter is chosen at random.
The probability that it is red is $\frac{8}{15}$.
Show that $7x^2 - 24x - 16 = 0$.

[2]

(c) Solve the equation $7x^2 - 24x - 16 = 0$ to find the value of x.

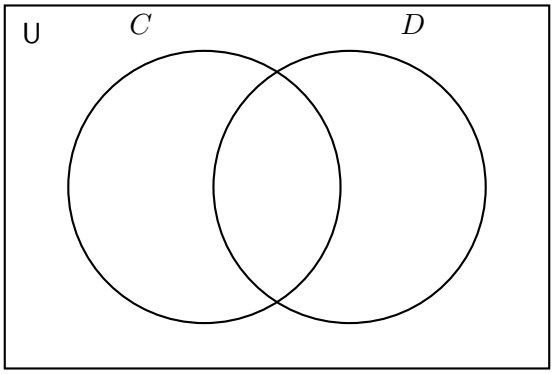
$x =$ ..... **[3]**

(d) Find the probability that a randomly chosen counter is not blue.
Give your answer as a fraction in its simplest form.

..... **[2]**

593 A group of 120 people were asked about the pets they own.
$C = \{\text{people who own a cat}\}$
$D = \{\text{people who own a dog}\}$
$n(\mathsf{U}) = 120$
$n(C) = 55$
$n(D) = 68$
18 people own neither a cat nor a dog.

(a) Complete the Venn diagram to show the number of people in each region.



[3]

(b) A person is chosen at random.
Find $P(C \cap D)$.

..... **[1]**

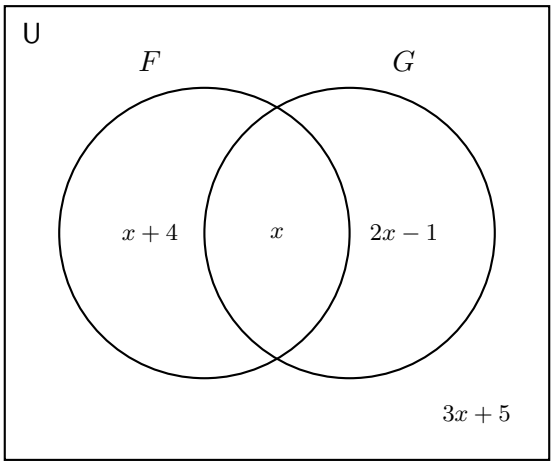
(c) Find $P(D')$.

..... **[2]**

(d) Find the probability that a person chosen at random owns a cat or a dog, but not both.

..... **[2]**

594 The Venn diagram shows information about the number of students in a class of 50 who study French (F) and German (G).



(a) Work out the value of x.

$x =$ ..... **[2]**

(b) A student is chosen at random.
Find $P(F')$.

..... **[2]**

(c) A student is chosen at random.
Find the probability that this student studies exactly one of these languages.

..... **[2]**

595 A biased five-sided spinner is numbered 1, 2, 3, 4, and 5.
The table shows the probability of the spinner landing on each number.

Number	1	2	3	4	5
Probability	0.15	0.25	x	0.2	$3x$

(a) Work out the value of x.

$x =$ ..... **[2]**

(b) The spinner is spun 250 times.
Calculate the expected number of times the spinner lands on a number greater than 3.

..... **[2]**

596 A biased four-sided die is rolled 60 times.
The scores are 1, 2, 3, and 4.
The frequencies of each score are shown in the table.

Score	1	2	3	4
Frequency	x	$2x + 5$	$3x - 2$	$x + 8$

(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find the relative frequency of rolling a 3.

..... **[1]**

(c) The die is rolled 1200 times.
Calculate the expected number of times it rolls a 2.

..... **[2]**

597 A spinner has four colours: Red, Blue, Green, and Yellow.
A student spins it 50 times and records the results.
The table shows the relative frequencies for three of the colours.

Colour	Red	Blue	Green	Yellow
Relative frequency	0.15	0.45	0.10	x

(a) Find the value of x.

$x =$ ..... **[1]**

(b) The spinner is spun 300 times in total.
Estimate the number of times it lands on Blue.

..... **[2]**

(c) The spinner is meant to be fair.
Comment on whether the results show it is fair or biased.

..... **[1]**

598 A researcher records the colours of cars passing a checkpoint.
Over one week, 800 cars pass the checkpoint.
The frequencies of the colours are shown in the table.

Colour	Silver	Black	White	Other
Frequency	260	180	w	$w + 40$

(a) Find the relative frequency of a car being Silver.
Give your answer as a fraction in its simplest form.

..... **[2]**

(b) Find the value of w.

$w =$ ..... **[2]**

(c) The researcher uses these results to estimate the probability that the next car passing the checkpoint is White.
Work out this estimate.

..... **[1]**

(d) In one month, 12000 cars are expected to pass the checkpoint.
Calculate the expected number of Black cars.

..... **[2]**

599 A biased four-sided spinner is numbered 1, 2, 3 and 4.
The table shows the probability of scoring each number.
The spinner is spun 400 times.
Let E_1 be the expected number of times the spinner lands on 1.
Let E_3 be the expected number of times the spinner lands on 3.
You are given that $E_1 \times E_3 = 5600$.
Also, the probability of scoring a 3 is greater than the probability of scoring a 1.

Score	1	2	3	4
Probability	x	$2x$	y	y

(a) Show that $300x^2 - 100x + 7 = 0$.

[3]

(b) Solve the equation to find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) Calculate the expected number of times the spinner lands on an even number.

..... **[2]**

600 Two biased coins, Coin A and Coin B, are tossed.
The probability of Coin A landing on Heads is p.
The probability of Coin B landing on Heads is $p + 0.15$.
Coin A is tossed 300 times and Coin B is tossed 200 times.
The total expected number of Heads from these 500 tosses is 255.

(a) Form an equation in p and show that $p = 0.45$.

[3]

(b) Both coins are tossed together 1000 times.
Work out the expected number of times they land on different outcomes.

..... **[3]**

(c) In a new experiment, Coin A is tossed N times.
The relative frequency of landing on Heads is 0.48 and it lands on Heads 72 times.

(i) Work out the value of N.

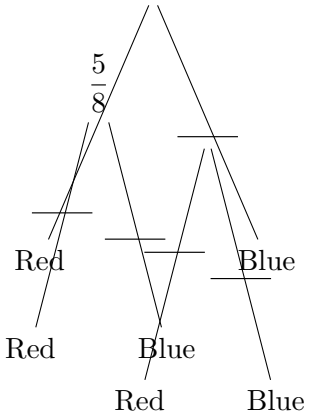
$N =$ ..... **[2]**

(ii) Calculate the expected number of Tails in these N tosses, using the theoretical probability $p = 0.45$.

..... **[2]**

601 A bag contains 5 red counters and 3 blue counters.
A counter is taken at random from the bag, its colour is noted and it is replaced.
A second counter is then taken at random.

(a) Complete the tree diagram.



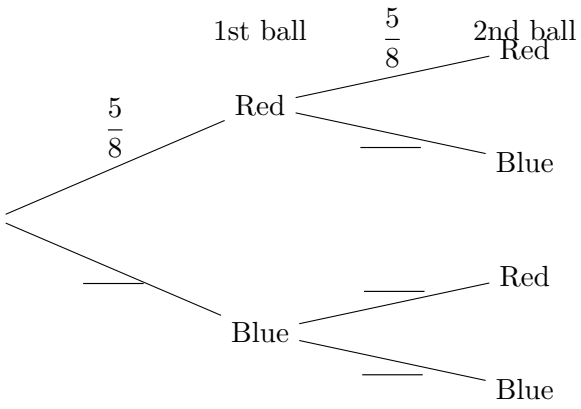
[2]

(b) Calculate the probability that both counters are red.

..... **[2]**

602 A bag contains 8 balls.
5 of the balls are Red and 3 of the balls are Blue.
A ball is taken at random from the bag, its colour is noted, and it is replaced.
A second ball is then taken at random from the bag.

(a) Complete the tree diagram.



[2]

(b) Calculate the probability that both balls are Red.

..... **[1]**

(c) Calculate the probability of picking one Red ball and one Blue ball in any order.

..... **[2]**

603 A box contains n counters.
3 of the counters are Red and all the remaining counters are Blue.
Two counters are taken at random from the box, without replacement.
The probability that both counters are Red is $\frac{1}{15}$.

(a) Show that $n^2 - n - 90 = 0$.

[3]

(b) Find the value of n.

$n =$ ..... **[2]**

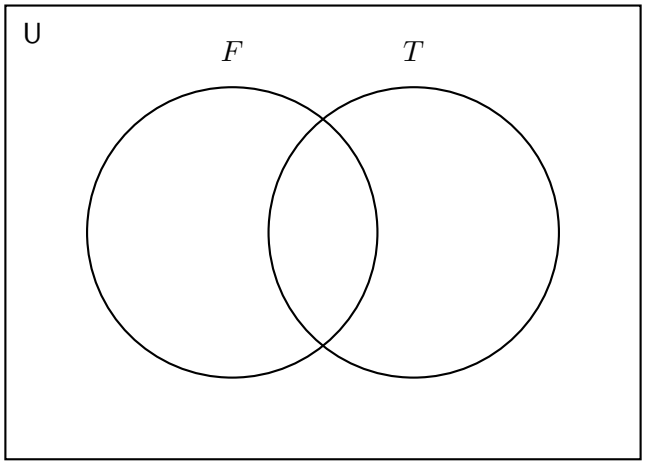
(c) Calculate the probability that exactly one of the two counters chosen is Red.

..... **[3]**

604 There are 30 students in a class.
U = {students in the class}
F = {students who play Football}
T = {students who play Tennis

18 students play Football. 14 students play Tennis. 4 students do not play Football and do not play Tennis.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

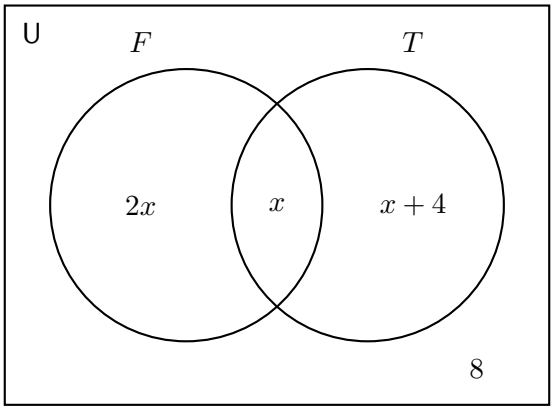
(b) A student is chosen at random from the class.
Find $P(F \cap T)$.

$P(F \cap T) =$ ..... **[1]**

(c) Two students are chosen at random from the class.
Calculate the probability that both students play Football but not Tennis.

..... **[3]**

605 A group of 40 students are asked if they play Football (F) or Tennis (T).
The Venn diagram shows the number of students in each region in terms of x.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find $P(F \cup T)$.

..... **[1]**

(c) Two students are chosen at random from the group of 40.
Calculate the probability that both students play Football only.

..... **[3]**

606 A bag contains 4 red discs and some blue discs.
Let the total number of discs in the bag be x.
Two discs are taken at random from the bag, without replacement.
The probability that both discs are red is $\frac{1}{11}$.

(a) Show that $x^2 - x - 132 = 0$.

[3]

(b) Find the number of blue discs in the bag.

..... **[3]**

(c) Calculate the probability that one disc is red and one disc is blue.

..... **[3]**

607 70 students are asked to choose their favourite sport from Football, Tennis, or Basketball.
The incomplete two-way table shows some information about their choices.

(a) Complete the two-way table.

	Football	**Tennis**	**Basketball**	**Total**
Boys	15		12	
Girls		14		30
Total	23		20	70

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Tennis.

..... **[1]**

608 50 students each choose one sport to play: Tennis, Football, or Swimming.
The students are classified as Boys or Girls.

(a) Complete the two-way table to show this information.

	Tennis	**Football**	**Swimming**	**Total**
Boys	7			28
Girls	8		6	
Total		20		50

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy who chose Football.

..... **[1]**

609 80 students were asked to choose their favourite sport from Football, Basketball, and Tennis.

- There are 35 girls in total.
- 38 students chose Football, of which 24 were boys.
- 12 girls chose Basketball.
- 20 students chose Tennis.

(a) Complete the two-way table.

	Football	**Basketball**	**Tennis**	**Total**
Boys				
Girls				
Total				

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Basketball.

..... **[1]**

(c) A girl is chosen at random.
Find the probability that her favourite sport is Tennis.

..... **[2]**

610 A survey of adults recorded their preferred hot drink, either Tea or Coffee.
Let x be the number of males who prefer Coffee.
The number of males who prefer Tea is $2x + 10$.
The number of females who prefer Coffee is $x + 5$.
The number of females who prefer Tea is $3x - 11$.

(a) Write down an expression, in terms of x, for the total number of adults.
Give your answer in its simplest form.

..... **[2]**

(b) The survey included 130 adults.
Find the value of x.

$x =$ ..... **[2]**

(c) Complete the two-way table.

	Coffee	**Tea**	**Total**
Male			
Female			
Total			

[3]

(d) One of the adults who prefers Coffee is chosen at random.
Find the probability that this adult is female.

..... **[2]**

611 A group of 120 adults were asked whether they prefer Tea or Coffee.

- 55 of the adults are women.
- 70 of the adults prefer Tea.
- 25 of the men prefer Coffee.

(a) Complete the two-way table to show this information.

	Tea	**Coffee**	**Total**
Men			
Women			
Total			

[3]

(b) Work out the number of women who prefer Tea as a percentage of the total number of adults.

..... % **[2]**

612 In a school of 400 students, the ratio of juniors to seniors is 3 : 2.
Each student travels to school by Bus, by Car, or Walks.

- 45% of the 400 students take the Bus.
- Of the seniors, 30 take the Car and 50 Walk.
- There are 80 juniors who take the Car.

(a) Complete the two-way table to classify this data.

	Bus	**Car**	**Walk**	**Total**
Juniors				
Seniors				
Total				400

[3]

(b) A student is chosen at random.
Find the probability that they are a junior who walks.

..... **[1]**

(c) Two seniors are chosen at random.
Find the probability that they both take the bus.

..... **[2]**

613 80 students travel to school either by Bus or by Walking.
The two-way table shows the number of boys and girls using each method of transport, in terms of x.

	Bus	**Walk**	**Total**
Boys	x	$2x$	
Girls	$x + 5$	$3x - 2$	
Total			80

(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Complete the two-way table with the number of students in each category.

	Bus	**Walk**	**Total**
Boys			
Girls			
Total			80

[2]

(c) A student is chosen at random from those who walk to school.
Find the probability that this student is a girl.

..... **[2]**

614 A company has 300 employees. Each employee is either a Manager, an Office Worker, or a Technician.
Each employee works either Full-Time or Part-Time.

- The ratio of Managers : Office Workers : Technicians is 2 : 5 : 3.
- 20% of the Managers work Part-Time.
- 40% of the Office Workers work Part-Time.
- The number of Part-Time Technicians is equal to the number of Part-Time Managers.

(a) Complete the two-way table for this information.

	Full-Time	**Part-Time**	**Total**
Managers			
Office Workers			
Technicians			
Total			

[4]

(b) A Part-Time employee is chosen at random.
Find the probability that they are an Office Worker.

..... **[2]**

(c) Two Managers are chosen at random.
Find the probability that exactly one of them works Part-Time.

..... **[3]**

615 A group of 120 students are classified by gender (Boys, Girls) and handedness (Left-handed, Right-handed).
The two-way table shows the number of students in each category in terms of x and y.
Two students are chosen at random from the 120 students.
The probability that both students are left-handed is $\frac{3}{34}$.

	Left-handed	**Right-handed**
Boys	x	y
Girls	$2x$	$y - 6$

(a) Show that $3x^2 - x - 420 = 0$.

[4]

(b) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) A right-handed student is chosen at random.
Find the probability that the student is a girl.

..... **[2]**

616 The number of goals scored by Team A in each of their last 6 matches is recorded below.

3 1 0 4 2 2

Team B also played 6 matches.
For Team B, the median number of goals scored was 1.5 and the range was 5.

(a) Work out the median and the range of the number of goals scored by Team A.

Median = ... , Range = ... **[2]**

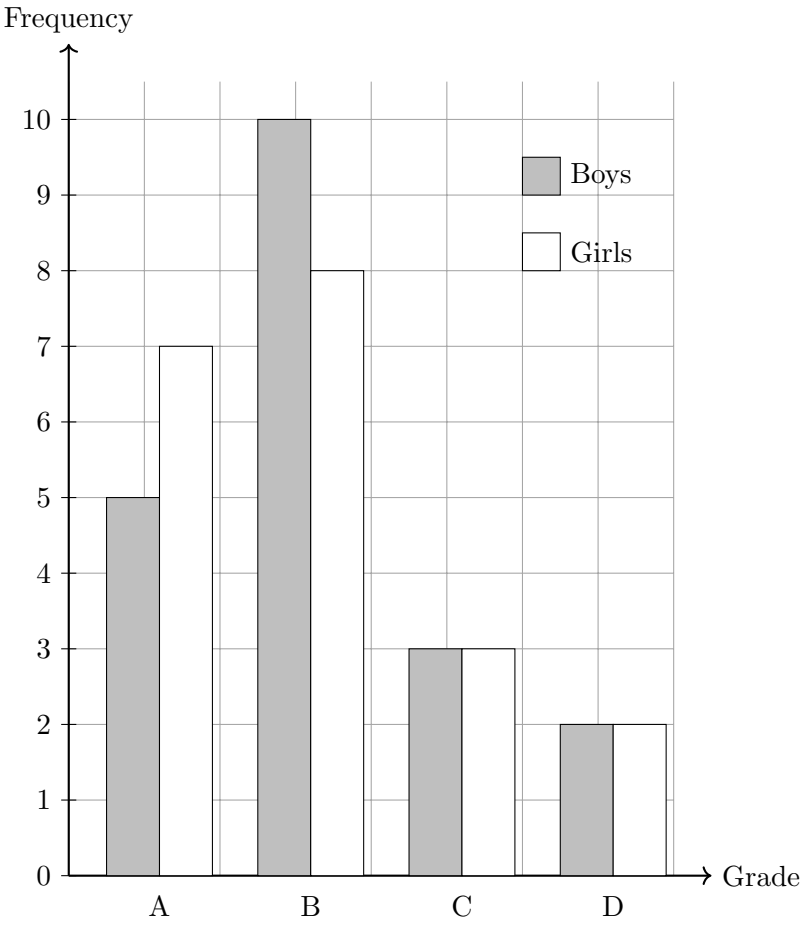
(b) Make two comments comparing the performance of Team A with the performance of Team B.

.....

.....

..... **[2]**

617 The dual bar chart shows the grades achieved by a group of boys and a group of girls in an examination.



(a) Write down the modal grade for the boys.

..... **[1]**

(b) Find the number of girls who scored Grade C.

..... **[1]**

(c) Calculate the percentage of all students who scored Grade A.

..... % **[2]**

618 The stem-and-leaf diagram shows the ages, in years, of 15 members of a chess club.

1	2 5 8
2	0 1 1 4 7 9
3	2 5 6 9
4	1 3

Key: 2 | 1 represents 21 years

(a) Find the median age.

..... years **[1]**

(b) Find the range of the ages.

..... years **[1]**

(c) Find the interquartile range.

..... years **[2]**

(d) A member is chosen at random.
Find the probability that they are older than 30 years.

..... **[1]**

619 A set of 5 integers has the following statistical properties:

- Mode = 3
- Median = 7
- Mean = 8
- Range = 11

(a) Find the 5 integers.

..... **[4]**

(b) A sixth integer, x, is added to the set.
The new mean is 9.
Work out the value of x.

$x =$ ..... **[2]**

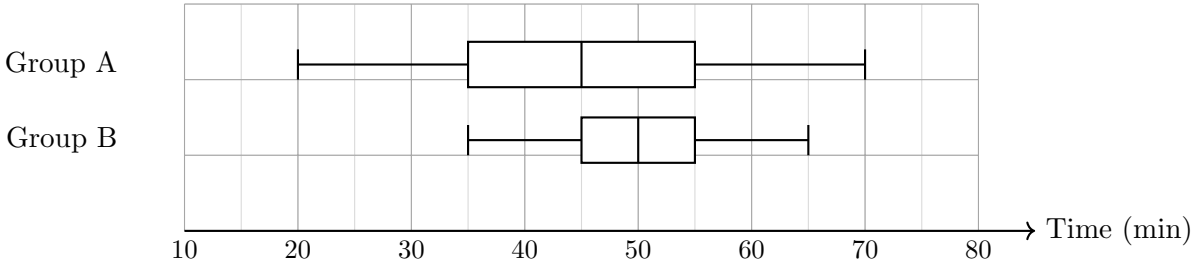
(c) A student claims: "If another completely different set of 5 integers has the same range of 11, it must also have the same standard deviation."

State whether the student is correct.
Give a statistical reason for your answer.

.....

..... **[1]**

620 The box-and-whisker plots show information about the times, in minutes, taken by two groups of students, Group A and Group B, to complete a mathematics test.



(a) Write down the median time for Group A.

..... minutes **[1]**

(b) Work out the interquartile range for Group B.

..... minutes **[1]**

(c) Make two comments comparing the distributions of the times taken by Group A and Group B.

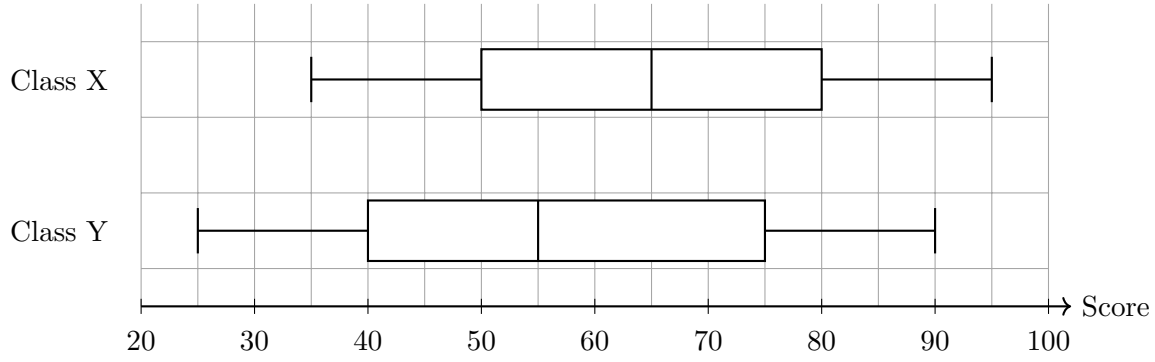
Comment 1:

Comment 2: **[2]**

(d) Group A has 80 students.
Estimate the number of students in Group A who took longer than 55 minutes.

..... students **[2]**

621 The box-and-whisker plots show the distribution of test scores for two classes, Class X and Class Y.



(a) Write down the median score for Class X.

..... **[1]**

(b) Work out the interquartile range for Class Y.

..... **[2]**

(c) Make two different comments comparing the distribution of test scores in Class X with Class Y.

[2]

(d) A student states: "Because the median score for Class X is higher, every student in Class X scored more than every student in Class Y."

Is this statement correct?
Give a statistical reason for your answer.

.....

..... **[1]**

622 The table shows information about the mass, m grams, of 60 mangoes collected by a farmer.

Mass (m grams)	$100 < m \leq 120$	$120 < m \leq 140$	$140 < m \leq 160$	$160 < m \leq 180$	$180 < m \leq 200$
Frequency	10	20	15	10	5

(a) State the modal class.

..... **[1]**

(b) Calculate an estimate of the mean mass of the mangoes.

..... g **[4]**

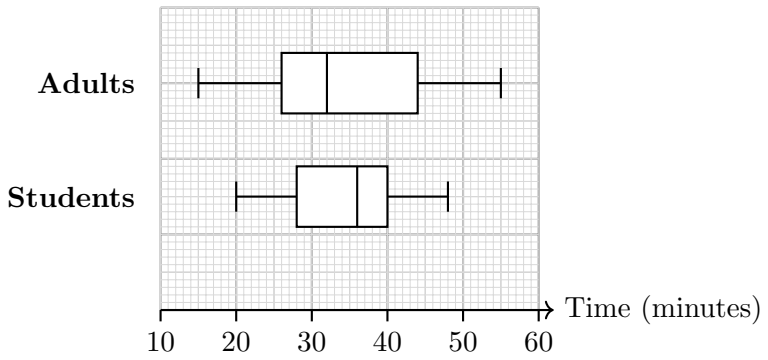
(c) The farmer states that exactly 15 mangoes have a mass of 150 grams because 150 is the midpoint of the class $140 < m \leq 160$.
Explain why the farmer's reasoning is incorrect.

.....

.....

..... **[2]**

623 The box-and-whisker plots show information about the times, in minutes, taken by a group of Adults and a group of Students to complete a logic puzzle.



(a) Use the diagram to find the interquartile range of the times taken by the Adults.

..... minutes **[2]**

(b) Make two different statements comparing the distribution of the times taken by the Adults with the times taken by the Students.

.....

.....

..... **[2]**

(c) A student claims: "Because the range of the Students' times is smaller, this proves that every student completed the puzzle faster than the slowest Adult."

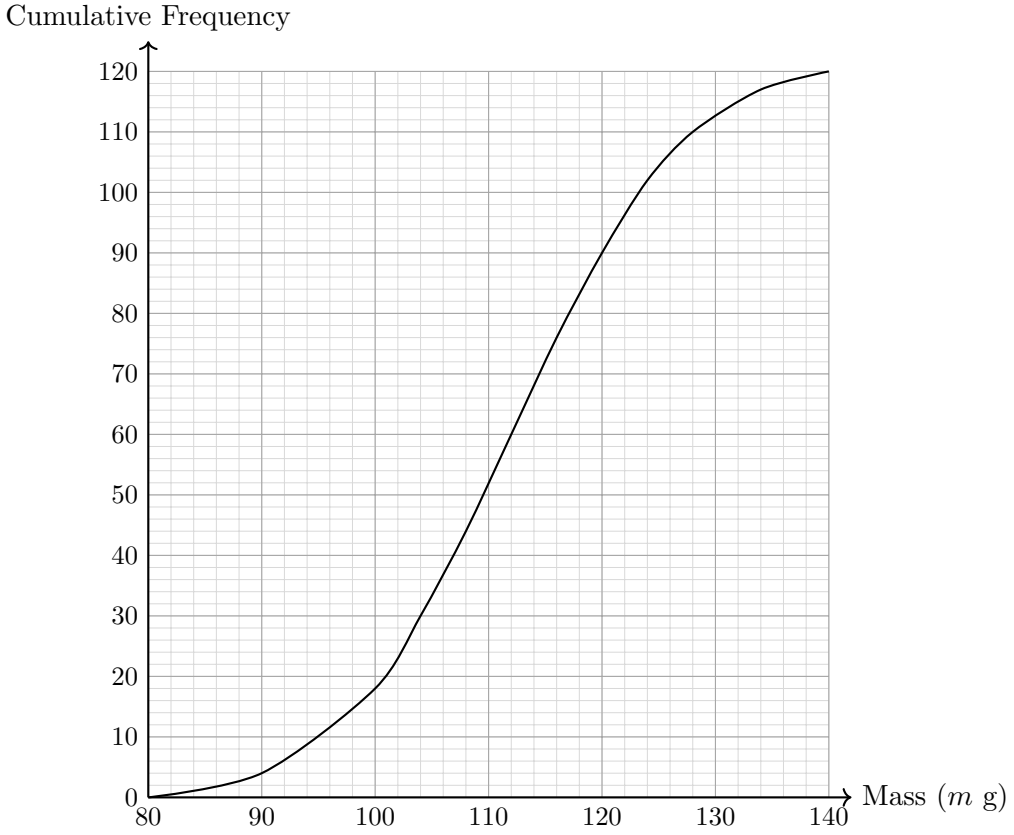
Comment on the validity of this reasoning.

.....

.....

..... **[2]**

624 The cumulative frequency curve shows information about the masses, m grams, of 120 apples from Farm X.



(a) Use the graph to find an estimate for the median mass.

..... g **[1]**

(b) Use the graph to find an estimate for the interquartile range.

..... g **[2]**

(c) Apples with a mass greater than 128 g are classed as Large.
Find the percentage of these apples that are Large.

..... % **[2]**

(d) 120 apples from Farm Y have a median mass of 105 g and an interquartile range of 22 g.
Compare the distribution of the masses of the apples from Farm X and Farm Y.

.....

..... **[2]**

625 A scientist records different types of weather data.
State whether each of the following types of data is discrete or continuous.

(a) The daily maximum temperature.

..... **[1]**

(b) The number of rainy days in a month.

..... **[1]**

(c) The amount of rainfall in one hour.

..... **[1]**

626 A sports analyst collects data during a football match.

(a) State whether the number of goals scored in the match is discrete or continuous.

..... **[1]**

(b) State whether the mass of the football is discrete or continuous.

..... **[1]**

(c) State whether the time taken to complete the match is discrete or continuous.

..... **[1]**

(d) State whether the temperature on the pitch is discrete or continuous.

..... **[1]**

627 A student collects data for a statistics project.
For each of the variables below, state whether the data collected is discrete or continuous.

(a) The number of goals scored in a football match.

..... **[1]**

(b) The mass of a football.

..... **[1]**

(c) The volume of water in a swimming pool.

..... **[1]**

(d) The number of people swimming in a pool.

..... **[1]**

628 Two sets of data are collected during a fitness test.
Data Set A is the exact time, t seconds, taken by athletes to run a race.
Data Set B is the number of push-ups, n, completed by the athletes.

(a) Data Set A is grouped into classes such as $10 \leq t < 12$.
Data Set B is grouped into classes such as $10 - 19$.
Explain why the classes for Data Set A use inequalities while the classes for Data Set B use a hyphen.

.....

..... **[2]**

(b) A student decides to draw a histogram to represent Data Set B. One of the classes is $20 - 29$.

(i) State the lower and upper class boundaries that must be used for this class on the horizontal axis of the histogram.

Lower = ... , Upper = ... **[1]**

(ii) Calculate the class width used for the frequency density calculation.

..... **[1]**

(c) Continuous data can theoretically take any value within a range.
Give an example of a continuous variable that is practically recorded as discrete data in everyday life.

.....

..... **[1]**

629 A student collects data about a local park for a science project.

(a) Explain the mathematical difference between discrete and continuous data.

.....

.....

..... **[2]**

(b) The student measures the lengths of 50 leaves to the nearest millimetre.

(i) State whether the true length of a leaf is discrete or continuous.

..... **[1]**

(ii) Explain why measuring the lengths to the nearest millimetre does not make the variable discrete.

.....

..... **[1]**

630 A farmer weighs 60 apples.
The masses, m grams, are recorded in the table.

Mass (m grams)	**Frequency**
$80 < m \leq 100$	12
$100 < m \leq 120$	26
$120 < m \leq 140$	15
$140 < m \leq 160$	7

(a) State whether the mass of an apple is discrete or continuous data.

..... **[1]**

(b) Calculate an estimate of the mean mass.

..... g **[3]**

(c) Explain why your answer to part (b) is an estimate.

..... **[1]**

631 A scientist measures the lengths of 50 worms and counts the number of spots on each worm.
The frequency table shows the data collected for the number of spots.

Number of spots	$0-4$	$5-9$	$10-14$	$15-19$
Frequency	12	18	14	6

(a) State which of the two variables, length or number of spots, is continuous.

..... **[1]**

(b) The length of one worm is recorded as 14.5 cm, correct to the nearest millimetre.

(i) Write down the lower bound of this length.

..... *cm* **[1]**

(ii) Write down the upper bound of this length.

..... *cm* **[1]**

(c) Using the frequency table, write down the class width for the $5-9$ class.

..... **[2]**

632 Data is collected about the members of an athletics club.

(a) State whether each of the following variables is discrete or continuous.

(i) The number of members in the club.

..... **[1]**

(ii) The exact distance a member jumps in the long jump.

..... **[1]**

(b) The shoe sizes of the members are recorded.
Shoe sizes are available in sizes such as $7, 7.5, 8, 8.5, 9$.
A student claims that because these shoe sizes include decimal values, the data is continuous.
Explain why the student is incorrect.

.....

.....

..... **[2]**

633 A factory packs identical books into boxes.
Each box contains exactly 24 books.

(a) State whether each of the following variables is discrete or continuous.

(i) The number of books in a box.

..... **[1]**

(ii) The mass of a book.

..... **[1]**

(b) The mass of one book is 850 g, correct to the nearest 10 g.
Find the upper bound for the total mass of the 24 books.
Give your answer in kilograms.

..... kg **[2]**

(c) An empty box has a mass of 1.2 kg, correct to 1 decimal place.
Calculate the lower bound for the total mass of a box containing 24 books.

..... kg **[3]**

634 A student records the number of points scored by a basketball team in 11 matches.

14 20 8 24 16 5 26 13 22 10 18

(a) Find the median.

..... **[1]**

(b) Find the range.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

(d) Calculate the mean.

..... **[2]**

635 The list shows the number of points scored by a basketball player in 7 games.

$$7 \quad 12 \quad 8 \quad 7 \quad 12 \quad x \quad 10$$

The mode of the number of points is 7.

(a) Write down the value of x.

$x =$ ..... **[1]**

(b) Find the range of the number of points.

..... **[1]**

(c) Find the median.

..... **[1]**

(d) Calculate the mean.

..... **[2]**

(e) In the 8th game, the player scores y points.
The new mean number of points for all 8 games is 9.5.
Find the value of y.

$y =$ ..... **[2]**

636 The table shows the heights, h cm, of 80 plants.

Height (h cm)	**Frequency**
$10 < h \leq 20$	12
$20 < h \leq 30$	25
$30 < h \leq 40$	28
$40 < h \leq 50$	11
$50 < h \leq 60$	4

(a) Write down the modal class.

..... **[1]**

(b) Calculate an estimate of the mean height.

..... cm **[4]**

637 The table shows information about the time, t seconds, taken by 100 students to complete a puzzle.

Time (t seconds)	Frequency
$10 < t \leq 20$	14
$20 < t \leq 30$	28
$30 < t \leq 40$	32
$40 < t \leq 50$	18
$50 < t \leq 60$	8

(a) Write down the modal time interval.

..... **[1]**

(b) Calculate an estimate of the mean time.

..... seconds **[4]**

(c) Calculate the percentage of students who took strictly more than 40 seconds to complete the puzzle.

..... % **[2]**

(d) Write down the interval that contains the upper quartile.

..... **[1]**

638 A factory produces 60 items.
The masses, m kg, of these items are recorded in the table.
The estimate of the mean mass is 48 kg.

Mass (m kg)	**Frequency**
$0 < m \leq 20$	p
$20 < m \leq 40$	14
$40 < m \leq 60$	16
$60 < m \leq 80$	q
$80 < m \leq 100$	5

(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[5]**

(b) State the class interval which contains the median mass.
You must show your working.

..... **[2]**

639 The table shows information about the masses, m kg, of 50 parcels.

Mass (m kg)	Frequency
$0 < m \leq 2$	15
$2 < m \leq 4$	x
$4 < m \leq 6$	y
$6 < m \leq 8$	8
$8 < m \leq 10$	2

The estimate of the mean mass of the parcels is 3.8 kg.

(a) Use the total number of parcels to write down an equation in x and y, and show that it simplifies to $x + y = 25$.

[1]

(b) Use the estimate of the mean to show that $3x + 5y = 101$.

[4]

(c) Solve the simultaneous equations to find the value of x and the value of y.
You must show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(d) Write down the interval that contains the median mass.

..... **[1]**

640 The number of text messages received by 15 students in one hour is shown below.

12	23	15	18	21	14	17	15	22	19	15	24	11	16	20

(a) Find the mean.

..... **[1]**

(b) Find the median.

..... **[1]**

(c) Find the interquartile range.

..... **[2]**

641 A survey was conducted to find the number of books read by 21 students during a month.
The results are shown in the table.

Number of books	1	2	3	4	5
Frequency	2	5	7	4	3

(a) Use a graphic display calculator to find the following statistics for this data.

(i) Calculate the mean number of books read.
Give your answer correct to 3 significant figures.

Mean = ... **[2]**

(ii) Find the median.

Median = ... **[1]**

(iii) Find the interquartile range.

Interquartile range = ... **[2]**

642 The table shows information about the time, t minutes, taken by 100 people to travel to work.

Time (t minutes)	Frequency
$0 < t \leq 10$	14
$10 < t \leq 20$	28
$20 < t \leq 30$	35
$30 < t \leq 40$	18
$40 < t \leq 60$	5

(a) Write down the modal class.

..... **[1]**

(b) Use your graphic display calculator to calculate an estimate of the mean time.

..... minutes **[3]**

(c) Explain why your answer to part (b) is an estimate.

..... **[1]**

643 The table shows the time, t minutes, taken by 80 people to complete a crossword puzzle.

Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 60$
Frequency	12	25	28	11	4

(a) Calculate an estimate of the mean time.

..... minutes **[3]**

(b) Explain why your answer to part (a) is an estimate.

..... **[1]**

(c) An 81st person completes the puzzle in 12 minutes.
Without calculating the new mean, explain how this affects the mean time.

..... **[1]**

644 The table shows the number of goals scored by a football team in 35 matches.

Number of goals	0	1	2	3	4	5
Frequency	3	x	y	6	4	1

(a) The mean number of goals scored is 2.2.
Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(b) Find the median number of goals.

..... **[1]**

645 A game is played where players can score $1, 2, 3, 4$, or 5 points.
The table shows the frequencies of the scores obtained by a group of players.
The mean score is 2.8.

Score	1	2	3	4	5
Frequency	10	k	$2k + 3$	7	5

(a) Show algebraically that $k = 5$.

[4]

(b) Use your graphic display calculator to find the following statistics for the scores.

(i) Find the median.

Median = ... **[1]**

(ii) Find the lower quartile.

Lower quartile = ... **[1]**

(iii) Find the upper quartile.

Upper quartile = ... **[1]**

646 The stem-and-leaf diagram shows the masses, in kg, of 15 parcels.

1	2	4	7		
2	0	1	1	1	5
3	0	2	6	8	
4	1	4	5		

Key: 1 | 2 represents 12 kg

(a) Find the mode of the masses.

..... kg **[1]**

(b) Find the median mass.

..... kg **[1]**

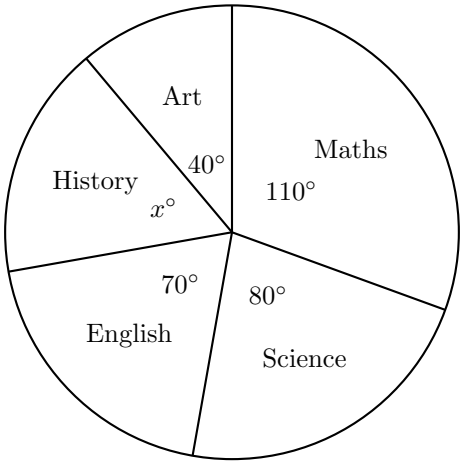
(c) Work out the range of the masses.

..... kg **[1]**

(d) A parcel is chosen at random.
Find the probability that its mass is strictly greater than 30 kg.

..... **[2]**

647 The pie chart shows information about the favourite subjects of 180 students.



(a) Find the value of x.

$x =$ ..... **[1]**

(b) Find the number of students whose favourite subject is Science.

..... **[2]**

(c) Write down the fraction of students whose favourite subject is Maths.
Give your fraction in its simplest form.

..... **[2]**

648 The masses, m kilograms, of 15 parcels are recorded.

3.2	1.5	4.6	2.1	1.8	3.5	2.9	4.1	3.2	1.9	2.4	3.7	4.0	2.6	1.7

(a) Complete the ordered stem-and-leaf diagram to show this information.

1	
2	
3	
4	

Key: 1 | 5 represents 1.5 kg

[3]

(b) Find the median mass.

..... kg **[1]**

(c) A parcel is chosen at random.
Find the probability that its mass is greater than 3.5 kg.

..... **[1]**

649 A survey asked 120 students about their favourite sport.
The results are shown in the table.

Sport	**Frequency**
Football	45
Basketball	35
Tennis	25
Swimming	15

(a) A pie chart is drawn to show this information.
Calculate the sector angle for Basketball.

..... ° **[2]**

(b) The pie chart has a radius of 6 cm.

Calculate the area of the sector representing Tennis.

..... cm^2 **[3]**

(c) In a different school, the pie chart for favourite sports has a sector angle of 108° for Football.
This sector represents 54 students.
Calculate the total number of students in this school.

..... **[2]**

650 A pie chart is drawn to show the favourite sport of a group of students.
The angles of the sectors for each sport are given in terms of x.

Football: $(2x + 30)^\circ$ Basketball: x° Tennis: $(x + 50)^\circ$ Swimming: $(2x - 20)^\circ$

(a) Find the value of x.

$x =$ ..... **[2]**

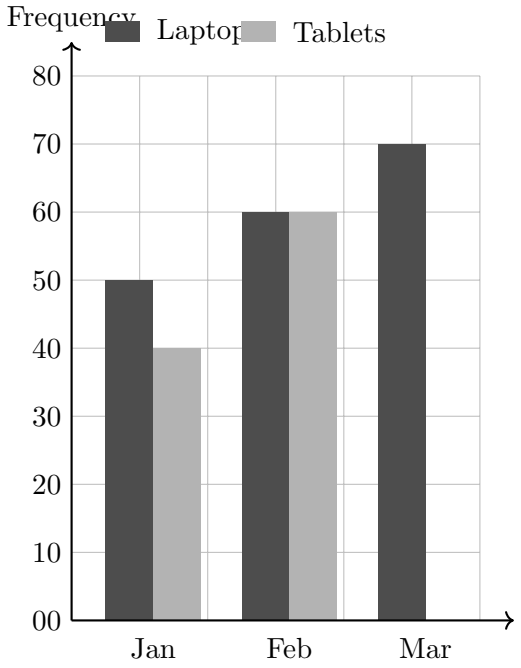
(b) 10 students chose Tennis as their favourite sport.
Calculate the total number of students in the group.

..... **[2]**

(c) Two students are chosen at random from the group.
Calculate the probability that one student chose Football and the other student chose Basketball.

..... **[3]**

651 A company records the number of Laptops and Tablets sold over three months.
The dual bar chart shows the sales for January, February, and March.
The bar for Tablets sold in March is incomplete and not shown.



(a) The total number of Laptops sold over the three months is 20% more than the total number of Tablets sold.
Calculate the number of Tablets sold in March.

..... **[4]**

(b) In April, the total number of items sold is $\frac{1}{3}$ of the total number of items sold in January, February, and March combined.
The ratio of Laptops to Tablets sold in April is 6 : 5.
Work out the number of Laptops sold in April.

..... **[4]**

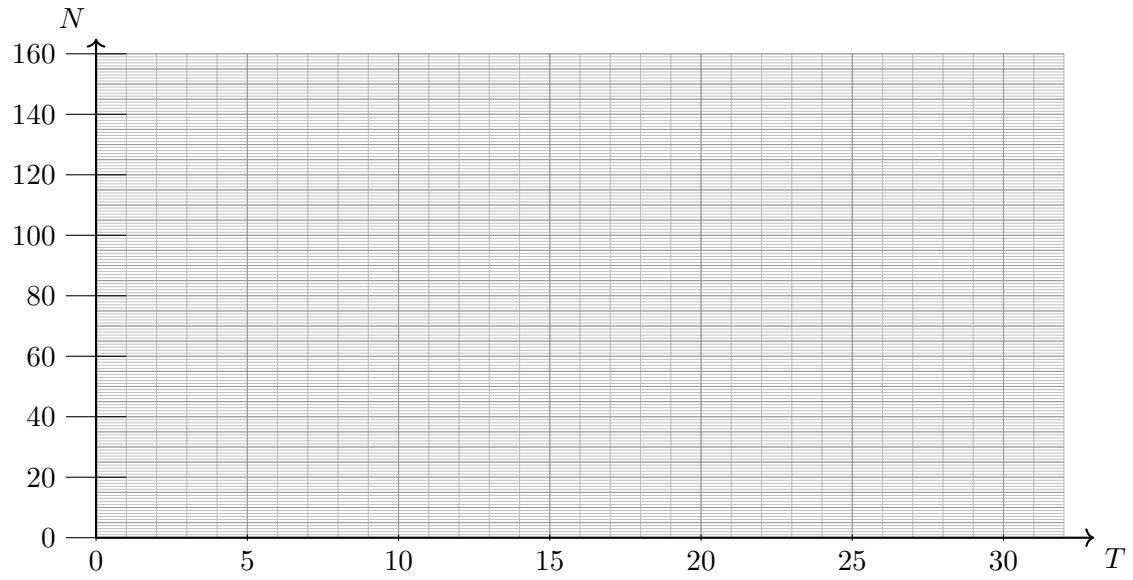
(c) The price of a Laptop is \$450 and the price of a Tablet is \$200.
Calculate the percentage of the total revenue in January that came from Tablet sales.

..... % **[3]**

652 The table shows the outside temperature, T°C, and the number of hot drinks, N, sold by a cafe on 10 different days.

Temperature (T°C)	5	8	12	15	18	20	22	25	28	30
Number of hot drinks (N)	140	125	100	95	70	65	50	40	25	15

(a) Complete the scatter diagram.
The first four points have been plotted for you.



[2]

(b) What type of correlation is shown by the scatter diagram?

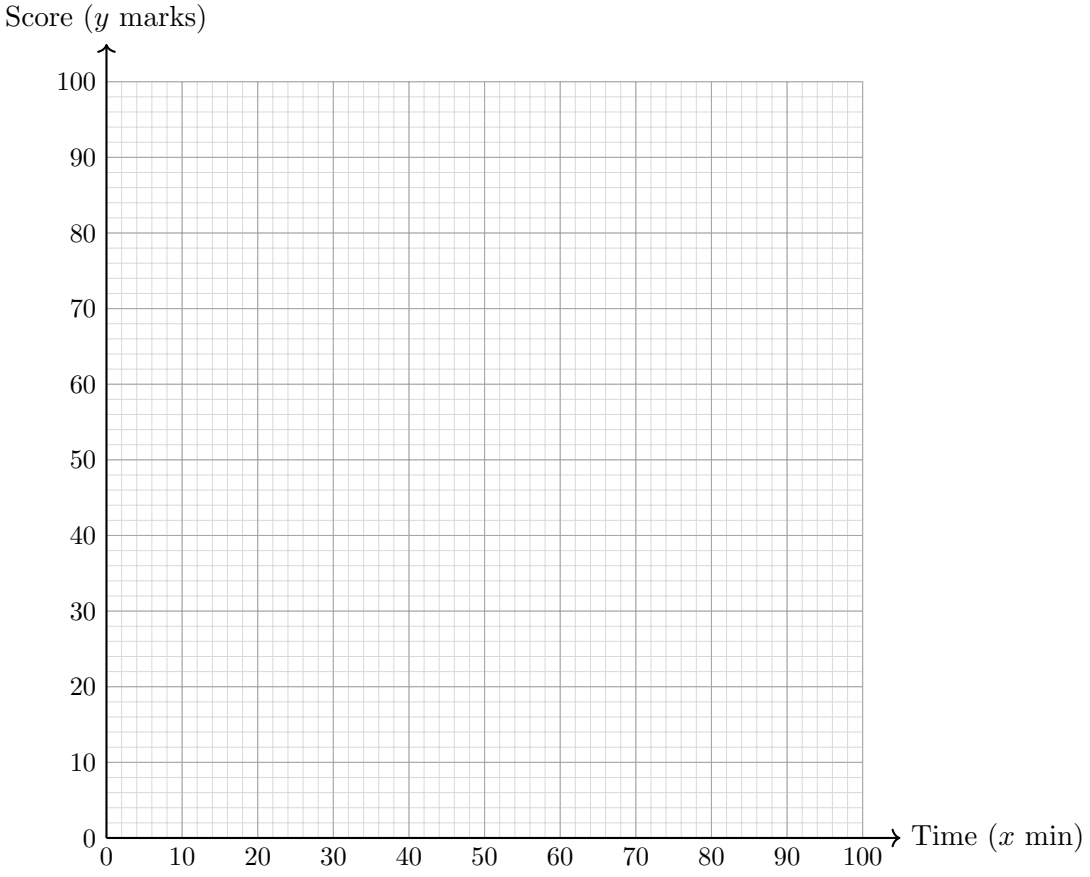
..... **[1]**

(c) Draw a line of best fit on the scatter diagram. **[1]**

653 A teacher records the time, x minutes, spent studying and the score, y marks, for 8 students.
The table shows the results.

Time (x min)	10	20	30	40	50	60	70	80
Score (y marks)	15	25	20	40	50	70	60	80

(a) On the grid, complete the scatter diagram to show this information.



[2]

(b) What type of correlation is shown by the scatter diagram?

..... **[1]**

(c) On the scatter diagram, draw a line of best fit. **[1]**

(d) Another student spent 45 minutes studying.
Use your line of best fit to estimate their score.

..... **[1]**

654 The table shows the time spent studying, x hours, and the test score, y marks, for a group of 8 students.

Study time (x hours)	2	3.5	4	5	6.5	8	9	10
Test score (y marks)	45	52	60	68	75	82	85	95

(a) Use your graphic display calculator to find the equation of the linear regression line, giving y in terms of x.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

(b) Use your equation to estimate the test score for a student who studies for 7 hours.

..... **[2]**

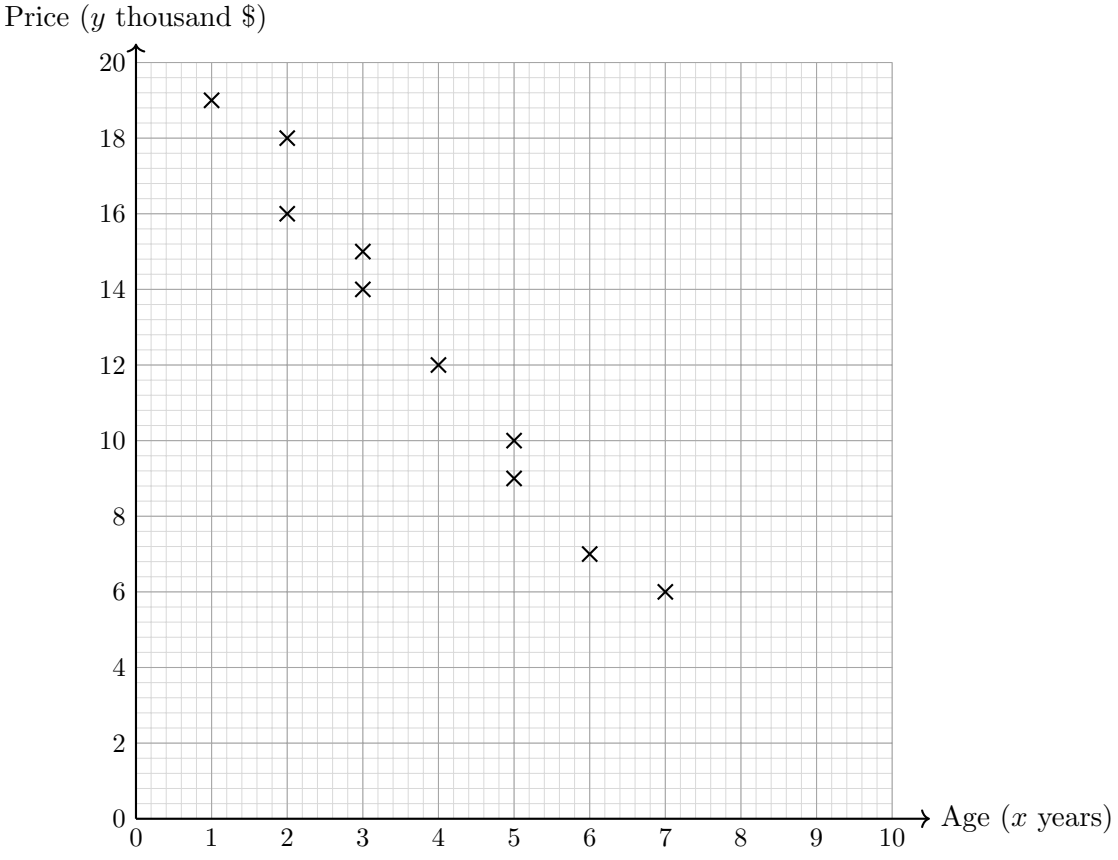
(c) Explain why it may not be reliable to use this equation to estimate the test score for a student who studies for 15 hours.

..... **[1]**

655 A car dealer records the age, x years, and the price, y thousand dollars, of 12 used cars of the same model.
The table shows the data.

Age (x years)	1	2	2	3	3	4	5	5	6	7	7	9
Price (y thousand \$)	19	18	16	15	14	12	10	9	7	6	5	1

(a) The scatter diagram shows the first 10 points from the table.
Plot the remaining 2 points on the scatter diagram.



[1]

(b) Calculate the mean age and the mean price.

Mean age = ... , Mean price = ... **[2]**

(c) On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

(d) State the type of correlation shown.

..... **[1]**

(e) Use your line of best fit to estimate the price of a car that is 8 years old.

..... thousand dollars **[1]**

656 A researcher measures the height, x cm, and the mass, y kg, of 12 adult males.
The results are summarised as follows:

$$\sum x = 2100 \qquad \sum y = 924$$

The equation of the linear regression line is $y = 0.85x - c$, where c is a constant.

(a) Calculate the mean height, $\bar{x}$, and the mean mass, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

(b) The point $(\bar{x}, \bar{y})$ lies on the linear regression line.
Find the value of c.

$c =$ ..... **[2]**

(c) A new adult male is added to the study.
His height is 180 cm.
His mass is exactly equal to the value predicted by the regression line.
Calculate the new mean mass for all 13 men.

..... kg **[2]**

657 The table shows the temperature, x°C, and the number of ice creams sold, y, at a shop over 8 days.

Temperature (x°C)	12	15	18	20	22	25	28	30
Ice creams sold (y)	40	52	68	75	84	98	110	115

(a) Calculate the mean temperature and the mean number of ice creams sold.

Mean temperature = ... , Mean ice creams sold = ... **[2]**

(b) Use a graphic display calculator to find the equation of the linear regression line for this data. Give your answer in the form $y = mx + c$, with values of m and c correct to 3 significant figures.

$y =$ ..... **[2]**

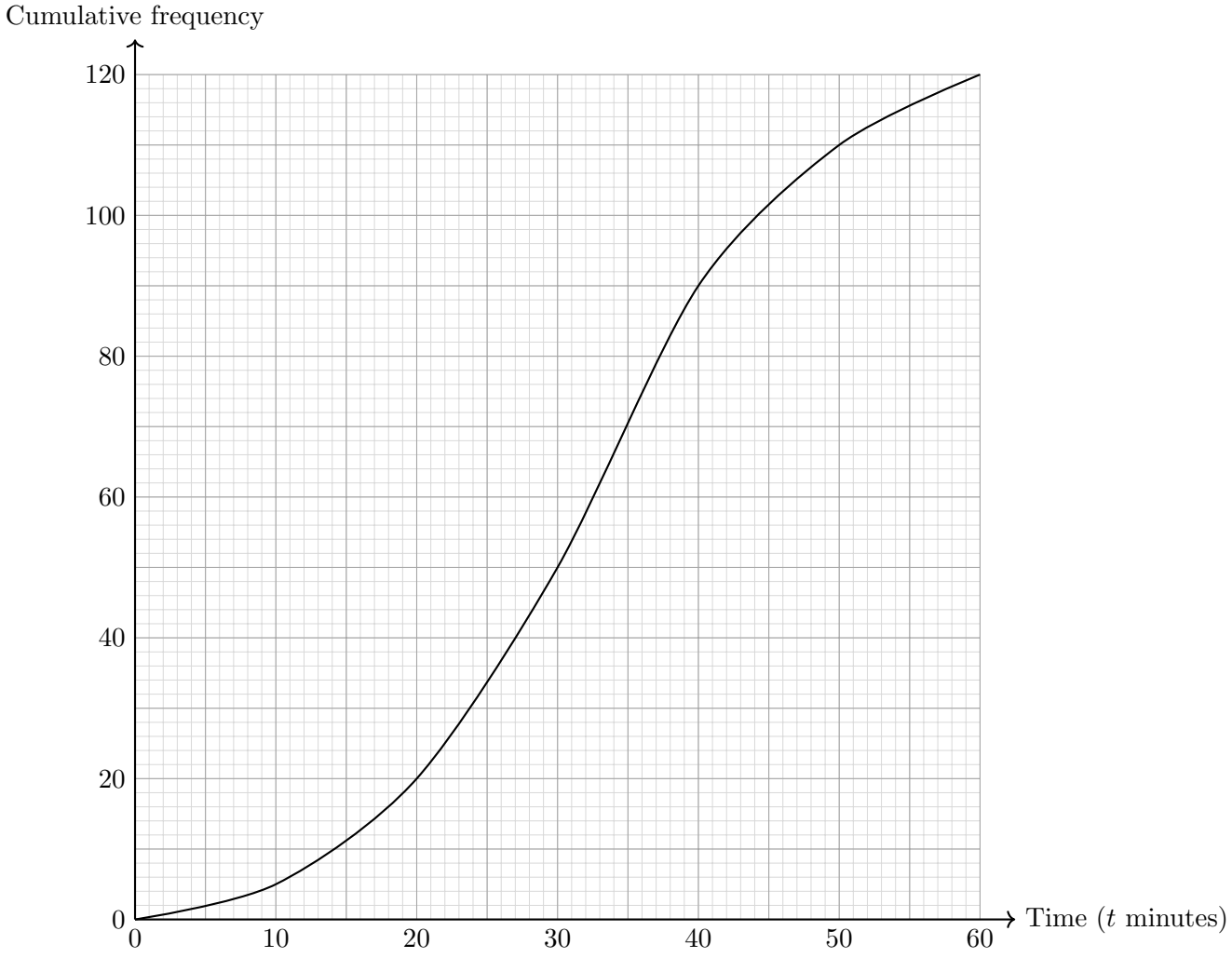
(c) The shop owner predicts the number of ice creams sold on a day when the temperature is 24°C. Use your equation to estimate this number.

..... **[2]**

(d) Comment on the reliability of using your equation to predict the number of ice creams sold on a day when the temperature is 42°C.

..... **[1]**

658 The cumulative frequency diagram shows the times taken, t minutes, by 120 students to complete a test.



(a) Use the diagram to estimate the median time.

..... minutes **[1]**

(b) Use the diagram to estimate the interquartile range.

..... minutes **[2]**

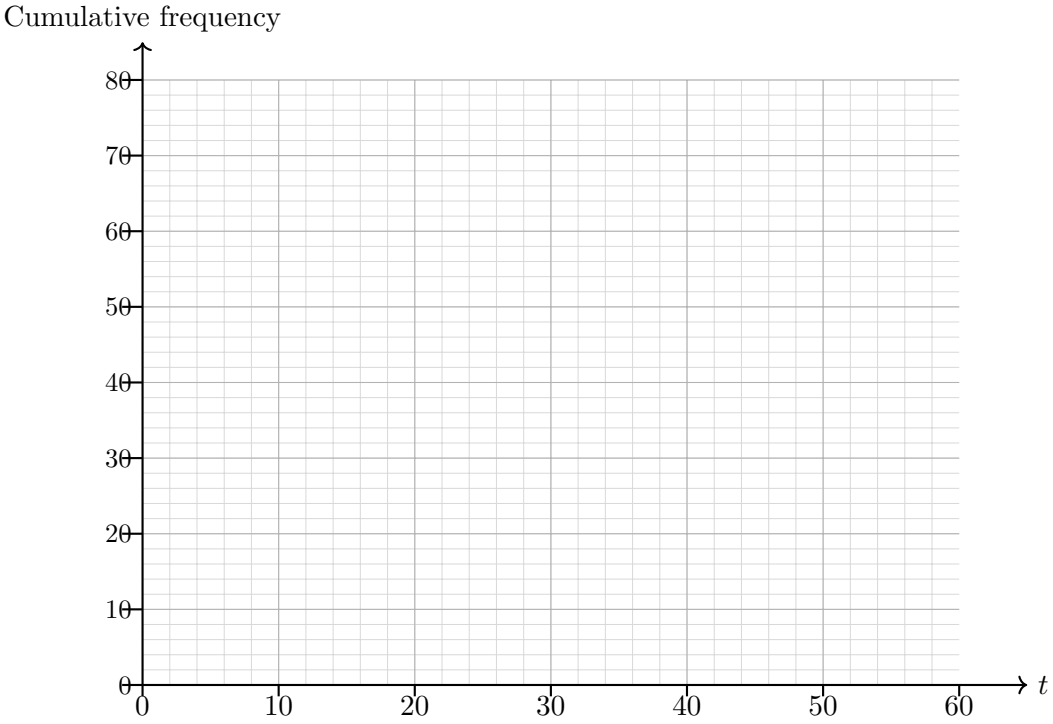
(c) Students who take longer than 45 minutes are given a penalty.
Estimate the number of students who are given a penalty.

..... **[2]**

659 The table shows the cumulative frequencies for the times, t minutes, taken by 80 students to complete a puzzle.

Time (t minutes)	$t \leq 10$	$t \leq 20$	$t \leq 30$	$t \leq 40$	$t \leq 50$	$t \leq 60$
Cumulative frequency	6	22	48	68	78	80

(a) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(b) Use your diagram to estimate the median time.

..... minutes **[2]**

660 A company records the salaries, s thousand dollars, of its 200 employees.
The frequency table shows the results.

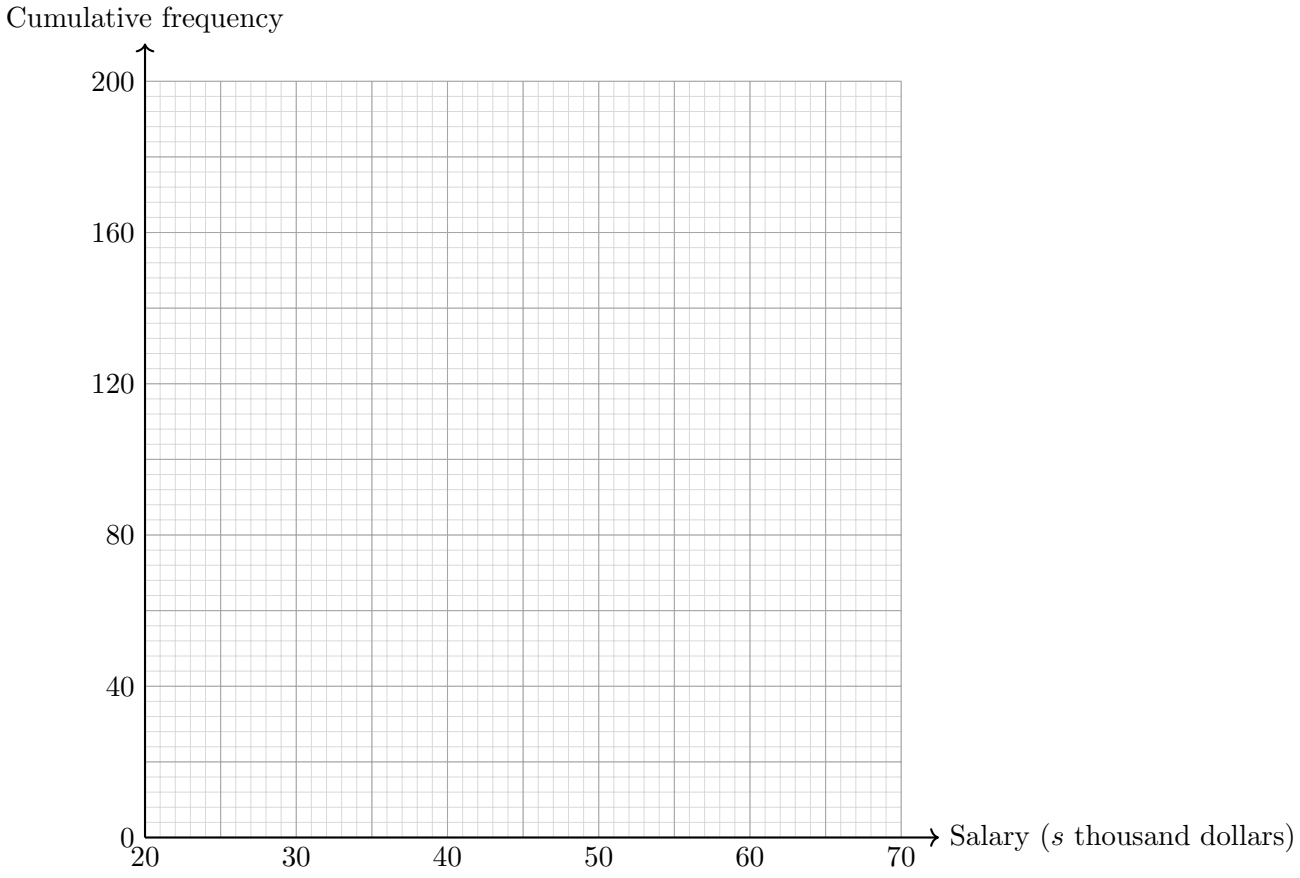
Salary (s thousand dollars)	$20 < s \leq 30$	$30 < s \leq 40$	$40 < s \leq 50$	$50 < s \leq 60$	$60 < s \leq 70$
Frequency	24	66	70	28	12

(a) Complete the cumulative frequency table.

Salary (s thousand dollars)	$s \leq 30$	$s \leq 40$	$s \leq 50$	$s \leq 60$	$s \leq 70$
Cumulative frequency	24				

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.

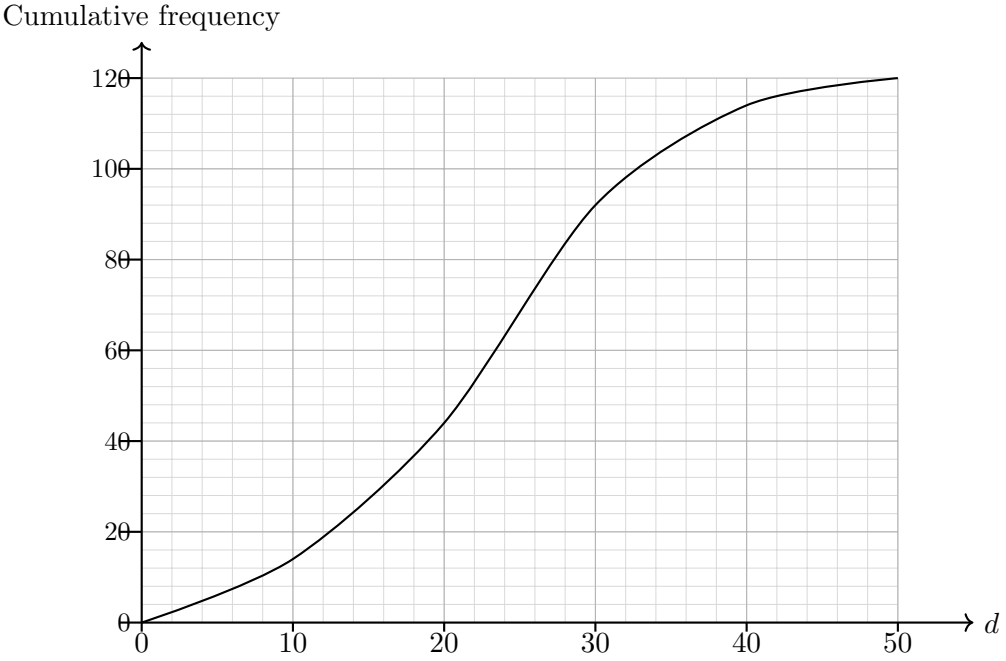


[3]

(c) Use your diagram to estimate the 70th percentile of the salaries.

..... thousand **[2]**

661 The cumulative frequency diagram shows information about the distances, d km, travelled by 120 people to work.



(a) Use the diagram to estimate the median distance.

..... km **[2]**

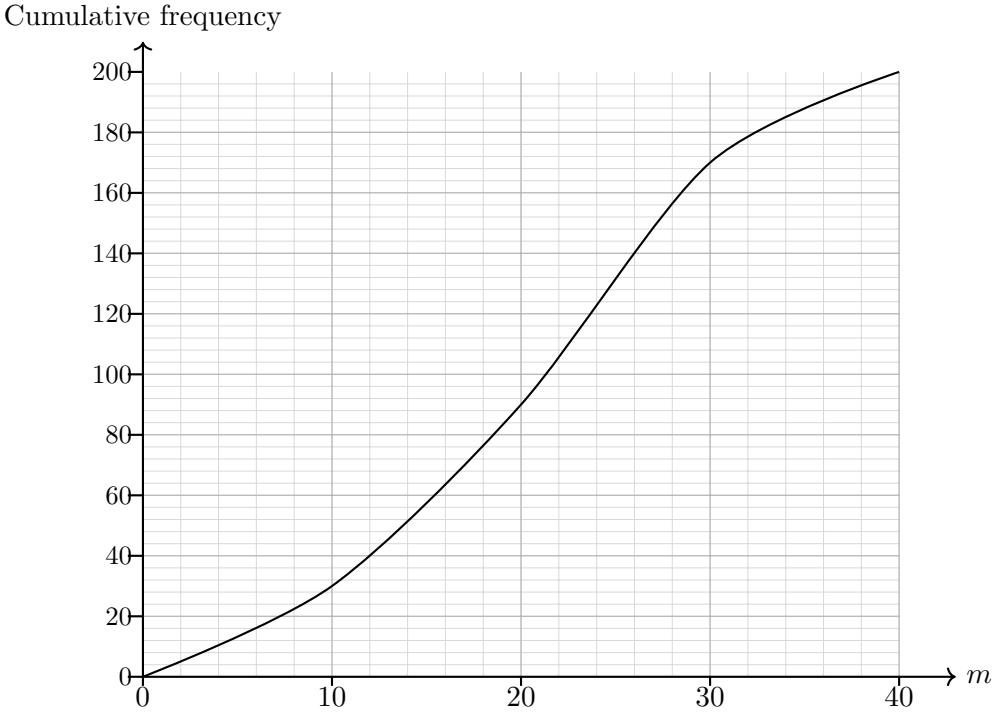
(b) Use the diagram to estimate the interquartile range.

..... km **[3]**

(c) Estimate the number of people who travelled more than 35 km to work.

..... **[3]**

662 The cumulative frequency diagram shows information about the masses, m grams, of 200 parcels.



(a) Find an estimate for the 80th percentile.

..... grams **[2]**

(b) Use the cumulative frequency diagram to complete the frequency table.

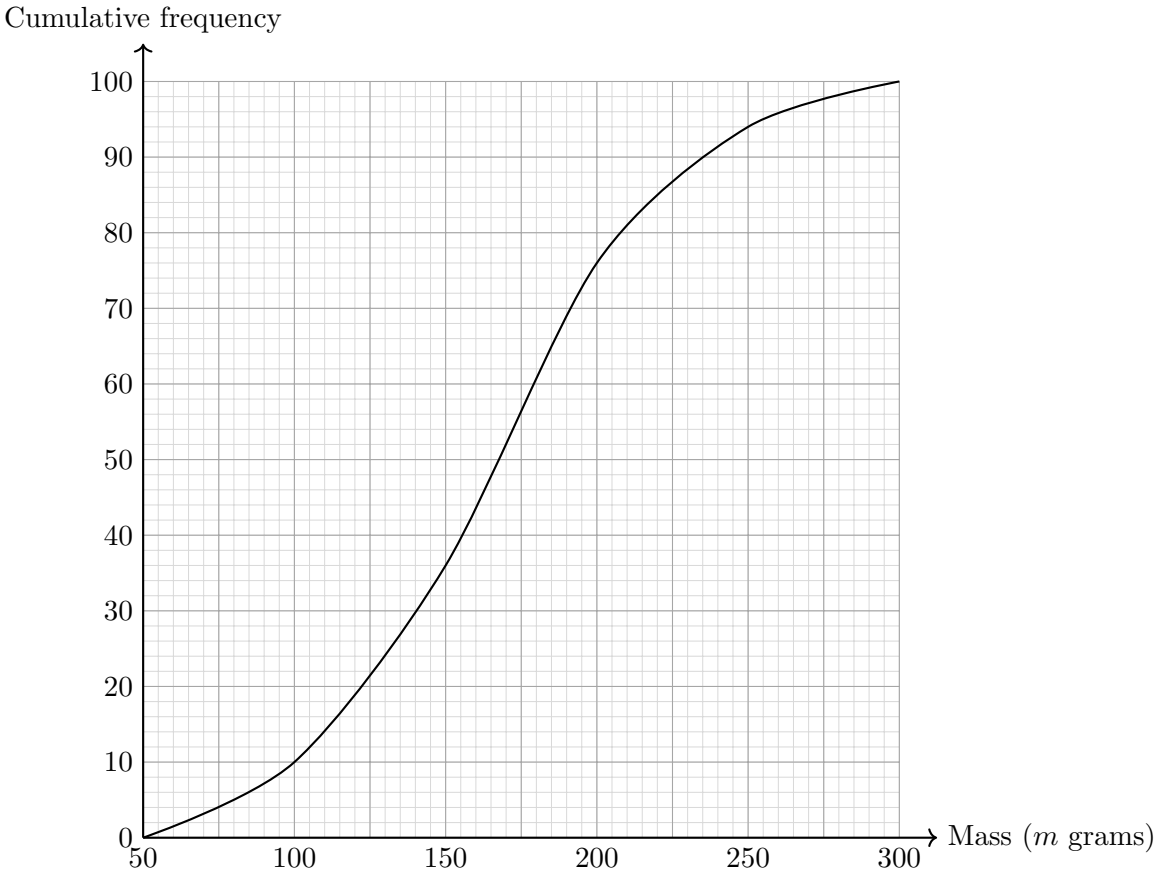
Mass (m grams)	$0 < m \leq 10$	$10 < m \leq 20$	$20 < m \leq 30$	$30 < m \leq 40$
Frequency				

[2]

(c) Calculate an estimate of the mean mass.

..... grams **[4]**

663 The cumulative frequency curve shows information about the masses, m grams, of 100 potatoes.



(a) Use the curve to estimate the number of potatoes with a mass greater than 220 grams.

..... **[3]**

(b) The curve was drawn using grouped data.
Find the values of p, q, and r.

Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	p	q	40	r	6

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

(c) Calculate an estimate of the mean mass of the potatoes.

..... g **[3]**